SDU
University of Southern Denmark

Applications to support cross-sectoral, person-centred and empowering stroke rehabilitation

Development, implementation, test, and evaluation of two co-designed applications

Marwaa, Mille Nabsen

DOI:
10.21996/g4zx-hr50

Publication date:
2024

Document version:
Final published version

Citation for pulished version (APA):
Marwaa, M. N. (2024). *Applications to support cross-sectoral, person-centred and empowering stroke rehabilitation: Development, implementation, test, and evaluation of two co-designed applications*. [Ph.D. thesis, SDU]. Syddansk Universitet. Det Sundhedsvidenskabelige Fakultet. https://doi.org/10.21996/g4zx-hr50

Go to publication entry in University of Southern Denmark's Research Portal

Terms of use
This work is brought to you by the University of Southern Denmark.
Unless otherwise specified it has been shared according to the terms for self-archiving.
If no other license is stated, these terms apply:

- You may download this work for personal use only.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying this open access version

If you believe that this document breaches copyright please contact us providing details and we will investigate your claim.
Please direct all enquiries to puresupport@bib.sdu.dk

Download date: 07. Nov. 2026

Applications to support cross-sectoral, person-centred and empowering stroke rehabilitation

Development, implementation, test, and evaluation of two co-designed applications



PhD Thesis

Mille Nabsen Marwaa

Center for Innovative Medical Technologies (CIMT), Odense University Hospital, Odense, Denmark; Department of Clinical Research, University of Southern Denmark, Odense, Denmark
Department of Physiotherapy Education, University College South Denmark, Esbjerg, Denmark.

December 2023

SDU

PhD Thesis

Applications to support cross-sectoral, person-centred and empowering stroke rehabilitation

Development, implementation, test, and evaluation of two co-designed applications

Mille Nabsen Marwaa, PhD student, Master of Science (MSc) in Health Science, Senior Lecturer, Physiotherapist, Department of Physiotherapy Education, University College Southern Denmark, Esbjerg, Denmark; Center for Innovative Medical Technologies (CIMT), Odense University Hospital, Odense, Denmark; Department of Clinical Research, University of Southern Denmark, Odense, Denmark.

E-mail: mnma@ucsyd.dk

Submission date: 29.12.2023

Supervisors

Hanne Kaae Kristensen (Main supervisor), Professor, Center for Innovative Medical Technologies, Odense University Hospital, Odense, Denmark; Department of Clinical Research, University of Southern Denmark, Odense, Denmark; Health Sciences Research Centre, UCL University College, Odense, Denmark.

Susanne Guidetti, Professor, Department of Neurobiology Care Sciences and Society, Karolinska Institutet, Huddinge, Sweden; Women's Health and Allied Health Professionals Theme, Karolinska University Hospital, Stockholm, Sweden.

Charlotte Ytterberg, Docent, Department of Neurobiology Care Sciences and Society, Karolinska Institutet, Huddinge, Sweden; Women's Health and Allied Health Professionals Theme, Karolinska University Hospital, Stockholm, Sweden.

Assessment Committee

Per Kjær (chairman), Professor, Head of Health Research at UCL University College and Professor of Musculoskeletal Research, University of South Denmark

Kirsten Schulz Petersen, associate professor, Public Health and Epidemiology Group, Faculty of Medicine and Health Technology, Aalborg University, Aalborg, Denmark.

Christina Brogårdh, Professor, Rehabilitation and Sustainable Health, Lunds University, Lund, Sweden.

Acknowledgements

Completing this PhD project has been an enriching journey that I couldn´t have undertaken alone. The opportunity to delve deeper into my field of interest, stroke rehabilitation and technologies supporting it, has been an honor for which I am profoundly grateful.

Foremost, I extend my heartfelt gratitude to my dedicated main supervisor, Hanne Kaae Kristensen. You unwavering guidance, support, and empathy have been invaluable in shaping my path as a researcher. Your ongoing encouragement has been pivotal in every step of this endeavor.

To my co-supervisors, Charlotte Ytterberg and Susanne Guidetti, I am thankful for your valuable insights and continuous encouragement throughout this PhD project.

My sincere appreciation extends to the stroke rehabilitation settings, therapists, stroke survivors, significant others, volunteer organizations and app-developers. Your active engagement and curiosity during the research phases of this PhD project have breathed life into its progress. Your collaboration has truly been the heart of this endeavor.

I extend my gratitude to *University College Southern Denmark* (UC Syd), for providing financial support, enabling the realization of this project. To my colleagues in the physiotherapy department, your curiosity and heartfelt interest in my project meant a lot. Special recognition goes to Bodil Bjørnshave Noe and Heidi Klakk, who consistently maintain connections with us PhD students, and offering unwavering support.

An appreciation to *Danske fysioterapeuter* for funding my research, and to the *University of Southern Denmark* (SDU) for grating me a scholarship, making this academic pursuit possible. To my fellow researchers at *The Person-centered Rehabilitation Research Group*, your emotional and professional support have been invaluable throughout the research journey.

Lastly, but no means least, my heartfelt gratitude and love extend to my cherished family and friends, especially my husband and my three children. Your unwavering support, positive distractions, and endless love have been my anchor, both during moments of triumph and during challenges. Your presence has made this journey possible and meaningful.

With heartfelt appreciation,
Mille Nabsen Marwaa, December 2023.

List of content

Supervisors 1
Assessment Committee 1
Acknowledgements 2
English summary 6
Dansk resumé 9
List of original articles included in this PhD thesis 12
Financial disclosure 12
List of abbreviations 12
List of Figures 13
List of Tables 13
Preface 14
Introduction 15
Chapter 1: Background 15
1.1 Stroke 15
1.2 Rehabilitation 16
1.2.1 Person-centred and empowering stroke rehabilitation 17
1.3 Challenges in current stroke rehabilitation 17
1.4 Information and communication technologies and mobile applications 18
1.4.1 *Mit Sygehus* and *Genoptræn.dk* 19
1.5 Rationale of this PhD project 19
Chapter 2: Research phases and aims 20
Phase 1: Identification/exploration 20
Phase 2: Development 21
Phase 3: Implementation, testing and process evaluation 21
Chapter 3: Research design 21
3.1 Design: The complex intervention framework 21
3.1.1 Stakeholder engagement 22
3.1.2 Elements of the MRC framework included in this PhD project 23
3.2 Theory 23
3.2.1 Philosophical and theoretical influences 24
3.2.2 Empowerment theory – a more detailed description and operationalization 24
3.3 Understanding context 27
3.3.1 Organization of Danish stroke rehabilitation 28
Chapter 4: Methods and findings from the three research phases 29
4.1 Phase 1 of the PhD project: Identification/exploration 30

4.1.1 Understanding the problem and context, by engaging stakeholders in three pre-studies 30
4.1.2 Reviewing the current evidence base by conducting a scoping review 32
4.2 Phase 2 of the PhD project: Development 33
4.2.1 Understanding context: Stroke rehabilitation in the region of Southern Denmark 34
4.2.2 Developing the intervention using Experience-Based Co-design 37
4.3 Phase 3 of this PhD project: Implementation, testing and process evaluation 44
4.3.1 Evaluation design 44
4.3.2 Methods 44
4.4 My role as a researcher 55
4.5 Ethical considerations 56
Chapter 5: Discussion 56
5.1 Summary and discussion of main findings across the articles 57
5.1.1 Use of ICT and apps to support known challenges in stroke rehabilitation 57
5.1.2. Summarizing empowering mechanisms across the three research phases 58
5.1.3 Implementation of the apps throughout the rehabilitation settings 59
5.1.4 Long-term needs and follow-up 60
5.1.5 Person-centred self-training activities supported by an app 61
5.1.6 Important mechanisms to benefit from app-supported stroke rehabilitation 64
5.2 Methodological discussion including strengths and limitations 64
5.2.1 Addressing uncertainties 65
5.2.2 Using a framework or methodology to guide the development 69
5.2.3 Choosing a relevant evaluation design: Process evaluation and program theory 69
5.2.4 Considering future implementation barriers and facilitators 73
5.2.5 Transferability 76
5.2.6 Economic considerations 76
Chapter 6: Conclusion 76
Chapter 7: Perspectives 77
7.1 Implications for practice 77
7.2 Implications and early preparations for future research 78
References 80
List of appendices 91
Appendix A: Links between epistemology, frameworks, theory, methodology and methods used in this PhD project 91
Appendix B: The empowerment model used in the EBCD process 91
Appendix C: Information letter for the development phase – first workshop 91
Appendix D: Information letter for the development phase – second workshop 91

Appendix E: Consent form to participate in the EBCD process (stroke survivors and significant others) .. 91
Appendix F: Instructions on how to download and use Mit Sygehus 91
Appendix G: Instructions on how to download and use Genoptræn.dk..... 91
Appendix H: Information letter for the test phase..... 91
Appendix I: Consent form for the test phase (significant others) 91
List of original articles..... 112
Article I: 112
Use of mobile/tablet and web-based applications to support rehabilitation after stroke: A scoping review 112
Article II..... 112
Using Experience-based Co-design to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation..... 112
Article III
Acceptability of two mobile applications to support cross-sectoral, person-centred and empowering stroke rehabilitation – a process evaluation 112

English summary

Introduction

Rehabilitation after stroke is crucial for enabling stroke survivors to develop the essential skills needed to enhance their functioning, physical activity, independence, and participation in everyday life activities. Central to achieving these goals are person-centred rehabilitation approaches that empower stroke survivors and their significant others by tailoring interventions to their individual needs and resources. Ensuring easy access to evidence-based stroke information, rehabilitation plans, person-centred interventions, and seamless, coordinated, cross-sectoral stroke rehabilitation are vital mechanisms of these approaches. Furthermore, the integration of information and communication technologies (ICTs) and applications (apps) holds promising potential in supporting person-centred stroke rehabilitation. Notably, the platforms in two known app-solutions, *Mit Sygehus* and *Genoptræn.dk*, operating within the Southern region of Denmark, have the potential to support cross-sectoral, person-centred, and empowering stroke rehabilitation.

Aim

The overall aim of this PhD thesis was to develop, implement, test and evaluate the co-designed content of two known app-solutions, *Mit Sygehus* and *Genoptræn.dk,* to support cross-sectoral, person-centred and empowering stroke rehabilitation for stroke survivors and their significant others.

Design and methods

The overall design was the British Medical Research Council's (MRC) framework for developing and evaluating complex interventions within health care.

	Aim	Methods
Phase 1 Identify/ explore	*Pre-studies:* To gain insight into stroke survivors', significant others', and therapists' perspectives on current cross-sectoral stroke rehabilitation, including challenges and possibilities of integrating ICT to support cross-sectoral, person-centred stroke rehabilitation	Qualitative individual and focus group interviews with stroke survivors (n=18), significant others (n=13) and therapists (n=9). A cross-analysis was performed to identify unmet needs and areas that needed improvement within stroke rehabilitation, also called "emotional touchpoints".

	Scoping review: The subsequent aim was to describe and review current evidence of mobile/tablet and web-based apps to support stroke rehabilitation. This included describing participants' stroke severity, and use of applications in relation to the setting and phase of the rehabilitation process	A scoping review was undertaken guided by the five-stage framework described by Arksey & O'Malley and Levac et al. Data extraction was supported by Wade' s elements of rehabilitation.
Phase 2 Development	To investigate and describe the process of using Experience-based Co-design (EBCD) to develop the content of two known mobile/tablet app solutions, *Mit Sygehus* and *Genoptræn.dk* to support cross-sectoral, person-centred, and empowering stroke rehabilitation	Participant and non-participant observations were performed in five out of six included stroke rehabilitation settings. An EBCD methodology guided the development phase, supported by the identified "emotional touchpoints".
Phase 3 Process Evaluation	To implement, test, and evaluate stroke survivors' and significant others' acceptability of the co-designed content of the two known app solutions, *Mit Sygehus* and *Genoptræn.dk* to support cross-sectoral, person-centred, and empowering stroke rehabilitation.	A program theory was developed before the test phase. Participant observations were performed during the test phase. Qualitative individual or dyadic interviews were performed with stroke survivors (n=10) and significant others (n=7) post-rehabilitation. A constructivist grounded theory analysis was performed.

Findings

Phase 1

Among the 46 studies included in the scoping review, the predominant focus was on stroke survivors in the chronic stage of recovery. Only three studies applied apps that supported more than one part of the complex rehabilitation process. This clearly underscored the necessity for developing apps that support greater part of the stroke rehabilitation process, including acute and subacute stroke rehabilitation, and the transfers between different rehabilitation settings.

Phase 2

The content of two known app-solutions, namely, *Mit Sygehus* and *Genoptræn.dk*, was co-designed based on stakeholders' needs, prioritization, and research knowledge regarding existing apps, by using a systematic Experience-based Co-design methodology.

Phase 3

The developed prototypes of the apps were implemented, tested and evaluated in 10 stroke survivors and seven significant others throughout cross-sectoral stroke rehabilitation. Stroke survivors and significant others found the content of *Mit Sygehus* and *Genoptræn.dk* acceptable to support cross-sectoral, person-centred, and empowering stroke rehabilitation. However, *Mit Sygehus* was most accepted by significant others in the acute rehabilitation setting, while *Genoptræn.dk* was most accepted by stroke survivors in the subacute rehabilitation setting. More stringent implementation of both apps was needed; *Genoptræn.dk* should be implemented during

acute stroke rehabilitation for stroke survivors being ready for self-training activities, and *Mit Sygehus* should be integrated more continuously during subacute and municipal rehabilitation.

Conclusion

The overall aim of this PhD project was to develop, implement, test, and evaluate the co-designed content of two known app-solutions, *Mit Sygehus* and *Genoptræn.dk,* to support cross-sectoral, person-centred and empowering stroke rehabilitation for stroke survivors and their significant others. The iterative and flexible framework of the MRC framework for developing and evaluating complex interventions overall guided the research phases within this PhD project.

Stroke survivors and their significant others found the content of *Mit Sygehus* and *Genoptræn.dk* acceptable to support cross-sectoral, person-centred and empowering stroke rehabilitation. The following facilitating intra -and interactional mechanisms were identified and accommodated to support experiencing empowerment: access to free-of-charge apps, access to relevant knowledge, gaining insights into and an overview of the rehabilitation process, support of coordinated and coherent transfers between rehabilitation settings, person-centred exercises and guidelines using video recordings, support from health professionals to create positive self-training habits, and reduced significant other burden. Additionally, stroke survivors and their significant others felt an ongoing sense of progress through use of the apps, which supported motivation and engagement throughout the rehabilitation process.

However, *Mit Sygehus* was most accepted by significant others in the acute rehabilitation setting, while *Genoptræn.dk* was most accepted by stroke survivors in the subacute rehabilitation setting. More stringent implementation of both apps was needed; *Genoptræn.dk* should be implemented during acute stroke rehabilitation for stroke survivors being ready for self-training activities, and *Mit Sygehus* should be integrated more continuously during subacute and municipal rehabilitation. Furthermore, using the apps to support cognitive rehabilitation and follow-up support post-rehabilitation needs further exploration.

Dansk resumé

Introduktion

Rehabilitering efter stroke er afgørende for at stroke overleverne kan udvikle nødvendige færdigheder for at forbedre deres funktion, fysiske aktivitet, uafhængighed og deltagelse i hverdagslivet. Person-centret og empowering tilgange er væsentlige for at opnå disse mål. Sikring af let adgang til evidensbaseret viden om stroke, rehabiliteringsplanen, person-centrerede interventioner, og sammenhængende rehabilitering på tværs af rehabiliteringssektorer er nødvendige mekanismer i disse tilgange. Integrering af informations- og kommunikations teknologier (ICT) og applikationer (apps) viser lovende potentiale til at støtte person-centreret stroke rehabilitering. Platformene af to kendte app-løsninger i region Syddanmark, *Mit Sygehus* and *Genoptræn.dk*, har formentlig potentialet til at understøtte tværsektoriel, person-centreret, og empowering stroke rehabilitering.

Formål

Det overordnede mål med dette PhD projekt var at udvikle, implementere, teste og evaluere det co-designede indhold i to kendte app-løsninger; *Mit Sygehus* og *Genoptræn.dk,* for at støtte tværsektoriel, person-centreret og empowering stroke rehabilitering for stroke overlevere og deres pårørende.

Design og metoder

Det overordnede design var British Medical Council's (MRC) ramme for udvikling og evaluering af komplekse interventioner indenfor sundhedsvæsenet.

	Formål	Metoder
Fase 1 Undersøgelse	*For-studier:* At undersøge stroke overleveres, deres pårørendes, og terapeuters oplevelser af nuværende stroke rehabilitering, herunder udfordringer og potentialer for at inddrage ICT til at støtte tværsektoriel, person-centreret stroke rehabilitering	Kvalitative individuelle og fokusgruppe interviews blev gennemført med stroke overlevere (n=18), deres pårørende (n=13) og terapeuter (n=9). En tværgående analyse blev udført med henblik på at identificere umødte behov og områder der kan forbedres indenfor stroke rehabilitering, også kaldet "emotional touchpoints".
	Scoping review: Dernæst, at beskrive og undersøge nuværende evidens indenfor mobil/tablet og web-baserede apps til at støtte stroke rehabilitering. Herunder, at beskrive deltagernes grad af stroke, brugen af apps i relation til rehabiliterings-setting og fase.	Et Scoping review blev blev udført, som var guided af fem skridt, beskrevet af Arksey & O'Malley and Levac et al. Data blev kategoriseret med støtte fra Wade' s rehabiliterings elementer.

Fase 2 Udvikling	At undersøge og beskrive Experience-based Co-design (EBCD) processen, til at udvikle indholdet i to kendte mobil/tablet app-løsninger, *Mit Sygehus* and *Genoptræn.dk,* til at støtte tværsektoriel, person-centreret og empowering stroke rehabilitering.	Deltager- og ikke-deltager observationer blev udført i fem ud af seks inkluderede stroke rehabiliterings-settings. EBCD-metodologien guidede udviklingsfasen, støttet op af de identificerede "emotional touchpoints".
Fase 3 Proces-evaluering	At implementere, teste og evaluere stroke overleveres og pårørendes accept af det co-designede indhold i de to kendte app-løsninger, *Mit Sygehus* and *Genoptræn.dk,* til at støtte tværsektoriel, person-centreret og empowering stroke rehabilitering.	En programteori blev udviklet inden testfasen. Deltager observationer blev udført under testfasen. Ved rehabiliterings ophør blev der gennemført kvalitative individuelle- og dyade interviews med stroke overlevere (n=10) og deres pårørende (n=7) En konstruktivist grounded theory analyse blev anvendt.

Resultater

Fase 1

Blandt de 46 inkluderede studier i dette scoping review, var det primære fokus på deltagere med kronisk stroke. Kun tre studier anvendte apps til at støtte mere end et enkelt element af den komplekse tværsektorielle stroke rehabilitation. Dette understreger nødvendigheden af at udvikle apps der omfavner større dele af den tværsektorielle stroke rehabilitering, herunder akut og subakut faserne, samt overgangene mellem rehabiliteringsfaserne.

Fase 2

Med udgangspunkt i metodologien Experience-based Co-design, blev indholdet i to kendte app-løsninger, *Mit Sygehus* og *Genoptræn.dk*, prioriteret og udviklet, med udgangspunkt i stroke overleveres, deres pårørendes og terapeuters behov og ønsker, samt eksisterende litteratur omkring apps.

Fase 3

De udviklede prototyper af de to apps blev implementeret, testet og evalueret i tværsektoriel stroke rehabilitering. 10 stroke overlevere og 7 pårørende deltog. Stroke-overleverne og deres pårørende evaluerede *Mit Sygehus* og *Genoptræn.dk* acceptabel til at støtte tværsektoriel, person-centreret og empowering stroke rehabilitering. Dog var *Mit Sygehus* mest acceptabel blandt pårørende under den akutte rehabilitering, mens *Genoptræn.dk* var mest acceptabel blandt stroke overlevende under sub-akut rehabilitering. Der er behov for mere stringent implementering af begge apps; *Genoptræn.dk* bør implementeres allerede under akut stroke rehabilitering for stroke overlevere, som er klar til at påbegynde selvtrænings aktiviteter, og *Mit Sygehus* bør integreres mere kontinuerligt gennem rehabiliteringsfaserne.

Konklusion

Det overordnede formål med dette PhD projekt, var at udvikle, teste og evaluere det co-designede indhold i to kendte apps, *Mit Sygehus* and *Genoptræn.dk,* til at støtte tværsektoriel, person-centreret

og empowering stroke rehabilitering for stroke overlevere og deres pårørende. MRC´s fleksible ramme for udvikling og evaluering af komplekse interventioner guidede forskningsprocesserne i dette projekt.

Stroke overleverne og deres pårørende evaluerede *Mit Sygehus* and *Genoptræn.dk* til at være acceptabel til at støtte tværsektoriel, person-centreret og empowering stroke rehabilitering. Følgende faciliterende intra- og interaktionelle mekanismer for oplevelsen af empowerment blev identificeret og imødekommet: adgang til gratis apps, let adgang til relevant viden om stroke, overblik over rehabiliteringsproces og plan, sammenhængende tværsektoriel stroke rehabilitering, person-centrerede interventioner understøttet af videooptagelser, støtte fra terapeuterne til at opnå positive vaner i forbindelse med selvtræning, og reduceret pårørende byrde. Yderligere viste resultaterne, at stroke overleverne og deres pårørende blev støttet i at opleve fremgang ved brugen af disse apps, hvilket støttede deres motivation, engagement, og deltagelse i hverdagslivet.

Dog var *Mit Sygehus* mest acceptabel blandt pårørende under akut stroke rehabilitering, mens *Genoptræn.dk* var mest acceptabel blandt stroke overlevere under sub-akut rehabilitering

Der er behov for mere stringent implementering af begge apps; *Genoptræn.dk* bør implementeres allerede under akut stroke rehabilitering for stroke overlevere som er klar til at påbegynde selvtræning aktiviteter, og *Mit Sygehus* bør integreres mere kontinuerligt gennem rehabiliteringsfaserne.

Inddragelse af apps til at støtte kognitiv rehabilitering og follow-up støtte efter rehabiliteringsophør, bør yderligere undersøges.

List of original articles included in this PhD thesis

This thesis is based on the following three articles:

1. Marwaa NM, Guidetti S, Ytterberg C, Kristensen HK: Use of mobile/tablet and web-based applications to support rehabilitation after stroke: A scoping review.
   Published in Journal of Rehabilitation Medicine. DOI: 10.2340/jrm.v54.452. January 2022.
   Open access.

2. Marwaa NM, Guidetti S, Ytterberg C, Kristensen HK: Using Experience-based Co-design to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation.
   Published in Research Involvement and Engagement at Springer Nature.
   DOI: 10.1186/s40900-023-00472-z. August 2023.
   Open access.

3. Marwaa NM, Guidetti S, Ytterberg C, Kristensen HK: Acceptability of two mobile applications to support a cross-sectoral, person-centred stroke rehabilitation – a process evaluation.
   Accepted for publication in Annals of Medicine, December 2023.
   Taylor and Francis have permitted the inclusion of the Accepted Manuscript in this thesis.

Financial disclosure

This PhD project has been funded by *University College South Denmark* (UC Syd), *University of Southern Denmark* (SDU) and *The Association of Danish Physiotherapists* (Danske Fysioterapeuter).

List of abbreviations

PCR: Person-centred rehabilitation

ICT: Information and communication technology

Apps: Applications

MRC: Medical Research Council's framework

OT: Occupational therapist

PT: Physiotherapist

SSS: Scandinavian Stroke Scale

NIHSS: National Institutes of Health Stroke Scale

List of Figures

Figure 1: Overview of the three research phases in this PhD project

Figure 2: PCR mechanisms that may support experiencing empowerment

Figure 3: The Danish stroke rehabilitation trajectory

Figure 4: Rehabilitation settings where observations were made

Figure 5: The content and interface of Mit Sygehus

Figure 6: The content and interface of Genoptræn.dk

Figure 7: The final logic model of the PhD project – inputs, activities, outputs, and outcomes

Figure 8: Included rehabilitation settings and stroke survivors

Figure 9: Providing exercises in *Genoptræn.dk* with numbers to create easier overview

Figure 10: Main and sub-categories of stroke survivors' and significant others' acceptability of *Mit Sygehus* and *Genoptræn.dk*

Figure 11: Analytical steps used in the research phases

List of Tables

Table 1: A list of elements of the MRC framework applied in this PhD project

Table 2: Participant characteristics of the three pre-studies

Table 3: Observational guide used to understand context, use of ICT and apps, and empowerment

Table 4: Overview of purposes and activities in the EBCD process

Table 5: Participant characteristics and participation in the EBCD process

Table 6: Research questions – on participants' acceptability of *Mit Sygehus* and *Genoptræn.dk*

Table 7: Stroke survivor and significant other characteristics

Table 8: Acceptability across the rehabilitation process

Table 9: Summarizing mechanisms for experiencing empowerment across studies

Preface

Working as a stroke rehabilitation physiotherapist, mostly in an acute rehabilitation setting, for about eight years, and lecturing physiotherapy students since 2015 at the University College Southern Denmark (UC Syd), my focus has increasingly been captivated by the challenge of fostering cross-sectoral stroke rehabilitation. Also, the last decade rehabilitation duration has decreased, especially the acute rehabilitation is very short and stroke survivors, and their significant others have only limited number of months to undergo rehabilitation and adapt to their altered lives. In my 2015 master's degree I explored how information and communication technologies (ICT) could aid stroke rehabilitation from stroke survivors' perspectives. This interest led me to investigate how significant others and therapists perceive cross-sectoral stroke rehabilitation and the potential for integrating ICT. Findings revealed a shared interest among stakeholders in using ICT to support stroke rehabilitation, focusing on evidence-based knowledge, coherence across rehabilitation settings, and tailored interventions. Also, the fact of increased use of ICT and apps within the public, including stroke survivors, underscored the importance of further investigating how ICT and apps can serve as a support mechanism for cross-sectoral person-centred stroke rehabilitation and formed the basis of my PhD project.

My time as a PhD student has been rewarding, inspiring and has enhanced my role as a lecturer and researcher.

Introduction

The overall aim of this thesis was to develop, implement, test and evaluate the co-designed content of two known application (app) platforms, *Mit Sygehus* and *Genoptræn.dk* to support cross-sectoral, person-centred and empowering stroke rehabilitation for stroke survivors and their significant others. Medical Research Councils (MRC) flexible and pragmatic framework for developing and evaluating complex interventions (1,2) guided the research process. First, unmet needs regarding stroke rehabilitation from the perspectives of stroke survivors, their significant others and therapists employed in different stroke rehabilitation settings were identified in three pre-studies. This knowledge was supplemented with a scoping review on existing apps to support stroke rehabilitation. Then, the content of two known app-solutions was co-designed based on stakeholders' priorities and tested in cross-sectoral stroke rehabilitation. Finally, a process evaluation of the acceptability of the content of the two apps to support cross-sectoral, person-centred and empowering stroke rehabilitation was undertaken (1,2). Stroke survivors, their significant others, therapists from various rehabilitation settings, volunteer organizations, app-developers and researchers were engaged throughout the research process. The purpose of stakeholder engagement was to facilitate and ensure the development of more acceptable apps, which support their implementation and uptake in different stroke rehabilitation settings (1,3,4).

Chapter 1: Background

In this section, updated knowledge on stroke and current stroke rehabilitation is presented, comprising the following topics: stroke, rehabilitation, person-centred and empowering stroke rehabilitation, challenges in current stroke rehabilitation, information and communication technologies (ICT) and apps to support cross-sectoral stroke rehabilitation, which finally leads to the rationale for this PhD project.

1.1 Stroke

Stroke is recognized as a major cause of disability worldwide (5). In Denmark about 90.000 people are affected by stroke and mostly people over 65 years (70-90%) (6) and more than 12,000 people suffer a stroke every year in Denmark (6,7). As a consequence of improved medical care, with decreasing mortality rates and an ageing population, the number of people living with the consequences of stroke is likely to increase (8–10). Stroke occurs when blood flow to the brain is disrupted; infarct (85-90%%) or hemorrhage (10-15%) with clinical symptoms lasting more than 24 hours and resulting in damage to the brain cells (11). The consequences of stroke are complex and

may include physical and sensory deficits such as paresis (75-85%), cognitive deficits (10-50%) and aphasia (20-25%), as well as social, and emotional deficits; which require rehabilitation to improve functioning and increase participation in everyday life (i.e., activities, routines that takes place in the persons' everyday life context) (6,9,12–16).

1.2 Rehabilitation

Rehabilitation after stroke is crucial to improve quality of life for the stroke survivor and their significant other. Rehabilitation supports the patient in acquiring the skills needed to improve functioning, physical activity, independence, and participation in everyday life (17) as well as optimizing their ability to adapt and respond to changing circumstances (14,18–20). Further, rehabilitation may decrease health care needs (14,18).

Although most stroke survivors' recovery takes place largely during the first 3-6 months, the impact of the stroke and the need for rehabilitation may continue for a long time (8,21). Stroke leaves approximately one-third of survivors with long-term disabilities. Thus, stroke can also affect the stroke survivors' significant others, that is spouses, family and close friends, who might experience physical, emotional and social strain and require rehabilitation themselves (13,22).

It is recommended that stroke rehabilitation should take place in specialized stroke units comprising an interdisciplinary team of healthcare professionals, including physicians, nurses, physiotherapists (PTs), occupational therapists (OTs), speech therapists and psychologists (14,17,19,23–25). According to Derrick Wade (19) rehabilitation is a cyclic process defined as "*an educational, problem-solving process focusing on activity limitations and aims to optimize the patients social participation and well-being, and to reduce stress on family"*. Wade, as well as national and international rehabilitation guidelines, further recommend that rehabilitation should start as early as possible (with regard to the needs, abilities and resources of patients' and their significant others'), be coherent (i.e. coordinated, with smooth transitions between different rehabilitation settings) (14,18,19,25,26) and comprise the following elements: 1) assessment of limitations faced by the stroke survivor, 2) goal-setting and ongoing evaluation, 3) intervention, 4) support of social participation, and 5) discharge support (18,19,25,27,28).

The main goals of stroke survivors and their significant others for the rehabilitation process are often to experience empowerment and be able to participate in meaningful everyday life activities (14,25). Person-centred rehabilitation (PCR) and empowering approaches may be essential to achieve these goals (14,25,29) and are further elaborated on in the following section.

1.2.1 Person-centred and empowering stroke rehabilitation

In line with Wade's suggestions (19), research and policies on rehabilitation have increasingly emphasized coherent (i.e., cross-sectoral), person-centred and empowerment approaches within health care, vital for independent living and high quality of life (25,29–37). Empowerment achieved during acute and subacute stroke rehabilitation is also posited to persist as rehabilitation transitions to the municipality/home (38).

Given the complexity of stroke, a "one-size-fits-all" "approach to rehabilitation is insufficient (14,29,39). PCR emerges as an approach that empowers stroke survivors and their significant others, shifting from "for" to "with" (14,29,40). Such empowerment relies on holistic perspectives, tailoring support to individual needs and preferences (34,41,42). Person-centred and empowerment approaches are closely intertwined, with the former seen as a precondition for the latter within healthcare and rehabilitation research (19,28,29,31–35,37,41–43).

However, research on person-centred and empowerment approaches, including operationalization and implementation, remains underdeveloped in health rehabilitation (29,43,44). This may partly be due to unclear definitions and synonyms (17,29,31,33,35,45) exemplified by the presence of various synonyms for person-centred approaches, such as client-centred, user-centred and individual focus (46). Additionally, there are many suggested outcomes of rehabilitation, such as *coping*, *autonomy*, *recovery* and/or *empowerment* (17,45).

1.2.1.1 The understanding of PCR and empowerment within this thesis

In this thesis, PCR and empowerment approaches are understood as multi-dimensional holistic approaches that, through co-construction, and based on the participants' needs, strengths, abilities, resources, preferences and goals for the contextually grounded rehabilitation process, support stroke survivors and their significant others in developing skills and capacities to participate in meaningful everyday life activities (18,23,29,33–35,42,47–52). Furthermore, PCR is considered a precondition to facilitate empowerment for the stroke survivor and their significant others (19,28,29,31–33,41–43). For a more detailed description and operationalization of empowerment, see theory section 3.2.2.

1.3 Challenges in current stroke rehabilitation

Stroke survivors and their significant others comprise a heterogeneous group and may have various needs for accomplishing their rehabilitation goals (14,17). Thus, stroke rehabilitation is complex, and an uncoordinated, isolated, and fragmented rehabilitation process (between rehabilitation services and sectors) from the perspectives of stroke survivors and their significant others is well-recognized as a general challenge in stroke rehabilitation. Furthermore, challenges such as a lack of

sufficient knowledge, coordinated communication across rehabilitation sectors, insight into and overview of the rehabilitation process, person-centred approaches, support for significant others, shared decision making and goal-setting hinder optimal stroke rehabilitation (12–14,26,39,53–55).

On the other hand, person-centred approaches, such as providing sufficient and easy access to evidence-based information on stroke, the rehabilitation plan (including short and long-term goals), ways to continue rehabilitation after discharge, and knowledge on returning to everyday life (e.g., work, social activities, driving) improves stroke survivor and significant other involvement, satisfaction and empowerment (56–58). However, to facilitate acting on any information, knowledge and educational material provided, the stroke survivors' and their significant others' health competencies must be supported (59). New technologies, co-designed with relevant stakeholders and rehabilitation settings, to address these challenges and facilitate cross-sectoral, person-centred, and empowering stroke rehabilitation are in demand (33,60,61) in the Danish stroke rehabilitation context.

1.4 Information and communication technologies and mobile applications

Information and communication technologies (ICT), such as mobile phones, computers and tablets are becoming more frequently owned and used by the public and stroke survivors (62,63). ICTs and apps are increasingly integrated into stroke rehabilitation (10), and studies have shown that stroke survivors, their significant others and therapists are interested in implementing technology-based solutions to support stroke rehabilitation (12,13,55,64–69). Implementing ICTs and apps to support stroke rehabilitation has shown the potential to support PCR by accommodating various unmet needs identified in current stroke rehabilitation. ICTs and apps can improve the level and timing of information provided to stroke survivors and their significant others (12,13,23,57,70–72), accessibility to rehabilitation (68,73), travel time and costs (64), participation in and adherence to therapeutic activities, the independent performance of exercises post-discharge (12,13,74), continuity and coherence (23,55,75,76), follow-up after discharge, and the extension of the rehabilitation process (12,13,23,66,75,77,78). Consequently, ICTs and apps can support PCR and empowerment by facilitating participation in everyday life (12,13,23,39,55,77,79–83) and social activities (74), whereby motivation, engagement, quality of life (49,85) and empowerment are increased (70,71,86–90).

Implementation of ICTs and apps is seen as both a local and national vision and a strategy to achieve equality, optimize the quality of health care, and enhance independence and participation in everyday life (14,17,60,91). However, ICTs and apps to support known challenges in stroke rehabilitation are still lacking (21,64,88,90), since many existing solutions focus mainly on isolated

aspects of stroke rehabilitation (92–94). A lack of stakeholder engagement in the development and testing phases, training on the use of the solutions, and person-centred solutions (23,83,95,96) is also known barriers to the acceptability and uptake of the solutions (26,92,93,97). In Denmark there are no overarching ICTs/apps that support the entire rehabilitation process and that accommodates the need for coherent and person-centred initiatives (94). However, two known app-platforms in the region of Southern Denmark, namely *Mit Sygehus* and *Genoptræn.dk*, have the potential to address many of the beforementioned challenges and needs. These apps are briefly described below.

1.4.1 *Mit Sygehus* and *Genoptræn.dk*

Mit Sygehus has been used in over 300 diagnoses in several hospital departments (e.g. gynaecology, oncology, orthopeadics, surgery, endocrinology, rheumatology) in the Southern region of Denmark to provide information relevant to the specific diagnosis (98). The existing modules and headlines in *Mit Sygehus* have shown the potential to accommodate many of the above-mentioned unmet needs of stroke survivors, their significant others and therapists in relation to the stroke rehabilitation process; however, the content of *Mit Sygehus* does not support stroke rehabilitation yet. The *Mit Sygehus* platform was made available for use within this PhD project; however, the headlines and content required development.

Genoptræn.dk is a self-training app currently used in both hospital and municipal rehabilitation settings in the Southern region of Denmark (99). *Genoptræn.dk* currently only comprises generic video-recorded exercises and has not yet been designed, developed and tested to support person-centred stroke rehabilitation. *Genoptræn.dk* was likewise made available for use within this PhD project.

1.5 Rationale of this PhD project

PCR and empowerment approaches based on the needs, goals and resources of stroke survivors and their significant others may support cross-sectoral, person-centred and empowering stroke rehabilitation. However, research on the conceptualization, implementation and evaluation of cross-sectoral, person-centred and empowering approaches is lacking.

While ICTs and apps have demonstrated potential in supporting stroke rehabilitation, currently no ICTs or apps that comprehensively support cross-sectoral, person-centred and empowering stroke rehabilitation exist. Additionally, existing ICT and app solutions lack stakeholder engagement in their development.

Mit Sygehus and *Genoptræn.dk* may have the potential to support cross-sectoral, person-centred and empowering stroke rehabilitation. However, to facilitate and increase the acceptability of the content of the apps, stakeholder engagement is essential.

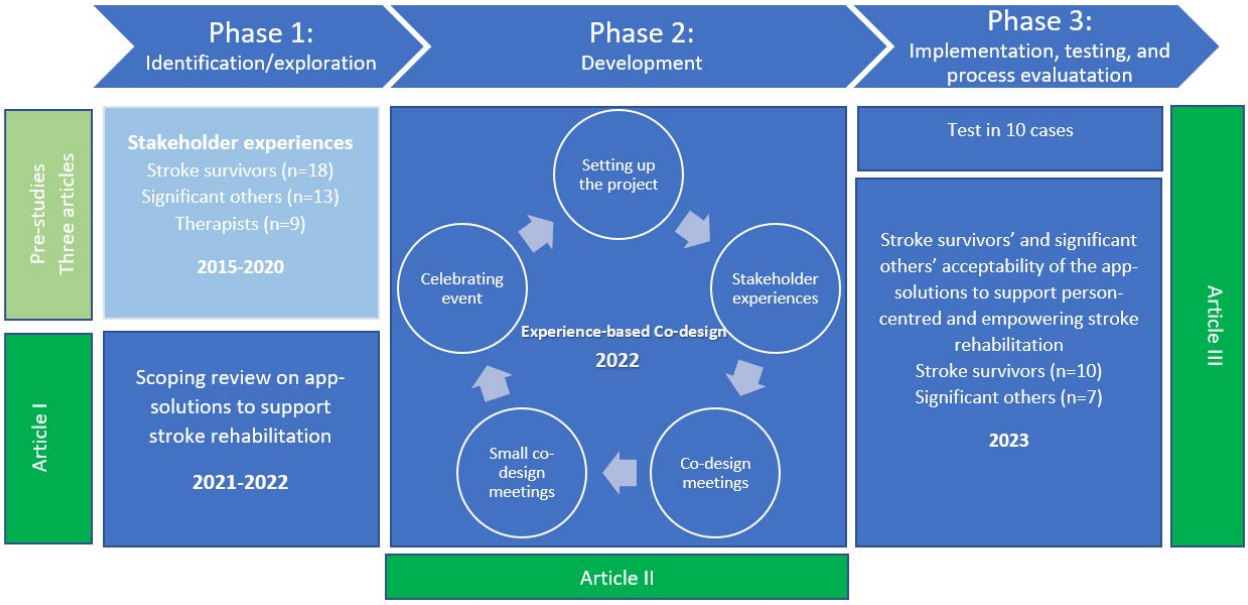
In light of these considerations, the subsequent chapter outlines the overall and specific aims of the research phases within this PhD project.

Chapter 2: Research phases and aims

As mentioned above, PCR and empowering approaches are recommended to support stroke rehabilitation, and ICTs and apps show potential in this regard. Therefore, the overall aim of this PhD project was to develop, implement, test and evaluate the co-designed content of two known app-solutions, *Mit Sygehus* and *Genoptræn.dk,* to support cross-sectoral, person-centred and empowering stroke rehabilitation for stroke survivors and their significant others.

When developing new complex interventions, the current issues in stroke rehabilitation need to be identified and explored as a point of departure. The interventions should then be co-designed with relevant stakeholders to ensure their acceptability (1). Therefore, as recommended by the MRC framework (1,2), this PhD project was divided into three overall research phases with separate aims, namely 1) identification/exploration, 2) development, and 3) implementation, testing and process evaluation, with three corresponding articles (Articles I, II and III). See Figure 1.

Figure 1: Overview of the three research phases in this PhD project



Phase 1: Identification/exploration

As shown in Figure 1, this phase had two steps, namely exploring stakeholder experiences, and reviewing current evidence base.

Initially, before undertaking this PhD project, three pre-studies were conducted and reported in three articles (12,13,55). The aim of the pre-studies was to gain insight into the perspectives of

stroke survivors, their significant others and therapists on current cross-sectoral stroke rehabilitation, including challenges in and possibilities of integrating ICTs into the process to support cross-sectoral, person-centred stroke rehabilitation.

Subsequently, a scoping review was undertaken with an aim to describe and review current evidence base on mobile/tablet and web-based apps to support stroke rehabilitation. This included describing participants' stroke severity, and use of apps in relation to the setting and phase of the rehabilitation process (Article I).

Phase 1 is to be seen as a (non-linear) point of departure for the development phase, wherein the identified unmet needs, along with knowledge on existing apps, inform the content development of the two apps to support cross-sectoral, person-centred, and empowering stroke rehabilitation.

Phase 2: Development

Here, the aim was to investigate and describe the process of using Experience-based Co-design (EBCD) to develop the content of two known mobile/tablet app-solutions, *Mit Sygehus* and *Genoptræn.dk* to support cross-sectoral, person-centred and empowering stroke rehabilitation (Article II). Based on the co-design phase, the prototypes of the two apps were to be implemented, tested and evaluated in cross-sectoral stroke rehabilitation.

Phase 3: Implementation, testing and process evaluation

Here, the aim was to implement, test and evaluate the acceptability of the co-designed content of the two apps *Mit Sygehus* and *Genoptræn.dk* to support cross-sectoral, person-centred and empowering stroke rehabilitation. Article III had focused on stroke survivors' and their significant others' acceptability of the intervention. Therapists' acceptability of the intervention will be reported in a later manuscript.

Chapter 3: Research design

This chapter describes the overall research design and theoretical considerations within this thesis.

3.1 Design: The complex intervention framework

The complexity of this PhD project lies especially in the multiple stroke rehabilitation settings in which the apps are to be implemented, the involvement of multiple therapists, the diversity of stroke survivors and their significant others (e.g., deficits, motivation, resources, skills), and the multiple interdependent and non-linear components and interventions over time in which cause-effect

relationships are difficult to establish (18). Furthermore, research on stroke rehabilitation is generally considered to be complex (100). Thus, the flexibility and tailoring of the developed intervention or its components are recommended to fit specific contexts (1,18,29).
Therefore, to increase the value of the research and minimize 'research waste', this PhD project was guided by the flexible and pragmatic framework of the MRC for developing and evaluating complex interventions in healthcare (1,2,101). The framework consists of four phases: 1) development, 2) testing, 3) evaluation, and 4) implementation (1). Based on these phases, this PhD project was divided into the three overall research phases shown in Figure 1: 1) identification/exploration, 2) development, and 3) implementation, testing and process evaluation. Initial implementation took place during the testing phase to support decisions on advancing to larger-scale implementation (102).
The MRC framework suggests that the following non-linear elements be considered when developing and evaluating complex interventions; 1) *understanding the problem*, 2) *reviewing the current evidence base*, 3) *understanding context* (including uncertainties), 4) *drawing on existing theories*, 5) *developing a program theory*, 6) *developing the intervention*, 7) *refining the intervention*, 8) *generating data*, 9) *designing the evaluation,* 10) *considering future implementation*, and 11) *economic elements* (1). Stakeholder engagement and reporting are also recommended throughout the research project (1,103). Stakeholder engagement is briefly elaborated on below.

3.1.1 Stakeholder engagement

The engagement of stakeholders (i.e. those who are targeted by the intervention, involved in its development or delivery, or other persons or professionals who have a stake in the topic) (1) is recommended according to national health policy and legislation and is expected to improve healthcare services. It secures the democratic representation and empowerment of disadvantaged groups and presumably increases the quality and integration of research in clinical practice.
A lack of stakeholder engagement, including prioritization of research focus, and a lack of person-centeredness are major reasons for 'research waste' (1,102,104). Thus, the purpose of stakeholder engagement is to ask relevant research questions about relevant issues and identify relevant knowledge gaps in current stroke rehabilitation *with* the stakeholders rather than *for* them. Active engagement may also support experiencing empowerment (1,2,105,106).

3.1.2 Elements of the MRC framework included in this PhD project

The non-linear and pragmatic elements suggested by the MRC framework above, apart from economic elements, are included in this design chapter and/or the next chapter, describing the methods and findings of the three research phases, and/or in the discussion chapter of this thesis (see Table 1). The research methods used, as well as the participants, settings, and findings, are elaborated on in the context of each research phase of this PhD project in the next chapter.

Table 1: A list of elements of the MRC framework applied in this PhD project

Elements in the MRC framework / **The PhD project**	**Design**	**Phase 1**	**Phase 2**	**Phase 3**	**Discussion**
Stakeholder engagement	x	x	x		x
Understanding the problem		x			x
Reviewing the current evidence base		x			x
Understanding context (including uncertainties)		x	x	x	x
Drawing on existing theories	x				x
Developing a program theory				x	x
Developing the intervention			x		x
Refining the intervention				x	x
Generating data		x	x	x	x
Designing the evaluation				x	x
Considering future implementation					x

3.2 Theory

The MRC element of drawing on existing theory is described in this section. Theory is included throughout the research process to ensure that the developed interventions are grounded in a solid understanding of the problem by building on existing research knowledge and stakeholders' views (1). Further, theory guides the development phase by identifying and predicting the underlying mechanisms (facilitators/barriers) for implementation and for achieving the desired outcomes. This section describes the overall philosophical and theoretical influence in this thesis. Furthermore, the multidimensional and complex empowerment theory and its operationalization is described. In the description of Phase 2, further theory is included, for example, how the content of the apps is prioritized and developed based on prior stakeholder knowledge and known empowering mechanisms. As part of Phase 3, a program theory is presented that supports the entire research process, including the evaluation.

3.2.1 Philosophical and theoretical influences

Constructivist Grounded Theory (GT) guided the phases of this project. Constructivist GT epistemological builds on constructivism. A constructivist philosophy emphasizes that knowledge and meaning, rather than being discovered, are actively constructed through experiences, interactions and reflections that are contextually grounded. From this perspective, the researcher is a co-constructor who brings assumptions and prior (research) knowledge with them, that can neither be deleted/ignored, nor should control the construction of knowledge. Rather, prior knowledge and assumptions should be transparent, and their influence on knowledge construction should be reflected on. Charmaz describes *sensitizing concepts* as giving the researcher initial ideas to pursue and questions to raise about the topic of interest. These *sensitizing concepts* should be seen as a starting point for studying the empirical world while retaining an openness for exploring it. In Constructivist GT, research questions and analysis are iterative, influencing the continuous knowledge construction, where new questions are added to the interview guide/observational guide continuously throughout data construction. Additionally, ongoing knowledge and meaning construction may also subsequently support the development of more acceptable interventions. The analytical steps in constructivist GT analysis are; 1 transcribing and reading, 2) open coding, where data is studied closely line-by-line or sentence-by-sentence, 3) focused coding, where data is sorted and synthesized, 4) theoretical coding (constant comparation), which is a dynamic process that involves a constant back-and-forth between codes, data and theory within and across the data material to enable a more in-depth and contextually grounded understanding, 5) theoretical sampling to make the constructed themes more robust and, 6) theoretical sensitivity, where the researcher stays open to and involves theoretical perspectives that might complement the data. Also, memo-writing is performed throughout the construction and analysis of data. Constructivist GT analysis should be seen as flexible steps and guidelines rather than strict rules (107,108).

3.2.2 Empowerment theory – a more detailed description and operationalization

Given the complexity of the concept of empowerment (109), an elaborate description of the close (interrelated) connections of PCR and empowerment and operationalization of the concept, including the PCR mechanisms that support the experience of empowerment, are given in this section to support a shared understanding (1).

3.2.2.1 Overall understanding of empowerment

Empowerment-based interventions are well-recognized to encompass both process and outcome elements (31,34,48,109,110). Both PCR and empowering approaches signify continuous collaboration between health professionals, the stroke survivor and the significant other, ensuring

that the rehabilitation plan evolves in response to individual needs, goals and resources through *co-construction*, shared power, responsibility, and ownership (14,17,18,26,26,28,31,42,46,52,56,100). Creating shared power relations between the stroke survivor, the significant other and the health professional is considered a human rights approach (32,111,112). Power distribution is not seen as one gaining power and one submitting power. Rather, the ideal is to create a win-win situation by increasing the sum of power through participation and collaboration between the health professional, stroke survivor and significant other (111,112). However, for the integration of ICTs and apps to support person-centred and empowering stroke rehabilitation, health professionals must establish new working routines (14,34,113). They must also possess skills to support and educate stroke survivors (and their significant others) in utilizing these solutions (113).

3.2.2.2 Operationalizing the components of individual empowerment

Mark Zimmermann (1995), a key figure in psychological empowerment, has made substantial contributions across research domains, including health research. He asserts that empowerment can be observed at the individual, organizational and community levels (32,34,43).

Given that PCR primarily centers on the individual stroke survivor and their significant other within a clinical setting, this PhD project focus on individual PCR and empowerment (32,41,109). Zimmermann identifies three pivotal components for experiencing individual empowerment, providing a framework for measuring intervention outcomes and the underlying empowering mechanisms: 1) **The intrapersonal component** concerns an individual's self-perception (e.g., sensing control, mastery, competence), intrinsic *motivation and initiatives* to engage in behaviours to influence desired outcomes. This component has close relations with self-efficacy (114). 2) **The interactional component** addresses how individuals comprehend, interact with and navigate within their social environment and take responsibility to become independent. This component involves *education, support* gaining *adequate relevant knowledge, meaningful* interventions, and *building skills* based on needs and goals. Within this component, relational mechanisms such as collaboration, co-construction, shared power and ongoing support from health professionals are identified as critical elements for achieving individual empowerment. 3) **The behavioural component** includes empowered *behaviours and actions taken to master or influence specific outcomes* (i.e., taking initiative and managing specific components of rehabilitation independently, living independently, participating in groups/associations, adapting to change) (109).

To achieve empowerment, interventions should therefore focus on *building motivation*, *confidence*, and *equipping patients with tools* (knowledge and skills) to *set personalized goals* and *develop effective strategies* for achieving them, fostering *independence* from health professionals (47). These components together provide a comprehensive view of experienced empowerment (31). This

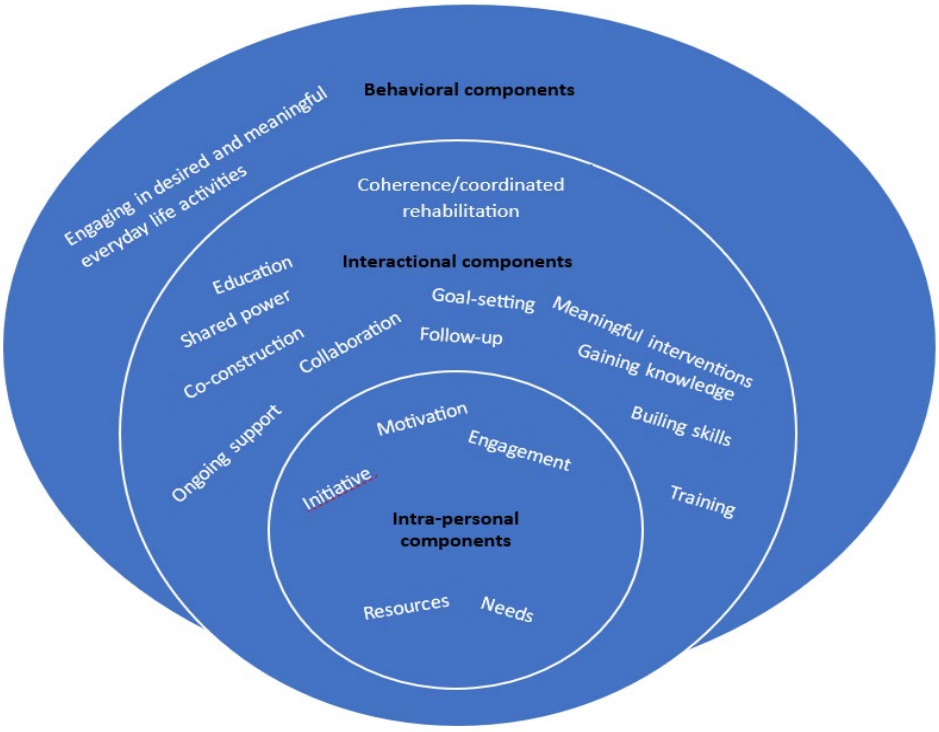
operationalization of empowerment fits well with the literature on mechanisms for achieving individual empowerment from the perspectives of stroke survivors and their significant others, which are briefly described below.

3.2.2.3 Empowering mechanisms identified by stroke survivors and their significant others

Empowerment research focuses on identifying mechanisms that facilitate desired outcomes (109). Key mechanisms identified in the literature for individual empowerment from the perspectives of stroke survivors and their significant others include opportunities to acquire needed *knowledge*; the development of needed *skills*; support and an adequate *power* balance based on respectful and sufficient communication; and *collaboration* and *coordination* between stroke survivors, their significant others and healthcare professionals (14,34,41). On the contrary, dis-empowering factors have also been identified and include a lack of knowledge and skills among stroke survivors and their significant others; a lack of involvement in rehabilitation decisions; a sense of loss of control over their lives; and feeling powerless, hopeless, and helpless. Furthermore, these negative aspects extend to dependency, a feeling of being "written off", unconstructive communication with health professionals, and being told "this is as good as it gets" (26,41).

To support the assessment of empowerment in Phase 3 of this PhD project, the (non-exhaustive) mechanisms identified for achieving empowerment are illustrated on a self-constructed figure (Figure 2), followed by a description of methods appropriate for assessing empowerment.

Figure 2: PCR mechanisms that may support experiencing empowerment



3.2.2.4 Using qualitative methods to explore PCR and empowerment within this PhD project

A variety of definitions and understandings of empowerment, such as *self-efficacy, self-management, self-care, patient participation, patient engagement, patient involvement, expert patient, shared decision-making, and health literacy* exist (47,48,115). Further, a variety of questionnaires to measure empowerment are available (41,48), however due to the aforementioned lack of a unified definition, overlap with similar concepts, different contextual factors, and different dimensions of empowerment measured, no universal questionnaire on empowerment exists, also within the field of stroke rehabilitation (41,48). Among 50 questionnaires reviewed, only one had good reliability and validity (116), and thus, debate on how empowerment interventions should be measured persists, with the parameters such as patient behaviours, health-related measures, and patient satisfaction being discussed (35). Additionally, Zimmermann (1995) states that empowerment is an open-ended and dynamic construct that may vary over time and therefore cannot easily be reduced to a universal set of operational rules and definitions. Rather, the measurement of empowerment must be connected to the individuals' experiences, which are contextually grounded in their life experiences (31,109).

Therefore, for assessing person-centred and empowering approaches, qualitative methods are suggested, since PCR and empowerment center on how rehabilitation is experienced by the participants engaged (i.e. the individual level) (1,29,32,117,118), thus making them appropriate to understand needs, experiences and acceptability (1,119,120). Additionally, qualitative approaches may more often lead to actions to improve patient experiences and healthcare services (119), whereas quantitative methods do not indicate what needs to be done to improve any situation (3). Furthermore, qualitative approaches supports exploring the complex, often interrelated diversity of needs, the context in which the intervention will be implemented, as well as the participants' perceptions of the interventions (29,101,102,118,120). This aligns well with a constructivist GT philosophy, the MRC framework, and the EBCD approach (the EBCD approach is described in section 4.2.2). In-depth stakeholder perspectives as a point of departure for developing interventions (described in Phase 1), which could not have been gathered using quantitative approaches, also align well with the MRC framework and constructivist GT philosophy (29,31,109,120). Consistency between the philosophy, frameworks, theories, methodologies and methods applied in the research process strengthens the trustworthiness of this PhD project (121), as illustrated in Appendix A.

3.3 Understanding context

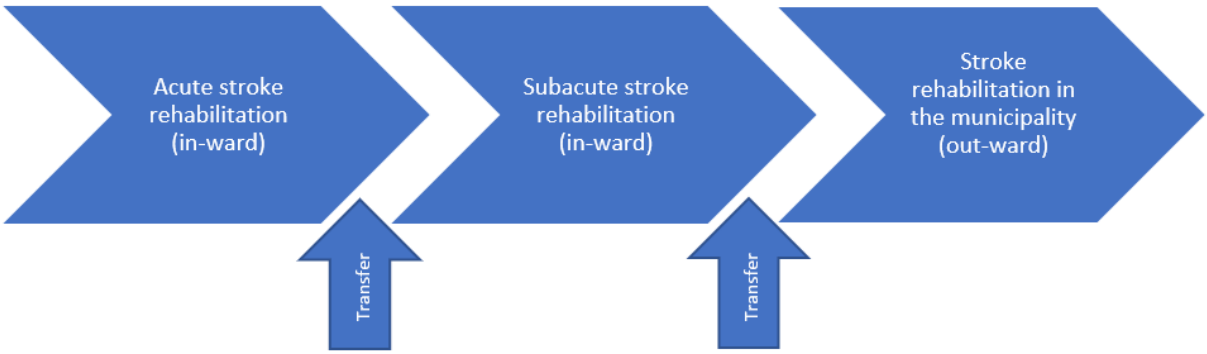
Understanding context throughout the research phases is suggested by the MRC framework to be crucial to theorize on *how* the intervention will work and for whom it will work/not work; this

includes the identification of barriers/facilitators of implementation in specific settings (1,105,117). Thus, taking the context into account throughout the research process will allow interventions to be developed and refined so that they fit the contexts in which they are to be implemented. Understanding context is also central to the transferability of an intervention across different contexts (1,102). Therefore, stroke rehabilitation within the Danish context is described below.

3.3.1 Organization of Danish stroke rehabilitation

In Denmark stroke rehabilitation may take place in different stroke rehabilitations settings. Initially, the stroke survivor is admitted to an acute hospital stroke unit. The typical admission in this setting last 1–3 days (17,122,123) and comprises an assessment of the stroke survivors' functional capacity, rehabilitation needs, potential, motivation and resources (17). When required, rehabilitation continues in a subacute hospital stroke unit and/or rehabilitation continues in the municipalities (i.e., home-rehabilitation or rehabilitation centres). Stroke survivors in need of rehabilitation in all three rehabilitation settings most often have more complex difficulties (i.e. cognitive, emotional, mental and/or physical issues) (124). Figure 3 shows the stroke rehabilitation trajectory as well as the transfers the stroke survivor and their significant others experience when in need of rehabilitation in all three rehabilitation settings.

Figure 3: The Danish stroke rehabilitation trajectory



Discharge meetings between hospital and municipal health professionals are held to support coherence in the rehabilitation process (17). The municipality must, within 4 days after hospital discharge, arrange the startup of rehabilitation, and the actual rehabilitation process must take place within 7 days after discharge (17,27). The intensity of rehabilitation is recommended according to the stroke survivors' deficits, resources, motivation and goals. However, physical rehabilitation in particular should be of high intensity and preferably daily for stroke survivors who are in need of and can engage in frequent interventions and where progress is made (17,27). Voluntary organizations such as "*Hjernesagen*" and "*Hjerneskadeforeningen*" offer activities that supplement or carry on after the termination of rehabilitation in the hospital and municipality (17). The severity

of stroke, regarding cognitive and motor deficits, can be categorized as mild, moderate or severe (125). In the Danish context stroke severity is assessed in the acute setting (on the first day) using the Scandinavian Stroke Scale (SSS) (11) or National Institutes of Health Stroke Scale (NIHSS) (125).

In this PhD project, the focus is on all three rehabilitation settings, that is cross-sectoral stroke rehabilitation, and stroke survivors in need of cross-sectoral rehabilitation in all three settings are perceived as having moderate to severe stroke. This topic is further elaborated on in the discussion.

3.3.1.1 Physiotherapy and occupational therapy within Danish stroke rehabilitation

Physiotherapists (PT) and occupational therapists (OT) are essential part of interdisciplinary stroke rehabilitation as per both national and international stroke rehabilitation guidelines (17,23–25). PTs and OTs have an important role throughout the rehabilitation process in promoting the stroke survivors' recovery and prognosis through activity and participation (17,23–25) and supporting coherent cross-sectoral stroke rehabilitation (53). PTs competences include, among others, support in improving physical function and mobility, while OTs competences include improving the ability to perform activities of daily living and engaging in social life. Strategies to manage and compensate for (remaining) cognitive deficits may also be a focus of OTs (17).

In the acute phases of rehabilitation, the main focus is on diagnosing and treating the stroke, although early mobilization and physical rehabilitation also are essential for recovery and survival (126). Functioning and social participation is, however, a key goal for stroke survivors and their significant others; thus, the focus gradually changes to more complex and meaningful skills and activities, to manage everyday life, including work, social and leisure activities. This requires collaboration with other sectors such as the employment sector. Some municipalities have employed "brain coordinators" to support stroke survivors and their significant others in engaging and coordinating with the different systems and sectors (17,22,28).

Due to OTs and PTs being an essential part of interdisciplinary and cross-sectoral stroke rehabilitation, they are the main health professionals included throughout this PhD project, together with stroke survivors and their significant others.

Chapter 4: Methods and findings from the three research phases

This chapter describes the methods used within the three research phases as well as the main findings.

4.1 Phase 1 of the PhD project: Identification/exploration

In describing Phase 1 of this PhD project the methods and main findings of the three pre-studies and the scoping review are presented, including the following elements of the MRC framework: *understanding the problem, understanding context*, *stakeholder engagement*, and *reviewing the current evidence base* (see Table 1).

4.1.1 Understanding the problem and context, by engaging stakeholders in three pre-studies

Prior to the intervention development, researchers need to understand the problem and the context in which the interventions are to be implemented (1,102,127). Researchers may have ideas some assumptions about the *problem* and the interventions that might address it; however, knowledge of the problem and the possibilities for an intervention should also be based on stakeholders' views (stroke survivors, significant others, therapists) and published research or theory. Nevertheless, early ideas and assumptions are likely to be refined and challenged throughout the research project (1,108,127). Stakeholder experiences, regarding rehabilitation barriers, unmet needs and priorities throughout the rehabilitation trajectory should be a point of departure (i.e., *sensitizing concepts)* for intervention development (1,3,100), which aligns with the rationale of developing interventions based on stakeholders' needs and experiences.

The purpose of the three pre-studies was therefore to understand the perspectives of stroke survivors, their significant others, and therapists on current cross-sectoral stroke rehabilitation, including challenges and possibilities to integrate ICTs into the rehabilitation process to support cross-sectoral, person-centred and empowering stroke rehabilitation.

4.1.1.1 Methods and participants

As mentioned earlier, qualitative methods are recommended to explore individual participants' experiences of cross-sectoral stroke rehabilitation (1,29,32,117–119,128). Therefore, qualitative in-depth interviews were used to support the understanding of the problem in the early stages of the research process, as well as the context and the stakeholders involved (1,102). Stroke survivors 6–12 months post-stroke who had participated in cross-sectoral stroke rehabilitation in all three rehabilitation settings, i.e. moderate to severe stroke (n=18) (12), their significant others (n=13) (13) and therapists (n=9) (55) working in cross-sectoral stroke rehabilitation were interviewed individually or in focus groups. Data were analyzed by using a constructivist GT approach. The participant characteristics are shown in Table 2 and the findings have been reported in three articles (12,13,55).

These pre-studies provided an important foundation for understanding the problem and unmet needs

from stakeholders' perspectives within cross-sectoral stroke rehabilitation as it is organized today in the region of Southern Denmark. They inspired the first ideas and assumptions about apps having the potential to support and accommodate the identified unmet needs of stakeholders. The three pre-studies also supported the development of "emotional touchpoints" as point of departure for the development phase (i.e., *sensitizing concepts)*, which is further presented in the below section.

Table 2: Participant characteristics of the three pre-studies

Stroke survivors (n=18)	**Sex and age** Male (n=11) ages: 41, 51, 56, 59, 61, 64, 67, 68, 68, 71, 74 Female (n=7) ages: 52, 54, 55, 56, 64, 72, 79,	
Significant others (n=13)	**Affiliation and age** Wife (n=5), 42,47, 49, 60, 68 Husband/cohabitants (n=3), 51, 57, 66 Mother, 65 Sister, 64 Daughter, 30 Son, 24 Female neighbor, 42	
Therapists (n=9)	**Setting of employment**	**Occupation, sex, age**
	Acute setting (n=4)	PT, females, (n=2), 40,42 OT, females (n=2), 39 and unknown
	Subacute setting (n=4)	PT, male and female (n=2), 30, 33 OT, females (n=2), 33, 42
	Municipal setting (n=1)	OT, female, 42

4.1.1.2 Main findings of the three pre-studies

An analysis was performed across the three pre-studies. This cross-analysis identified unmet needs and areas that needed improvement within stroke rehabilitation, also called "emotional touchpoints" (3,129), which could be considered dis-empowering mechanisms within current cross-sectoral stroke rehabilitation. Important mechanisms for achieving cross-sectoral, person-centred and empowering stroke rehabilitation (19,28,30,31) derived from the three pre-studies included easy access to relevant information, coherent stroke rehabilitation, an overview of the goals and planned activities (rehabilitation plan), assessments of the ability to use ICTs, the need for learning to use ICTs and apps, the development of user-friendly apps, and the need for provision of person-centred guidelines and daily exercises/training. Furthermore, a need for support from therapists to use ICTs was identified. Coherence between rehabilitation settings regarding information and communication was additionally suggested to be supported through apps comprising videos and/or pictures in addition to or instead of text. This cross-analysis also served as *theoretical sensitivity*, an analytical step by Charmaz (2014) where theoretical perspectives might support understanding and complement the data (108) which in this sense was the theory of PCR and empowerment.

4.1.2 Reviewing the current evidence base by conducting a scoping review

In line with the recommendations of the MRC framework, the element of identifying relevant evidence should be addressed prior to intervention development (127). Based on the assumptions from the pre-studies suggesting that apps may have the potential to accommodate many of the unmet needs in current stroke rehabilitation, the next step of this PhD project involved reviewing existing apps to support stroke rehabilitation.

Apps are increasingly being used to support stroke rehabilitation. Yet, given the limited and diverse nature of the existing literature (21,88,90), and recognizing the significance of identifying knowledge gaps to pave the way for future developments (1,2), a scoping review methodology was deemed appropriate.

The purpose of the scoping review was to identify and map existing literature on mobile/tablet apps being used to support stroke rehabilitation. This encompassed detailing the stroke severity of participants as well as the use of apps in relation to the setting and phase of the rehabilitation process.

4.1.2.1 Methods

The scoping review included published scientific and grey literature, and followed the five-stage framework described by Arksey and O'Malley (130) and Levac et al. (131): 1) identifying the research question; 2) searching for relevant studies; 3) selecting studies; 4) charting the data; and 5) collating, summarizing, and reporting the results (130,131). All studies obtained from the database searches were compiled into an Excel spreadsheet. Subsequently, title and abstract screening, and full text reading was performed independently by myself and my main supervisor. Ultimately, data extraction was undertaken independently by myself and my main supervisor, using a table that was iteratively developed for this purpose. Wade's elements of rehabilitation guided the structure of this stage (i.e., assessment, goal-setting, intervention/training, support of social participation, and discharge support) (18,19,28). For further details, see Article I (94).

4.1.2.2 Main findings of the scoping review

Article I describe and provides an overview of the evidence concerning: 1) mobile/tablet and web-based apps used to support the rehabilitation process after stroke and 2) participants' stroke severity and use of applications in relation to the setting and phase of the rehabilitation process. The systematic search resulted in 12,065 records. After title, abstract and full text reading, 46 articles were included in the scoping review. The included studies were categorized according to Wade's elements regarding the rehabilitation process: 1) assessment of limitations faced by the stroke

survivor (n=16); 2) goal-setting (n=0); 3) interventions/training (n=25); 4) support of social participation (n=0) and 5) discharge support (n=2). Three studies included more than one element of the rehabilitation model (e.g., assessment, goal-setting, training and/or discharge support).
Mostly, the included studies involved participants with mild or mild/moderate stroke (n=33), and four studies included participants with severe stroke. In nine studies, the stroke severity was not reported.
Participants in the included studies were mainly suffering from chronic stroke, i.e., more than 3 months post-stroke (n=23), while participants with acute and/or subacute stroke were included in 16 studies. Five studies included participants with acute and/or subacute and/or chronic stroke, while two studies lacked details on stroke onset. The apps included in the 46 studies were used in either a rehabilitation setting (*n*=21), home setting (*n*=17), or both settings (*n*=3). In five studies the setting was not reported.

The discussion highlights the need for apps that focus on a greater part of the rehabilitation process, including acute/subacute stroke rehabilitation, and support greater elements throughout the rehabilitation process, including the transfers between rehabilitation settings. Future apps should also address the stroke survivors' and significant others' needs for customized interventions regardless of stroke severity.

The identification/exploration phase of this PhD project, including the cross-analysis from the pre-studies, and the findings of the scoping review, was used as point of departure for the development phase of this PhD project. The findings from Phase 1 also supported the development of a program theory, which is presented in Phase 3 of this thesis.

4.2 Phase 2 of the PhD project: Development

Phase 2 of this PhD project comprised the development phase, describing the process of systematically developing the content of two known apps based on the rationale of engaging stakeholders to develop more acceptable solutions.
As recommended by the MRC framework (1), understanding the context in which the interventions are to be implemented is crucial before initiating app development. When developing the apps, stakeholder engagement is recommended to ensure that acceptable solutions are developed and to support the implementation (1). To systematically prioritize and develop the content of the two apps to support cross-sectoral stroke rehabilitation in two stroke rehabilitation trajectories in the region of Southern Denmark, the EBCD methodology was used. The following elements of the MRC framework are presented in this section: *understanding context, stakeholder engagement* and

developing the intervention (see Table 1).

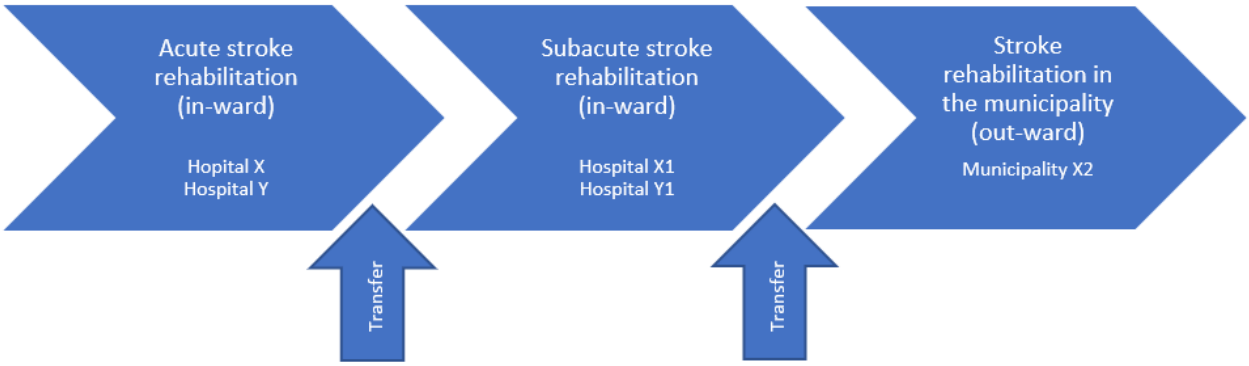
4.2.1 Understanding context: Stroke rehabilitation in the region of Southern Denmark

For this PhD project, six different stroke rehabilitation settings were engaged based on new (i.e., two larger municipalities having stroke rehabilitation units) and existing (i.e., settings that were engaged already in the pre-studies) networks and collaborations. The functions and organization of stroke rehabilitation in Denmark have been described earlier. However, to understand the six chosen contexts for this PhD project, observations were made prior to the co-design phase, which are described below. Observations were also made during the test and evaluation phase, which are described in Phase 3 of this thesis.

4.2.1.1 Methods and setting

Prior to engaging stakeholders in the co-design process in Phase 2 of this PhD project, participant and non-participant observations were made in five of the six included rehabilitation settings (two acute hospitals X and Y, two subacute hospitals X1 and Y1, and one rehabilitation setting in the municipality X2). Observations in municipality Y2 were not performed since this municipality was included a bit later in the research process. Figure 4 presents the rehabilitation settings where observations were made.

Figure 4: Rehabilitation settings where observations were made



Participant observations involve the presence of the researcher to some extent in the field studied, whereas non-participant researchers observe without involvement (108,132).

The purpose of performing these observations was to gain a deeper understanding of the settings in which stroke rehabilitation takes place, more precisely OT and PT interventions across different rehabilitation settings, and where the apps were to be implemented. Insights into similarities in context and organizational processes, interventions, and interactions between therapists, stroke survivors and their significant others were also obtained. Understanding the context in which the apps are to be implemented, and taking contextual considerations into account support the

development and the subsequent implementation (1). The next section describes the development of the observational guide.

4.2.1.1.1 The developed observational guide

An observational guide was developed to support the observations and avoid the gathering of large amounts of unfocused data (108,133–135). However, I stayed open to observing incidences/situations outside the observational guide.

My observation guide was inspired by Spradley (1980), Charmaz (2014), the concept of PCR, empowerment and the *sensitizing concepts* from earlier research (the pre-studies). It contained both more general and descriptive topics like "What is going on when therapists engage with stroke survivors and their significant others", as well as a more *sensitizing focus* (108) like "How, when and where do actions occur that may support empowerment for the stroke survivor and the significant other" and "How, when and where are ICT/apps used to support the rehabilitation process". Informal interviews were also conducted during these observations, for example by asking "What severity of stroke do you see most in this setting, and has this changed over time?", "When and how do you communicate with the next sector?", and "How do you see your role in this setting as a therapist?" (See Table 3).

Table 3: Observational guide used to understand context, use of ICT and apps, and empowerment

Stroke rehabilitation process	What does "a first" assessment in acute setting contain? What does a PT and OT intervention consist of in the different settings? How is the focus divided on bodily, activity and participation level? Which element does the sectoral transfer contain, who is involved? How is the home context a part of the rehabilitation? How is the focus on the maintenance phase after municipal rehabilitation is about to end?
ICT/app use	How, when, where and for which purposes is ICT/apps used to support the rehabilitation process?
Empowerment	When is education a part of the intervention to support knowledge and skills (for whom stroke survivor/significant other)? (interpersonal component) How and when is motivation facilitated? (intrapersonal component) How is goal-setting a part of rehabilitation? (interpersonal component) How is the stroke survivor actively engaged in the rehabilitation? (interpersonal component) How is the significant other actively engaged in the rehabilitation? (interpersonal component) When does the stroke survivor express perceived control and mastery? (intrapersonal component) How is the communication/construction/interaction between therapist and stroke survivors/significant other? (interpersonal component) Which actions does the stroke survivor and/or significant other take to influence desired outcomes? (behavioral component)

4.2.1.1.2 Observations in the stroke rehabilitation settings

I visited stroke rehabilitation settings for 5 full days. I clarified my research questions and my role from the beginning (134) by supplying information on the observational purpose via e-mails. I followed both OTs and PTs when they intervened with stroke survivors. I was also present when they documented their interventions. My observations were dictated after each therapeutic session (which lasted between 30 and 45 minutes). The dictation app immediately translated my speech to written text, and before forwarding the anonymized notes to my e-mail, I erased any wrong words or sentences included by the dictation app. Field notes were also written at the end of each day or the next day. Both non-participant observations (standing in a place where I could see the intervention taking place, and where the patient could see me but not getting in the way of the intervention), and participant observations (handing materials to the therapists and supporting intervention when needed, or engaging in conversation with the stroke survivor and/or therapist) were undertaken (108,132). Observations on similarities and differences in the local organizational processes were valuable. For example, knowledge regarding the short hospitalization duration in acute setting and the number of therapists involved in the stroke survivors' rehabilitation process made it clear that, especially in acute rehabilitation settings, continuity may be difficult to achieve. This could potentially challenge the implementation of the apps in this setting. Implementation should, according to the MRC framework, be considered throughout the intervention development, test and evaluation phase (100,101) and is further explored in Phase 3 and the discussion section of this PhD project.

The observations from the five settings were also used to support, contribute to, and validate the identified "emotional touchpoints" from the pre-studies and the scoping review, which are elaborated on in the development section below. For further details see Article II (136).

4.2.1.1.3 Reflections on my role during observations

As a former neurological therapist employed in an acute rehabilitation setting, I was familiar with the rehabilitation activities in especially the acute setting but less familiar with those in the subacute and municipality settings. However, my knowledge was from some years ago, which created many opportunities to ask elaborating questions on the therapists 'doings' and the organization of stroke rehabilitation today. Coming to an unknown setting has the advantages of bringing a fresh view to the phenomenon observed and gaining analytical distance from the data. On the other hand, the advantages of being an "insider" are that access is much easier and that the researcher is already aware of many insider perspectives and may more easily ask the "right" questions (108,134).

My observations did not notably seem to disturb or affect the stroke survivor or the therapist during

their interventions. I tried to blend in as much as I could by using my prior experience as a PT in stroke rehabilitation settings rather than merely standing in a corner (108,133–135). The observations had several purposes and benefits. As an "outsider" in all rehabilitation settings, the obvious benefit was that relations and connections were made with all the therapists who participated in the later co-design process. The therapists also used the opportunity to ask more about the project and gain greater insight into the purpose of the next stage of the research process, which was the development phase.

4.2.2 Developing the intervention using Experience-Based Co-design

EBCD has often been used as a quality improvement methodology (137) based on stakeholders' experiences to address their priorities, needs and wishes in a collaborative process (119,138). After "*understanding the problem and context"* in Phase 1 of this PhD project, through stakeholder experiences, which are understood as the reconstructed and recalled experiences of the stroke rehabilitation process (3), we used two known, free-of charge apps, *Mit Sygehus* and *Genoptræn.dk* as a starting point. The content to be developed for the apps was prioritized using an EBCD approach.

EBCD involves a structured, facilitated, two-phased process, namely a discovery phase and a co-design phase, each comprising three stages (134,139). An overview of the purposes and activities in the EBCD process is shown in Table 4.

Integrating an EBCD process into the development process of the MRC framework has a number of benefits including co-constructed and more acceptable person-centred apps based on lived-experience perspectives; increased knowledge and skills for contributing to the research process; a sense of pride, accomplishment and empowerment when stakeholders share the power; and the experiences of all participants are being valued (104,106). On the stakeholder influence continuum, an EBCD process is seen a participatory and engaging process rather than one in which stakeholders merely advise and consult (128,140). Addressing issues that are of relevance for stroke survivors, their significant others and therapists may also facilitate development, implementation, and maintenance through owner-ship of the developed apps (3). Based on the recommendations of the MRC framework and the EBCD methodology, stakeholders were engaged throughout a co-design process in the development phase (1,119,128) of this PhD project, which is described below.

Table 4: Overview of purposes and activities in the EBCD process

	EBCD phases	Activities		Purpose
Discovery phase	**Stage 1** Setting up the project 2014-2015	**Engaging stroke rehabilitation settings (n=6):** Acute stroke units (n=2) Subacute stroke units (n=2) Stroke units in the municipalities (n=2)		Engage different stroke rehabilitation settings
	Stage 2+3 Engaging stroke survivors, significant others, and staff 2014-2017 2021	**Interviews (focus group or individual)** Stroke survivors (n=18) Significant others (n=13) PTs (n=4) OTs (n=5)		Explore experiences of stroke rehabilitation across sectors, identification of unmet needs and use of ICT in the rehabilitation process and everyday life
		Observations of stroke rehabilitation settings (5 settings)		
Co-design phase	**Stage 4** Co-design meetings 2021	**Workshop 1 (Jutland) (n=14)** Stroke survivors (n=2) Significant others (n=2) PTs (n=3) OTs (n=3) App developers (n=2) Researchers (n=2)	**Workshop 1 (Funen) (n=11)** Stroke survivors (n=2) Significant others (n=2) PTs (n=2) OT (n=1) App developers (n=2) Researchers (n=2)	Present emotional "Touchpoints" and the modules in "Mit Sygehus". Additionally, to generate concrete input on the content of each module, to meet stakeholders' needs and to prioritize the modules.
		Scoping review on existing app solutions to support stroke rehabilitation		
	Stage 5 Small co-design meetings 2021	**Step 1: Co-design meetings in Jutland working with "knowledge" module (n=4)** PT (n=1) OT (n=1) Head therapist (n=1) Representative from patient organization (n=1)	**Step 1: Co-design meetings on Funen working with "knowledge" module (n=3)** PT (n=1) Speech therapist (n=1) Physician (n=1)	Co-design content in "Mit Sygehus" to meet stakeholders' prioritized needs. Furthermore, to assess if the "training module" was suitable to meet stakeholders' needs and applicable in the different rehabilitation settings.
		Step 2: Co-design meetings in Jutland working with "training" module (n=4) PT (n=1) OT (n=1) App developer (n=1) Researcher (n=1)	**Step 2: Co-design meetings on Funen working with "training" module (n=4)** PT (n=1) OT (n=1) App developers (n=2) Researcher (n=1)	
	Stage 6 Celebration event 2021	**Workshop 2 (Jutland) (n=10)** PTs (n=2) OTs (n=3) Representative from a patient organization (n=1) App developers (n=2) – one attended online Researchers (n=2)	**Workshop 2 (Funen) (n=9)** Stroke survivor (n=1) Significant other (n=1) PT (n=1) OTs (n=2) App developers (n=2) Researchers (n=2)	Presentation of app solution and "hands-on" test of all the content in the app, as well as feedback on written instructions to be used in later a testing period
			Workshop 2 (Funen) (n=4) Online PT (n=1) OT (n=1) App developer (n=1) Researcher (n=1)	

Table 4 is slightly altered from article II: Using Experience-based Co-design to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation. Published in Research Involvement and Engagement, August 2023 (136).

4.2.2.1 Participants – stakeholder engagement

The co-design phase took place from June to December 2021. The rehabilitation leaders of the six stroke rehabilitation settings each identified one PT and one OT to participate throughout Phases 2 and 3 of this PhD project. Some therapists had also participated in the pre-studies. Acute setting X, however, did not identify a PT to participate in the co-design phase and the municipal setting X2

engaged another OT in the second workshop due to the initial OT being unavailable. In total, 12 therapists employed in acute settings (n=3), subacute settings (n=4) and municipal settings (n=5) participated in the co-design phase. Therapists from the municipality assisted in identifying stroke survivors who had undertaken rehabilitation throughout the stroke rehabilitation trajectory (acute, subacute and municipal rehabilitation) i.e., moderate to severe stroke and had recently terminated their rehabilitation process (n=5) and one of their significant others (n=5) to participate in the co-design process. Also, one representative from the patient organization "Hjerneskadeforeningen", three app-developers, my main supervisor and I participated (see Table 4 and 5).

Table 5: Participant characteristics and participation in the EBCD process

Participants	Age, years	Gender	Workshop 1	Workshop 2	Rehabilitation setting	Neurological work experience, years
Stroke survivor[A] and husband	63/66	Female/Male	X			
Stroke survivor[A] and wife	85/79	Male/Female	X			
Stroke survivor[A] and wife	57/55	Male/Female		X		
Stroke survivor[A] and cousin	37/35	Female/Female	X			
Stroke survivor and husband	69/74	Female/Male	X			
OT	44	Female		X	Acute setting X	18
OT	47	Female		X (online)	Subacute setting X1	19
OT	38	Female		X	Municipality X2	10
OT	46	Female	X		Municipality X2	13
PT	35	Female	X	X (online)	Subacute setting X1	11
PT	37	Female	X	X	Municipality X2	6
OT	46	Female	X	X	Acute setting Y	4
OT	29	Male	X	X	Subacute setting Y1	6
OT	36	Female	X	X	Municipality Y2	3
PT	43	Female	X		Acute setting Y	14
PT	25	Female	X	X	Subacute setting Y1	2
PT	36	Female	X	X	Municipality setting Y2	8
Representative from a patient organization	48	Female		X		
App developer from *Mit Sygehus*	-	Female	X			
App developers from Genoptræn.dk (n=2)	-	Males	X	X		
Main supervisor	62	Female	X	X		
PhD student	42	Female	X	X		

The white boxes illustrate participants from setting X, while the light grey boxes illustrate participants from setting Y.
Table 5 is lightly altered from article II: Using Experience-based Co-design to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation. Published in Research Involvement and Engagement, August 2023 (136).

4.2.2.2 Methods – using workshops in the EBCD process

4.2.2.2.1 Workshop 1 – stakeholder engagement

When engaging stroke survivors, their significant others and therapists in a co-design process, it is suggested that the following reflections should be made by the researchers: *why*, on *what* and *how* stakeholders should collaborate, and *how* trust, empowerment and engagement are facilitated and moderated to ensure that power is shared and all voices are expressed and heard (104). To accommodate these factors, the first workshops were initiated by giving participants an overview of the "emotional touchpoints" across the three pre-studies and findings from the scoping review. Furthermore, participants were provided with an overview of the potential of the two apps *Mit Sygehus* and *Genoptræn.dk* to support cross-sectoral, person-centred stroke rehabilitation. They were also made attentive to the fact that *Mit Sygehus* comprised several "empty" modules/headlines but lacked content to support stroke rehabilitation and that this was to be prioritized and developed during the EBCD process. Additionally, they were introduced to *Genoptræn.dk*, which only comprised generic exercises at the time. Finally, they were introduced to the background of the app-developers developing a "video recording" function in *Genoptræn.dk* for this PhD project, namely to accommodate previously identified needs for more person-centred exercises and guidelines (12,13,55). Then, the participants were divided into smaller working groups to create a safe place for collaboration and engagement (119,128). They discussed and prioritized the content to be developed for the apps. Also, as part of the first workshop, participants were asked to give their perspectives on what they would experience as empowering during the stroke rehabilitation process by marking their perspectives on a self-constructed "empowerment model" (see Appendix B). As mentioned by Charmaz (2014) theoretical perspectives (i.e., *theoretical sensitivity*) is an analytical step to support understanding and complement the data (108).

4.2.2.2.2 Findings of the first workshop

Presenting the "emotional touchpoints" and conducting the empowerment exercise (Appendix B) had a validating effect, since participants agreed that relevant evidence-based knowledge, and person-centred exercises and guidelines using video recordings were the most important to be developed in the apps for supporting cross-sectoral, person-centred, and empowering stroke rehabilitation. Furthermore, easy and coherent communication between rehabilitation settings as well as ongoing support from therapists were considered important mechanisms.

4.2.2.2.3 Smaller workshops – stakeholder engagement

In the next step of the EBCD process (see Table 4), the content was developed in several smaller workshops as suggested by Bate and Roberts (2009). Working in smaller groups made it possible to

engage more deeply and thoroughly (128) with creating evidence-based material and knowledge in *Mit Sygehus* according to the participants' prioritization. I initially created the content of *Mit Sygehus* using a "test-patient" and then shared the "tests-patients" access to *Mit Sygehus* with health professionals from subacute setting X1 and Y1. One OT, two PTs, one head therapist, one chief physician, one speech therapist and one representative from the patient organization gave feedback on the content of *Mit Sygehus*. Mostly, the feedback suggested using more reader-friendly and concise language, giving more information on cognitive deficits, and including links to the municipal "brain coordinators" in the app. Further, one OT and PT from subacute setting Y1 and one OT and PT from municipality X2 tested and gave feedback on the video-recording function developed for *Genoptræn.dk*. The app-developers from *Genoptræn.dk* also participated and subsequently engaged with the feed-back before the second workshop was held. The subacute and municipal settings were included in these smaller co-design workshops due to stroke rehabilitation being longer in these settings, thus giving stroke survivors more time to learn to use the apps.

4.2.2.2.4 Workshop 2 – stakeholder engagement

In the last workshop of the EBCD cycle (see Table 4), the prototypes of the two apps and written instructions for their download and use, were tested by all participants to support the subsequent initial implementation. Due to the Covid-19 pandemic and other factors, the workshop in setting Y was held without stroke survivors and their significant others. Their voices were however represented through the participant from the patient organization.

The development phase was considered terminated after the last workshop, which only brought about small adjustments to the written instructions for downloading and using the apps (127). Analysis and results from research Phase 1 of this PhD study and an ongoing more focused and theoretical analysis, as well as reflections and learnings throughout the EBCD process supported the actions in the next stages, in line with a constructivist GT approach (108,141).

4.2.2.2.5 Reflections and learnings from the EBCD process

Participants perceived that their contribution was considered important and valued throughout the co-design development phase. To support and facilitate participants' active engagement throughout the EBCD process, the following actions were useful during the workshops: 1) choosing a setting known to the stroke survivor and their significant other, 2) introducing all participants, 3) clearly communicating the purpose and timeframes of the workshop, 4) dividing participants into smaller groups for discussion, 5) pairing stroke survivors and their significant others with a therapist they knew and felt confident with, 6) designing smaller activities/tasks that fostered the active engagement of all participants, 7) facilitating the activities to ensure that participants achieved all

tasks within the time they were given, and 8) summarizing the insights and learnings from each workshop. Similar facilitating actions have been reported elsewhere (106,119,129,139).

4.2.2.3.1 Findings: Prototypes of *Mit Sygehus* and *Genoptræn.dk* - the content and interfaces

Through a structured and facilitated EBCD approach (119,128,137,138) with stroke survivors, their significant others, PTs and OTs from all rehabilitation settings, a patient organization, and app developers, prototypes of *Mit Sygehus* and *Genoptræn.dk* were developed, including the content of *Mit Sygehus* and new functions in *Genoptræn.dk* to address the priorities and needs of all stakeholders. The apps were subsequently implemented, tested and evaluated in Phase 3 of this PhD project. The developed content and interfaces of *Mit Sygehus* and *Genoptræn.dk* are described briefly below.

4.2.2.3.1.1 Mit Sygehus

Fourteen modules/headlines were co-designed in *Mit Sygehus* each with "sub-headlines" and informational text: 1) what is stroke, 2) consequences of stroke, 3) the rehabilitation process, 4) significant others, 5) driver's license, 6) my hospital journal, 7) training, 8) coordinator of the municipalities, 9) patient organizations, 10) the research project, 11) guide for videocalls, 12) my appointments, 13) personal notes and 14) contact (i.e., to the different rehabilitation settings) (136). See Figure 5.

Figure 5: The content and interface of Mit Sygehus

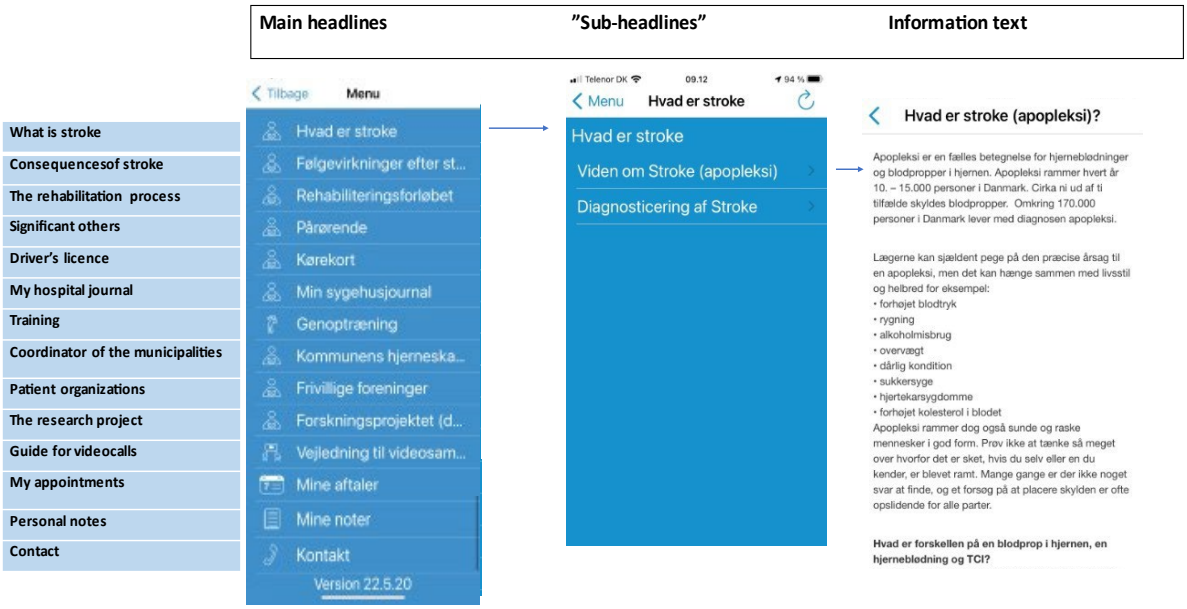


Figure 5 is from article II: Using Experience-based Co-design to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation.
Published in Research Involvement and Engagement. August 2023 (136).

4.2.2.3.1.2 Genoptræn.dk

Genoptræn.dk contains more than 700 generic video-recorded exercises, that can be assigned by the therapists to the stroke survivors' personal app. Content of the app is accessible for therapists in all rehabilitation settings, thus supporting cross-sectoral coherence, communication and documentation (99). As mentioned earlier, a "video recording" function in *Genoptræn.dk* was developed for this PhD project to accommodate the previously identified need for more person-centred exercises and guidelines (12,13,55). Figure 6 shows the content and interface of *Genoptræn.dk*.

Figure 6: The content and interface of Genoptræn.dk

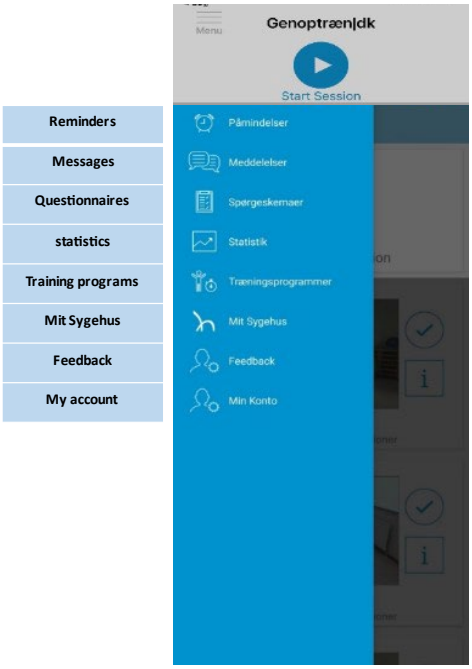


Figure 6 is from article II: Using Experience-based Co-design to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation.
Published in Research Involvement and Engagement. August 2023 (136).

As suggested by the MRC framework, we published the development phase, including reflections on stakeholder engagement, to support other researchers working in similar areas or who wish to develop interventions (1,102,104,142). A full description of the intervention development process and reflections/learnings throughout the EBCD process can be found in Article II (136).

4.2.2.4 Ethical considerations

Prior to attending each workshop, all participants were informed about the purpose and duration of the workshops via written information letters (Appendix C+D). They signed a form consenting to participate and to have pictures taken during each workshop (see Article II (136) and Appendix E).

4.3 Phase 3 of this PhD project: Implementation, testing and process evaluation

The following elements of the MRC framework are described in this section: *designing the evaluation*, *developing a program theory*, *understanding context, refining the intervention, and data generation* (1). Additionally, the setting and participants, data analysis, findings, my role as a researcher, and ethical considerations are described.

4.3.1 Evaluation design

As suggested by the MRC framework the development phase was followed by an implementation, testing and process evaluation phase (1,2,117,127). Initial testing on a smaller scale and smaller refinements during the test phase are recommended when evaluating newly developed solutions where existing literature is scarce (102,117,143) and before deciding on larger-scale implementation and evaluation (1,118,144). Additionally, developing a program theory is recommended when conducting process evaluations (1,2,105,117).

4.3.2 Methods

4.3.2.1 Developing a program theory

A program theory, also called *theory of change* or *intervention logic*, is an explicit statement, theory or model of how an intervention is expected to lead to a set of specific outcomes through a series of intermediate results by asking ***what***, ***why, how, for whom,*** and under ***what*** circumstances and contexts an intervention works, and defining unexpected consequences and if refinements were required (1,2,105,117). Researchers are increasingly interested in not only treatment effectiveness ("does this work"), which is assessed *after* the project is completed, but also the *process* during the implementation and testing phase and mechanisms that support/hinder the acceptability of the intervention. However, it may be challenging to choose and identify relevant and meaningful outcome measures within ICT and app-based approaches. In a review by Freund et al. (2022), only three out of 17 studies included outcomes for significant others, and the studies lacked outcomes measuring if the ICTs and apps supported meaningful outcomes for stroke survivors, namely functional independence and participation in everyday life, which often are the ultimate key goals of stroke rehabilitation (14,22,29).

When conducting process evaluations of newly-developed complex interventions, the evaluation of ***acceptability*** is essential to improve the quality of larger-scale projects (1,118,120,143). Acceptability refers to *how* the intervention is perceived by the involved stakeholders and meets their needs, and how it fits the treatment setting/context (29,101,118). Methods such as rating scales to assess acceptability are perceived as insufficient; however, the use of qualitative research methods for a more in-depth understanding of mechanisms affecting the acceptability of the

intervention is common (118,144,145) to provide feedback on the intervention (144).

Logic models, also called *program theory* or *theory of change*, are often used to provide a better understanding of what intermediate outcomes need to be achieved for a complete and acceptable delivery of the program (145). Moreover, implementation failure (i.e., not done right) and/or theory failure (i.e., done right but still did not work) are more easily identified by investigating the relationship between the blocks in the logic model (1,105). The refinements needed before up-scaling the project to secure a more optimal implementation and maintenance of change are also more easily defined (105).

Optimally, the program theory and logic model should be developed at the beginning of the research project, based on stakeholders' perspectives, and the evidence and theory within the research field and should be refined continuously throughout the research process. A refined program theory will support the transferability of the intervention across settings, and identify if the intervention is less likely to be acceptable when implemented in other contexts (101).

In this PhD project, the first logic model was developed prior to intervention development, where input and activities (*why* and *how* the apps will work), outputs (*why* did it work), and outcomes (the theory of change) were identified and addressed (101). The inputs (*why* will it work) from the logic model were supported by the pre-studies and the scoping review (i.e., identification of stakeholders' needs and knowledge of existing apps). The activities (*how* will it work) were supported by evidence on co-design and literature on integrating apps in stroke rehabilitation (the scoping review), that is sufficient education in using the apps for all stakeholders, (co-constructed) meaningful content supplied in the apps, and the facilitation of implementation. Outputs (*why* did it work) were identified with support from evaluation research, that is the acceptability of the apps. Outcomes were identified through knowledge from the pre-studies and through the aim of the process evaluation, that is the *acceptability* of the content of the apps in supporting cross-sectoral, person-centred and empowering stroke rehabilitation. Logic models may have a fifth box to assess social and organizational long-term impacts (105), however, these impacts were not assessed in this PhD project due to a lack of long-term follow-up. The logic model was refined throughout the project, and the final version can be seen in Figure 7.

Figure 7: The final logic model of the PhD project – inputs, activities, outputs, and outcomes

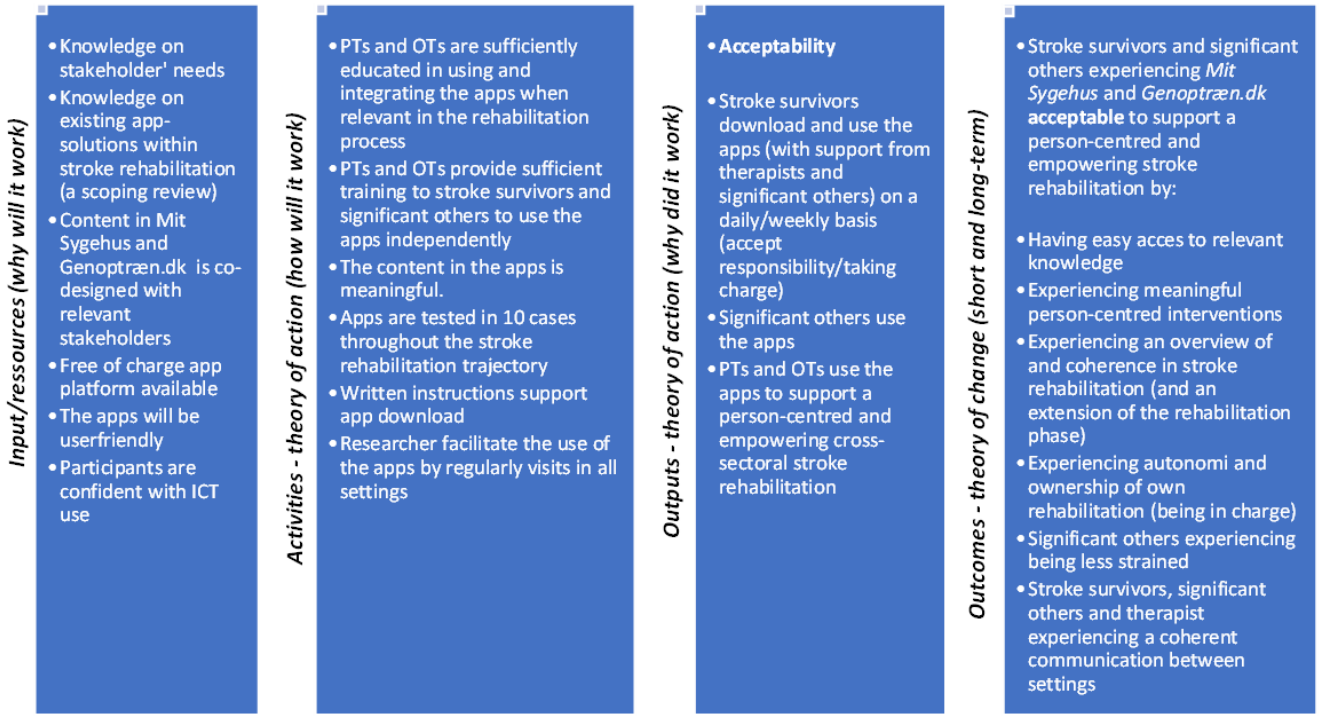


Figure 7 is from article III: Acceptability of two mobile applications to support cross-sectoral, person-centred and empowering stroke rehabilitation – a process evaluation.
Accepted for publication in Annals of Medicine, December 2023.

4.3.2.2 Setting and participants

Mit Sygehus and *Genoptræn.dk* were initially implemented in two stroke rehabilitation trajectories, comprising two acute settings (hospitals X and Y), two subacute settings (hospitals X1 and Y1) and two municipal settings (municipalities X2 and Y2) in the region of Southern Denmark. Written descriptions on how to download and use the apps supported the implementation (see Appendix F and G). From each rehabilitation setting one PT and OT participated (who had also participated in the co-design process), except for the PT from acute setting X and a second PT from setting Y2, who were engaged later in the research process. The gatekeepers from the acute settings were attentive to including participants who were affected by moderate stroke (measured by either SSS or NIHSS) and/or in need of cross-sectoral rehabilitation (acute, subacute, and municipal rehabilitation) and had previous experience of using ICT. Due to stroke survivors not always being ready to consent to participate in research projects during acute rehabilitation, it was decided that inclusion could also take place in the subacute settings. The rehabilitation settings, transfers between them and the inclusion of participants are shown in Figure 8. Of the stroke survivors invited to participate, only one participant withdrew his consent to participate since he later decided to skip subacute rehabilitation and wished to be discharged directly to municipal rehabilitation. The inclusion of participants took place from February to October 2022. Participant characteristics are described in Table 7.

Figure 8: Included rehabilitation settings and stroke survivors

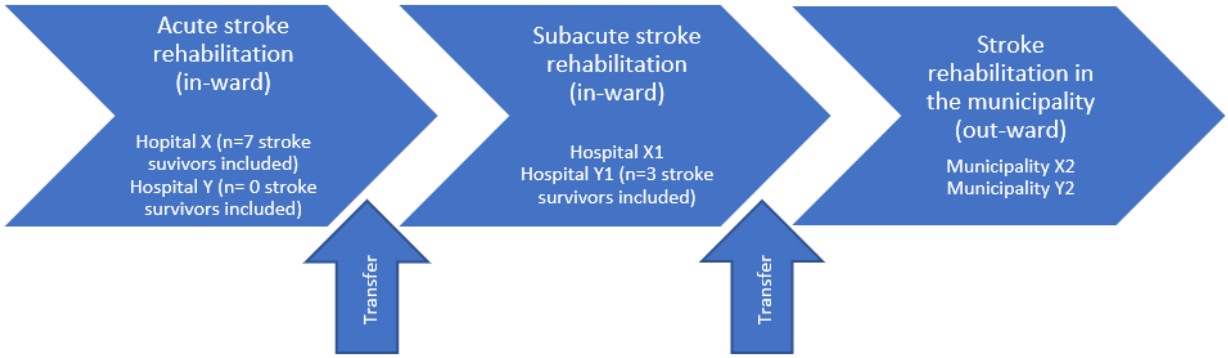


Figure 8 is from article III: Acceptability of two mobile applications to support cross-sectoral, person-centred and empowering stroke rehabilitation – a process evaluation.
Accepted for publication in Annals of Medicine, December 2023.

4.3.2.3 Understanding context – observations made during the test phase

Charmaz suggests that GT observations should record significant processes and actions occurring in the specific setting/context and address what participants find interesting and/or problematic within their contexts (108). Therefore, during the test phase, I visited each stroke survivor once during the rehabilitation process, alternately visiting the participants in subacute and municipal settings. Visits in the acute settings could not be accomplished, as the stroke survivors were admitted only for a few days. These visits and observations had four purposes: 1) to gain ongoing knowledge of each context and continue to maintain a relationship with the participating therapists, 2) to observe and facilitate the implementation of the two apps, for example, by supporting the use of the apps and providing education when required, 3) to get acquainted with the stroke survivors and their significant other (if they were present that day), and 4) to obtain knowledge (including barriers and facilitators) of their current use of the apps. Field notes on the therapists' and stroke survivors' use of the apps in the specific settings/context were made immediately after the observations and were used to support the post-rehabilitation interviews (108,134,135). For further details, see Article III.

4.3.2.4 Refining the intervention and facilitating implementation

In process evaluations, data construction and analysis are suggested to be carried out simultaneously to facilitate necessary smaller refinements and thus, the implementation, and acceptability of the intervention before a potential scale-up (1,101,102,117). Simultaneously data construction and analysis aligns well with the constructivist GT approach (146).

Early and "open-minded" analysis may identify elements of the intervention (strengths and/or weaknesses) that may lead to refinements of the intervention, as well as the mechanisms of action of the intervention (i.e., refining the program theory), to optimize the intervention (144).

Smaller refinements were made regarding the written instructions for therapists on how to use the functionalities in *Genoptræn.dk*. Additionally, regular newsletters were shared with participating therapists, the therapist leaders, app-developers and the volunteer organization.

The purpose of the newsletters was to ensure ongoing engagement and communication with participants and facilitate the implementation of the two apps (147) by sharing information regarding 1) status on inclusion; 2) education on using functionalities in the apps that had been identified as challenging, including new functions; and 3) describing stroke survivors' and significant others' perspectives on what worked and why, as well as their wishes and additional needs (derived from the ongoing analysis), which could then be considered when including more participants. For example, participants wanted more status recordings to be made and stored in *Genoptræn.dk* during the rehabilitation process. In the newsletters I also continuously offered to present the PhD project, including the purpose of implementing the two apps, and ongoing results, which two of the six settings accepted (settings X and Y2). Furthermore, the process and preliminary results were presented at national conferences.

The smaller refinements included labelling each exercise with a number to support an overview of the content of each exercise program (see Figure 9), and more easily determine if more than one exercise program was present (e.g., strength, balance, upper motor and/or facial exercise).

Figure 9: Providing exercises in Genoptræn.dk with numbers to create easier overview



To ensure transparency (1,2,127), the refinements made to *Mit Sygehus* and *Genoptræn.dk* throughout the testing and evaluation phase were reported in Article III.

4.3.2.5 Data generation: stroke survivors' and significant others' acceptability

As mentioned above, when conducting process evaluations of newly developed complex interventions, evaluating participants' ***acceptability*** is essential before larger-scale implementation and evaluation. Based on the logic model and the literature on acceptability (1,118,120,143–145), the following measures were chosen to evaluate the potentials of *Mit Sygehus* and *Genoptræn.dk* to support cross-sectoral, person-centred and empowering stroke rehabilitation: 1) acceptability, that is, how was the intervention perceived in relation to practicality and barriers/facilitators in using the apps and how did it support the rehabilitation process, required adaptations and unexpected consequences, 2) what/which part of the apps were used. Table 6 shows the related research questions.

Table 6: Research questions – on participants' acceptability of Mit Sygehus and Genoptræn.dk

Research questions		
	Theory of action	**Theory of change**
Acceptability assessed qualitatively	Which inputs and actions were required for the apps to be used throughout the cross sectoral rehabilitation processes? To what extent do the apps need to be refined or adapted to make them more acceptable, more relevant, or more useful to the specific contexts in which they are delivered? Were there any positive or negative unexpected consequences? Has implementation varied by settings/ contextual factors that needs to be considered?	What are the stroke survivors´ and significant others' perspectives of the content of *Mit Sygehus* and *Genoptræn.dk*? What were stroke survivors´ and significant others' experiences of *Mit Sygehus* and *Genoptræn.dk* to support a person-centred and empowering stroke rehabilitation?
Acceptability assessed quantitively	Which parts of the apps were used by participants (usability)	

Table 6 is from article III: Acceptability of two mobile applications to support cross-sectoral, person-centred and empowering stroke rehabilitation – a process evaluation.
Accepted for publication in Annals of Medicine, December 2023.

To assess the acceptability of the apps, qualitative interviews were conducted with stroke survivors (n=10) and their significant others (n=7) when rehabilitation in the municipality had just ended (or was very close to ending). The stroke survivor and their significant other decided if the post-rehabilitation interview should be individual or dyadic. Dyadic interviews were performed in seven cases in the participants' home (n=6) or the municipal rehabilitation setting (n=1). Individual interviews were performed in three cases (not married, divorced or widowed) in the stroke survivors' home (n=1), in the municipal rehabilitation setting (n=1) and online (n=1). The participant characteristics are shown in Table 7.

Table 7: Stroke survivor and significant other characteristics

	Patient, age (years)	Diagnoses and SSS	Significant other, age (years), relation	Duration of rehabilitation in each setting	Interview type and duration
1. Setting X	Beate, 65	Right side infarct SSS 46	Jonas, 72 Husband	Acute: 9 days Sub-acute: 7 days Municipal: 149 days (30 interventions)	Dyadic interview 97 minutes
2. Setting Y	Peter, 74	Left side infarct SSS 39*	Ella, 72 Wife	Acute: 10 days Sub-acute: 20 days Municipal: 91 days (3 interventions)	Dyadic interview 99 minutes
3. Setting X	Liam, 86	Right side infarct SSS: 43	Jane, 71 Wife	Acute: 2 days Sub-acute: 14 days Municipal: 123 days (2 interventions)	Dyadic interview 49 minutes
4: Setting Y	Jamie, 61	Left side cerebellum infarct SSS: 46	Betty, 56 Wife	Acute: 6 days Sub-acute: 14 days Municipal: 212 days (79 interventions)	Dyadic interview 47 minutes
5. Setting X	Robert, 58	Right side infarct SSS: 42	Ida, 61 Wife	Acute: 4 days Sub-acute: 15 days Municipal: 8 days (5 interventions)	Dyadic interview 83 minutes
6. Setting X	David, 50	Right side infarct SSS: 47	None Divorced	Acute: 7 days Sub-acute: 10 days Municipal: 28 July – **	Individual interview 49 minutes
7. Setting X	Erin, 80	Right side haemorrhage SSS: 29	Berry, 78 Husband	Acute: 6 days Sub-acute: 24 days Municipal: 29 days (unknown number of interventions)	Dyadic interview 17 minutes (telephone)
8. Setting X	Noah, 63	Right side infarct SSS: 31	None Divorced	Acute: 10 days Sub-acute: 57 days Municipal: continued in a private rehabilitation clinic	Individual interview 70 minutes (online)
9. Setting X	Edvin, 54	Right side infarct SSS: 54	Carol, 50 Wife	Acute: 8 days Sub-acute: 14 days Municipal: 211 days (39 interventions)	Dyadic interview 62 minutes
10. Setting Y	Anne, 84	Right side infarct SSS:49	None Widower	Acute: 3 days Sub-acute: 21 days Municipal: unknown duration (21 interventions)	Individual interview 44 minutes

*NIHSS score converted to SSS score (148)
**The stroke survivor had a second stroke just after being interviewed and just before being terminated from municipal rehabilitation, and thus continued his rehabilitation. Duration in municipal rehabilitation was therefore not obtained.
Table 7 is from article III: Acceptability of two mobile applications to support cross-sectoral, person-centred and empowering stroke rehabilitation – a process evaluation.
Accepted for publication in Annals of Medicine, December 2023.

Dyadic interviews are used to interview pairs who share a pre-existing relationship, such as married couples, to facilitate interactions and reactions to comments from each other (149). Thus, dyadic interviews are ideal to generate data from both perspectives, and to support participants' recollections (especially when having cognitive and/or speech deficits). Moreover, in dyadic

interviews, sharing allows each participant to compare, extend or contrast what the other states, which may move the discussion in alternative directions or lead to the construction of new ideas/issues (149).

In many ways, the interactions in the dyadic interviews were similar to those in the focus groups (150) in the sense that participants picked up on each other's statements and experiences, elaborating on them, contrasting them, and triggering ideas. In the dyadic interviews, I also saw that the stroke survivor and significant other engaged directly with each other and asked elaborating or exploratory questions. In those cases, I just listened and only engaged when elaboration was needed or to keep the discussion within the broad topics of the interview.

Charmaz recommends that the processes and actions within the rehabilitation process and context (and how they may change over time and across contexts) should be assessed to obtain detailed and rich data. Furthermore, she suggests paying attention to "taken-for-granted and hidden assumptions" (108). Investigating processes supports an understanding of *why* and *how* (which mechanisms bring about the intended change) and *how* context affects implementation and outcomes (101,108,135).

Therapists' descriptions in the stroke survivors' journals of the parts of the apps, that were introduced and used during rehabilitation also supported the post-rehabilitation interviews by triggering participants' memories. The interviews lasted 17 –99 minutes.

4.3.2.6 Data analysis: stroke survivors' and significant others' acceptability

Interviews were analysed using Charmaz' Constructivist GT approach (108). In the ongoing analysis, the concepts of person-centred and empowering rehabilitation (18,29–31,47,52) (i.e., *theoretical sensitivity*) was included after the initial and focused coding steps, to support development of sub-themes describing the construct of ***acceptability*** of *Mit Sygehus* and *Genoptræn.dk* to support person-centred and empowering stroke rehabilitation, from the perspective of stroke survivors and significant others. Comparing, contrasting and (re)organizing the codes, code groups, themes and sub-themes was an iterative process. As recommended by Charmaz (2014), the aforementioned *sensitizing concepts* were incorporated into my interview guide as point of departure but kept flexible and open and were altered if needed. For example, the importance of providing "a sense of progress" and "therapist support" was evident early in the analysis and was thus incorporated into subsequent interviews.

For further details, see Article III.

4.3.2.6 Findings

Acceptability of the content of the two apps *Mit Sygehus* and *Genoptræn.dk* from the perspectives of stroke survivors (n=10) and significant others (n=7), is presented in two overall themes: "***acceptability*** of *Mit Sygehus"* and "**acceptability** of *Genoptræn.dk"* to support person-centred and empowering stroke rehabilitation. Mechanisms facilitating acceptability as well as contextual factors (i.e., rehabilitation phases) affecting acceptability, reasons for "non-use" and suggestions for adaptions are elaborated on in respectively two and eight sub-themes (Figure 10) and is further elaborated on in Article III.

Figure 10: Main and sub-themes of stroke survivors' and significant others' acceptability of Mit Sygehus and Genoptræn.dk

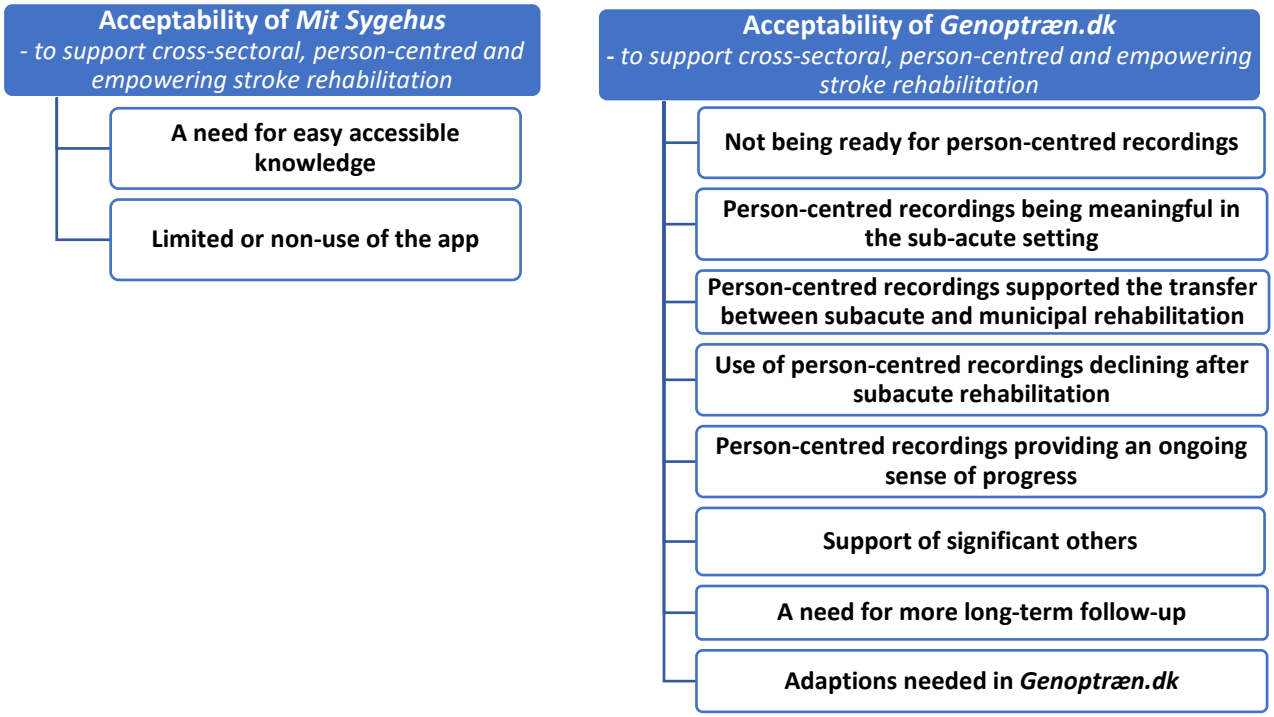


Figure 10 is from article III: Acceptability of two mobile applications to support cross-sectoral, person-centred and empowering stroke rehabilitation – a process evaluation.
Accepted for publication in Annals of Medicine, December 2023.

4.3.2.6.1 Mechanisms found to support cross-sectoral, person-centred and empowering stroke rehabilitation

The following mechanisms were found to support the acceptability of *Mit Sygehus* and *Genoptræn.dk* to support cross-sectoral, person-centred and empowering stroke rehabilitation: 1) being accessible (free of charge), user-friendly, and inspiring confidence in their use; 2) support of a greater part of the stroke rehabilitation process; 3) easy and sufficient access to relevant and evidence-based information and education when needed regarding stroke, deficits, the rehabilitation process; 4) meaningful and person-centred content based on needs and goals through personalized

video recordings; 5) person-centred recordings supporting motivation and engagement, a sense of "clarification" of what activities needs to be trained to improve functioning, as well as a sense of "ableness" and ongoing progress; 6) support of coherence when transferring between different rehabilitation settings, especially between subacute and municipal rehabilitation; and 7) facilitation of continuous relationships with and frequent support from therapists to make self-training activities meaningful and habitual.

In the following two sections and in Table 8 the acceptability is visualized in relation to the rehabilitation process, including reasons for limited/declining/non-use of the apps.

Table 8: Acceptability across the rehabilitation process

	Acute rehabilitation	**Subacute rehabilitation**	**Municipal rehabilitation**	**Maintenance phase**
Mit Sygehus	Accepted, however mostly by significant others.	Acceptability declining.	Acceptability declining.	Acceptability declining.
Mit Sygehus *Reasons for limited/non-use*	Stroke survivors found the content "confronting". Significant others and health professionals provided stroke survivors the needed education and information. One stroke survivor and her husband were not interested in app-supported rehabilitation.			
Genoptræn.dk	Low acceptability. Implemented in only one case by therapists. This stroke survivor was satisfied with the content of the app (facial exercises)	High acceptability. Implemented in 10 cases. Support from and collaboration with therapists facilitated meaningful and motivating content, that also fostered engagement and positive self-training habits.	Acceptable to support the transfer to the municipality but used less by stroke survivors later in this setting.	Accepted for sensing an ongoing progress, and to get an idea of "how it was". Support of habits, to keep on going and to maintain quality in the self-training exercises.
Genoptræn.dk *Reasons for limited/non-use*	Most stroke survivors did not feel ready to engage in self-training activities, however two stroke survivors suggested earlier integration for stroke survivors expressing being ready for self-training.		No need to access the app every time, the content could be remembered. Everyday life takes up more time. Therapists including the app less. One stroke survivor did not use the app due to pain in her knee, and due to hassle with the order of the exercises. Another stroke survivor could remember instructions and did not need further app-support.	

4.3.2.6.2 Acceptability of Mit Sygehus and reasons for limited/non-use

Stroke survivors and significant others expressed *Mit Sygehus* being acceptable to support cross-sectoral, person-centred and empowering stroke rehabilitation. However, *Mit Sygehus* seemed to be most acceptable in the acute phases of rehabilitation, and acceptability declined throughout the rehabilitation process as shown in Table 8. Furthermore, significant others seemed to value the content of *Mit Sygehus* higher than stroke survivors. Participant also suggested that content on "return-to-work" and "re-gaining a driver's license/permission to drive" was further developed in *Mit Sygehus*. Five stroke survivors stated not having used *Mit Sygehus* at all, while four stroke

survivors stated having only visited the app a few times. Reasons for limited or non-use were for example that the content was perceived as being "confronting" and reminding them of deficits they had/could have had.

In these cases, *Mit Sygehus* did not directly support stroke survivors in gaining knowledge and experiencing empowerment; however, they expressed being sufficiently educated and informed by significant others and/or health professionals. In contrast, data from the back-end of *Mit Sygehus* showed that six stroke survivors, who explained a non-use or limited use of *Mit Sygehus,* had actually visited *Mit Sygehus* several or a few times throughout the rehabilitation process, typically in the acute phase of rehabilitation. One stroke survivor and her husband were overall not "keen on" ICT and app use.

4.3.2.6.3 Acceptability of Genoptræn.dk and reasons for limited/non-use

Genoptræn.dk was perceived as acceptable to support cross-sectoral, person-centred and empowering stroke rehabilitation, especially in the subacute setting. *Genoptræn.dk* was only integrated in one case during acute rehabilitation (hospital X) and found acceptable by this stroke survivor to support training of facial paresis, however, most participants did not feel ready to engage in self-training in the acute setting. Although, two male stroke survivors thought that *Genoptræn.dk* was or could have been acceptable in the acute setting, to support daily training activities and waiting time before being transferred to a sub-acute setting, or before sub-acute therapists engaged in using the app to support rehabilitation.
Acceptability of *Genoptræn.dk* declined during municipal rehabilitation since everyday activities gradually took up more time. Also, therapists seemed to include *Genoptræn.dk* less in the municipal setting. The content of *Genoptræn.dk* was mostly facial, strength, balance, lower and upper limb, transfer guidelines (for example transfer from wheelchair to bed) and walking/balance exercises. Cognitive rehabilitation did not seem to be explicitly included in *Genoptræn.dk* during this test phase. However, receiving reminders, and using personalized recordings and verbal cues from therapist, as well as having the opportunity to re-read and re-watch the content of *Mit Sygehus* and *Genoptræn.dk* when needed, was perceived as cognitive support by participants.

Two stroke survivors explained that they had not used *Genoptræn.dk* after discharge from sub-acute rehabilitation. Reasons given were that one stroke survivor could remember the instructions received by the therapists, and thus he performed self-training without any other support. The other stroke survivor explained that fatigue and pain in her knee prevented her from performing self-training (waiting for knee replacement). She also explained that there were some challenges with *Genoptræn.dk* during the sub-acute rehabilitation, e.g., when having performed the first exercise

and trying to move on to the next, the app repeatedly returned to the first exercise. Two other male stroke survivors also mentioned that obtaining an overview of how far they had come in their self-training program was challenging. This was adjusted before their rehabilitation ended by supplying each exercise with a number, which solved the issue for these two participants. However, some participants expressed a need for more flexibility in what exercises to perform. *Genoptræn.dk* supplies the exercises in the same order each time, and stroke survivors had to "tick-off" each exercise before they could continue to the next or shift to another exercise program (if they had two or more programs, e.g., balance, lower limb strength, facial, upper motor exercises, etc.). Finally, some stroke survivors called for a more long-term follow-up after rehabilitation termination, for example 3-6 months post-rehabilitation.

Lack of reading abilities, fine motor difficulties, and/or cognitive deficits, such as trouble remembering codes or navigating within the apps, were only mentioned in one case as barriers of using the apps. In that case, the significant other had installed both apps on her own mobile phone, and she assisted her husband in gaining the required knowledge and performing his self-training. The other significant others found it sufficient to have the opportunity to watch the content on the stroke survivors' mobile phone, and this also relieved the significant others of "being in charge".

In summary this process evaluation showed that *Mit Sygehus* and *Genoptræn.dk* may be acceptable to support person-centred and empowering stroke rehabilitation. However, implementation of *Genoptræn.dk* in an acute setting, for stroke survivors expressing being ready for this, and more systematical integration in the municipal setting need to be further investigated. Also, a greater implementation of *Mit Sygehus* across rehabilitation settings could potentially support especially significant others even more. Finally, an investigation of how cognitive rehabilitation can play a greater role in app-supported stroke rehabilitation and how more long-term follow-up can be supported is needed.

4.4 My role as a researcher

As mentioned earlier, being a former therapist in stroke rehabilitation (albeit several years ago) supplied me with some prior assumptions on stroke rehabilitation from a therapist's perspective. Moreover, being a teacher at a University College and teaching about stroke rehabilitation requires me to be updated on current stroke rehabilitation organization. However, not being employed in any of the six rehabilitation settings included in this PhD project allowed me to be "a stranger" and ask about "taken-for-granted" features when visiting the settings and performing observations. However, it is worth noting that my presence may have affected the data constructed in accordance

with the understanding of constructivist GT. I tried to be explicit and reflexive about assumptions, knowledge from earlier studies and theories used to support the construction of the observational and interview guides. I also tried to be open minded during the data construction and analysis steps (151).

4.5 Ethical considerations

Overall, this PhD project followed the ethical guidelines for health research, the code of conduct of the Association of Danish Physiotherapists and the Declaration of Helsinki (152,153).
Participants received oral and written information before consenting to participate in the test phase and in the following interviews (see Appendix H). Stroke survivors consented directly in *Mit Sygehus*, while their significant others signed a written consent form (see Appendix I). The therapist leaders of the six rehabilitation settings gave written consent for the apps being tested in their setting, and for 2–3 therapists being engaged throughout the project.
The testing of the interventions implied no risk, harm, or negative unexpected consequences, but was rather perceived as beneficial for most participants involved.
To continue being transparent about this PhD project and the ongoing process of data construction and analysis, as mentioned earlier, I submitted monthly newsletters to all involved stakeholders.
Ethical approval was acquired from the Danish Data Protection Agency (18/60280) and data were stored safely in the Open Patient data Explorative Network (OPEN) through the University of Southern Denmark (OP_1378).

Chapter 5: Discussion

Through a co-design process, the content of two known apps were developed based on stakeholders' needs and literature on existing apps to support stroke rehabilitation. Subsequently, the apps were implemented, tested and evaluated in relation to their acceptability to support cross-sectoral, person-centred and empowering stroke rehabilitation.

This section initially summarizes and discusses the main findings across the research phases, including the mechanisms that were found to be essential for experiencing empowerment. A more detailed discussion on person-centred self-training follows given that *Genoptræn.dk*, in particular, was assessed as being acceptable among stroke survivors within this PhD project. Finally, methodological considerations are discussed, including the strengths and limitations of this PhD project.

5.1 Summary and discussion of main findings across the articles

5.1.1 Use of ICT and apps to support known challenges in stroke rehabilitation

Known challenges within current stroke rehabilitation in Southern Denmark, from the perspectives of stroke survivors, significant others and therapists working within cross-sectoral stroke rehabilitation, have been found to be an uncoordinated, isolated, and fragmented rehabilitation process. Also, lack of sufficient knowledge, coordinated communication across rehabilitation sectors, insight into and overview of the rehabilitation process, person-centred approaches, support for significant others, shared decision making and goal-setting have been found to hinder optimal stroke rehabilitation (12,13,55) which aligns with other national and international literature (14,26,39,53,54). Stroke survivors, significant others and therapists in this PhD study called for more person-centred approaches, including ICT and apps to support cross-sectoral and person-centred stroke rehabilitation, since ICT such as mobile phones is an essential and important part of everyday life even after experiencing a stroke. But, developed ICT solutions needs to be user-friendly and flexible in that sense that the content can be individualized to changing needs and goals (12,13,55).

In an overview from 2019 of "the good and bad" within ICT´s to support rehabilitation highlights that mobile based apps have the advantage of being portable, socially acceptable, multifunctional, familiar and easy to use, also for people with cognitive deficits (92). ICTs and apps have also shown potential to support PCR and empowering approaches (12,13,23,39,55,57,70–72,74–80,82,83,92), including many of the challenges within stroke rehabilitation mentioned earlier, such as sufficient knowledge and skills to manage everyday life (38,92), coordinated and coherent rehabilitation (including discharge and follow-up support) (29,37,38,92), self-management (92) and significant other support (154). However, the scoping review in this PhD study showed that most existing apps to support rehabilitation are narrowly focused in supporting one or a few elements of stroke rehabilitation (94), which have also been discussed elsewhere (92,93). Also, apps are primarily tested on stroke survivors suffering from mild or mild/moderate stroke and mainly in the chronic rehabilitation phase (i.e., more than 3 months post-stroke). Further, it was discussed in the scoping review that lack of stakeholder engagement in the app development and test phases, lack of evaluation on their acceptability and long-term effects, as well as lack of health professional support in using ICT solutions are barriers for development and use of person-centred ICT solutions (94), which was also discussed elsewhere (92,155–158).

There are thousands of apps available for download (92,159,160), however, less than one percent of them are grounded in research evidence (92). A Google Scholar search on "mobile applications", "stroke", and "rehabilitation" resulted in more than 125,000 hits, and the same search in diverse

databases resulted in 1529 hits in a recent review (160). Considering the huge number of apps available to support stroke rehabilitation, it is understandable that therapists are challenged when trying to navigate the app landscape (55). As a support tool, the National Health Service in England has created a database, MyTherAppy[1], where apps that may benefit individuals suffering from brain injuries are reviewed in a standardized manner by trained clinical staff and patients (92). Among the thousands of apps, five are recommended for stroke survivors: 1) "Language Therapy 4-in1", a speech supportive app; 2) "Dexteria" to support fine motor and upper motor rehabilitation; 3) "Peak" to support cognitive functions such as memory; 4) "Stroke Patient" to support knowledge and information gathering; and 5) "Headspace" to support managing mood after stroke. Of the five recommended apps, only the app "Stroke Patient" is completely free of charge and already available in acute rehabilitation settings. Some of the other apps offer free trial periods. A comprehensive app that supports the entire process of cross-sectoral stroke rehabilitation may seem over ambitious; however, for therapists to implement these solutions and stroke survivors and significant others to feel engaged and motivated to use them, the apps must be acceptable across rehabilitation settings and support more than one or a few elements of rehabilitation (92,94,100).

Therefore, the overall aim of this PhD project was to develop, implement, test and evaluate the co-designed content of two known apps to support cross-sectoral, person-centred and empowering stroke rehabilitation for stroke survivors and their significant others. The mechanisms that were considered important in the identification/exploration and development phases (pre-studies, Article I and II) and the needs that were accommodated in the implementation, testing and process evaluation phase (Article III) are summarized below.

5.1.2. Summarizing empowering mechanisms across the three research phases

The following intra- and interactional mechanisms for experiencing empowerment were accommodated during the research process: easy access to relevant knowledge when needed, the provision of insight into and an overview of the rehabilitation process, support of cross-sectoral transfers (i.e., coherent stroke rehabilitation), support from health professionals, ICT integration to support stroke rehabilitation, easy access to free-of charge apps, and reduction of significant other burden. Additionally, stroke survivors and their significant others were provided with an ongoing "sense of progress" that promoted motivation and engagement. ICTs and apps also supported the behavioural component of empowerment by accommodating the need for activity and participation through active engagement in person-centred self-training activities.

Having easier access to health professionals and compensating for remaining deficits (including

[1] https://www.my-therappy.co.uk/news/2020/top-5-apps-for-stroke-recommended-by-nhs-clinicians

cognitive deficits) need to be further investigated (see Table 9). More long-term follow-up was also called for in this PhD project, which is further discussed below. Also, the implementation of the apps could have been more stringent, which is also discussed below.

Table 9: Summarizing mechanisms for experiencing empowerment across studies

Mechanisms for experiencing empowerment		**Pre-studies**	**Article I**	**Article II**	**Article III**
Intrapersonal component and Interactional components	ICT use provide a sense of security	x		x	x
	ICT may support in gaining relevant and sufficient knowledge	x	x	x	x
	ICT may support gaining insight and overview of the rehabilitation plan and process	x	x	x	x
	ICT/apps may support physical rehabilitation	x	x	x	x
	ICT/apps may support cognitive rehabilitation	x	x		
	ICT/apps may support continuous goal-setting	x	x		x
	ICT may support coherence and transfer between rehabilitation settings (communication, documentation (video recordings/pictures)	x	x	x	x
	ICT may support follow-up and extension of the rehabilitation process	x	x		
	ICT/apps may support access to peers	x	x	x	x
	ICT requires ongoing support from health professionals	x	x		x
	Apps must be user-friendly	x	x	x	x
	ICT use is necessary	x	x	x	x
	Apps must be free-of charge	x	x		x
	ICT may relive significant other burden	x	x		x
	Gaining a sense of progress				x
Behavioural components	ICT promotes activity and participation	x	x	x	x
	ICT may support adherence to self-training	x	x	x	x
	ICT support compensating for remaining deficits	x			
	ICT may support easier access to health professionals	x			
	An individualized app-solution may support PCR	x	x	x	x
	Apps may support active engagement	x	x	x	x

5.1.3 Implementation of the apps throughout the rehabilitation settings

In this PhD project *Mit Sygehus* was implemented in the acute rehabilitation setting X and subacute setting Y. However, as mentioned earlier, integration and use of *Mit Sygehus* declined during the rehabilitation process and stroke survivors seemed less occupied with the content of *Mit Sygehus* than significant others. Although, stroke survivors stated limited or non-use of *Mit Sygehus* in general it also seemed that they might have forgotten their use of *Mit Sygehus* as data from the back-end revealed that many had been using the apps.

Regarding acceptability of *Genoptræn.dk* stroke survivors experienced implementation in the subacute setting, which likewise declined later in the rehabilitation process, partly due to the content being remembered and everyday life activities taking up more time. *Genoptræn.dk* was only in one case implemented in the acute setting. Reasons given were that stroke survivors did not feel ready to engage in self-training in this setting, due to a short hospitalization and due to many other things going on in acute setting, thus rehabilitation and self-training was limited until subacute rehabilitation was initiated. However, three stroke survivors felt ready to engage in rehabilitation

and self-training in the acute setting, especially when acute hospitalization was prolonged, or therapists did not initiate self-training activities during the first days in the subacute rehabilitation setting.

In the literature it is recommended that empowerment approaches start with the support of stroke survivors and their significant others in the acute stroke rehabilitation setting and continue on to the stroke survivors' homes, since empowering processes and outcomes achieved within acute and subacute stroke rehabilitation may persist when rehabilitation continue in the municipality/home (38). This knowledge could be important since a major part of stroke rehabilitation in Denmark takes place in the municipalities (17,27); thus, being empowered to engage in rehabilitation activities independently and/or supported through apps may be important when considering stroke survivors having more long-term needs (8,9,13,21,22). PCR and empowerment approaches must, however, not be seen as "tick-off boxes", and it is considered a misconception that once communicated, knowledge and information will be understood and remembered (37), especially given that stroke survivors may suffer from cognitive deficits, such as an impaired memory or ability to understand (9,12–14). Therefore, a more stringent and ongoing implementation and integration of the two apps is needed, which is further discussed in section 5.2.4.

5.1.4 Long-term needs and follow-up

Focusing on long-term follow-up was called for by participants throughout this PhD project to support person-centred, empowering stroke rehabilitation. This is in line with stroke survivors having more long-term rehabilitation needs and deficits (8,9,13,21,22). Follow-up support has been found to be an important mechanism for empowerment in other studies as well. Studies have shown that mobile phone-based interventions after discharge, varying from three to14 phone calls within 6 months post-stroke, in addition to one personal visit from health professionals, can reduce stroke survivors' and their significant others' depression and burden and improve their health and satisfaction with the rehabilitation process (38,39,68,75,78,161–163). However, evaluating long-term effects of ICT and app use to support stroke rehabilitation and the maintenance of outcomes are lacking (92).

In the context of Danish stroke rehabilitation, follow-up post-rehabilitation is challenging; however, some municipalities have employed "brain coordinators" to support stroke survivors and their significant others in engaging and coordinating with the complex system (14,17,22,28). Having a designated "contact person/coordinator" and support material, such as the information in *Mit Sygehus*, may support the continuity of care after discharge (29) and post-rehabilitation. Information on how to contact the local "brain coordinator" was provided in *Mit Sygehus,* but a lack of

knowledge of the role of a "brain coordinator" may have caused limited use of this service and contact with that person. It seems important that stroke survivors and their significant others are supported in accessing and contacting the "brain coordinator" to support more long-term follow-up post-rehabilitation.

5.1.5 Person-centred self-training activities supported by an app

Genoptræn.dk was among stroke survivors experienced as acceptable, especially in the subacute setting to support cross-sectoral, person-centred and empowering stroke rehabilitation. The availability of person-centred self-training activities through *Genoptræn.dk* was found to be meaningful and motivating for stroke survivors, fostering adherence, coherence and aiding in the memorization of prescribed exercises.

Although primarily used in the subacute setting, *Genoptræn.dk* seemed sufficient to establish habits that persisted during the transfer to municipal rehabilitation through ongoing therapist support. This correlates with previous studies indicating that therapists supporting and encouraging positive behavioural patterns within the initial 3 months post-stroke may dictate long-term physical activity habits (38,164). These patterns encompass exercise prescription, educational instructions, feedback and updated content tailored to evolving goals, fostering improved therapist-stroke survivor communication. Continuous support, training and supervision to effectively operate and utilize interventions further support empowerment (164–167). This aligns with the interactional component of empowerment, where stroke survivors, through adequate education and support, gain the *skills and knowledge* needed to independently engage with ICTs and apps to undertake self-training (31). ICTs and apps have in other studies also shown to be acceptable from the perspectives of stroke survivors, their significant others and therapists to support stroke rehabilitation, including self-training (66,67,92,166), where correct exercise execution, that is, the quality of performance, is considered as important as its dose and duration (168).

The American Stroke and Heart Association advises stroke survivors to engage in physical activity for 20–60 minutes 2 –5 days per week as a preventive measure against future stroke and cardiovascular diseases. However, the actual levels of physical activity among stroke survivors tend to be lower (56,164).

Szeto et al. (2023) argues in a recent systematic review that most stroke rehabilitation apps target motor deficits (160), which is not surprising given that paresis is a common deficit after stroke (75-85%) (6,15,16). Mobile technology holds promise for facilitating self-training, physical activity and overall participation in everyday life. Self-training is not understood as being aimed only at the deficit level in this PhD study, but rather as being aimed at activities perceived important by the

stroke survivor to participate in everyday life, based on their needs and goals, which aligns well with the understanding in other self-training studies (67,164).

Challenges to adhering to self-training and home exercises include pain, fatigue, exercise difficulty, cognitive impairment, lack of clear instructions, non-meaningful exercises, difficulty recalling specific exercises, and difficulty integrating exercises into daily routines. Intrapersonal and interactional dimensions like limited knowledge of stroke, motivation, perceived recovery, sense of control, mastery and competence also contribute to adherence issues (31,165,167). Conversely, mechanisms encouraging adherence include stroke survivors' desire to improve functional abilities, increased confidence, ongoing supervision, and social support, which, in turn, positively impact empowerment (31,165). These acceptability mechanisms for self-training activities closely resonate with the findings of this PhD project.

Stroke survivors in this PhD study agreed that self-training activities delivered though person-centred recordings supported adherence and probably more than if self-training activities had been supported only by written or verbal instructions. Szeto et al. (2023) investigated effectiveness of apps to support stroke rehabilitation and likewise found increased adherence regarding self-training. Four out of seven RCTs and quasi-experimental studies included in their systematic review showed improvement on adherence regarding exercises among stroke survivors suffering from stroke (less than 40 days post-stroke) and when intervention persisted for 3 months. On the other hand, in one RCT no improvement was not found in stroke survivors suffering from chronic stroke (more than 120 post-stroke) when intervention lasted only 1 month. The systematic review concluded that increased adherence, especially during early rehabilitation, where functional recovery is highest, may lead to increased therapy time, which may subsequently lead to increased functional gain for stroke survivors (160). This therefore highlights that early interventions that is continued across rehabilitation settings after stroke is important to obtain more long-term and stabile self-training habits.

A 2019 systematic review of self-training activities delivered through ICTs likewise found increased adherence and acceptability when compared to conventional methods like written or verbal instructions (169). Realistic personalized interventions, the promotion of motivation, and the ease of following instructions given through ICTs, were reasons for increased adherence. However, while nine of the 14 included studies used generic exercises (169), only two studies in the review (170,171) recorded the participant completing their exercises/activities, like we did in this study using *Genoptræn.dk*. One study used a tablet for upper motor home exercises (170) and one used a portable digital media player for voice training (171). In these two studies, the ICT devices were lent to the participants, and participants' own ICT devices were not used, as was the case in this PhD project. The findings of these two studies revealed that person-centred self-training enhanced

motivation. Additionally, being able to show the content to family members, maintaining hope and a sense of forward momentum according to individual goals increases adherence and thus, the acceptability of the intervention (55,170), mirroring the findings of this PhD project. In contrast, a study using generic upper and lower extremity exercises for self-training in a chronic stage of stroke, noted a post-intervention reduction in adherence. The authors highlighted the importance of including exercises that target the activity level rather than solely addressing deficits, fostering long-term adherence (163). Notably, the content of *Genoptræn.dk* in this PhD project encompasses exercises on both a deficit and an activity level. Additionally, the content aligned with stroke survivors' needs and goals, which are often more focused on functioning and social participation during later rehabilitation (17,22,28). This might also explain the decline in app usage during municipal rehabilitation and post-rehabilitation in this PhD project, where everyday life activities overtake the self-training activities provided in *Genoptræn.dk*.

One study showed that significant others can also experience being empowered to support stroke survivors in managing self-training, although generic exercise programs were used (154). While this aligns with the experience of one significant other, mostly, the significant others interviewed in this PhD project did not take charge of self-training activities or wished to do so. The stroke survivors were mostly themselves empowered to be in charge. This could have to do with the fact that the recordings were personalized, that is, themselves being on the recordings, making self-training activities easier to comply with, more meaningful, and motivating, and therefore, more acceptable compared to generic exercises, subsequently relieving the burden of significant others. Although, some stroke survivors called for more flexibility to choose themselves which exercises and training programs to perform from day to day, which is currently not an option in *Genoptræn.dk* (further refinements needed).

Further, as mentioned in the results, two stroke survivors explained not using *Genoptræn.dk* after discharge from subacute rehabilitation. One stroke survivor could remember the self-training instructions without further app-support (primarily walking on uneven surfaces to challenge his balance). The other stroke survivor explained that pain in her knee overall prevented her from performing self-training activities, and she was also disturbed by the order of the exercises in *Genoptræn.dk* being unclear. In addition, her husband explained that she had never been keen on doing exercises or training, which could explain her lack of willingness to use *Genoptræn.dk*. Her husband himself also explained a resistance towards and lack of confidence in using ICT as part of everyday life.

5.1.6 Important mechanisms to benefit from app-supported stroke rehabilitation

In this PhD study I found that to benefit from app-supported stroke rehabilitation, the users must; be motivated to use ICT/apps and receive sufficient and individualized support to use ICT/apps independently or supported by health professionals and/or significant others. The stroke survivors included in this PhD study, aside from one case, managed to navigate their mobile phone sufficiently, including remembering codes and obtaining skills to enter and navigate the apps. Health professional support was required at all phases of the rehabilitation process. In the acute setting the therapists supported in download of both *Mit Sygehus* and *Genoptræn.dk* (setting X). some stroke survivors needed a lot of support while others managed supported by written instructions and/or significant other support. However, the apps were not further integrated in the acute setting. For stroke survivors expressing being ready to engage in self-training activities more support is needed from health professionals in this setting. To create more stable habits regarding self-training activities, frequent integration of the apps and health professional support is needed in the subacute setting (where rehabilitation duration typically is 3-4 weeks), which is then customized to the stroke survivors' needs in the municipal setting. Stroke survivors explained that intensive support from the therapists, especially during subacute rehabilitation, supported them in gaining the needed skills to use the apps independently.

Meaningful content provided in the apps is equally important to benefit from app-supported stroke rehabilitation, thus therapists must continuously adjust the content to match the stroke survivors' actual needs and goals.

5.2 Methodological discussion including strengths and limitations

In this section methodological considerations are discussed, including the strengths and limitations of this PhD project.

This PhD project adopted the MRC framework, an structured approach designed to guide the systematic development and evaluation of the two apps (1,2). The MRC framework recommends using the following elements of its checklist to assess the quality of the PhD project: *addressing uncertainties* (i.e., understanding the intervention in relation to the context and current evidence base), *engaging stakeholders* (i.e., in the identification and the development phases), *using a formal framework or methodology to guide development*, *choosing a relevant evaluation design* and *outcomes, developing and refining the program theory*, *considering future implementation barriers and facilitators,* and *economic considerations* (1). These elements are discussed below, along with *trustworthiness*, which is a quality criterion within qualitative research. *Trustworthiness*, encompassing, *credibility*, *dependability* and *confirmability,* is strengthened overall by being

transparent and *reflexive* about the researcher's own role, decisions, alterations and conclusions made throughout the research process. Furthermore, *transferability* is discussed. Transferability is strengthened through detailed and thick descriptions of the contexts, sampling procedure and participants, as well as a comparative analysis of research finding with existing literature (146,172).

5.2.1 Addressing uncertainties

Stakeholder engagement, reviewing the current evidence base and understanding context were undertaken to address uncertainties regarding the use of apps to support cross-sectoral, person-centred and empowering stroke rehabilitation (1). However, engaging therapists from the acute setting even more could potentially have supported implementation better, which is further discussed in section 5.2.4. Reviewing the current evidence base is discussed further below.

5.2.1.1 Reviewing the current evidence base by conducting a scoping review

A solid evidence base is recommended (1) to support the development of solutions that accommodate the needs of stakeholders. Undertaking a review to investigate the effectiveness of existing interventions, contextual variation, relevant outcome measures, acceptability, feasibility, and mechanisms of action are recommended, as well as a critical appraisal of the quality of the studies (102). A scoping review methodology was deemed appropriate to obtain a solid evidence base and identify knowledge gaps regarding the use of apps to support stroke rehabilitation. However, due to the heterogeneity of existing studies on apps, the aim of the scoping review was to map the evidence rather than assess its quality (130,131,172,173).

5.2.1.2 Engaging stakeholders

While acknowledging the complexity of encompassing the entire rehabilitation process within a single project, rehabilitation research strongly advocates for the exploration of stakeholders' needs and cross-sectoral processes (100). Carefully designed interventions and systematic stakeholder engagement, with acceptability tested on a minor scale before full-scale implementation and evaluation, are recommended to prevent 'research waste' (1,2,102).

Engaging relevant stakeholders throughout the research process is a major strength of the MRC framework to support implementation and bridge the gap between theory and practice (1,102,174). Stakeholder engagement was complied with in all three research phases of this PhD study. Known challenges in stakeholder engagement include an unequal power balance and trust issues (106,129,137,142,175,176); however, participants in this PhD project felt that their voices were heard and valuable and may have been empowered through this process, which aligns with other literature (2,26,104,129,137,175). Facilitating activities during the EBCD process such as; smaller

working groups, being paired with a therapists they had a relation with, designing smaller activities/tasks fostered active engagement and ensured that all participants were heard. The co-design process may also support bridging theory and practice and may be considered an important step for improving intervention acceptability (26). However, stakeholder engagement could have been done even more systematically, for example, by having an advisory board and continuously engaging it in the ongoing testing, process evaluation, and refinement of the program theory and logic model. Engaging stakeholders in the analysis and reporting of the findings could also have been an additional strength (1,140,144).

5.2.1.3 Sampling

We were dependent on therapists voluntarily registering to participate in this PhD project. Actively engaged therapists may find it easier to convince stroke survivors and their significant others to participate in the project, thus facilitating the implementation process. However, they also bear a huge responsibility for deciding who to include, even when supported by inclusion criteria. As mentioned in the methods section, the SSS was used as a criterion for inclusion. SSS scores of 43–58 indicate mild stroke, scores of 26– 42 indicate moderate stroke, and scores below 25 indicate severe stroke. The SSS scores of six stroke survivors in this PhD project categorized them as having mild stroke (scores of 43, 46, 46, 47, 49, and 54). Four stroke survivors were categorized as suffering from a moderate stroke (scores of 29, 31, 39, and 42). Initially, we thought that by using the SSS score, we could include participants with moderate stroke. However, this proved to be insufficient. Therefore, the SSS scores were made secondary as we concluded that participants in needs of cross-sectoral rehabilitation in all three settings, were clinically diagnosed as having moderate stroke. This is in line with the definitions used in the Danish stroke rehabilitation context (177). Stroke survivors suffering from mild stroke can be discharged from the acute setting to the municipal setting, whereas stroke survivors clinically assessed as having more complex difficulties (cognitive, physical, emotional), that is, moderate or severe stroke, are transferred to a regional subacute setting, since they need specialized interdisciplinary and intense stroke rehabilitation (177) before moving on to rehabilitation in the municipalities.

It seems important to consider stroke survivors' clinical characteristics when including participants for rehabilitation research. Moreover, the characteristics of stroke survivors influencing the potential outcomes of person-centred and empowering stroke rehabilitation could be further investigated.

5.2.1.3.1 Sampling – who was represented and who was not

Stroke is caused by infarct (85-90%%) or hemorrhage (10-15%) and mostly affect men and women over 60-65 years (70-90%) although men are slightly higher represented than women. In Denmark about 90.000 people are affected by stroke, with deficits such as paresis (75-85%), aphasia (20-25%) and cognitive deficits (10-50%) (6,15,16).

As mentioned above therapists held a major role in identifying and including stroke survivors for this PhD study throughout the research phases. In the first two research phases stroke severity was not assessed, the only inclusion criteria were stroke survivors who had been engaged in cross-sectoral stroke rehabilitation in all three settings, were smartphone users, and could engage and communicate sufficiently in an interview or workshop. Further, stroke survivors in the pre-study had to be able to score the Stroke Impact Scale 3.0, which comprise 59 items (12). Thus, some cognitively impaired stroke survivors may have been excluded. However, in the pre-studies one stroke survivor with aphasia was included, and stroke survivors suffering from cognitive deficits such as fatigue, impaired concentration and emotional symptoms (e.g., laughter and crying lability) was represented throughout the research phases.

Most included stroke survivors in the third research phase (test and evaluation) were affected by paresis compromising their gait and balance. Also, most stroke survivors indicated having cognitive impairments such as compromised memory and concentration, fatigue and problems with executive functioning (on activity and participation levels). Only one stroke survivor in the third research phase had a left hemisphere stroke and was affected by aphasia. Regarding representativity of included stroke survivors we succeeded in including both men and women and their deficits were representative, even though aphasia was only represented in 10% of the cases. The included men were younger than the women (men 50, 54, 58, 61, 63, 74, 86 years and women 65, 80, 84 years). This could simply have to do with the fact that men typically are five years younger than women when affected by stroke (178). Although, research has shown that ICT and apps are increasingly accepted and used by healthy and chronically ill elderly people, albeit some subgroups may require more support (e.g., living alone, lower levels of education, lower ICT literacy and people with disabilities) (12,13,66,93,158,165,166). This aligns with the stroke survivor suffering from cognitive deficits such as impaired memory and aphasia in the third research phase of this PhD study, who was dependent on significant other support to use the apps, as well as the stroke survivor and her husband who both had low ICT literacy. The need for increased support for these subgroups is important knowledge when implementing ICT/apps to support stroke rehabilitation. However, in this study we did not find that stroke survivors living alone needed more support than those living with a significant other.

Even though not being an exclusion criterion, stroke survivors suffering from more severe deficits, such as frontal lobe symptoms, were not included in this study. In the scoping review of the first research phase (94) we also found that stroke survivors with more severe deficits are often not included in research. Reasons in this study could be that the stroke survivor with more severe deficits lacked resources to engage in app-supported stroke rehabilitation. However, *Mit Sygehus* as well as person-centred recordings in *Genoptræn.dk* could potentially be valuable for the significant others to support communication with health professionals within and across rehabilitation settings, which was also suggested by therapists in the pre-studies (55). Furthermore, therapists engaged in the third phase of this PhD study indicated during the test phase and in the following post-rehabilitation interviews (which will be reported in a later manuscript) that they saw potential for using especially *Genoptræn.dk* even more to support stroke survivors with severe deficits. They suggested recordings of how to support wheelchair-users when transferring to or from bed, recordings of how to support activities of daily living, such as getting dressed or eating. These recordings subsequently may also serve as cognitive support for the stroke survivor, as well as a communication tool for significant others and other health professionals engaged in the rehabilitation process (e.g., nursing staff). Therefore, in an upcoming project, proceeding this PhD study, called ADICOR (ADvancing ICT interventions to support COherent person-centered and empowering stroke Rehabilitation), the aim is to include also stroke survivors with more severe deficits. We plan to engage one of two highly specialized rehabilitation level in Denmark in the further development and testing of the two apps, for stroke survivors with more severe motor, sensory and/or cognitive deficits (179). The ADICOR project will also have a more explicit focus on support of cross-sectoral goal-setting and on how cognitive rehabilitation can play a greater part of ICT and app-supported stroke rehabilitation. Even though studies on ICT/app-supported cognitive rehabilitation is limited, existing studies show that compensatory strategies to support memory through reminders and calendars are the most used to support stroke related cognitive deficits (90,92,160,180). It is also known that mechanisms such as; education and knowledge on stroke and management; as well as stroke survivor and significant other engagement and collaboration impact other cognitive dimensions such as increased satisfaction and reduced depression (180). *Mit Sygehus* and *Genoptræn.dk* have shown to provide support on education, knowledge and management (through continued access to the content of *Mit Sygehus)*, and through meaningful co-constructed personalized content (with verbal cues from therapists) in *Genoptræn.dk*, thus the apps may have indirectly supported also more cognitive dimensions of the stroke rehabilitation. However, cognitive deficits such as neglect may be difficult to train by using small screens (160).

5.2.2 Using a framework or methodology to guide the development

The MRC´s flexible and pragmatic framework for developing and evaluating complex interventions (1,2) guided the research process. The EBCD methodology proved to be an acceptable methodology to support the systematic development of the apps. Carefully planned activities and actions throughout the EBCD process ensured the real engagement of stroke survivors, their significant others and therapists, thus promoting the development of more acceptable solutions (128).

The MRC framework highlights the iterative stages as important for learning and refinements being made, subsequently resulting in more acceptable solutions being developed (1,2,117). Reflexivity, an iterative approach through an ongoing analysis and learnings throughout the research process align well with the constructivist GT approach (108), as well as with the EBCD methodology (119,137). Being transparent about these reflections and learnings throughout the research process increases the overall trustworthiness of this study (108); therefore, all phases of this PhD project were published, including the reflections and learning during the development and evaluation phases.

5.2.3 Choosing a relevant evaluation design: Process evaluation and program theory

A number of evaluation designs exist for complex interventions, as shown in a systematic review (181). However, among the 105 included evaluation designs, 80% used a process evaluation design, as chosen for this PhD project.

A program theory was developed to support anticipating how the apps would lead to cross-sectoral, person-centred and empowering stroke rehabilitation. As suggested by the MRC (1) and supported by Sekhon (2017) (120), **acceptability** was assessed from the perspectives of the stroke survivors and their significant others to evaluate and understand mechanisms such as ***what***, ***why, how, for whom,*** and under ***what*** circumstances and contexts the intervention works, as well as unexpected consequences (1,2,105,117). Moreover, using person-centred and individual empowerment concepts supported the understanding of individual perspectives on how the apps supported acceptability. The ongoing analysis and smaller refinements made during the test phase subsequently supported the development of more acceptable and transferable solutions (1,108,117).

5.2.3.1 Gathering data: Reflections on the timing of post-rehabilitation interviews

The assessment of stroke survivors' and significant others' acceptability occurred post-rehabilitation, usually around 3–4 months post-stroke. Remembering the entire rehabilitation process may have been challenging in the retrospective interviews. However, by conducting observations during the rehabilitation process and reading the stroke survivors' journals upfront, I

could incorporate this knowledge during the interviews. This approach helped jog the participants' memories and obtain a more in-depth understanding of their experiences.

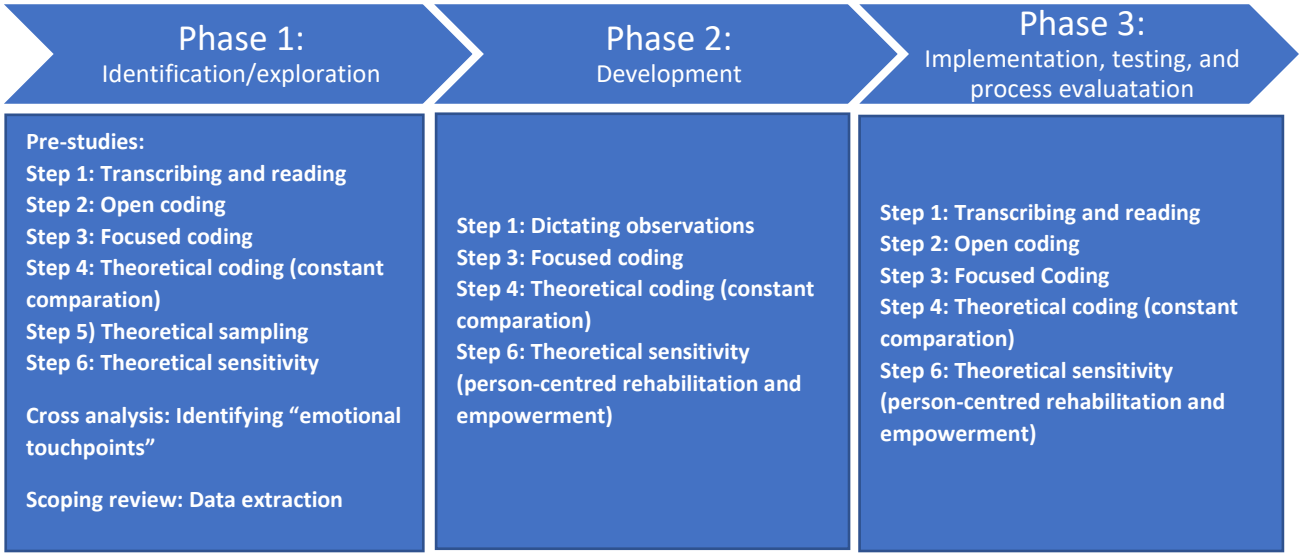
5.2.3.2 Gathering data: Reflections on performing observations

Familiarization with the contexts in which the intervention was to be implemented was a further strength of this study. Visiting the therapists in their "home base" in the different rehabilitation settings, proved to be a strength for positive engagement and collaboration. However, more than one observational and facilitating visit during the testing phase could have strengthened implementation, especially in the acute setting, where observations and facilitating visits were not undertaken.

5.2.3.3 Reflections on the analysis

During the research phases the analytical steps described by Charmaz (2014) and mentioned in section 3.2.1 were used and shown in Figure 11. However, the analysis made during the development phase (the EBCD process) was somewhat different. The "emotional touchpoints", (i.e., sensitizing concepts which should be understood as a point of departure) and theoretical sensitivity (146) from the first research phase supported the development phase and therefore also the analysis of the data. Data from the EBCD process was obtained from observations made in five stroke rehabilitation settings, notes on big flip over papers (according to important headlines derived from the first research phase) and markings made on a self-constructed empowerment model, as well as memos/notes made during the co-design process. The data analysis in the second research phase, thus, were more deductive, however, we tried to stay open for new insights during the process, as suggested by Charmaz (2014) by adding questions (by the moderator and/or on the flip-over papers) that could further enrich the development phase, such as "are there any unmet needs or positive experiences in relation to the rehabilitation process that you could share, which have not already been discussed (i.e., "emotional touchpoints")? The analysis and results were therefore also more descriptive, where focused and theoretical coding (constant comparative method) were used to validate and contrast knowledge from the first research phase and to further understand important mechanisms for experiencing empowerment (*theoretical sensitivity*). This is in line with the understanding of using focused and theoretical coding to support the emerging theory (141,146), which is also mentioned and shown in the beforementioned program theory and logic model (Figure 7).

Figure 11: Analytical steps used in the research phases



In the ongoing analysis of the dyadic and individual interviews in the third research phase, the open coding was kept close to the data and were made in Danish to make quick and spontaneous codes. Focused and theoretical coding followed, through a constant comparative analysis, and supported the following interviews to make the constructed categories more robust. The theoretical concepts of person-centred and empowering rehabilitation (18,29–31,47,52) were then included (*theoretical sensitivity*), to support development of sub-themes describing the construct of ***acceptability*** of *Mit Sygehus* and *Genoptræn.dk* to support person-centred and empowering stroke rehabilitation, from the perspective of stroke survivors and significant others. Reflections (written memos) after each interview and observations during the test phase also supported the analysis and the following construction of data. Comparing, contrasting and (re)organizing the code groups, themes and sub-themes was an iterative process, and throughout the analysis, the findings were discussed and reviewed with my supervisors.

The steps of the analysis and insights into the observational and interview guides were made transparent, which strengthens the study´s trustworthiness (146). Saturation is another critical component within trustworthiness of qualitative research (135,146), to ensure that the participants' diverse and in-depth perspectives and experiences are adequately represented and the findings are grounded in data. In this PhD project, I found that no new themes emerged during the last individual and dyadic interviews. Nevertheless, had we included more participants during the testing and evaluation phases, additional suggestions for improvement may have emerged. Although, 10 participants may seem like a small sample size, it is considered sufficient when testing new interventions since at least 80% of the problems with the apps will be identified. However, had 20 participants been included, this number would have increased to 95% (144).

5.2.3.4 Reflections on choosing relevant outcomes – a discussion of empowerment

One criticism of empowerment evaluations is that some focus on standard patient outcomes, such as disease management and interactions with healthcare providers, rather than also encompassing patients' holistic values. However, empowerment was assessed holistically in this study, by staying open to understanding the diverse mechanisms that were perceived as empowering by stroke survivors and their significant others during data construction. For example, the relational mechanisms stood out more than was first anticipated during the analysis. A further critique is that empowering approaches may result in the transfer of responsibility from health providers to patients and may be used to reduce healthcare costs. This may be problematic and potentially dis-empowering when patients do not have the ability, skills or desire to take on the responsibility for their rehabilitation (41). Questions such as "Should healthcare systems work to empower all patients?", "Is the goal empowerment for all patients?" and "Can all patients manage power and how much power are they capable of obtaining?", have been raised (35).

However, distributing power is only empowering if the stroke survivor is ready and capable of collaborating and accepting this power. If power is shared and transferred regarding elements the stroke survivor is not ready for or does not understand, for example, handling apps independently, the result would be dis-empowerment. In this PhD study, stroke survivors and their significant others expressed being empowered in different ways, some through sufficient access to knowledge and others through support in relation to self-training activities, where the content supplied depended on their needs and goals. Therefore, it is essential to bear in mind that while empowerment is both an educational process and an outcome, it is not a dichotomous variable, in that one is or is not empowered. Instead, empowerment is a continuous variable, where the process and outcomes differ by individual (37). Empowerment is thus a constant collaboration on each element of the rehabilitation process, with attention to the local context (35).

Finally, empowerment-based approaches are sometimes criticized for solely aiming to increase patient "compliance or adherence" (i.e., willingness to follow the instructions of health professionals) (37). However, as shown in this PhD project, PCR and empowerment approaches are more than that. Specifically, engagement, collaboration, and adjustments according to the participants' needs, goals, and resources, seemed to have been accommodated in this PhD study when using the apps to support the rehabilitation process.

5.2.3.5 Reflections on my role as a researcher

It is suggested that the researcher take a personal, critical stance on their own role during the research process (128,140,146). One could be critical of the same researcher who oversaw the identification/exploration and development phases also performing the evaluation. This might be

perceived as a challenge in relation to participants being honest about their perceptions regarding the apps and may affect trustworthiness (151). However, I estimate that I overcame this challenge, by engaging the same therapists in the development and evaluation phases, which seemed to create an ownership of the solutions and an interest in refining the apps to fit their context and daily routines. Regarding the acceptability of stroke survivors and their significant others, I explicitly communicated that the observations conducted during the test phase were aimed at assisting them in case they encountered any difficulties while using the apps. I remained receptive, ensuring that participants understood that their experiences held great value for future stroke survivors and their significant others. This approach seemed to encourage participants to provide open and honest feedback.

The following actions aided in tackling the challenges regarding my simultaneous engagement in developing and evaluating the intervention: providing transparency and reflexivity regarding the constructed data, involving my main supervisor in data construction, and engaging all supervisors in the analysis to obtain external perspectives. Additionally, the publication of all research phases underwent a peer-review process to ensure quality. These actions ultimately enhanced the overall trustworthiness of this PhD project (146,151).

5.2.4 Considering future implementation barriers and facilitators

In this PhD project, implementation took place as part of the testing and evaluation phase. Stroke survivors, significant others and therapists were willing and motivated to co-design and use ICTs and apps to support stroke rehabilitation in this PhD project (12,13,55,136), similar to another study (113).

In this PhD project, it seemed that therapists, especially in the subacute setting, provided sufficient training and education on using *Genoptræn.dk* for stroke survivors to manage the app independently. Health professionals' competency to offer app-supported stroke rehabilitation, that is, education and training, is, considered an important mechanism in successful intervention uptake (113). On the other hand, known barriers to research implementation are lack of time, high caseloads, staff turnover, organizational constraints and poor intervention descriptions (29,182), which often lead to health professionals falling back to old routines (29,100). In this study therapists in acute setting Y did not succeed in including participants and thus in implementing the apps, which could be considered a possible implementation failure. This was unexpected, especially since therapists from all settings had participated through all research phases. Also, we performed additional meetings with setting Y (leaders and therapists) during the test phase, to support the inclusion of participants, which were also expected to lead to inclusion of stroke survivors.

Therefore, identifying barriers for inclusion of relevant stroke survivors in this acute setting, and for using the apps in both acute and municipal settings could have been better. For example, the acute setting could have been more engaged in the smaller workshops in the EBCD development phase. The implementation of *Mit Sygehus* could also have been more stringent across the rehabilitation settings, which should be considered in future research. However, in this PhD study two municipal therapists, one from setting X2 and one from setting Y2, was replaced with two other therapists during the test phase, and one therapist from acute setting Y terminated her involvement in the project early in the test phase, due to other job engagement, and she was not replaced. These staff related changes and turn-overs could potentially also have hindered optimal implementation in these settings.

5.2.4.1 Frameworks to support implementation

The Integrated-Promoting Action on Research Implementation in Health Services framework (I-PARIHS) states that successful implementation is dependent on 1) evidence (i.e., consensus between research-based knowledge, clinical experience, and patient preference), 2) context (i.e., organizational context regarding willingness to implement new knowledge and interventions, learning and competence development culture, and evaluation practices) and 3) facilitation (i.e., education, reminders, audit, feedback, identifying facilitators and barriers of implementation and outcomes (183,184). These elements align well with the phases of the research process and the elements described in the aforementioned program theory. However, using the I-PARIHS framework already in the planning phase and during the test phase might have benefitted this project even more. Retrospectively, a more in-depth discussion with therapists on implementation challenges and facilitators might have further benefitted implementation in the acute and municipal settings.

The I-PARISH framework, like the MRC framework, highlights the importance of understanding the organizational contexts in which the apps were to be implemented (1,183). As mentioned previously, I failed to make observational visits in the acute settings during the test phase. Therapists in the acute rehabilitation setting are mostly focused on the assessment of rehabilitation needs, since the duration is most often only a few days (17,122,123); thus, the implementation of the apps was challenging in this setting. However, setting X succeeded in including and informing participants about the project, instructing the stroke survivor to download the two apps, and overall explaining the purpose of the content of the apps. Setting Y did not implement any of the apps, this step only took place in setting Y1. To facilitate implementation in the acute setting, more insight into barriers to implementation was needed, especially in setting Y, that might have to do with factors other than short hospitalization duration. Also, in-depth discussions with therapists could

support in identifying relevant areas of app-supported rehabilitation for stroke survivors suffering from more severe stroke. As suggested by therapists in the pre-study and mentioned in section 5.2.1.3.1, *Mit Sygehus* and *Genoptræn.dk* may have the potential to support stroke survivors suffering from more severe stroke indirectly; the apps could provide significant others with needed education and knowledge on stroke and rehabilitation, as well as coherence, intervention and communication on how to support the stroke survivor best during activities such as transfer from or to bed or wheelchair, dressing, eating and other every life activities.

5.2.4.2 Frameworks to support the evaluation of implementation

To evaluate implementation, the commonly used Reach, Effectiveness, Adoption, Implementation, and Maintenance **(RE-AIM)** framework could have provided structure for assessing the uptake of the intervention (185,186). However, the RE-AIM framework is more interested in evaluating reach, effectiveness, and fidelity, that is, quantitative measurements, which was not the aim of this project, which had a more processual evaluation purpose. However, RE-AIM literature also suggests that ***acceptability*** should be assessed along with other measures. RE-AIM provides suggestions on how to maximize implementation, that is, developing educational materials to support implementation, which, in this PhD project, was done through written instructions for downloading and using the apps. It is also suggested to appoint local facilitators to lead the implementation of the intervention (147,185,186). However, in this project, only 10 stroke survivors and seven significant others were included in two different stroke trajectories; therefore, it made little sense to appoint local facilitators, and local gatekeepers were appointed instead (in the acute and subacute settings to ensure that eligible participants were identified and asked to participate). This gatekeeper role was managed differently in the two stroke rehabilitation trajectories. During the testing phase, we had a meeting with the two therapists (gatekeepers) and the therapist leaders in setting Y to identify why inclusion was challenging. Additionally, to support participant identification and save time during this process, we suggested that they initially screen participants according to their home municipality rather than initiating with stroke severity. We also decided that inclusion could take place in setting Y1. However, despite these decisions, it is possible that eligible participants were missed in both stroke rehabilitation trajectories.

A final suggestion in the RE-AIM framework to support implementation, is providing feedback to the therapists on their use of the intervention over time, which, in this PhD project, was done by updating all therapists and leaders through the aforementioned monthly newsletters.

5.2.5 Transferability

Because complex interventions are context-dependent, transferability may be challenging (1,108). However, by thoroughly describing the research context in which the intervention was implemented, the included participants, data construction, and the analysis process, the transferability of the findings of this project to other similar contexts was strengthened. Additionally, acquiring multiple participants' perspectives (i.e., stroke survivors, significant others, and therapists), with the stakeholders being different at each stage of the research process, as well as comparing and contrasting the findings with other literature, showed that needs in relation to supporting cross-sectoral, person-centred and empowering stroke rehabilitation are universal. This may further support transferability (101,135,146). Finally, being transparent about my own role throughout the research process and continuously refining the program theory and logic model, support the future implementation and transferability of the intervention across settings (101,102). By developing solutions that accommodate various needs reported by the stakeholders in this and other studies, it is possible that solutions like *Mit Sygehus* and *Genoptræn.dk* could be acceptable if the stroke rehabilitation settings have organizational access to the apps. However, even in a small country like Denmark, several different IT systems exist within the healthcare system. Moreover, the implementation of solutions like *Mit Sygehus* and *Genoptræn.dk* outside the region of Southern Denmark would require their compatibility with the solutions used in other regions.

5.2.6 Economic considerations

All elements of the MRC checklist (1) have been discussed herein apart from the economic element, which was excluded due to the small sample size. This could be seen as a limitation when deciding on larger-scale implementation. An economic evaluation is, however, planned in the ADICOR project that follows this PhD project. Among others, therapists' time resources will be assessed when using apps to support stroke rehabilitation, which was also suggested in a recent systematic review on ICT-based interventions aimed at stroke survivors and their significant others (22).

Chapter 6: Conclusion

The overall aim of this PhD project was to develop, implement, test, and evaluate the co-designed content of two known apps to support cross-sectoral, person-centred and empowering stroke rehabilitation for stroke survivors and their significant others.

The content of the two apps *Mit Sygehus* and *Genoptræn.dk* was co-designed based on stakeholders' needs, prioritization, and research knowledge regarding existing apps. The developed

prototypes of the apps were then implemented, tested and evaluated in 10 stroke survivors and seven significant others throughout cross-sectoral stroke rehabilitation. The iterative and flexible framework of the MRC for developing and evaluating complex interventions overall guided the research phases within this PhD project. Additionally, the EBCD methodology guided the development phase.

Stroke survivors and their significant others found *Mit Sygehus* and *Genoptræn.dk* acceptable to support cross-sectoral, person-centred and empowering stroke rehabilitation. The following intra - and interactional facilitating mechanisms were identified and accommodated: access to free-of-charge apps, access to relevant knowledge, gaining insights into and an overview of the rehabilitation process, support of coherent transfers between rehabilitation settings, person-centred exercises and guidelines using video recordings, support from health professionals to create positive self-training habits, and reduced significant other burden. Additionally, stroke survivors and their significant others felt an ongoing sense of progress through use of the apps, which supported motivation and engagement throughout the rehabilitation process.

However, more stringent implementation of *Mit Sygehus* and *Genoptræn.dk*, especially in the acute setting, as well as using the apps to support cognitive rehabilitation and follow-up support post-rehabilitation needs further exploration.

Chapter 7: Perspectives

7.1 Implications for practice

This PhD study supports clinical practice in stroke rehabilitation by providing an evidence-based app-supported intervention to support cross-sectoral, person-centred and empowering stroke rehabilitation.

The findings of this PhD project highlight the importance of engaging stakeholders throughout the research process to ensure more acceptable person-centred solutions based on stakeholder needs and the specific stroke rehabilitation context. Further this PhD project shows that PCR and empowerment approaches should be applied already in acute and subacute stroke rehabilitation settings and continue on to the stroke survivors' homes, since empowering processes and outcomes achieved within acute and subacute stroke rehabilitation may persist when rehabilitation continue in the municipality/at home (38). This could be important, since a major part of stroke rehabilitation in Denmark takes place in municipalities, thus, being empowered to engage in rehabilitation activities independently and/or supported through apps, in addition to face-to-face delivery, may be important when considering stroke survivors having more long-term needs (8,9,13,21,22,160).

7.2 Implications and early preparations for future research

Content in *Mit Sygehus* and *Genoptræn.dk* were developed to support cross-sectoral, person-centred and empowering stroke rehabilitation in the region of Southern Denmark. However, when evaluating newly developed interventions, it is essential to assess ***acceptability*** of the recipients (i.e., stroke survivors and their significant others) as well as the interventionists (i.e., therapists). Therefore, a natural next step of this research project is to assess therapists' acceptability of the content of the two apps to support cross-sectoral, person-centred and empowering stroke rehabilitation. Three focus group interviews have been conducted post-rehabilitation including all therapists engaged within the implementation, testing, and evaluation phase. The evaluation of therapists' acceptability may further support the understanding of implementation issues and if additional refinements are needed for acceptability in each rehabilitation context. Moreover, therapists' suggestions on the transferability of the apps to other neurological areas within their clinical settings needs to be investigated. These findings will be reported in an upcoming article. Furthermore, future research should focus on facilitating implementation in each stroke rehabilitation setting. Even though the acute setting may seem challenging to perform research in, the relationship between stroke survivors, their significant others and therapists established within the acute setting is essential for PCR (29,38). Furthermore, goal-setting is important throughout stroke rehabilitation, which may also be challenging in the early phases of rehabilitation. However, empowerment is always based on meeting the patients' needs (i.e., person-centered). If needed, the stroke survivor may give the health professional the power to make specific recommendations (37). The health professional must be aware of the power im-balance during rehabilitation and share and collaborate with the stroke survivor and their significant other to make the best use of their resources. This demands good relational competencies from the health professional (33,41). Further, organizational arrangements such as the same health professionals providing rehabilitation supports the development of a stroke survivor-significant other-health professional relationship, continuity, shared knowledge and goal-setting and thus promotes the possibility of providing person-centred and empowering stroke rehabilitation (29,37). These considerations will therefore be investigated when evaluating therapists' acceptability of the apps, as well as in the upcoming research project (ADICOR).

This PhD project mainly identified barriers and facilitators of individual empowerment; however, structural and organizational barriers that affect PCR and individual empowerment should be explored before larger-scale implementation and evaluation (32). The European Patients Forum (EPF) divides empowering systems into 1) a **micro level**, the individual patient and the healthcare professional participating as partners, 2) a **meso level**, the rehabilitation setting, and 3) a **macro level**, i.e., a political level. All these levels can facilitate or hinder the experience of empowerment

(34). In this PhD project, the **micro level** was the focus (i.e., the empowerment of stroke survivors and their significant others). The **meso level** was also investigated through ongoing observations made in each rehabilitation setting. The upcoming article on therapists' acceptability will further add to this level. However, for wider implementation, it could be necessary to engage not only local therapist leaders, like in this PhD project, but also policy and decision makers, since the implementation of these apps requires cross-sectoral agreements and compatible IT systems. Finally, engaging stroke survivors with more severe stroke and engaging more health professionals within the interdisciplinary rehabilitation team (29) to facilitate the implementation of the apps, especially *Mit Sygehus*, needs further investigation.

References

1. Skivington K, Matthews L, Simpson SA, Craig P, Baird J, Blazeby JM, m.fl. Framework for the development and evaluation of complex interventions: gap analysis, workshop and consultation-informed update. Health Technol Assess. september 2021;25(57):1–132.

2. Schultz Petersen K, Maindal HT, Ledderer L, Overgaard C. Komplekse interventioner: Udvikling, test, evaluering og implementering [in english: Complex Interventions: Development, test, evaluation and implementation]. Aalborg Universitetsforlag; 2022.

3. Bate P, Robert G. Experience-based design: from redesigning the system around the patient to co-designing services with the patient. Qual Saf Health Care. 1. oktober 2006;15(5):307–10.

4. Hawkins J, Madden K, Fletcher A, Midgley L, Grant A, Cox G, m.fl. Development of a framework for the co-production and prototyping of public health interventions. BMC Public Health. december 2017;17(1):689.

5. Ciou SH, Hwang YS, Chen CC, Luh JJ, Chen SC, Chen YL. Football APP based on smart phone with FES in drop foot rehabilitation. Technol Health Care. 28. juni 2017;25(3):541–55.

6. Sundhedsstyrelsen. Factsheet: stroke in Denmark [Faktaark: Stroke i Danmark] [Internet]. 2020. Available at: https://www.sst.dk/-/media/Udgivelser/2020/Stroke/Faktaark-stroke.ashx?la=da&hash=5A3D3EF6673A8FE92810AED1827BF527C54826A4

7. Mairey I, Rosenkilde S, Klitgaard MB, Thygesen LC. Sygdomsbyrden i Danmark - sygdomme [the burden of illness in Denmark] [Internet]. Statens Institut for Folkesundhed, Syddansk Universitet. København: Sundhedsstyrelsen; 2022 s. Chapter 15. Available at: file:///C:/Users/mnma/AppData/Local/Microsoft/Windows/INetCache/Content.Outlook/Y52SG9XJ/Sygdomme%20Sygdomsbyrden%202023.pdf

8. Camak DJ. Addressing the burden of stroke caregivers: a literature review. J Clin Nurs. september 2015;24(17–18):2376–82.

9. Sumathipala K, Radcliffe E, Sadler E, Wolfe CD, McKevitt C. Identifying the long-term needs of stroke survivors using the International Classification of Functioning, Disability and Health. Chronic Illn. marts 2012;8(1):31–44.

10. Donnan GA, Fisher M, Macleod M, Davis SM. Stroke. The Lancet. maj 2008;371(9624):1612–23.

11. Wæhrens EE, Winkel A, Jørgensen HS. Neurologi og neurorehabilitering. Kbh.: Munksgaard; 2013.

12. Gustavsson M, Ytterberg C, Nabsen Marwaa M, Tham K, Guidetti S. Experiences of using information and communication technology within the first year after stroke – a grounded theory study. Disabil Rehabil. 15. december 2016;1–8.

13. Marwaa MN, Ytterberg C, Guidetti S. Significant others' perspectives on person-centred information and communication technology in stroke rehabilitation – a grounded theory study. Disabil Rehabil. 16. januar 2019;1–8.

14. Maribo T, Ibsen C, Thuesen J, Nielsen CV, Johansen CV, Vind AB. Hvidbog om rehabilitering. Aarhus, Denmark: Rehabiliteringsforum Danmark; 2022.

15. Sundhed.dk. Apopleksi og TCI [Internet]. 2023. Available at: https://www.sundhed.dk/sundhedsfaglig/laegehaandbogen/hjerte-kar/tilstande-og-sygdomme/apopleksi-og-tia/apopleksi-og-tia-tci/

16. Rigshospitalet. Stroke statistics [Tal og statistik - Hjerneksade som følge af apopleksi] [Internet]. 2020.Available at: https://www.rigshospitalet.dk/afdelinger-og-klinikker/neuro/vnr/viden/statistik/Sider/default.aspx

17. Danish Health Authority. Anbefalinger for tværsektorielle forløb for voksne med erhvervet hjerneskade - apopleksi og transitorisk cerebral iskæmi (TCI) - traume, infektion, tumor, subarachnoidalblødning og encephalopati [Internet]. 2020. Available at: https://www.sst.dk/-/media/Udgivelser/2020/Hjerneskade/Anbefalinger-forloeb-hjerneskade.ashx?la=da&hash=A7A96AC766D6AA68D26F32B96C0015BF828C93FF

18. Wade DT. What is rehabilitation? An empirical investigation leading to an evidence-based description. Clin Rehabil. maj 2020;34(5):571–83.

19. Wade DT. Describing rehabilitation interventions. Clin Rehabil. december 2005;19(8):811–8.

20. Solvang PK, Hanisch H, Reinhardt JD. The rehabilitation research matrix: producing knowledge at micro, meso, and macro levels. Disabil Rehabil. 11. september 2017;39(19):1983–9.

21. Kotzian ST, Haider S, Grabovac I, Schubert N, Josef S. Successful performance of basic gestures in tablet technology in post stroke patients: A feasibility study. Technol Health Care. 7. november 2019;27(6):613–22.

22. Freund M, Carey M, Dilworth S, Waller A, Mansfield E, Rose A, m.fl. Effectiveness of information and communications technology interventions for stroke survivors and their support people: a systematic review. Disabil Rehabil. 14. august 2022;44(17):4563–78.

23. Brouns B, Meesters JJL, Wentink MM, de Kloet AJ, Arwert HJ, Vliet Vlieland TPM, m.fl. Why the uptake of eRehabilitation programs in stroke care is so difficult—a focus group study in the Netherlands. Implement Sci. december 2018;13(1).

24. Dansk Apopleksiregister. Dansk Apopleksiregister. Årsrapport 2021 [Internet]. Report No.: 2022. Available at: https://www.sundhed.dk/content/cms/69/4669_dap_aarsrapport-2021_270622.pdf

25. Platz T. Evidence-Based Guidelines and Clinical Pathways in Stroke Rehabilitation—An International Perspective. Front Neurol. 8. marts 2019;10:200.

26. Krieger T, Floren M, Feron F, Dorant E. Optimising a complex stroke caregiver support programme in practice: a participatory action research study. Educ Action Res. 1. januar 2021;29(1):37–59.

27. Anbefalinger for tværsektorielle forløb for voksne med erhvervet hjerneskade: apopleksi og transitorisk cerebral iskæmi (TCI) - traume, infektion, tumor, subarachnoidalblødning og encephalopati. Sundhedsstyrelsen; 2020.

28. Wade D. Rehabilitation – a new approach. Overview and Part One: the problems. Clin Rehabil. november 2015;29(11):1041–50.

29. Jesus TS, Papadimitriou C, Bright FA, Kayes NM, Pinho CS, Cott CA. Person-Centered Rehabilitation Model: Framing the Concept and Practice of Person-Centered Adult Physical Rehabilitation Based on a Scoping Review and Thematic Analysis of the Literature. Arch Phys Med Rehabil. januar 2022;103(1):106–20.

30. Barr PJ, Scholl I, Bravo P, Faber MJ, Elwyn G, McAllister M. Assessment of Patient Empowerment - A Systematic Review of Measures. Bond K, redaktør. PLOS ONE. 13. maj 2015;10(5):e0126553.

31. Zimmerman MA. Psychological empowerment: Issues and illustrations. Am J Community Psychol. oktober 1995;23(5):581–99.

32. Pulvirenti M, McMillan J, Lawn S. Empowerment, patient centred care and self-management: Empowerment, patient centred care and self-management. Health Expect. juni 2014;17(3):303–10.

33. Region Hovedstaden. Patient Empowerment i Region Hovedstaden 2012 [Internet]. 2012. Available at: https://www.regionh.dk/til-fagfolk/Sundhed/kvalitet/Documents/Patient_empowerment_RegionH.pdf

34. European Patients Forum. Toolkit for Patient Organisations on Patient Empowerment [Internet]. 2017. Available at: https://www.eu-patient.eu/globalassets/library/publications/patient-empowerment---toolkit.pdf

35. Michelsen LP. Patient-empowerment [Internet]. Ugeskrift for læger; 2014. Available at: https://ugeskriftet.dk/videnskab/patient-empowerment

36. Wolfe A. Institute of Medicine Report: Crossing the Quality Chasm: A New Health Care System for the 21st Century. Policy Polit Nurs Pract. august 2001;2(3):233–5.

37. Anderson RM, Funnell MM. Patient empowerment: Myths and misconceptions. Patient Educ Couns. juni 2010;79(3):277–82.

38. Deyhoul N, Vasli P, Rohani C, Shakeri N, Hosseini M. The effect of family-centered empowerment program on the family caregiver burden and the activities of daily living of Iranian patients with stroke: a randomized controlled trial study. Aging Clin Exp Res. juli 2020;32(7):1343–52.

39. Bakas T, Clark PC, Kelly-Hayes M, King RB, Lutz BJ, Miller EL. Evidence for Stroke Family Caregiver and Dyad Interventions: A Statement for Healthcare Professionals From the American Heart Association and American Stroke Association. Stroke. september 2014;45(9):2836–52.

40. Dean SG, Siegert RJ, Taylor WJ, redaktører. Interprofessional rehabilitation: a person-centred approach. Chichester, West Sussex ; Ames, Iowa: Wiley-Blackwell; 2012. 195 s.

41. Hartford W, Lear S, Nimmon L. Stroke survivors' experiences of team support along their recovery continuum. BMC Health Serv Res. december 2019;19(1):723.

42. Martín-Sanz MB, Salazar-de-la-Guerra RM, Cuenca-Zaldivar JN, Salcedo-Perez-Juana M, Garcia-Bravo C, Palacios-Ceña D. Person-centred care in individuals with stroke: a qualitative study using in-depth interviews. Ann Med. 31. december 2022;54(1):2167–80.

43. Castro EM, Van Regenmortel T, Vanhaecht K, Sermeus W, Van Hecke A. Patient empowerment, patient participation and patient-centeredness in hospital care: A concept analysis based on a literature review. Patient Educ Couns. december 2016;99(12):1923–39.

44. Anderson RM, Funnell MM. Patient empowerment: reflections on the challenge of fostering the adoption of a new paradigm. Patient Educ Couns. maj 2005;57(2):153–7.

45. Hjortbak BR, redaktør. Udfordringer til rehabilitering i Danmark. 1. udg. Aarhus: Rehabiliteringsforum Danmark : [Eksp. Marselisborgcentret]; 2011.

46. Kaae Kristensen H, Pallesen H. Personcentreret neurorehabilitering - viden og praksis [Person-centred neurorehabilitation -knowledge and practice]. Gad; 2022. 19–32 s.

47. Cerezo PG, Juvé-Udina ME, Delgado-Hito P. Concepts and measures of patient empowerment: a comprehensive review. Rev Esc Enferm USP. august 2016;50(4):667–74.

48. Fumagalli LP, Radaelli G, Lettieri E, Bertele' P, Masella C. Patient Empowerment and its neighbours: Clarifying the boundaries and their mutual relationships. Health Policy. marts 2015;119(3):384–94.

49. Wildevuur SE, Simonse LWL. Information and communication technology-enabled person-centered care for the "big five" chronic conditions: scoping review. J Med Internet Res. 27. marts 2015;17(3):e77.

50. Kristensen HK, Tistad M, Koch L von, Ytterberg C. The Importance of Patient Involvement in Stroke Rehabilitation. Quinn TJ, redaktør. PLOS ONE. 10. juni 2016;11(6):e0157149.

51. Ekstam L, Johansson U, Guidetti S, Eriksson G, Ytterberg C. The combined perceptions of people with stroke and their carers regarding rehabilitation needs 1 year after stroke: a mixed methods study. BMJ Open. 12. februar 2015;5(2):e006784–e006784.

52. Bergström A, Koch L, Andersson M, Tham K, Eriksson G. Participation in everyday life and life satisfaction in persons with stroke and their caregivers 3-6 months after onset. J Rehabil Med. 2015;47(6):508–15.

53. Rolving N, Gregersen L, Østergaard DS, Nielsen CV, Maribo T. Perspektiver på rehabilitering 2021. Fysioterapeuten. 2021;(6):45–50.

54. Levack WM, Weatherall M, Hay-Smith EJC, Dean SG, McPherson K, Siegert RJ. Goal setting and strategies to enhance goal pursuit for adults with acquired disability participating in rehabilitation. Cochrane Consumers and Communication Group, redaktør. Cochrane Database Syst Rev [Internet]. 20. juli 2015 [henvist 14. februar 2023];2015(7). Available at: http://doi.wiley.com/10.1002/14651858.CD009727.pub2

55. Marwaa MN, Kristensen HK, Guidetti S, Ytterberg C. Physiotherapists' and occupational therapists' perspectives on information and communication technology in stroke rehabilitation. Disabil Rehabil. 2020;

56. Norrving B, Barrick J, Davalos A, Dichgans M, Cordonnier C, Guekht A, m.fl. Action Plan for Stroke in Europe 2018–2030. Eur Stroke J. december 2018;3(4):309–36.

57. Davoody N, Koch S, Krakau I, Hägglund M. Post-discharge stroke patients' information needs as input to proposing patient-centred eHealth services. BMC Med Inform Decis Mak. december 2016;16(1).

58. Zawawi NSM, Aziz NA, Fisher R, Ahmad K, Walker MF. The Unmet Needs of Stroke Survivors and Stroke Caregivers: A Systematic Narrative Review. J Stroke Cerebrovasc Dis. august 2020;29(8):104875.

59. Sundhedsstyrelsen. Sundhedskompetence i det danske sundhedsvæsen. 2022.

60. Bech M, Bonde Klausen M, Buch MS. Tværsektoriel sundhedsforskning i Danmark: aktører, fokusområder og forskningsbehov med udgangspunkt i Region Syddanmark. VIVE - Det Nationale Forsknings- og Analysecenter for Velfærd; 2022.

61. DANSKE REGIONER. EN KLAR RETNING FOR SUNDHEDSFORSKNING I DANMARK [Internet]. 2023. Available at: https://www.regioner.dk/media/23170/udspil-en-klar-retning-for-sundhedsforskning-i-danmark.pdf

62. Iosa M, Morone G, Fusco A, Bragoni M, Coiro P, Multari M, m.fl. Seven Capital Devices for the Future of Stroke Rehabilitation. Stroke Res Treat. 2012;2012:1–9.

63. Jang SH, Jang WH. The effect of a finger training application using a tablet PC in chronic hemiparetic stroke patients. Somatosens Mot Res. 2. april 2016;33(2):124–9.

64. Edgar MC, Monsees S, Rhebergen J, Waring J, Van der Star T, Eng JJ, m.fl. Telerehabilitation in Stroke Recovery: A Survey on Access and Willingness to Use Low-Cost Consumer Technologies. Telemed E-Health. maj 2017;23(5):421–9.

65. Mallet K, Shamloul R, Pugliese M, Power E, Corbett D, Hatcher S, m.fl. RecoverNow: A patient perspective on the delivery of mobile tablet-based stroke rehabilitation in the acute care setting. Int J Stroke. februar 2019;14(2):174–9.

66. Gustavsson M, Ytterberg C, Guidetti S. Exploring future possibilities of using information and communication technology in multidisciplinary rehabilitation after stroke – a grounded theory study. Scand J Occup Ther. 23. september 2019;1–8.

67. Kamwesiga JT, Tham K, Guidetti S. Experiences of using mobile phones in everyday life among persons with stroke and their families in Uganda – a qualitative study. Disabil Rehabil. 27. februar 2017;39(5):438–49.

68. Johansson T, Wild C. Telerehabilitation in stroke care – a systematic review. J Telemed Telecare. januar 2011;17(1):1–6.

69. Pugliese M, Ramsay T, Johnson D, Dowlatshahi D. Mobile tablet-based therapies following stroke: A systematic scoping review of administrative methods and patient experiences. Graetz I, redaktør. PLOS ONE. 23. januar 2018;13(1):e0191566.

70. Kang YN, Shen HN, Lin CY, Elwyn G, Huang SC, Wu TF, m.fl. Does a Mobile app improve patients' knowledge of stroke risk factors and health-related quality of life in patients with stroke? A randomized controlled trial. BMC Med Inform Decis Mak. december 2019;19(1):282.

71. Dubey D, Amritphale A, Sawhney A, Amritphale N, Dubey P, Pandey A. Smart Phone Applications as a Source of Information on Stroke. J Stroke. 2014;16(2):86.

72. Lundberg S. The results from a two-year case study of an information and communication technology support system for family caregivers. Disabil Rehabil Assist Technol. juli 2014;9(4):353–8.

73. Forducey PG, Glueckauf RL, Bergquist TF, Maheu MM, Yutsis M. Telehealth for persons with severe functional disabilities and their caregivers: Facilitating self-care management in the home setting. Psychol Serv. 2012;9(2):144–62.

74. Silveira TM, Tamplin J, Dorsch S, Barlow A. Let's Improvise! iPad-based music therapy with functional electrical stimulation for upper limb stroke rehabilitation. Aust J Music Ther [Internet]. 2018;29. Available at: https:// www.austmta.org.au/journal/article/let%E2%80%99s-improvise-ipad-based-music-therapyfunctional- electrical-stimulation-upper

75. Tistad M, von Koch L, Sjöstrand C, Tham K, Ytterberg C. What aspects of rehabilitation provision contribute to self-reported met needs for rehabilitation one year after stroke - amount, place, operator or timing? Health Expect. september 2013;16(3):e24–35.

76. Uddannelses- og forskningsministeriet. It-understøttet patient empowerment [Internet]. 2013. Available at: https://ufm.dk/publikationer/2013/inno-det-innovative-danmark/inno/modtagede-indspil/afsender/patient-home/it-understottet-patient-empowerment

77. Saywell N, Taylor D. Focus group insights assist trial design for stroke telerehabilitation: a qualitative study. Physiother Theory Pract. 3. april 2015;31(3):160–5.

78. Steiner V, Pierce LL, Windnagel F, Martincin K, Pawlak R, Salvador D. The Impact of a Web-Based Caregiver Intervention on the Reasons Stroke Survivors Use Health Care Services During the First Year Post Treatment. Top Stroke Rehabil. marts 2009;16(2):122–32.

79. Solli H, Bjørk IT, Hvalvik S, Hellesø R. Like an extended family: Relationships that emerge when older caregivers use written messages to communicate in an ICT-based healthcare service. Inform Health Soc Care. 3. april 2018;43(2):207–17.

80. Grau-Pellicer M, Lalanza J, Jovell-Fernández E, Capdevila L. Impact of mHealth technology on adherence to healthy PA after stroke: a randomized study. Top Stroke Rehabil. 2. december 2019;1–15.

81. Kim J, Park S, Lim H, Park G, Kim M, Lee B. Effects of the Combination of Rhythmic Auditory Stimulation and Task-oriented Training on Functional Recovery of Subacute Stroke Patients. J Phys Ther Sci. 2012;24(12):1307–13.

82. Wingham J, Adie K, Turner D, Schofield C, Pritchard C. Participant and caregiver experience of the Nintendo Wii Sports ™ after stroke: qualitative study of the trial of Wii ™ in stroke (TWIST). Clin Rehabil. marts 2015;29(3):295–305.

83. Zonneveld M, Patomella AH, Asaba E, Guidetti S. The use of information and communication technology in healthcare to improve participation in everyday life: a scoping review. Disabil Rehabil. 9. april 2019;1–8.

84. Lindqvist E, Borell L. Computer-based assistive technology and changes in daily living after stroke. Disabil Rehabil Assist Technol. september 2012;7(5):364–71.

85. Allen Christensen K, Lund KM, Thuesen J. Evaluation of Person-Centredness in Rehabilitation for People Living with Dementia Is Needed: A Review of the Literature. J Aging Res. 2. maj 2019;2019:1–9.

86. Choi YH, Park HK, Ahn K hwan, Son Y joo, Paik NJ. A Telescreening Tool to Detect Aphasia in Patients with Stroke. Telemed E-Health. september 2015;21(9):729–34.

87. Paul L, Wyke S, Brewster S, Sattar N, Gill JMR, Alexander G, m.fl. Increasing physical activity in stroke survivors using STARFISH, an interactive mobile phone application: a pilot study. Top Stroke Rehabil. 2. april 2016;23(3):170–7.

88. McKay FH, Cheng C, Wright A, Shill J, Stephens H, Uccellini M. Evaluating mobile phone applications for health behaviour change: A systematic review. J Telemed Telecare. januar 2018;24(1):22–30.

89. Fell N, True HH, Allen B, Harris A, Cho J, Hu Z, m.fl. Functional measurement post-stroke via mobile application and body-worn sensor technology. mHealth. oktober 2019;5:47–47.

90. Zhou X, Du M, Zhou L. Use of mobile applications in post-stroke rehabilitation: a systematic review. Top Stroke Rehabil. 3. oktober 2018;25(7):489–99.

91. Esbjerg kommune. Fremtidens velfærd med ny teknologi [Future welfare with the use of new technology] [Internet]. Available at: http://dinsundhed.esbjergkommune.dk/om-sundhed/politikker-og-handleplaner/strategi-for-velfaerdsteknologi.aspx

92. Ramey L, Osborne C, Kasitinon D, Juengst S. Apps and Mobile Health Technology in Rehabilitation. Phys Med Rehabil Clin N Am. maj 2019;30(2):485–97.

93. Wentink M, van Bodegom-Vos L, Brouns B, Arwert H, Houdijk S, Kewalbansing P, m.fl. How to improve eRehabilitation programs in stroke care? A focus group study to identify requirements of end-users. BMC Med Inform Decis Mak. december 2019;19(1):145.

94. Marwaa MN, Guidetti S, Ytterberg C, Kristensen HK. The use of mobile and web-based applications to support rehabilitation after stroke: a scoping review. J Rehabil Med [Internet]. 17. februar 2022 [henvist 6. april 2022]; Available at: https://medicaljournalssweden.se/jrm/article/view/452

95. Gopal A, Sahyoun G, Stieglitz E, Torous J, Minen M. The Functionality, Evidence and Privacy Issues around Smartphone Apps for the Top Neuropsychiatric Conditions: A Comprehensive Study (P3.9-064). Neurology. 9. april 2019;92(15 Supplement):P3.9-064.

96. Hughes CML, Padilla A, Hintze A, Raymundo TM, Sera M, Weidner S, m.fl. Developing an mHealth app for post-stroke upper limb rehabilitation: Feedback from US and Ethiopian rehabilitation clinicians. Health Informatics J. 30. september 2019;146045821986835.

97. Cogollor JM, Rojo-Lacal J, Hermsdörfer J, Ferre M, Arredondo Waldmeyer MT, Giachritsis C, m.fl. Evolution of Cognitive Rehabilitation After Stroke From Traditional Techniques to Smart and Personalized Home-Based Information and Communication Technology Systems: Literature Review. JMIR Rehabil Assist Technol. 26. marts 2018;5(1):e4.

98. Region Syddanmark. MitSygehus i region Syddanmark [Internet]. 2019. Available at: https://www.regionsyddanmark.dk/wm509838

99. genoptræn.dk. Genoptræn|dk – din digitale træningsmakker [Internet]. 2019. Available at: https://www.syddansksundhedsinnovation.dk/projekter/genoptraen-dk-din-digitale-traeningsmakker/

100. Aadal L, Nielsen TL, Jensen AB, Bjerrum M, Nielsen CV, Angel S. What Does It Take for Research to Be Rehabilitation Research? Front Rehabil Sci. 11. juli 2022;3:823159.

101. Skivington K, Matthews L, Simpson SA, Craig P, Baird J, Blazeby JM, m.fl. A new framework for developing and evaluating complex interventions: update of Medical Research Council guidance. BMJ. 30. september 2021;n2061.

102. Bleijenberg N, de Man-van Ginkel JM, Trappenburg JCA, Ettema RGA, Sino CG, Heim N, m.fl. Increasing value and reducing waste by optimizing the development of complex interventions: Enriching the development phase of the Medical Research Council (MRC) Framework. Int J Nurs Stud. marts 2018;79:86–93.

103. Craig P, Dieppe P, Macintyre S, Michie S, Nazareth I, Petticrew M. Developing and evaluating complex interventions: the new Medical Research Council guidance. BMJ. 29. september 2008;a1655.

104. Slattery P, Saeri AK, Bragge P. Research co-design in health: a rapid overview of reviews. Health Res Policy Syst. december 2020;18(1):17.

105. Funnell SC, Rogers PJ. Purposeful program theory: effective use of theories of change and logic models. 1st ed. San Francisco, CA: Jossey-Bass; 2011. 550 s.

106. Greenhalgh T, Hinton L, Finlay T, Macfarlane A, Fahy N, Clyde B, m.fl. Frameworks for supporting patient and public involvement in research: Systematic review and co-design pilot. Health Expect. august 2019;22(4):785–801.

107. Rieger KL. Discriminating among grounded theory approaches. Nurs Inq. januar 2019;26(1):e12261.

108. Charmaz K. Constructing grounded theory. 2nd edition. London ; Thousand Oaks, Calif: Sage; 2014. 388 s. (Introducing qualitative methods).

109. Perkins DD, Zimmerman MA. Empowerment theory, research, and application. Am J Community Psychol. oktober 1995;23(5):569–79.

110. World Health Organization, WHO Patient Safety. WHO guidelines on hand hygiene in health care . Part V: Patient empowerment and health care. 2009;(WHO/IER/PSP/2009/01):190–2.

111. Kjærgaard H, redaktør. Social- og specialpædagogik - Empowerment, kritisk teori og aktionsforskning. 2. udgave. Kbh.: Akademisk Forlag; 2018. Chapter 4.

112. Eskildsen NB, Joergensen CR, Thomsen TG, Ross L, Dietz SM, Groenvold M, m.fl. Patient empowerment: a systematic review of questionnaires measuring empowerment in cancer patients. Acta Oncol. 1. februar 2017;56(2):156–65.

113. Jonasdottir SK, Thordardottir I, Jonsdottir T. Health professionals' perspective towards challenges and opportunities of telehealth service provision: A scoping review. Int J Med Inf. november 2022;167:104862.

114. Hempel PS, Zhang ZX, Han Y. Team Empowerment and the Organizational Context: Decentralization and the Contrasting Effects of Formalization. J Manag. marts 2012;38(2):475–501.

115. Warner G, Packer T, Villeneuve M, Audulv A, Versnel J. A systematic review of the effectiveness of stroke self-management programs for improving function and participation outcomes: self-management programs for stroke survivors. Disabil Rehabil. 6. november 2015;37(23):2141–63.

116. Herbert RJ, Gagnon AJ, Rennick JE, O'Loughlin JL. A Systematic Review of Questionnaires Measuring Health-Related Empowerment. Res Theory Nurs Pract. maj 2009;23(2):107–32.

117. Moore GF, Audrey S, Barker M, Bond L, Bonell C, Hardeman W, m.fl. Process evaluation of complex interventions: Medical Research Council guidance. BMJ. 19. marts 2015;350(mar19 6):h1258–h1258.

118. Gadke DL, Kratochwill TR, Gettinger M. Incorporating feasibility protocols in intervention research. J Sch Psychol. februar 2021;84:1–18.

119. Donetto S, Pierri P, Tsianakas V, Robert G. Experience-based Co-design and Healthcare Improvement: Realizing Participatory Design in the Public Sector. Des J. juni 2015;18(2):227–48.

120. Sekhon M, Cartwright M, Francis JJ. Acceptability of healthcare interventions: an overview of reviews and development of a theoretical framework. BMC Health Serv Res. december 2017;17(1):88.

121. Al-Ababneh MM. Linking Ontology, Epistemology And Research Methodology. Sci Philos [Internet]. 18. juli 2020 [henvist 19. juli 2023];8(1). Available at: https://doi.org/10.23756/sp.v8i1.500

122. Tistad M, Ytterberg C, Sjöstrand C, Widén Holmqvist L, von Koch L. Shorter Length of Stay in the Stroke Unit: Comparison Between the 1990s and 2000s. Top Stroke Rehabil. marts 2012;19(2):172–81.

123. Stroke sengeafsnit. Aarhus Universitetshospital [Internet]. 2023. Available at: https://www.auh.dk/afdelinger/neurologi/stroke-sengeafsnit/

124. Socialstyrelsen. Rehabilitering (English: Rehabilitation) [Internet]. 2018. Available at: (https://socialstyrelsen.dk/handicap/hjerneskade/om-hjerneskade/rehabilitering)

125. Lyden P. Using the National Institutes of Health Stroke Scale: A Cautionary Tale. Stroke. februar 2017;48(2):513–9.

126. Dansk Apopleksi Register. Dokumentalistrapport - Indikatorer, standarder og prognostiske faktorer [Internet]. 2021. Available at: https://www.rkkp.dk/siteassets/de-kliniske-kvalitetsdatabaser/databaser/apopleksi/ny-apopleksi/dokumentalistrapport-2020--version-1.-februar-2021_2.pdf

127. O'Cathain A, Croot L, Duncan E, Rousseau N, Sworn K, Turner KM, m.fl. Guidance on how to develop complex interventions to improve health and healthcare. BMJ Open. august 2019;9(8):e029954.

128. Bate P, Robert G. Bringing user experience to healthcare improvement: the concepts, methods and practices of experience-based design. Oxford: Radcliffe; 2009.

129. Green T, Bonner A, Teleni L, Bradford N, Purtell L, Douglas C, m.fl. Use and reporting of experience-based codesign studies in the healthcare setting: a systematic review. BMJ Qual Saf. januar 2020;29(1):64–76.

130. Arksey H, O'Malley L. Scoping studies: towards a methodological framework. Int J Soc Res Methodol. februar 2005;8(1):19–32.

131. Levac D, Colquhoun H, O'Brien KK. Scoping studies: advancing the methodology. Implement Sci. 2010;5(1):1–9.

132. Spradley JP. Participant observation. New York: Holt, Rinehart and Winston; 1980. 195 s.

133. Coelho C, de Lima CE. The seven deadly sins and ethnography. Qual Res Organ Manag Int J. 21. maj 2021;16(2):315–31.

134. Green J, Thorogood N. Qualitative methods for health research. 2. ed. London: SAGE; 2009. 304 s. (Introducing qualitative methods).

135. Brinkmann S, Tanggaard L. Kvalitative metoder: en grundbog [Qualitative research: a textbook]. Kbh.: Hans Reitzel; 2015.

136. Marwaa MN, Guidetti S, Ytterberg C, Kristensen HK. Using experience-based co-design to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation. Res Involv Engagem. 24. august 2023;9(1):69.

137. Gustavsson, Andersson T. Patient involvement 2.0: Experience-based co-design supported by action research. Action Res. december 2019;17(4):469–91.

138. Launsø L, Rieper O, Olsen L. Forskning om og med mennesker : forskningstyper og forskningsmetoder i samfundsforskning. 6. udgave. Kbh.: Nyt Nordisk Forlag; 2011. 250 sider.

139. Fylan B, Tomlinson J, Raynor DK, Silcock J. Using experience-based co-design with patients, carers and healthcare professionals to develop theory-based interventions for safer medicines use. Res Soc Adm Pharm. december 2021;17(12):2127–35.

140. Bradbury H, Glenzer K, Apgar M, Embury DC, Friedman V, Kjellström S, m.fl. *Action Research Journal* 's seven quality choicepoints for action oriented research for transformations. Action Res. marts 2020;18(1):3–6.

141. Qureshi HA, Ünlü Z. Beyond the Paradigm Conflicts: A Four-Step Coding Instrument for Grounded Theory. Int J Qual Methods. 1. januar 2020;19:160940692092818.

142. Clarke D, Gombert-Waldron K, Honey S, Cloud G, Harris R, Macdonald A, m.fl. Co-designing organisational improvements and interventions to increase inpatient activity in four stroke units in England: a mixed-methods process evaluation using normalisation process theory. BMJ Open. januar 2021;11(1):e042723.

143. Bowen DJ, Kreuter M, Spring B, Cofta-Woerpel L, Linnan L, Weiner D, m.fl. How We Design Feasibility Studies. Am J Prev Med. maj 2009;36(5):452–7.

144. O'Cathain A, Hoddinott P, Lewin S, Thomas KJ, Young B, Adamson J, m.fl. Maximising the impact of qualitative research in feasibility studies for randomised controlled trials: guidance for researchers. Pilot Feasibility Stud. december 2015;1(1):32.

145. Saunders RP, Evans MH, Joshi P. Developing a Process-Evaluation Plan for Assessing Health Promotion Program Implementation: A How-To Guide. Health Promot Pract. april 2005;6(2):134–47.

146. Charmaz K. Grounded Theory in Global Perspective: Reviews by International Researchers. Qual Inq. november 2014;20(9):1074–84.

147. Anne Marie Lyngsø, Julie Christina Grew, Michaela Schiøtz. Forskning i praksis – erfaringer fra forskning med brugerinddragelse og forskning i samarbejde med kommuner og almen praksis [Internet]. Centre for Clinical Research and Prevention, Bispebjerg and Frederiksberg Hospital; 2020. Available at: https://www.regionh.dk/tfe/videndeling/litteratur/Documents/Forskning%20i%20praksis%20-%20erfaringer%20fra%20forskning%20med%20brugerinddragelse%20og%20forskning%20i%20samarbejde%20med%20kommuner%20og%20almen%20praksis.pdf

148. Gray LJ, Ali M, Lyden PD, Bath PMW. Interconversion of the National Institutes of Health Stroke Scale and Scandinavian Stroke Scale in Acute Stroke. J Stroke Cerebrovasc Dis. november 2009;18(6):466–8.

149. Morgan DL, Ataie J, Carder P, Hoffman K. Introducing Dyadic Interviews as a Method for Collecting Qualitative Data. Qual Health Res. september 2013;23(9):1276–84.

150. Halkier B. Fokusgrupper [Focus groups]. 2. udg. Frederiksberg: Samfundslitteratur; 2012.

151. Patton MQ, Patton MQ. Qualitative research and evaluation methods. 3 ed. Thousand Oaks, Calif: Sage Publications; 2002. 598 s.

152. Danske Fysioterapeuter. Etiske retningslinjer [Internet]. Available at: https://www.fysio.dk/om-os/opbygning/rad-og-udvalg/etisk-rad/etiske-retningslinjer

153. Danske Fysioterapeuter. Politik for forskning [Internet]. Available at: https://www.fysio.dk/politik/politikker/politik-for-forskning

154. Vloothuis J, Mulder M, Nijland RHM, Konijnenbelt M, Mulder H, Hertogh CMPM, m.fl. Caregiver-mediated exercises with e-health support for early supported discharge after stroke (CARE4STROKE): study protocol for a randomized controlled trial. BMC Neurol. december 2015;15(1):193.

155. Nyman A, Zingmark M, Lilja M, Guidetti S. Information and communication technology in home-based rehabilitation – a discussion of possibilities and challenges. Scand J Occup Ther. 2. januar 2023;30(1):14–20.

156. Brouns B, Meesters J, Wentink M, Kloet A, Arwert H, Boyce L, m.fl. Factors associated with willingness to use eRehabilitation after stroke: A cross-sectional study among patients, informal caregivers and healthcare professionals. J Rehabil Med. 2019;51(9):665–74.

157. Epalte K, Tomsone S, Vētra A, Bērziņa G. Patient experience using digital therapy “Vigo” for stroke patient recovery: a qualitative descriptive study. Disabil Rehabil Assist Technol. 6. november 2020;1–10.

158. Czaja SJ. Usability of Technology for Older Adults: Where Are We and Where Do We Need to Be. J Usability Stud. februar 2019;14(2):61–4.

159. Cao W, Kadir AA, Wang Y, Wang J, Dai B, Zheng Y, m.fl. Description of apps targeting stroke patients: A review of apps store. Digit Health. januar 2023;9:20552076231181473.

160. Szeto SG, Wan H, Alavinia M, Dukelow S, MacNeill H. Effect of mobile application types on stroke rehabilitation: a systematic review. J NeuroEngineering Rehabil. 24. januar 2023;20(1):12.

161. Lindberg B, Nilsson C, Zotterman D, Söderberg S, Skär L. Using Information and Communication Technology in Home Care for Communication between Patients, Family Members, and Healthcare Professionals: A Systematic Review. Int J Telemed Appl. 2013;2013:1–31.

162. Bishop D, Miller I, Weiner D, Guilmette T, Mukand J, Feldmann E, m.fl. Family Intervention: Telephone Tracking (FITT): A Pilot Stroke Outcome Study. Top Stroke Rehabil. marts 2014;21(sup1):S63–74.

163. Tangnitipong S, Vuttanon N, Pothiban L, Chintanawat R. Effectiveness of a Home-Based Coping Enhancement Program for Stroke Survivors: A Randomized Controlled Trial. Pac Rim Int J Nurs Res. 12. september 2022;26(4):585–99.

164. Fini NA, Holland AE, Keating J, Simek J, Bernhardt J. How Physically Active Are People Following Stroke? Systematic Review and Quantitative Synthesis. Phys Ther. 1. juli 2017;97(7):707–17.

165. Kim H, Kim GJ. Attitudes and use patterns for mobile technology and upper extremity home exercises in stroke survivors in the United States. Br J Occup Ther. 14. februar 2022;030802262110705.

166. Sureshkumar K, Murthy G, Natarajan S, Naveen C, Goenka S, Kuper H. Evaluation of the feasibility and acceptability of the ‘Care for Stroke’ intervention in India, a smartphone-enabled, carer-supported, educational intervention for management of disability following stroke. BMJ Open. februar 2016;6(2):e009243.

167. Mahmood A, Nayak P, Kok G, English C, Manikandan N, Solomon JM. Factors influencing adherence to home-based exercises among community-dwelling stroke survivors in India: a qualitative study. Eur J Physiother. 2. januar 2021;23(1):48–54.

168. Frost R, Levati S, McClurg D, Brady M, Williams B. What Adherence Measures Should Be Used in Trials of Home-Based Rehabilitation Interventions? A Systematic Review of the Validity, Reliability, and Acceptability of Measures. Arch Phys Med Rehabil. juni 2017;98(6):1241-1256.e45.

169. Emmerson KB, Harding KE, Taylor NF. Providing exercise instructions using multimedia may improve adherence but not patient outcomes: a systematic review and meta-analysis. Clin Rehabil. april 2019;33(4):607–18.

170. Emmerson KB, Harding KE, Lockwood KJ, Taylor NF. Home exercise programs supported by video and automated reminders for patients with stroke: A qualitative analysis. Aust Occup Ther J. juni 2018;65(3):187–97.

171. van Leer E, Connor NP. Use of Portable Digital Media Players Increases Patient Motivation and Practice in Voice Therapy. J Voice. juli 2012;26(4):447–53.

172. Kristensen HK, Peoples H. Evidensbaseret praksis: Forskning, brugerperspektiv, praksisviden. Munksgaard; 2021. chapter 6+7.

173. Munn Z, Peters MDJ, Stern C, Tufanaru C, McArthur A, Aromataris E. Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. BMC Med Res Methodol. december 2018;18(1):143.

174. van Buul LW, Sikkens JJ, van Agtmael MA, Kramer MHH, van der Steen JT, Hertogh CMPM. Participatory action research in antimicrobial stewardship: a novel approach to improving antimicrobial prescribing in hospitals and long-term care facilities. J Antimicrob Chemother. 1. juli 2014;69(7):1734–41.

175. Bombard Y, Baker GR, Orlando E, Fancott C, Bhatia P, Casalino S, m.fl. Engaging patients to improve quality of care: a systematic review. Implement Sci. december 2018;13(1):98.

176. Lindblom S, Flink M, Elf M, Laska AC, von Koch L, Ytterberg C. The manifestation of participation within a co-design process involving patients, significant others and health-care professionals. Health Expect. juni 2021;24(3):905–16.

177. Severinsen K, Ravnborg M. Neurorehabilitering, strategidokument [Internet]. 2015. Available at: https://nnbv.dk/neurorehabilitering/

178. Red hjernen. Reasons for stroke [Hvad er årsagen til et stroke?] [Internet]. Available at: https://redhjernen.dk/hvad-er-et-stroke/stroke/aarsag-til-stroke/

179. Dansk selskab for NeuroRehabilitering. Neuro rehabilitation- how is ot organized? [Neurorehabilitering - hvordan foregår det?] [Internet]. 2023. Available at: https://www.neurorehabilitering.dk/neurorehabilitering/hvordan-foregaar-det/

180. Jørgensen LB, Brunner I. Evidence based recommendations for stroke rehabilitation [Evidensbaserede kliniske anbefalinger for forløb i stroke rehabilitering] [Internet]. 2022. Available at: Evidensbaserede kliniske anbefalinger for forløb i stroke rehabilitering

181. Minary L, Trompette J, Kivits J, Cambon L, Tarquinio C, Alla F. Which design to evaluate complex interventions? Toward a methodological framework through a systematic review. BMC Med Res Methodol. december 2019;19(1):92.

182. Juckett LA, Wengerd LR, Banhos M, Darragh AR. Conducting Implementation Research in Stroke Rehabilitation: A Case Example and Considerations for Study Design. Neurorehabil Neural Repair. december 2022;36(12):770–6.

183. Bergström A, Ehrenberg A, Eldh AC, Graham ID, Gustafsson K, Harvey G, m.fl. The use of the PARIHS framework in implementation research and practice—a citation analysis of the literature. Implement Sci. december 2020;15(1):68.

184. Harvey G, Kitson A. Implementing Evidence-Based Practice In Healthcare: A facilitation guide [Internet]. 1. udg. Harvey G, Kitson A, redaktører. Routledge; 2015 [henvist 15. september 2022]. Available at: https://www.taylorfrancis.com/books/9781136768088

185. Glasgow RE, Estabrooks PE. Pragmatic Applications of RE-AIM for Health Care Initiatives in Community and Clinical Settings. Prev Chronic Dis. 4. januar 2018;15:170271.

186. Gaglio B, Shoup JA, Glasgow RE. The RE-AIM Framework: A Systematic Review of Use Over Time. Am J Public Health. juni 2013;103(6):e38–46.

List of appendices

Appendix A: Links between epistemology, frameworks, theory, methodology and methods used in this PhD project

Appendix B: The empowerment model used in the EBCD process

Appendix C: Information letter for the development phase – first workshop

Appendix D: Information letter for the development phase – second workshop

Appendix E: Consent form to participate in the EBCD process (stroke survivors and significant others)

Appendix F: Instructions on how to download and use Mit Sygehus

Appendix G: Instructions on how to download and use Genoptræn.dk

Appendix H: Information letter for the test phase

Appendix I: Consent form for the test phase (significant others)

Appendix A: Links between epistemology, frameworks, theory, methodology and methods used in this PhD project

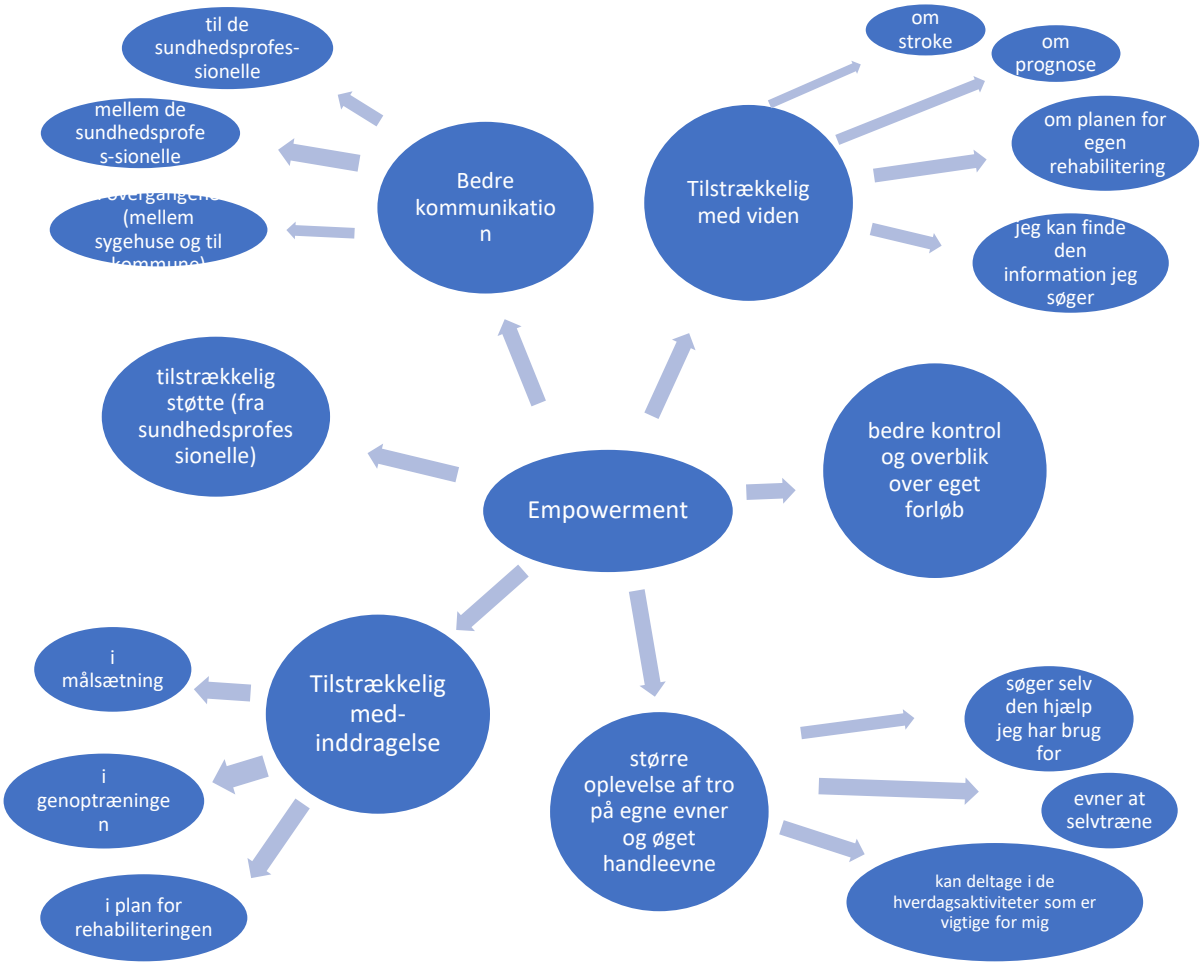
Epistemology, and theoretical philosophy	**Constructivism and constructivist GT** *Knowledge and meaning are actively co-constructed through experiences, interactions, and reflections*
Framework	**MRC** *Identify/explore, develop, implement, test, and evaluate complex interventions*
Theory	**PCR and empowerment** approaches are based on holistic perspectives on stroke rehabilitation by providing the appropriate support and tools according to each *person's individual needs and preferences.* Measurement of PCR and empowerment must be connected to the *persons experiences which are contextually grounded* in their life experiences
Methodology, Methods, and analysis	Article I: Five step approach by Arksey and O´Malley and Levac et al. Descriptive analysis, based on Wades elements of rehabilitation. Article II: Observations, cross-analysis to find "emotional touchpoints" and EBCD using workshops with shared power among participants. Article III: Observations, individual, dyadic and focus group interviews. Constructivist GT analysis.

Appendix B: The empowerment model used in the EBCD process

Empowerment: Hvordan kan vi vurdere om Mit Sygehus har støttet jeres behov i rehabiliteringsprocessen?

Sæt ring omkring de ord nedenfor som er medvirkende til, at I vil opleve jer mere "empowered" i et rehabiliteringsforløb

Navn:



Andet som er vigtigt for mig i rehabiliteringsforløbet for at opleve empowerment:

Hvordan har det været at deltage i workshoppen (din personlige oplevelse, noget du ønsker at tilføje, noget der overraskede dig?). Skriv 2-3 sætninger:

Appendix C: Information letter for the development phase – first workshop

Juni 2021

Information om deltagelse i workshop til borger og pårørende i forskningsstudiet – 2. workshop

Udvikling af en app løsning til at støtte rehabilitering efter hjerneskade (apopleksi)

Vi kontakter dig, fordi du har haft en blodprop eller blødning i hjernen (apopleksi) eller er pårørende til en borger med apopleksi. Vi vil med denne skrivelse spørge, om du vil deltage i et forskningsstudie, hvor vi ønsker at udvikle og tilpasse en app løsning til at støtte rehabilitering *på sygehus og i din egen kommune*. App løsningen hedder "Mit Sygehus" og anvendes i forvejen på andre sygdomsområder i region Syddanmark. Forskningsprojektet er et samarbejde mellem Rehabiliteringsafdelingen på Odense Universitetshospital, Svendborg Neurorehabilitering, Sydvestjysk Sygehus, Sydvestjysk Sygehus Grindsted og en række kommuner i Region Syddanmark.

Vi håber, du og én af dine nærmeste pårørende har lyst til og kan afse tid til at medvirke i vores undersøgelse. Hvis du kan, vil vi bede dig deltage i en workshop. I workshoppen deltager også fysioterapeuter og ergoterapeuter, som arbejder med rehabilitering i forskellige sektorer. Første workshop handler om at se på hvad "Mit Sygehus" bør indeholde så den imødekommer brugernes behov og så den understøtter den eksisterende rehabilitering. Hvis der gives tilsagn til det, vil vi gerne tage nogle billeder undervejs i workshoppen til brug i projektet. Vi vil yderligere gerne bede om journaladgang til informationer omkring dit sygdomsforløb.

Projektet er godkendt efter gældende forskningsetiske regler og er godkendt af Datatilsynet. Såfremt du og dine pårørende giver tilsagn om at medvirke, vil de oplysninger, projektet får kendskab til udelukkende blive behandlet af forskergruppen og af de projektmedarbejdere, vi har tilknyttet, og dine data vil blive anonymiseret i resultatopgørelser. Det er frivilligt at deltage i projektet, og du kan når som helst og uden at give en grund trække dit samtykke tilbage. Vi håber, at du med denne information har fået tilstrækkeligt indblik i, hvad det vil sige at deltage i projektet, og at du føler dig rustet til at tage beslutningen om en eventuel deltagelse. Hvis du vil vide mere om projektet, er du meget velkommen til at kontakte os.

Workshoppen vil foregå den 28. juni kl. 13-15 i kantinen på [redacted]

Ansvarlige kontaktpersoner i projektet

Mille Marwaa

Fysioterapeut, phd studerende

Forskningsenheden for Rehabilitering, SDU

mnma@ucsyd.dk

Tlf.: 72665127



Hanne Kaae Kristensen

Professor

Syddansk universitet

hkristensen@health.sdu.dk



Appendix D: Information letter for the development phase – second workshop

Dec. 2021

Information om deltagelse i workshop til borger og pårørende i forskningsstudiet – 2. workshop

Udvikling af en app løsning til at støtte rehabilitering efter hjerneskade (apopleksi)

Vi kontakter dig, fordi du har haft en blodprop eller blødning i hjernen (apopleksi) eller er pårørende til en borger med apopleksi. Vi vil med denne skrivelse spørge, om du vil deltage i et forskningsstudie, hvor vi ønsker at udvikle og tilpasse en app løsning til at støtte rehabilitering *på sygehus og i din egen kommune*. App løsningen hedder "Mit Sygehus" og anvendes i forvejen på andre sygdomsområder i region Syddanmark. Forskningsprojektet er et samarbejde mellem Rehabiliteringsafdelingen på Odense Universitetshospital, Svendborg Neurorehabilitering, Sydvestjysk Sygehus, Sydvestjysk Sygehus Grindsted og en række kommuner i Region Syddanmark.

Vi håber, du har lyst til og kan afse tid til at medvirke i vores undersøgelse. Hvis du kan, vil vi bede dig deltage i en workshop. I workshoppen deltager også fysioterapeuter og ergoterapeuter, som arbejder med rehabilitering i forskellige sektorer. Derudover deltager personale som arbejder med app-løsningen. Første workshop blev afviklet i juni 2021, og handlede om at arbejde med indholdet i "Mit Sygehus", så den imødekommer brugernes behov og så den understøtter den eksisterende rehabilitering. Næste workshop handler om at se på det indhold som app-løsningen indeholder nu, og drøfte om det stemmer overens med brugernes behov og ønsker, samt at afprøve løsningen i praksis. Hvis der gives tilsagn til det, vil vi gerne tage nogle billeder undervejs i workshoppen til brug i projektet. Vi vil yderligere gerne bede om journaladgang til informationer omkring dit sygdomsforløb i forbindelse med apopleksien.

Projektet er godkendt efter gældende forskningsetiske regler og er godkendt af Datatilsynet. Såfremt du og dine pårørende giver tilsagn om at medvirke, vil de oplysninger, projektet får kendskab til udelukkende blive behandlet af forskergruppen og af de projektmedarbejdere, vi har tilknyttet, og dine data vil blive anonymiseret i resultatopgørelser. Det er frivilligt at deltage i projektet, og du kan når som helst og uden at give en grund trække dit samtykke tilbage. Vi håber, at du med denne information har fået tilstrækkeligt indblik i, hvad det vil sige at deltage i projektet, og at du føler dig rustet til at tage beslutningen om en eventuel deltagelse. Hvis du vil vide mere om projektet, er du meget velkommen til at kontakte os.

Workshoppen vil foregå den 7. december kl. 13-14.30 i kantinen [redacted]

Ansvarlige kontaktpersoner i projektet

Mille Marwaa

Fysioterapeut, phd studerende

Forskningsenheden for Rehabilitering, SDU

mnma@ucsyd.dk

Tlf.: 72665127



Hanne Kaae Kristensen

Professor

Syddansk universitet

hkristensen@health.sdu.dk



Juni 2021

Samtykkeerklæring vedr. deltagelse i workshop (borger/pårørende)

Udvikling af en app løsning til at støtte rehabilitering efter hjerneskade (apopleksi/stroke)

Jeg bekræfter hermed, at jeg har fået skriftlig og mundtlig information, og at jeg ved nok om formål og metode til at sige ja til at deltage i forskningsprojektet.

Jeg ved, at det er frivilligt at deltage, og at jeg altid kan trække mit samtykke om deltagelse tilbage, uden dette vil få nogen konsekvenser for mig. Jeg er informeret om, at jeg er sikret anonymitet i forhold til, at jeg ikke nævnes ved navn eller citeres på en sådan måde at jeg kan genkendes i den videre anvendelse af materialet fra workshoppen og i projektets resultater.

Navn: _____

Sæt et kryds: borger ramt af apopleksi:_____ Pårørende_____

Bopæl (postnummer og by)_____

_____ _____ _____

Dato Underskrift Mailadresse

Må forskerne i projektet kontakte dig telefonisk efter workshoppen, i fald der måtte komme opfølgende spørgsmål?

Sæt et kryds: Ja_____ Nej_____ Tlf. nr.:_____

Ønsker du at deltage i næste workshop:

Sæt et kryds: Ja_____ Nej_____

Samtykke til brug af billeder i workshoppen?

Jeg giver hermed tilladelse til, at der må tages billeder under workshoppen, og at materialet herfra må anvendes i forbindelse med konferencer og præsentation af projektets resultater.

Sæt et kryds: Ja_____ Nej_____

Til borgerne ramt af apopleksi:

Jeg giver hermed tilladelse til, at der må indhentes oplysninger fra min journal i forbindelse med mit sygdomsforløb og rehabilitering såvel på sygehus som i min kommune.

Sæt et kryds: Ja_____ Nej_____

Ansvarlige kontaktpersoner i projektet

Mille Marwaa
Fysioterapeut, phd studerende
Forskningsenheden for Rehabilitering, SDU
mnma@ucsyd.dk / Tlf.: 72665127

Hanne Kaae Kristensen
Professor
SDU
hkristensen@health.sdu.dk

Appendix F: Instructions on how to download and use Mit Sygehus

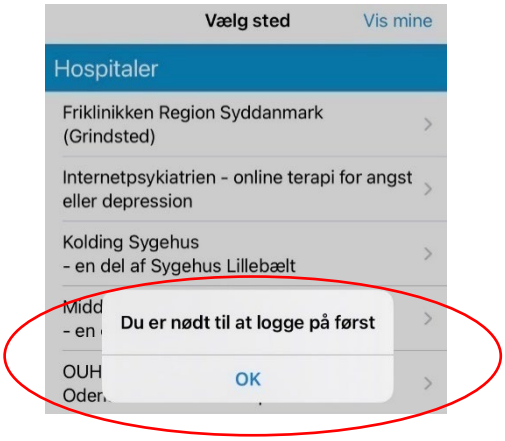
Vejledning til download og anvendelse af Mit Sygehus + installering af læse-højt funktion

1. **Download Mit Sygehus app**
    a. hent **Mit Sygehus** i google play eller app store. Og åben Mit Sygehus

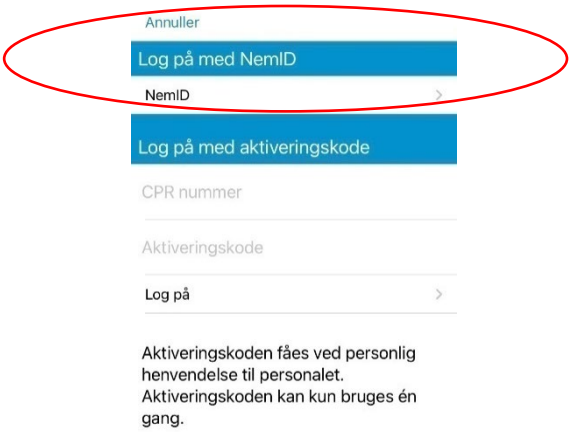


2. **Du kommer frem til "vælg sted". Tryk på "Vis mine" og derefter tryk "OK" når der står "Du er nødt til at logge på først"**





3. **Vælg "log på med NemID"**



4. **Vælg en pinkode til Mit Sygehus**



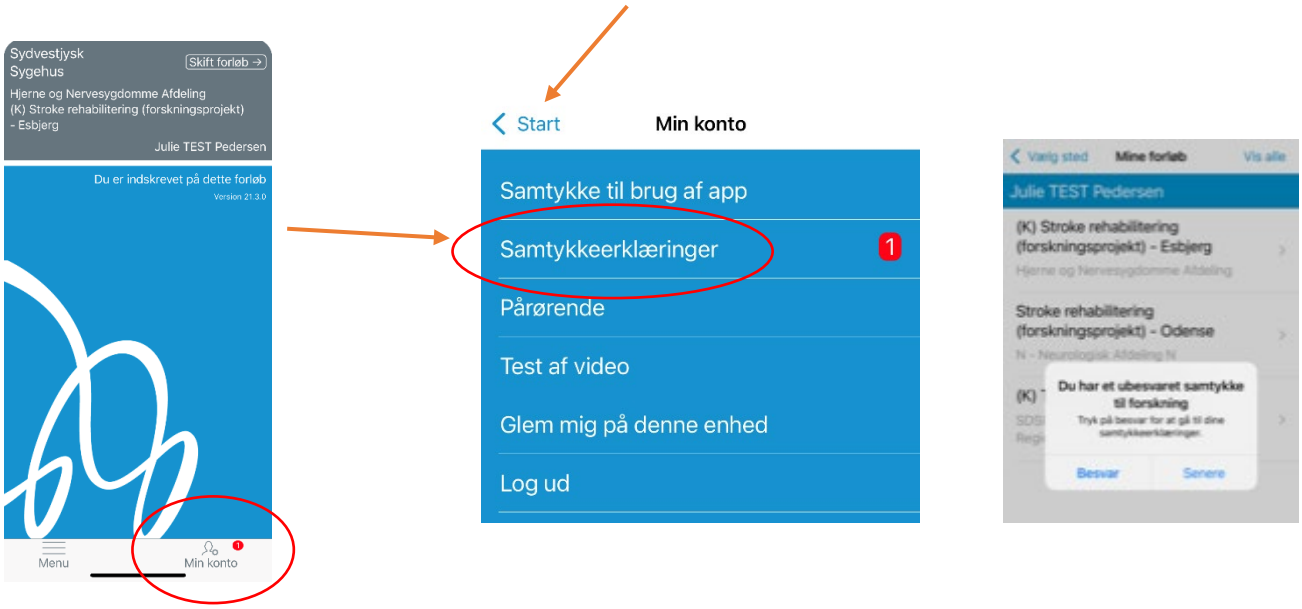
5. **Tryk ”Tillad” til at Mit Sygehus må sende dig notifikationer**



6. **Vælg ”Stroke rehabilitering (forskningsprojekt)- Odense”**



7. **Tryk på dig videre ind i App-løsningen:**
   a. Klik på ”MIN KONTO” (nederst i højre hjørne- OBS kan se anderledes ud på iPad). Læs samtykkeerklæringen og besvar samtykket
   b. Gå tilbage til start siden (tryk start i øverste venstre hjørne)



8. **Gå derefter ind i ”MENU”** (de 3 streger i bunden i venstre side - OBS kan se anderledes ud på iPad)
    a. Her ser du indholdet/alle modulerne i Mit Sygehus
    b. Læs de moduler igennem som er relevante.



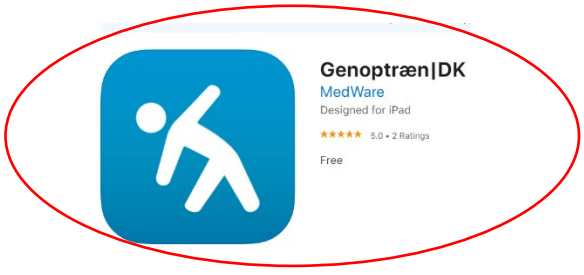


9. **For at få adgang til dine træningsøvelser/program, skal du trykke ind på ”Genoptræning”**



10. **Du vil blive bedt om at installere Genoptræn|DK inden du kan få adgang til dine øvelser/videoer.**



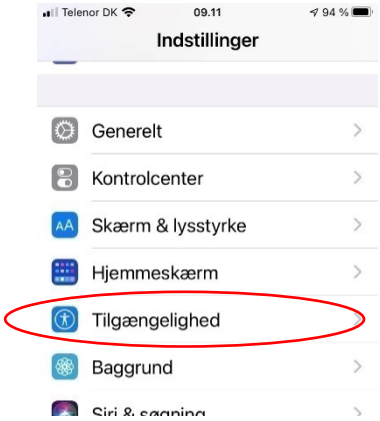


Vejledning til at installere "LÆSE OP" funktionen:
Patient og pårørende kan indstille telefonen, så den læser indholdet op (hele eller dele).

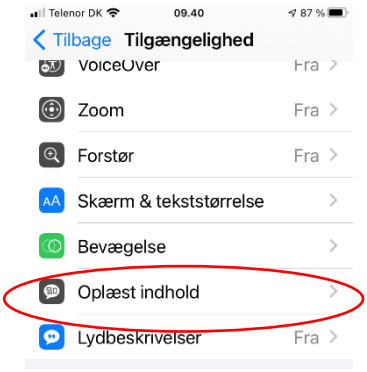
a. Gå i telefonens indstillinger



b. Tryk på "Tilgængelighed"



c. Tryk på "Oplæst indhold"



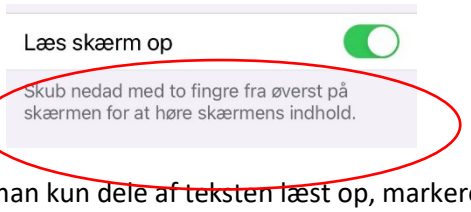
d. Aktivér "Læs det valgte op" og "Læs skærm op"

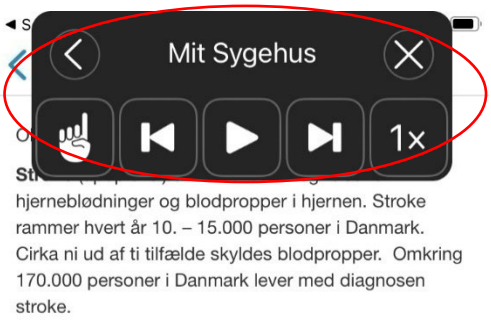


Når patient/pårørende har brug for at få læst noget højt

a. Gå ind i Mit Sygehus igen. Gå ind på det modul du ønsker at få læst højt.

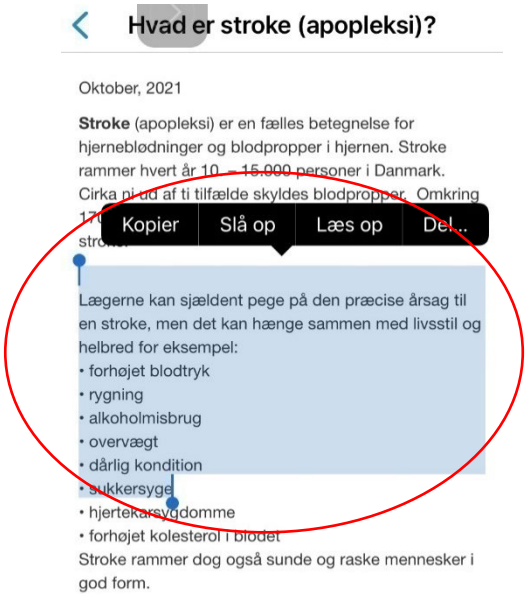
b. Skal hele afsnittet læses højt, kan man stryge 2 fingre fra toppen af skærmen og ned (OBS det kan være svært at ramme rigtigt med de 2 fingre).
Sorte knapper kommer frem. Tryk ”start” -så læses hele afsnittet højt



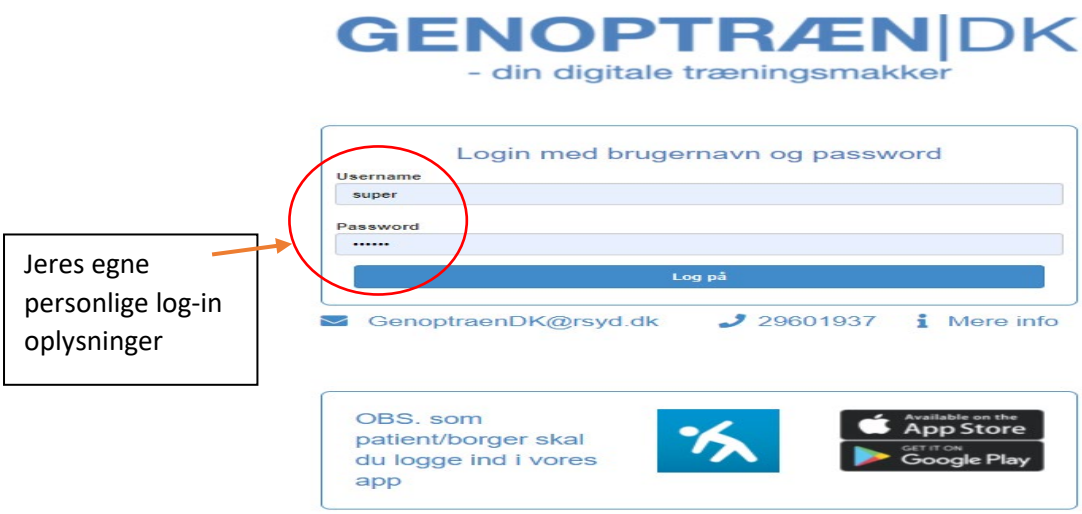


c. Ønsker man kun dele af teksten læst op, markeres det afsr området der skal læses op, højreklik og tryk læs op (og det





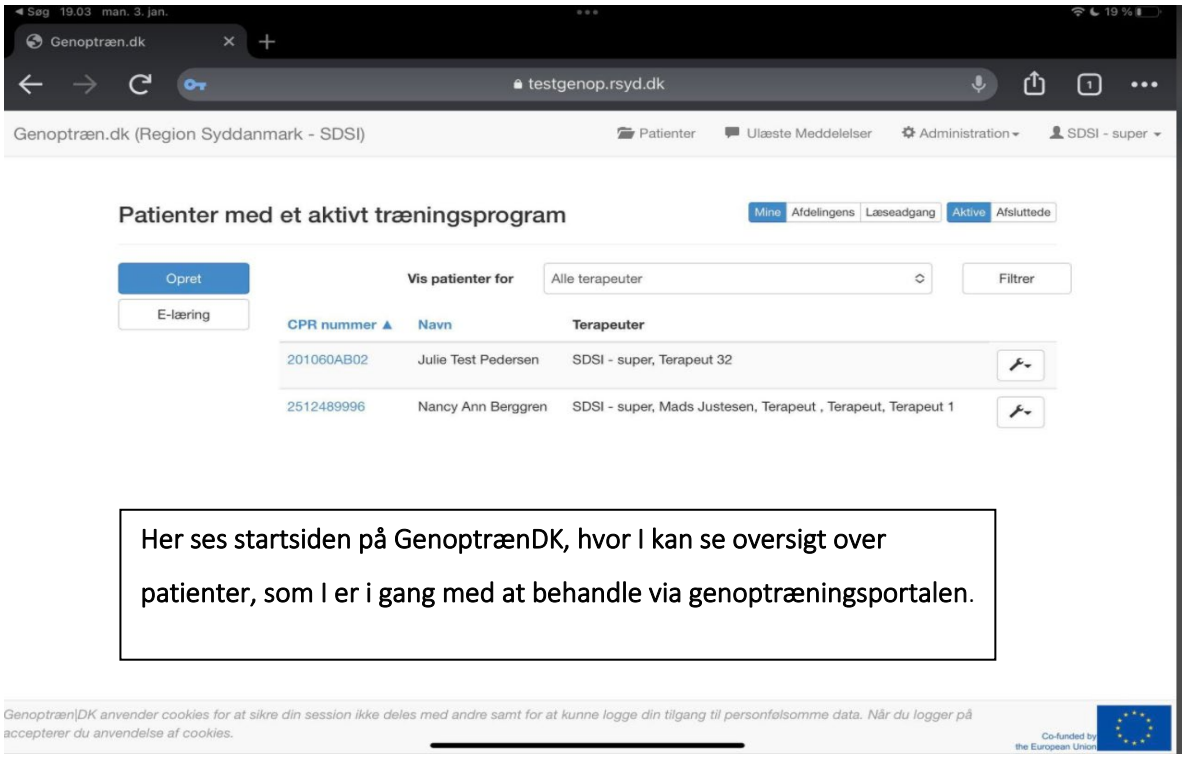
Appendix G: Instructions on how to download and use Genoptræn.dk



Klik jer ind på genoptræn.dk, vi anbefaler, at I udelukkende benytter jer af Google Chrome.

Når I har åbnet siden på Google Chrome, skal I indtaste jeres udleverede log-ind oplysninger.

Forside på Genoptræn|DK



Her ses startsiden på GenoptrænDK, hvor I kan se oversigt over patienter, som I er i gang med at behandle via genoptræningsportalen.



På forsiden kan I ændre jeres adgangskode, dette gør I ved at klikke øverst til højre på jeres navn og skift password.

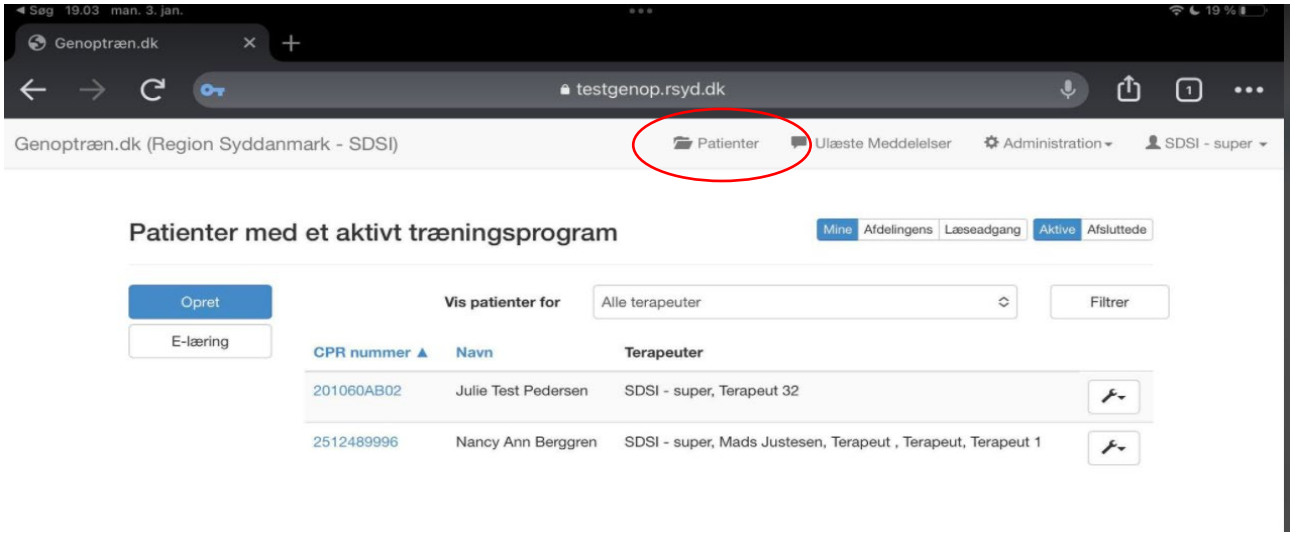
Oprette et træningsprogram: dette gøres ved at klikke på en borgers CPR-nummer, så I kan se, redigere og tilføje træningsprogrammer.

Sådan opretter I en patient

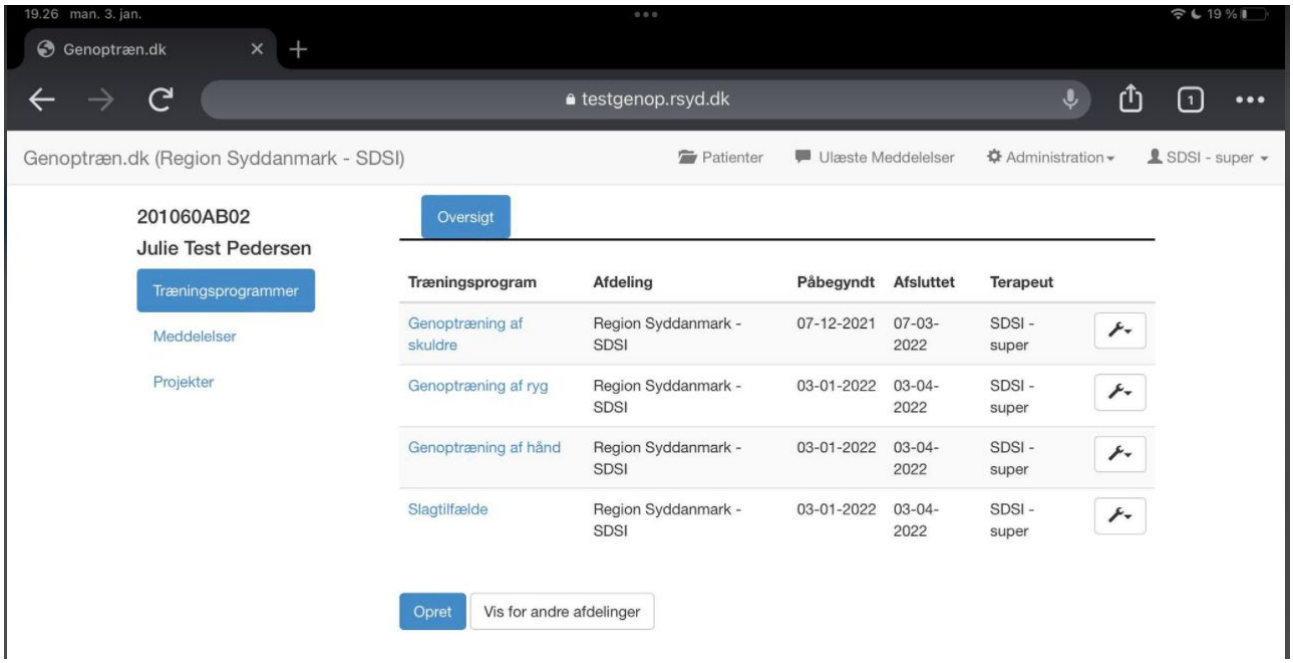


På venstre side af skærmen klikker I på bjælken *Opret* og herefter indtaster i CPR-nummeret på patienten og tilføjer et træningsprogram. Når patienten er oprettet med et træningsprogram på GenoptrænDK, kan patienten logge ind med CPR-nummer inklusiv de sidste 4 sidste cifre som brugernavn og password vil være de sidste 4-cifre af CPR-nummer som password.

Få adgang til træningsprogrammer



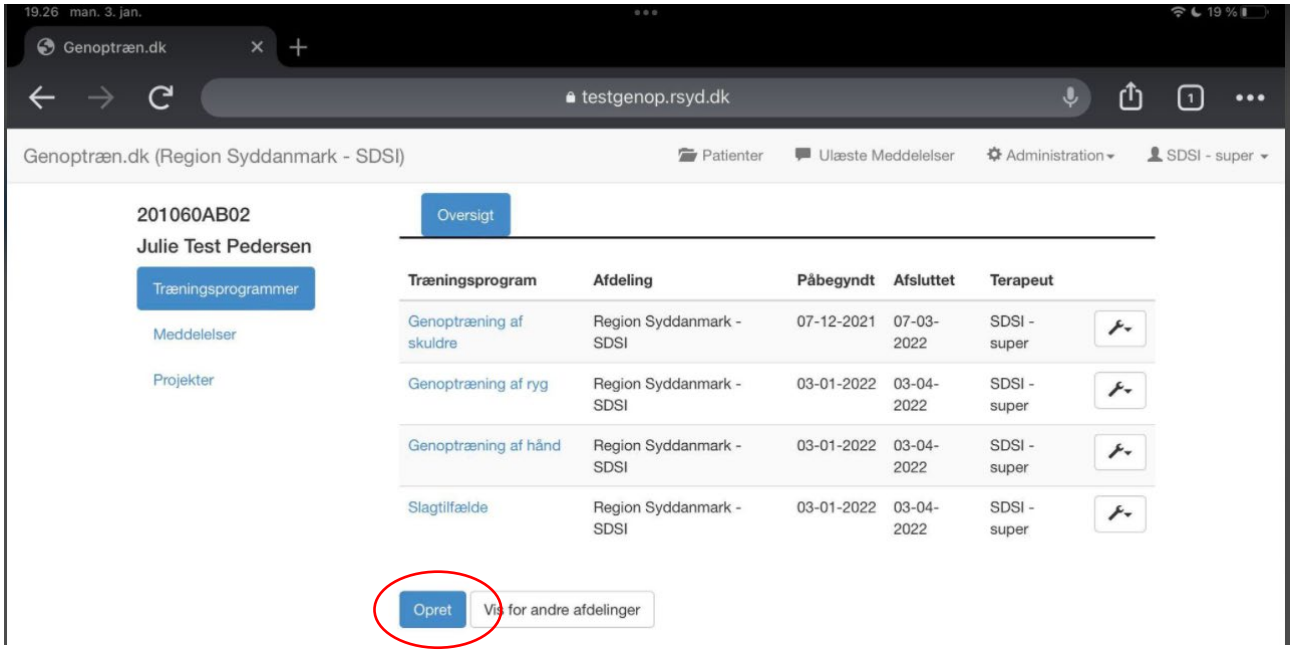
Klik på *patienter* i øverste bjælke, hvor du herefter vil få en oversigt over patienter med CPR-nummer, navn på patient samt hvilke terapeuter har adgang til patienten. Klik på CPR-nummeret for at få adgang til patientens træningsprogrammer.



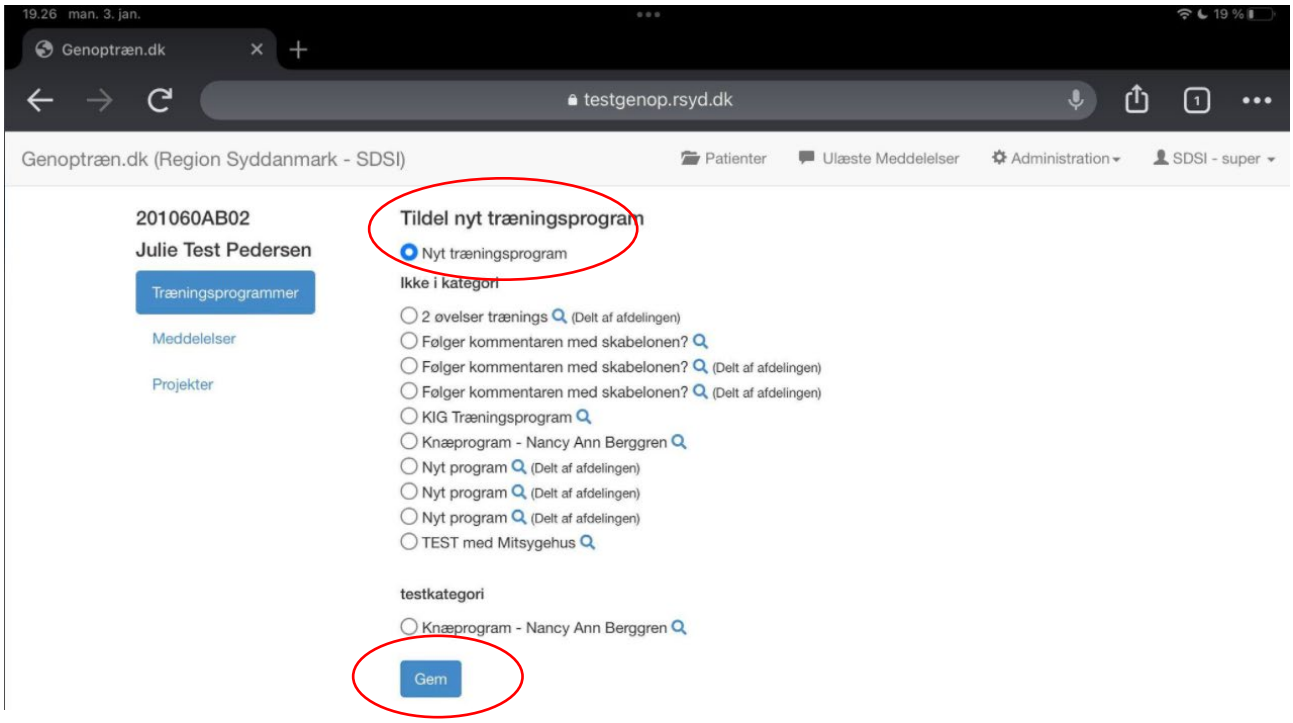
Herefter bliver du mødt af en oversigt over alle træningsprogrammer, afdeling, startdato på træningsprogrammet samt slutdato og hvilken terapeut der arbejder på træningsprogrammet.

Klik på pågældende træningsprogram for at få adgang til det.

Opret et nyt træningsprogram

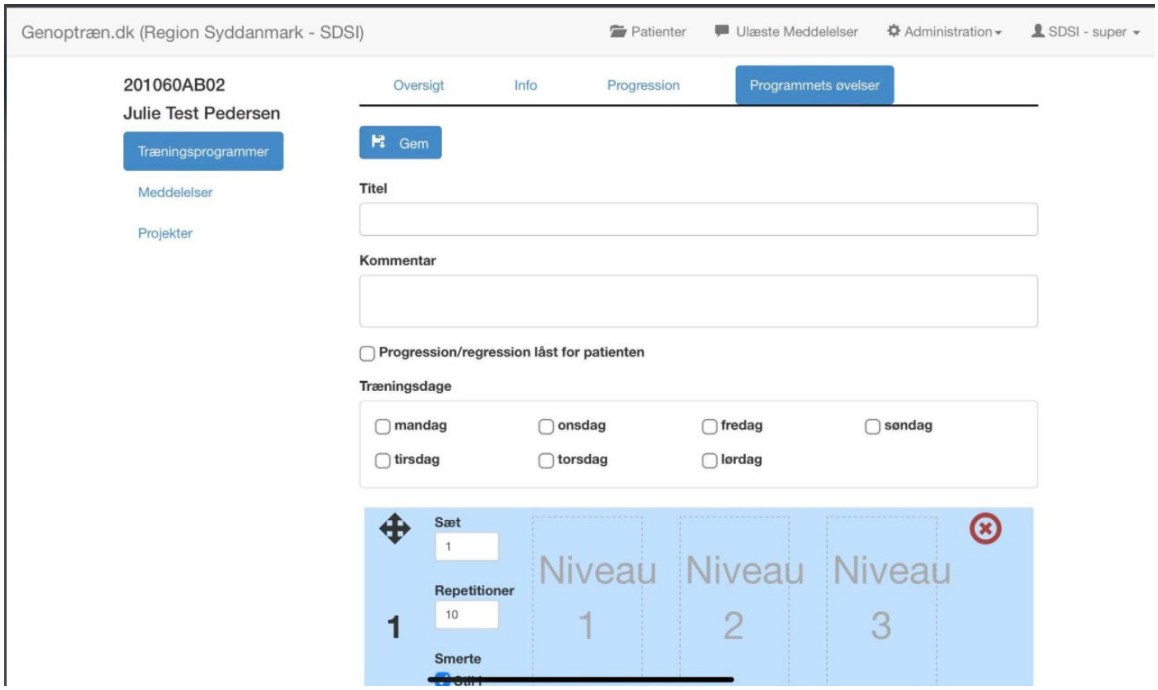


Vi går tilbage til overblikket over træningsprogrammer. For at tilføje et helt nyt træningsprogram klikkes der på den blå knap "opret" under træningsprogrammerne.

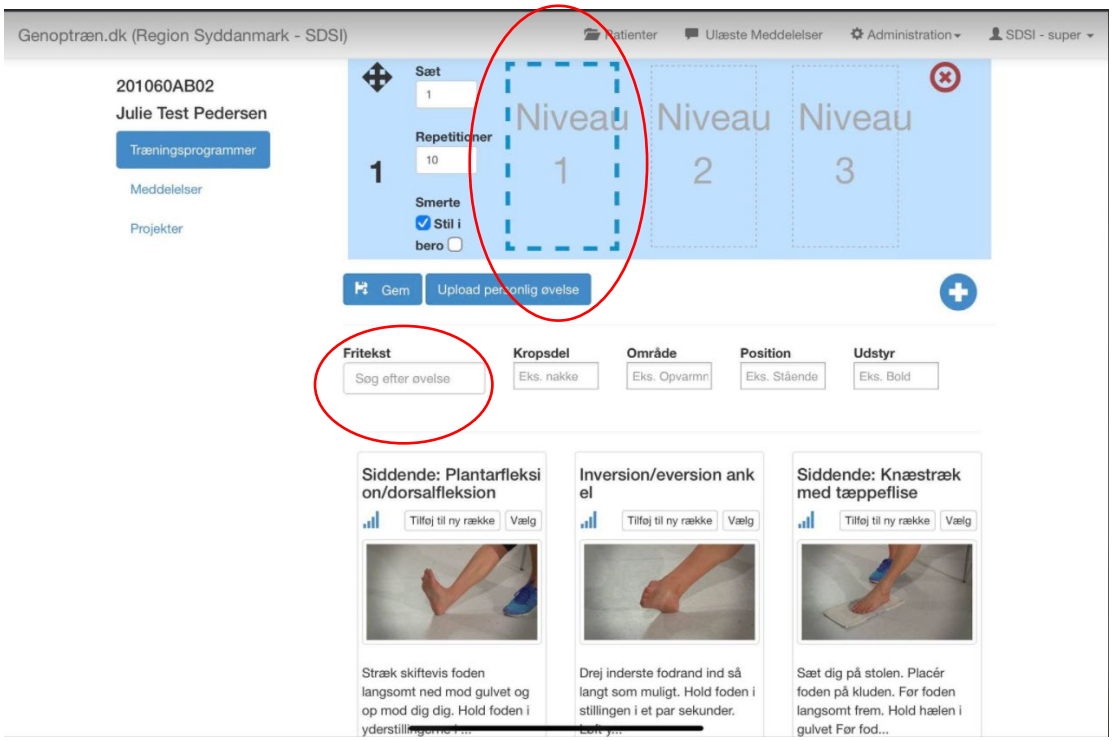


Det første i ser er en lang liste af kategorier med træningsprogrammer, men vi vil gerne oprette et helt nyt træningsprogram.
Klik og marker "*Nyt træningsprogram*" og derefter klikkes der på Gem nederst under listen.

Skabelon på træningsprogram

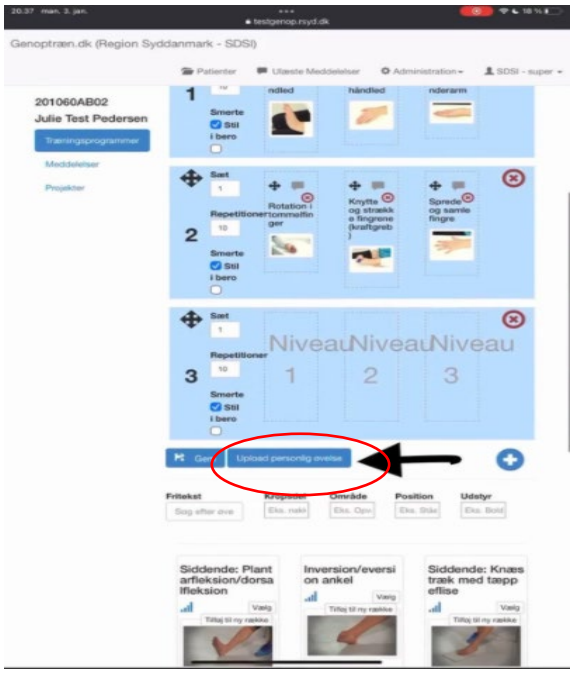


Det næste du ser, er selve skabelon på træningsprogram, hvor vi kan tilføje en titel på træningsprogrammet og efterfølgende en kommentar. Kommentaren kan være alt fra en position, de skal fokusere på under træningsprogrammet eller nogle opmærksomhedspunkter i sammen har talt om, inden påbegyndelse af genoptræningsforløbet. Herefter kan I vælge, hvilke ugedage træningsdagene skal finde sted.

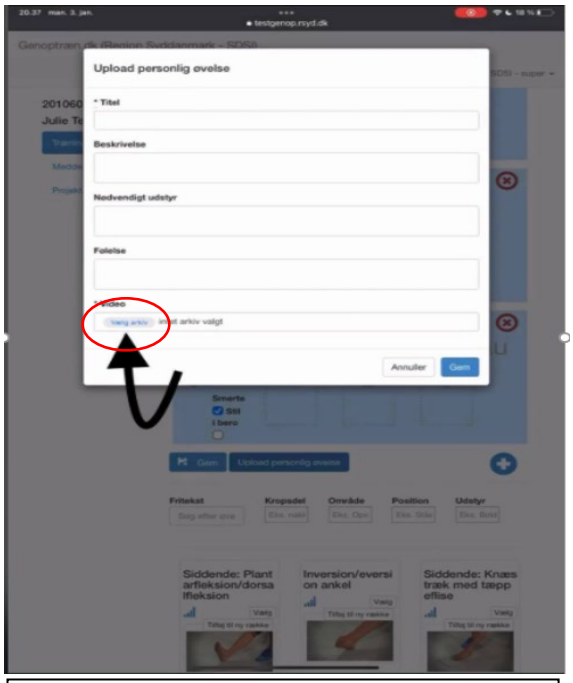


For at tilføje en øvelse til et program, så markerer I første kasse i rækken (niveau 1), når det er markeret korrekt, ser I en stiplet markering omkring. For at finde specifikke genoptræningsvideoer, kan I søge efter en øvelse, kropsdel, område, position eller udstyr. I kan også tilføje en personlig video-optaget øvelse (se næste side).

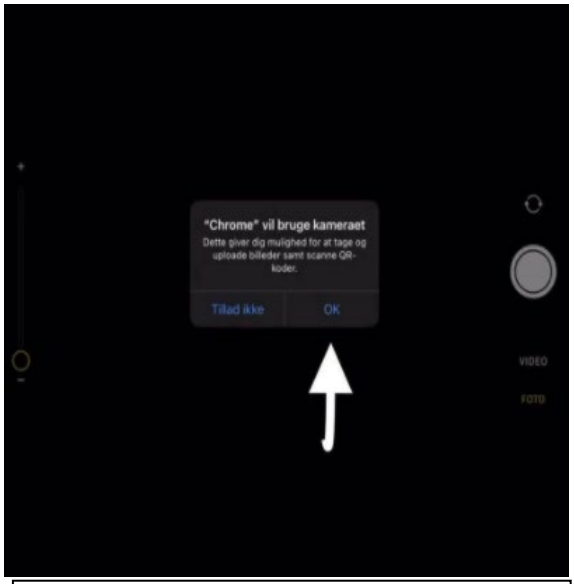
Optag-selv-funktion



For at oprette jeres egen video, klikker I på Upload personlig øvelse

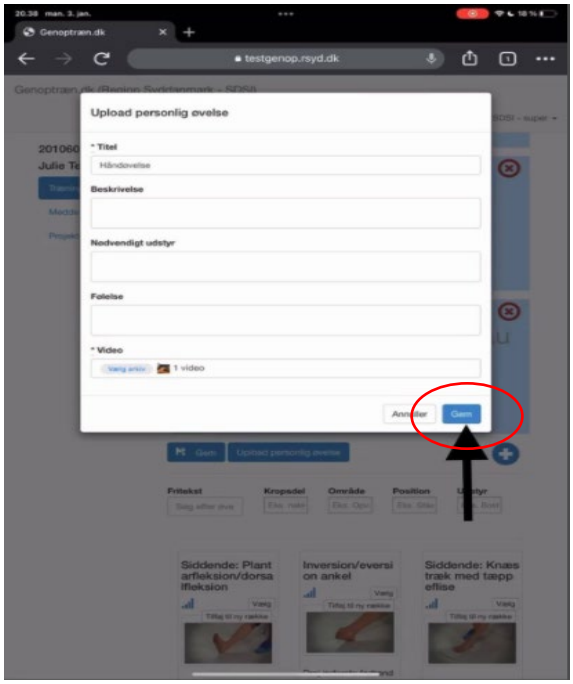


Herefter klikker på vælg arkiv og tag foto eller video

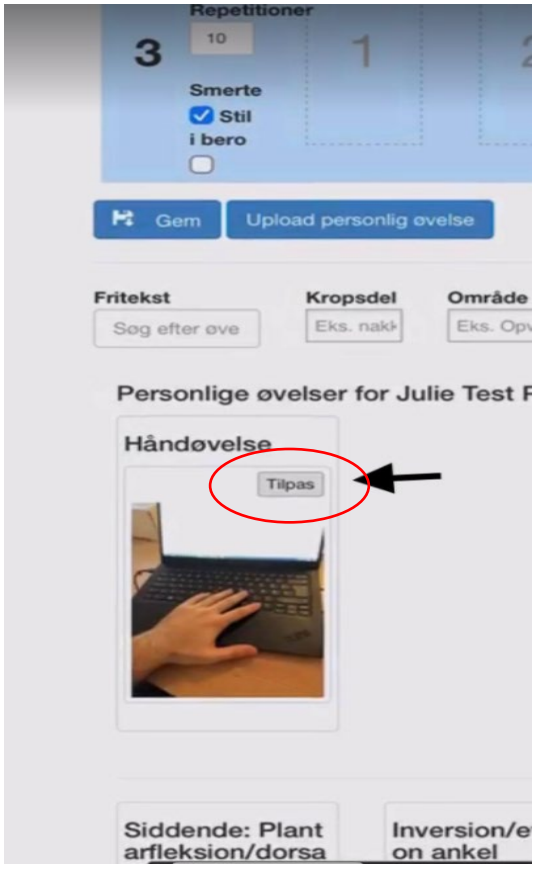


Giv Google Chrome adgang til at benytte kameraet (Dette skal kun gøres en gang).

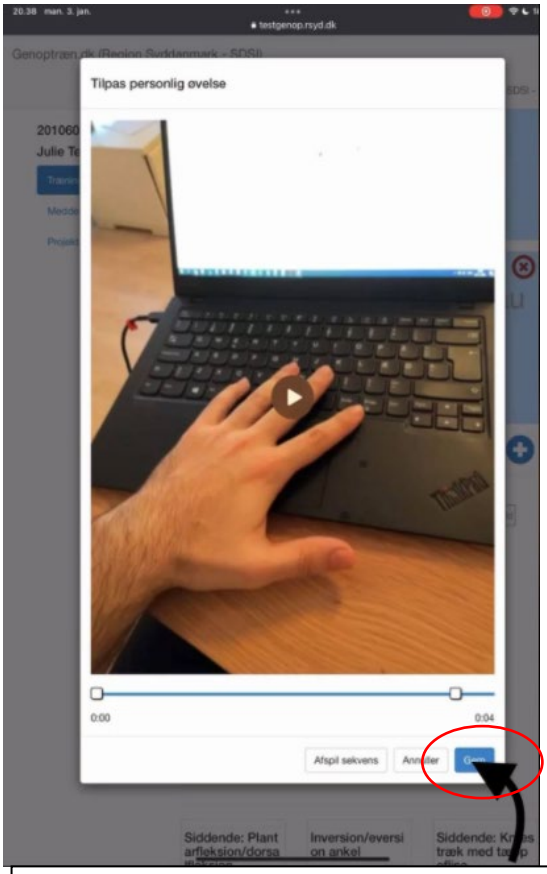
Optag herefter ved at klikke på den røde optag-knap og klik igen, når i er færdige og efterfølgende klik OK



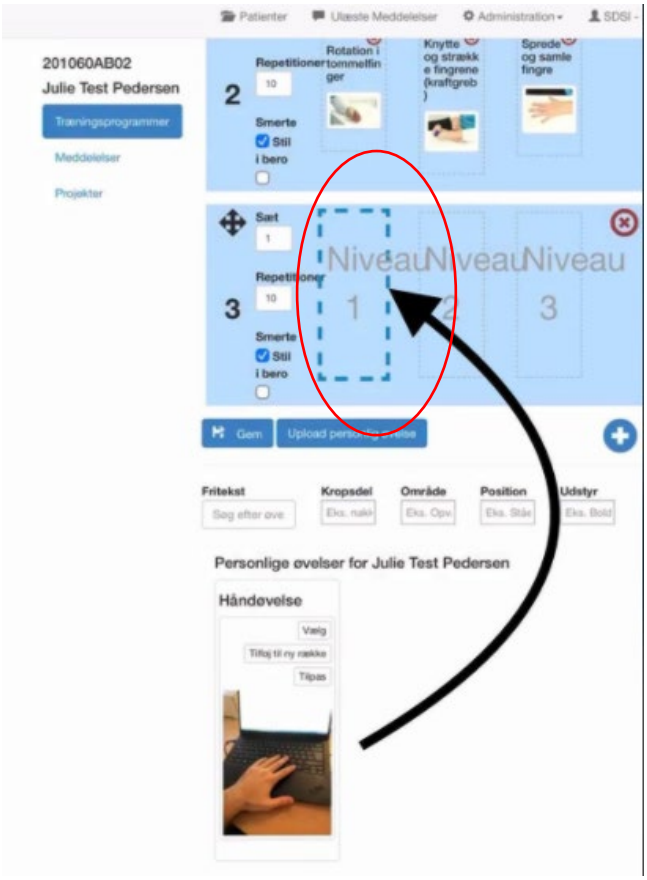
Sørg for at videoen ses i feltet og klik herefter på gem



Nu er videoen tilføjet i serveren, klik herefter på Tilpas.



Her kan du både afspille videoen, forkorte den ved at trække på de to hvide kasser, når du er færdig, klikkes der på Gem.



Nu er I klar til at tilføje jeres egen video i genoptræningsprogrammet. Dette gør I ved at markere Niveau 1 og så klikke på Vælg på jeres egen video.

Se evt. denne video omkring optage-selv funktionen:
https://video.qbrick.com/play2/embed/qbrick-player?accountId=AccjH5gwd9lLEaj5M7owJGgRQ&mediaId=275f9bfd-00090320-ac18d00a&configId=qbrick-player&pageStyling=adaptive&autoplay=true&repeat=true&sharing=true&volume

Support henvendelser

Supportmail: genoptraenDK@rsyd.dk
Support-nummer: 2434 6455

Bisljim Braimi, Syddansk Sundhedsinnovation,
Genoptræn|DK

Appendix H: Information letter for the test phase

Jan 2022

Deltagerinformation til borger og pårørende om forskningsprojektet

Test af app-løsningen (Mit Sygehus) til at støtte rehabilitering efter hjerneskade (Stroke)

Dette informationsbrev er henvendt til borgere ramt af en blodprop eller blødning i hjernen (Stroke) og til pårørende. Terapeuten på sygehuset har spurgt dig, om du vil medvirke til at teste en App-løsning som hedder Mit Sygehus.

Hvad er formålet med projektet?

Formålet med at teste brugen af Mit Sygehus, som supplement til den nuværende rehabilitering, er at give borgere ramt af Stroke og deres pårørende værktøjer og viden til at forbedre deres rehabiliteringsforløb og derved tilpasse behandlingen til deres personlige behov.

Hvad er mit Sygehus?

Indholdet i App-løsningen er indhold udviklet i samarbejde med borgere ramt af Stroke, deres pårørende, terapeuter fra sygehuse og kommuner, samt app-udviklere. Mit Sygehus indeholder aktuelt forskellige moduler, såsom; information om sygdommen, typiske følgevirkninger efter Stroke, information om genoptrænings- og rehabiliteringsforløbet, viden til pårørende, viden om frivillige støtteforeninger, samt et genoptræningsmodul. Genoptræningsmodulet, kaldet Genoptræn|DK, er en app-løsning integreret med Mit Sygehus og har funktioner som er særligt udviklet til borgere ramt af Stroke; at video-optage borgeren fx under håndtering (forflytninger), og/eller under træningsøvelserne, således at det bliver lettere at efterleve instruktionerne.

Hvordan behandler vi dine data?

Såfremt du samtykker til at medvirke i projektet, vil de relevante oplysninger knyttet til projektet blive behandlet fortroligt af forskergruppen og af de projektmedarbejdere vi har tilknyttet. Data vil blive opbevaret og behandlet sikkert på servere i Region Syddanmark og i henhold til den gældende lovgivning. Dine data vil blive anonymiseret i resultatopgørelser og videnskabelige artikler. Du kan læse mere om dine rettigheder i forhold til databehandlingen i bilaget "Dine rettigheder i projektet" længere nede i dette dokument.

Hvordan deltager jeg i projektet?

Det er frivilligt at deltage i projektet, og du kan når som helst og uden at give en grund trække dit samtykke tilbage. Vi håber, at du med denne information har fået tilstrækkeligt indblik i, hvad det vil sige at deltage i projektet, og at du føler dig rustet til at tage beslutningen om en eventuel deltagelse. Hvis du vil vide mere om projektet, er du meget velkommen til at kontakte os.

Ansvarlige kontaktpersoner i projektet

Mille Marwaa

Fysioterapeut, phd studerende

Forskningsenheden for Rehabilitering, SDU

University College Syddanmark (UC Syd)

mnma@ucsyd.dk

Tlf.: 72665127



Hanne Kaae Kristensen

Professor

Syddansk universitet

hkristensen@health.sdu.dk



Bilag: Dine rettigheder i projektet:

Test af app-løsningen (Mit Sygehus) til at støtte rehabilitering efter hjerneskade (Stroke)

Sådan bruger vi dine oplysninger

Oplysningerne behandles efter databeskyttelseslovens § 10, stk. 1-2.

Sletning og opbevaring af dine data

Samtykket til indsamling af dine journaloplysninger bortfalder senest et år efter, det er afgivet, hvorefter du igen vil blive spurgt om du vil give dit samtykke til indsamling af dine journaloplysninger til brug for projektet.

Du gøres endvidere opmærksom på

- Du har ret til at se de oplysninger, der er indsamlet i forbindelse med projektet/databasen samt ret til at få rettet eventuelle forkerte oplysninger
- Du har ret til at få begrænset behandlingen af dine oplysninger
- Du har ret til at klage til Datatilsynet over behandlingen af dine oplysninger.

Yderligere oplysninger

Behandling og opbevaring af personoplysninger sker i overensstemmelse med Databeskyttelsesforordningen og Databeskyttelsesloven. Hvis du har spørgsmål til Region Syddanmarks behandling af dine oplysninger, er du altid velkommen til at kontakte Region Syddanmarks databeskyttelsesrådgiver via din digitale postkasse (borger.dk eller e-boks.dk). Du kan også sende en e-mail til databeskyttelsesraadgiver@rsyd.dk

Januar 2022

Samtykkeerklæring vedr. deltagelse i forskningsprojekt (pårørende)

Afprøvning af app løsninger til at støtte rehabilitering efter stroke (apoleksi)

Jeg bekræfter hermed at have modtaget tilstrækkelig information om forskningsprojektet til at give samtykke til deltagelse. Jeg samtykker også til, at terapeuterne og forskerne knyttet til projektet, kan indhente relevante oplysninger for projektet fra App-løsningerne (Mit Sygehus og Genoptræn|DK).

Jeg ved, at det er frivilligt at deltage, og at jeg altid kan trække mit samtykke om deltagelse tilbage, uden dette vil få nogen konsekvenser for mig. Jeg er informeret om, at jeg er sikret anonymitet i forhold til, at jeg ikke nævnes ved navn eller citeres på en sådan måde at jeg kan genkendes i den videre anvendelse af materialet og i projektets resultater.

Navn: _____

Pårørende til: _____

Bopæl (postnummer og by)_____

_____ _____
Dato Underskrift

_____ _____
Mailadresse Tlf. nr.

Ansvarlige kontaktpersoner i projektet
Mille Marwaa
Fysioterapeut, phd studerende
Kompetence Center for Rehabilitering (KCR),
mnma@ucsyd.dk
Tlf.: 72665127

Hanne Kaae Kristensen
Professor
SDU Syddansk universitet
hkristensen@health.sdu.dk

List of original articles

Article I:

Use of mobile/tablet and web-based applications to support rehabilitation after stroke: A scoping review

Published Jan 13, 2022 in Journal of Rehabilitation Medicine
DOI: 10.2340/jrm.v54.452

Article II

Using Experience-based Co-design to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation

Published in Research Involvement and Engagement at Springer Nature.
DOI: 10.1186/s40900-023-00472-z. August 2023.

Article III

Acceptability of two mobile applications to support cross-sectoral, person-centred and empowering stroke rehabilitation – a process evaluation

Accepted for publication, Annals of Medicine, December 2023
This is an Original Manuscript of an article accepted for publication on the 5th of December 2023 in Annals of Medicine, by Taylor & Francis.
Permission to include the Accepted Manuscript in the thesis was obtained from Taylor and Francis on the 22th of December 2023.

JRM Journal of Rehabilitation Medicine JRM Journal of Rehabilitation Medicine JRM

REVIEW ARTICLE

USE OF MOBILE/TABLET AND WEB-BASED APPLICATIONS TO SUPPORT REHABILITATION AFTER STROKE: A SCOPING REVIEW

Mille Nabsen MARWAA, PT, MSc, PhD student[1,3], Susanne GUIDETTI, OT, PhD professor[2,4], Charlotte YTTERBERG, PT, PhD professor[2–4] and Hanne Kaae KRISTENSEN, OT, PhD professor[3]

From the [1]Department of Physiotherapy Education, University College Southern Denmark, Esbjerg, Denmark, [2]Department of Neurobiology Care Sciences and Society, Karolinska Institutet, Huddinge, Sweden, [3]Research Unit of Rehabilitation, Department of Clinical Research University of Southern Denmark, Odense, Denmark, and [4]Women's Health and Allied Health Professionals Theme, Karolinska University Hospital, Stockholm, Sweden

Objective: **To describe and review evidence of mobile/tablet and web-based applications to support the rehabilitation process after stroke. The secondary aim was to describe participants' stroke severity, and use of applications in relation to, respectively, the setting and phase of the rehabilitation process.**

Methods: **A scoping review methodology was used to identify studies, through databases such as PubMed, CINAHL, Embase and AMED. In addition, grey literature was searched. The studies were categorized according to Wade's model of rehabilitation.**

Results: **The literature search resulted in 12,065 records. Forty-six studies were included, in which applications were used to support: assessment (*n* = 16); training (*n* = 25); discharge from hospital (*n* = 2); and 2 studies were targeted at supporting discharge/education/information and training. One study targeted assessment, discharge support and goal-setting. No studies were related to the element "participation" using Wade's model of rehabilitation. Of the 46 studies, 33 studies included participants with mild to moderate stroke, and 4 studies included participants with severe stroke. In 9 studies the stroke severity was not reported. Twenty-three studies included participants with chronic stroke, 16 acute and/or subacute stroke, and 5 included participants with acute and/or subacute and/or chronic stroke. In 2 studies, stroke onset was not reported. Applications were used in a rehabilitation setting (*n* = 21), home setting (*n* = 17), and both settings (*n* = 3). In 5 studies the setting was not reported.**

Conclusion: **Most studies of applications developed to support the rehabilitation process after stroke have been explorative. They primarily include participants with mild or moderate stroke and focus on a limited aspect of the rehabilitation process, e.g. assessment or training. Future applications to support stroke rehabilitation should accommodate stroke survivors' and significant others' need for solutions, irrespective of stroke severity and throughout the entire rehabilitation process.**

LAY ABSTRACT

Studies have shown that the integration of applications (apps) can support the rehabilitation process after stroke. However, this scoping review found that most existing app solutions have limited usability, providing only 1 aspect of support in the rehabilitation process, such as clinical assessment or a narrow focus on exercises. Furthermore, currently available app solutions mainly target stroke survivors with mild stroke. Stroke survivors and significant others express a need for more comprehensive solutions to support the entire cross-sectorial rehabilitation process. Therefore, there is a need for development of app solutions that support a greater part of the stroke rehabilitation process regardless of stroke severity.

Key words: rehabilitation process; information and communication technology; apps; mobile health; technology support; physiotherapy; occupational therapy; stroke.

Accepted Jan 13, 2022; Epub ahead of print Feb 17, 2022

J Rehabil Med 2022; 54: jrm00269

DOI: 10.2340/jrm.v54.452

Correspondence address: Mille Nabsen Marwaa, Department of Physiotherapy Education, University College Southern Denmark, Esbjerg, Denmark. E-mail: mnma@ucsyd.dk

Globally, stroke is a major cause of disability (1). Furthermore, as a consequence of improved medical care, with decreasing mortality rates and an ageing population, the number of people living with the consequences of stroke is likely to increase (2–4). Although most stroke survivors' recovery takes place largely during the first 3–6 months, the impact of the stroke and the need for rehabilitation may continue for years (2, 5). Long-term consequences, such as physical, cognitive, social and emotional deficits, can result in dependence and reduced participation in everyday life (3, 6, 7). Stroke severity can be categorized as mild, moderate, or severe in relation to cognitive, motor, sensory, and speech-related impairments (8).

Published by Medical Journals Sweden, on behalf of the Foundation for Rehabilitation Information. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 International License (https://creativecommons.org/licenses/by-nc/4.0/)

JRM Journal of Rehabilitation Medicine JRM Journal of Rehabilitation Medicine JRM

Rehabilitation after stroke is important to improve functioning in daily activities and increase participation in everyday life (9). Furthermore, it is recommended that rehabilitation is person-centred, based on the needs and preferences identified by the stroke survivor and their significant others (9–13), and its provision should be based on a partnership between stroke survivor, significant others, and health professionals (9, 14). There is a lack of agreement on how to define rehabilitation (9). However, Wade (15) describes rehabilitation as "an educational, problem-solving process that focuses on activity limitations and aims to optimize patient social participation and well-being, and to reduce stress on family". He emphasizes that rehabilitation should start as early as possible, with a smooth transition between sectors, and that patient involvement is essential (9, 15).Wade (15) includes the following elements in his model of the rehabilitation process: *(i)* identify/screen for activity limitations faced by the patient (assessment: cognitive skills (memory, concentration, speech) and motor skills); *(ii)* establish the patient's goals and expectations, both short-term and long-term, especially regarding level of activity and/or participation (goal-setting); *(iii)* undertake the planned interventions (training: training cognitive skills, motor skills, motivation, and strategies to compensate for deficits); *(iv)* provide any support needed to maintain the patient's well-being, including focus on optimizing the patient's social participation; and *(v)* discharge: structure, planning, communication, rehabilitation plan (9, 15, 16). The 5 elements in this framework of a rehabilitation process will be used to categorize the included studies in this scoping review.

Information and communication technologies and mobile apps

Worldwide, information and communication technology (ICT), including mobile phones, tablets and computers are increasingly accessible (17), and society's demand for ICT is also increasing rapidly (18, 19). A myriad of mobile-based and web-based applications (apps), available on smartphones, tablets and computers allow daily activities to be done in novel ways, including reading the newspaper, playing games, and communicating with family and friends (18, 20, 21). Studies have also shown that the integration of apps can support the stroke rehabilitation process, e.g. to deliver health information to patients (22, 23), and to support stroke survivors in independently performing exercises post-discharge (24), thereby promoting their motor abilities (25). By integrating apps in rehabilitation, continuity can be enhanced (12, 26) and stroke survivors are supported to become more active in managing their own health through increased patient motivation, engagement and empowerment (22, 23, 27–31).

Although access to various app solutions has increased, knowledge regarding mobile/tablet apps to support rehabilitation after stroke is limited (5, 29, 31), including which elements, and in which phase and setting of the rehabilitation process the apps are used. Thus, identifying knowledge gaps is important to indicate directions for future development. The aim of this study was therefore to describe and review evidence of mobile/tablet and web-based applications being used to support the rehabilitation process after stroke. The secondary aim was to describe participants' stroke severity, and use of applications in relation to, respectively, the setting and phase of the rehabilitation process.

METHODS

A scoping review design was used to map existing evidence of the use of mobile/tablet and web-based apps in stroke rehabilitation. Scoping reviews include a range of study designs in both published scientific literature and grey literature and are ideal when the aim is coverage of a body of literature on a given topic, beyond those related to intervention effectiveness (32, 33). A scoping review study is less likely to assess the quality of included studies; whereas a systematic review focuses on a well-defined question and a relatively narrow range of quality assessed studies are included (32–34). A scoping review is thereby useful to guide the process of exploring and mapping the body of existing literature regarding apps used to support stroke rehabilitation (33–35).

This scoping review followed the stages, as described by the methodological framework devised by Arksey & O'Malley (34), and Levac et al. (32). The framework consists of 5 stages: *(i)* identifying the research question; *(ii)* searching for relevant studies; *(iii)* selecting studies; *(iv)* charting the data; and *(v)* collating, summarizing, and reporting the results (32, 34).

Stage 1: identifying the research question

First, a research question that was broad, but still specific enough to answer through a scoping review, was identified (32, 34): "What evidence is available regarding the use of mobile/tablet and web-based apps to support stroke rehabilitation, in which phase and setting of the rehabilitation process are the apps used, and what is the stroke severity of the participants?" These questions addressed the research gap that was discovered from initial searches and had clinical relevance. In line with recommendations made by Levac et al. (32), it was essential to define the concept of "rehabilitation", to establish a clear scope for the study,

JRM Journal of Rehabilitation Medicine JRM Journal of Rehabilitation Medicine JRM

help guide the search strategy and establish parameters around study selection in subsequent stages. Therefore, "rehabilitation" was defined using Wade's model (15), as described earlier.

Stage 2: searching for relevant studies

In the literature search the following electronic bibliographic databases were used: PubMed, CINAHL, Embase and AMED. In addition, grey literature was searched in the following databases and webpages; OpenGrey, Google Scholar, SveMed, Fysio.dk, Den Danske Forskningsdatabase, and bibliotek.dk. The search was conducted between October 2019 and March 2020, with a 10-year restriction to publication date (March 2010 to March 2020) (36). An updated search was performed in November 2021 and January 2022. The search terms and search strategy were customized for each database to ensure precision and accuracy. The search strategy included thesaurus terms (MeSH terms and subject headings) combined with free-text words. Examples of main search terms used were apps, mobile app, health app, mobile health app, smart phone app, mobile application, rehabilitation, stroke, apoplexy and therapeutics. To maximize the search results, multiple sets of search terms were used. The search was performed until an overlap in the identified studies was observed. The literature search was performed with support from an experienced research librarian (see supplementary material). All studies retrieved from the database searches were imported into an Excel spreadsheet.

Stage 3: selecting studies

The inclusion criteria for this scoping review were: *(i)* mobile/tablet and web-based app solutions used to support stroke rehabilitation; *(ii)* studies including stroke survivors; *(iii)* studies published between 2010 and 2021 (36); and *(iv)* studies written in the English language. Primary studies were included, e.g. randomized controlled trials (RCTs), case studies, pilot studies, feasibility studies; thus, systematic reviews were excluded, however, systematic reviews informed the background and discussion section. Criteria for exclusion were app solutions with a primary focus on pharmacology/medications, robot technology, virtual reality or other not easily accessible hardware, electromyography, brain surgery or diagnosis. Furthermore, studies were excluded if they only introduced the interface or internal structure of the apps (e.g. study protocols) or if studies contained several app solutions without defining them. Title and abstract screening were performed independently by 2 authors (MM and HK) and a research assistant using the Excel spreadsheet mentioned earlier. Full-text screening was performed by the first and last author. In case of disagreements, consensus was reached though discussion (32). A flow-chart of the search strategy and selection process is shown in Fig. 1 (39).

Stage 4: charting the data

Two authors (MM and HK) collectively determined which variables to extract from the included articles, with a view to answering the research question. This stage was an iterative process in which the authors continuously extracted data from the included articles and updated the data-charting form (32). The following variables were chosen: *(i)* author, year, country of publication; *(ii)* study design; *(iii)* objective/aim; *(iv)* participants (numbers, stroke onset, stroke severity); *(v)* target of the specific app solution (using Wade's model of rehabilitation); *(vi)* context (rehabilitation setting and duration of the intervention); and *(vii)* key findings. These are shown in Tables I–II. Two authors (MM and HK) independently extracted data from the first 5 included studies using a data extraction form (see Table II) and subsequently discussed and determined whether the approach to data extraction was consistent with the research question and aim (32).

Stage 5: collating, summarizing and reporting the results

This stage had 2 steps. First, a descriptive numerical summary of the variables mentioned in stage 4 was conducted. Secondly, a manifest qualitative content analysis with a deductive approach was performed to find common elements across the dataset and answer the research question (37). By applying the method content analysis, the researchers could include large amounts of text and systematically identify its properties, such as phase and setting of rehabilitation (rehabilitation centre or at-home rehabilitation), stroke severity (mild, moderate, severe), stroke onset of the included participants (acute: 0–24 h after stroke; subacute: 1–90 days after stroke; chronic stroke: 3 months after stroke) (38), and categorize the apps in the studies in relation to Wade's model of rehabilitation (15), thus providing an overview of which elements of the rehabilitation process the app solutions were intended to support.

RESULTS

The initial literature search resulted in 10,142 records, of which 65 were duplicates. Applying the inclusion criteria, all titles were initially screened for relevance to the research question by the first and last author (MM and HK) and a research assistant, and non-relevant titles were removed.

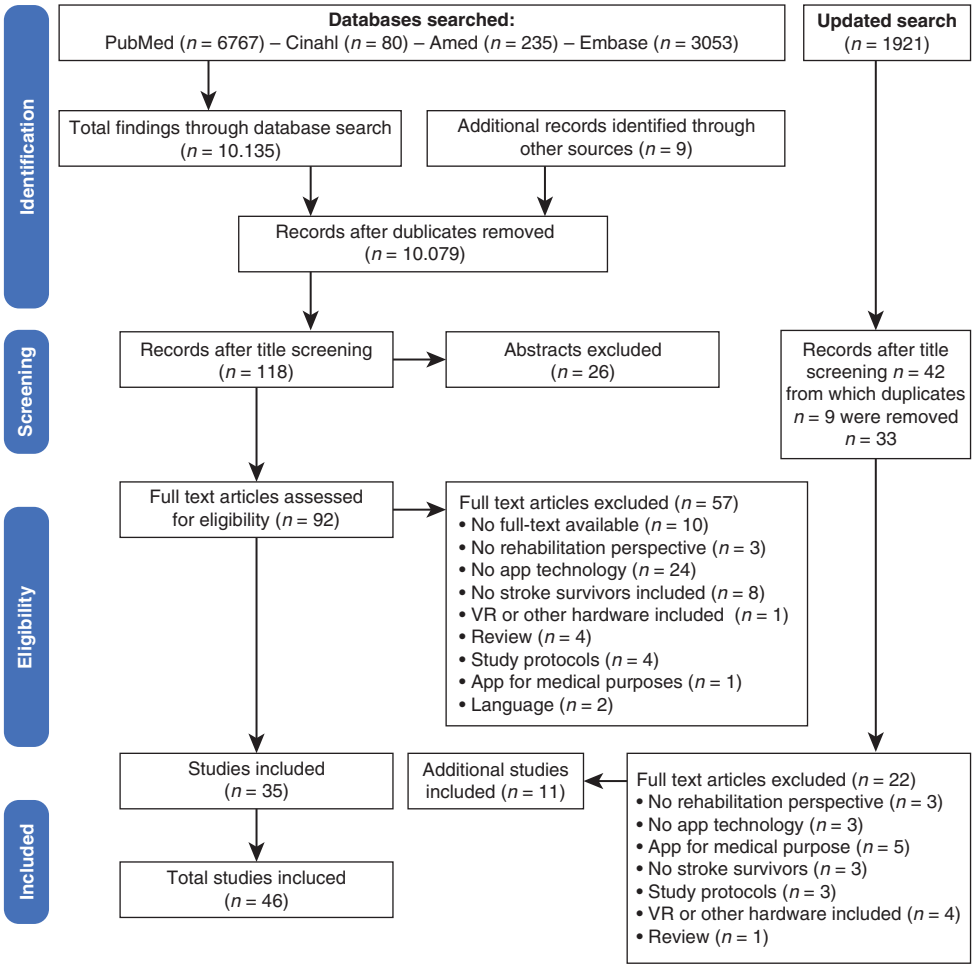


Fig. 1. Prisma flow-chart of search strategy and selection process.

After this selection, a total of 118 studies remained for an abstract review. Of these, 92 abstracts were considered to meet the inclusion criteria and were thus eligible for a full-text reading, which was conducted independently by 2 authors (MM, HK). However, 10 relevant studies were inaccessible and therefore excluded from this study, leaving 82 articles to be read in full text. A search alert was created for the latest published studies in PubMed and CINAHL, which resulted in 1 study (30). Finally, 35 articles were included, with consensus by two authors (MM and HK). Furthermore, an updated search was performed in November 2021 and January 2022 (see supplementary material). When searching only for studies published between 2020 and 2021, 1,921 hits were identified and screened by title, leaving 42 studies

Table I. Characteristics of the included articles (*n* = 46)

Items	Characteristic	Number of articles
Publication type	Primary articles	45
	Conference abstract	1
Study type/data collection method	Feasibility	9 (41–49)
	Pilot	10 (30, 50–58)
	Cohort	2 (40, 59)
	RCT	8 (19, 22, 25, 60–64)
	Case	5 (24, 65–68)
	Case-control	3 (27, 28, 69)
	Comparative	5 (70–74)
	Survey	1 (75)
	Mixed method	1 (76)
	Crossover design	1 (77)
	Qualitative design	1 (78)

(Continued)

Table I. (*contd.*)

Items	Characteristic	Number of articles
Place of origin	Australia	4 (24, 42, 45, 73)
	Belgium	1 (46)
	Brazil	3 (47, 70, 72)
	Canada	3 (49, 54, 66)
	China	2 (43, 62)
	England	3 (58, 59, 77)
	Ghana	1 (57)
	India	1 (76)
	Italy	2 (64, 75)
	Israel	1 (44)
	Latvia	1 (78)
	Netherland	1 (68)
	Korea	8 (19, 25, 27, 51, 56, 61, 63, 74)
	Scotland	1 (28)
	Singapore	1 (69)
	Slovenia	1 (41)
	Spain	2 (48, 60)
	Sweden	1 (40)
	Switzerland	1 (55)
	Taiwan	2 (22, 50)
	USA	6 (30, 52, 53, 65, 67, 71)
Stroke severity	Mild	15 (25, 28, 30, 41, 50, 51, 53, 61–64, 67, 68, 70, 71)
	Moderate	1 (24)
	Mild-moderate	17 (19, 22, 42, 44–46, 48, 54, 56, 57, 60, 65, 66, 72, 74, 76, 78)
	Severe (severe aphasia)	2 (52, 58)
	Mild-moderate-severe aphasia	2 (69, 73)
	Not reported	9 (27, 40, 43, 47, 49, 55, 59, 75, 77)
Stroke onset	Acute	12 (22, 24, 40, 43, 46, 54, 56, 62, 67–69, 76)
	Subacute	2 (25, 64)
	Acute/subacute	2 (30, 45)
	Acute and chronic	1 (73)
	Subacute and chronic	2 (44, 59)
	Chronic	23 (19, 28, 41, 42, 47–51, 53, 57, 58, 60, 61, 63, 65, 66, 70–72, 74, 75, 77)
	Acute, subacute and chronic	2 (27, 78)
	Not reported	2 (52, 55)
Setting of rehabilitation	Hospital/rehabilitation centre	21 (24, 25, 30, 41, 43, 44, 46, 47, 49–51, 54, 56, 60, 61, 63, 68, 70–72, 77)
	At-home	17 (22, 28, 40, 42, 45, 48, 53, 57–59, 62, 65–67, 74, 76, 78)
	Both	3 (69, 73, 75)
	Not reported	5 (19, 27, 52, 55, 64)
Target of app	A: Assessment	16
		Speech (aphasia) ($n=2$) (27, 69)
		Physical activity/function ($n=5$) (30, 41, 47, 70, 72)
		Upper limb ($n=1$) (44)
		Stroke severity ($n=4$) (40, 43, 62, 68)
		Cognitive function 2 (56, 73)
		Pulmonary function 1 (71)
		Unmet needs 1 (75)
	B: Goalsetting	0
	C: Training	25
		Lower limb and balance ($n=9$) (25, 28, 45, 50, 51, 57, 60, 61, 63)
		Fine motor/upper limb ($n=5$) (19, 24, 48, 53, 74)
		Speech (aphasia) ($n=8$) (42, 46, 52, 54, 55, 58, 66, 77)
		Visual therapy ($n=1$) (59)
		Cognitive therapy ($n=2$) (64, 65)
	D: Social participation	0
	E: Discharge (Health information)	2 (22, 67)
	C+E: Exercises (lower limb and balance) and discharge/health information	2 (76, 78)
	A-B-E (assessment, goal-setting, discharge support, information/ education)	1 (49)

for abstract reading. However, 9 studies were already included in the initial search, thus 33 studies were full-text read, from which 11 studies were included using the above-mentioned inclusion criteria. Thus, a total of 46 studies using 1 or more mobile/tablet and web-based app solutions, were included in this scoping review (see fig.1).

Of the 46 included studies, 43 were published from 2016 to 2020, 2 studies were published in 2015 (27, 40) and 1 in 2012 (25). The countries in which

Table II. Overview of main characteristics, target and context of the app solutions and key findings of the included studies

Study	First author, year, country	Study design	Objective/aim	Participants	Target of app	Context	Study
1	Ballard KJ et al., 2019 Australia (42)	Feasibility	To assess the feasibility of a newly developed speech therapy application using automatic speech recognition (ASR)-based feedback on performance to improve word production accuracy	5 participants Stroke onset: chronic stroke (more than 6 months) Stroke severity: mild-moderate apraxia of speech plus aphasia deficit	C: Language training	Setting: at-home Duration: 4 weeks, 4 sessions a week +1 face-to-face session each week, (in the clinic or via video calls)	Improvements were maintained 1 month post-treatment. Participants found the app engaging and enjoyed the app-based activities and would continue to use it. However, they wished to maintain regular contact with the speech pathologist.
2	Capela NA et al, 2016 Slovenia (41)	Feasibility	To compare the performance of a smartphone-based wearable mobility monitoring system (WMMS) between able-bodied participants and people who had had a stroke.	15 participants with stroke and 15 able-bodied participants Stroke onset: chronic (mean 9.6 months before) Stroke severity: mild stroke (mean FIM score was 107 points (max. point is 126).	A: Assessment of physical activity	Setting: rehabilitation centre Duration: 30 min	The algorithm performed reasonably well for both stroke and able-bodied participants when differentiating between sit, stand, lie, and walk and between mobile and immobile states. When stair-climbing and small movements were added to the classification, algorithm performance decreased.
3	Chang H et al., 2018 China (43)	Feasibility	To assess the accuracy of self-assessment for acute stroke patients via mobile phone application-based scales and determine the value and prospect of clinical use. The aim was also to compare the results of nurse assessment and self-assessment.	50 participants Stroke onset: acute (onset within 12 h before application on smartphone) Stroke severity: not reported	A: Assessment of stroke severity mRS and ADL	Setting: at the hospital, the day before release Duration: 1-time self-assessment. 27 patients self-assessed the scales. Caregivers of other 23 patients completed the assessment	There is a substantial consistency between self-assessment and nurse assessment. Therefore, caregivers can serve as the proper assessor when patients are out of hospital.
4	Chae SH et al., 2020 Korea (74)	Comparative study	To develop and evaluate a home-based rehabilitation system that can recognize and record the type and frequency of upper limb rehabilitation exercises using a smartwatch and smartphone app	23 participants (17 in intervention group and 6 in control group) Stroke onset: chronic stroke Stroke severity: mild-moderate stroke	C: physical therapy	Setting: at-home Duration: Both groups received education on upper limb exercises. Participants were asked to perform the home exercises for 12 weeks. Both groups received weekly phone-calls from a physiotherapist.	Using a commercial smartwatch can facilitate participation in home training and improve the functional score of the Wolf-Motor-Function-Test (WMFT) and shoulder ROM of flexion and internal rotation in the treatment of patients with chronic stroke. This strategy may be a cost-effective tool for the home stroke care treatment in the future.
5	Choi et al., 2015 Korea (27)	Case-control	To develop a valid, reliable mobile aphasia screening test (MAST) for patients in remote locations.	30 patients with aphasia and 30 persons without aphasia Stroke onset: Acute, subacute, chronic stroke (aphasia: 2–8,128 days) Stroke severity: not reported	A: Assessment of aphasia (screening tool)	Setting: not reported Duration: 1-time assessment	The MAST is a valid and reliable tool for detecting aphasia in patients with stroke

(Continued)

Table II. (*contd.*)

Study	First author, year, country	Study design	Objective/aim	Participants	Target of app	Context	Study
6	Ciou et al., 2017 Taiwan (50)	Pilot study	A drop foot rehabilitation training device based on the smart phone is presented. This device is combined with a football APP and feedback controlled functional electrical stimulation (FES).	5 stroke survivors Stroke onset: chronic stroke (5–19 months) Stroke severity: mild stroke (MMSE 24–30)	C: Physical training	Setting: rehabilitation centre Duration: 30 min per session, twice a week for 6 weeks	Significant performance improvement ($p < 0.05$) was found in the patient's ankle dorsiflexion strength, ankle dorsiflexion angle, control timing and Timed Up and Go.
7	Cock DE et al., 2021 Belgium (46)	Feasibility	To investigate the feasibility, usability and acceptability of the Speech Therapy App (STAPP) in the acute phase post-stroke	25 participants with aphasia Stroke onset: acute (2 weeks post-stroke) Stroke severity: mild-moderate (NIHSS 3–15)	C: Aphasia training	Setting: rehabilitation centre Duration: after 2 short training sessions (15–30 min) participants were asked to practice independently with "STAPP' as an add-on to standard care until discharge (median length of hospital stay 9 days)	STAPP is feasible to use as an additional rehabilitation tool along with standard of care in the acute phase post-stroke.
8	Cooray C et al., 2015 Sweeden (40)	Cohort	To investigate whether automatic assessment of the mRS based on a mobile phone questionnaire may serve as an alternative to mRS follow-up assessments at clinical visits after stroke	48 patients Stroke onset: acute stroke Stroke severity: not reported	A: Assessment of mRS (modified Rankin scale (stroke severity) after 3 months (20 questions).	Setting: at-home Duration: 1-time self-assessment:16 made by patient, 6 in collaboration with caregiver, 26 made by caregiver, and 1-time assessment by health professional.	Mobile phone-based questionnaires for automatic assessment of mRS after stroke performed good to excellent compared with clinical visit mRS assessments.
9	Costa et al., 2020 Brazil (47)	Feasibility	To investigate the validity and test-retest reliability og mHealth devices (Google Fit, Health, STEPZ, Pacer, and Fitbit Ultra) to estimate the number of steps and to compare if the number of steps is affected by their location on the body (paretic and non-paretic side)	55 participants Stroke onset: chronic stroke Stroke severity: mild stroke (>18 on the Mini-Mental State Examination)	A: Assessment of physical activity	Setting: rehabilitation centre Duration: 1-time assessment	mHealth devices ((Google Fit, Health, STEPZ, Pacer, and Fitbit Ultra)) are valid and reliable for step counting in chronic stroke survivors. Body location (paretic or non-paretic side) does not affect validity or reliability of the step count metric.
10	De Bartolo et al., 2018 Italy (75)	Survey	To validate the Post Soft Care-App (PSC), which was administered by physiotherapists to a sample of chronic stroke patients with the aim of highlighting their unmet needs.	53 patients Stroke onset: chronic stroke Stroke severity: not reported	A: Assessment of unmet needs (11-item questionnaire to highlight any changes since discharge or last check-up)	Setting: at-home or rehabilitation centre Duration: 1-time (96% were patient answered, 4% in collaboration with caregiver/staff	The PSC was judged to have helped therapists to improve their communication with patients and to have highlighted at least 1 unmet need. And was fast and easy to administer.
11	Epalte et al., 2020 Latvia (78)	Qualitative interviews	To describe insights about the app "Vigo" usability from a patients' perspective	12 participants Stroke onset: acute, subacute, chronic (9–1,087 days) Stroke severity: mild-moderate (minimum score of 18 on Montreal Cognitive Assessment (MoCA))	C+E Education/ information to promote participation/ adherence to training (music, text, pictures, videos, exercises)	Setting: at-home Duration: each participant tested the application for the duration of a month in their home.	Stroke survivors in general had a positive attitude towards using tablet technologies in their home environment. However, a greater variety of exercises provided in the solution and technical support were wanted.
12	Faria GS et al., 2019 Brazil (70)	Comparative study	To investigate the concurrent validity of the GT3X®ActiGraph accelerometer and Google Fit®smartphone application in estimating energy expenditure of stroke survivors during fast over ground walking.	30 participants Stroke onset: chronic stroke (mean time 98 months Stroke severity: Mild stroke (able to walk independently, no cognitive impairments)	A: Assessment of steps (physical activity)	Setting: rehabilitation centre Duration: 1-time	The GT3X®ActiGraph accelerometer and the Google Fit® smartphone application do not provide valid measures of energy expenditure in chronic stroke individuals during fast overground walking.

(Continued)

Table II. (*contd.*)

Study	First author, year, country	Study design	Objective/aim	Participants	Target of app	Context	Study
13	Fell et al., 2019 USA (30)	Pilot study	To examine correlations between clinician and mobile app scoring of real-time movement data in 4 common clinical tools used to assess stroke recovery at the body function and activity levels of the ICF model.	35 participants Stroke onset: acute/subacute stroke (stroke onset within the last 3 months). Stroke severity: Mild stroke (able to follow 3-step commands and ambulate 10 m with minimal support, and no aphasia or apraxia).	A: Assessment of movement	Setting: rehabilitation centre Duration: 1-time (physiotherapist assessment and mobile app assessment)	Significant correlation between clinician and mobile app scores for all 4 tests supports the authors' objectives to create a mobile health app that provides clinicians with objective movement data to inform and track recovery.
14	Grau-Pellicer M et al., 2019 Spain (60)	RCT	To investigate the effectiveness of a mobile-health (mHealth) app in improving levels of physical activity.	41 participants Intervention group (IG) $n = 24$ control group (CG) $n = 17$ Stroke onset: chronic stroke Stroke severity: Mild-moderate (Functional ambulation classification ≥ 3, Barthel Index ≥ 45).	C: Physical activity	Setting: rehabilitation centre Duration: IG: 8-week rehabilitation program (2 days/week, 1 h/session): aerobic, task-oriented, balance and stretching. CG: conventional program.	Community ambulation increased more in IG than in CG. Sitting time was reduced by 2.96 h/day in the IG and by 0.53 h in the CG. Support and guidance of researchers and careers are required to ensure completeness of clinical assessment data and protocol adherence.
15	Guo et al., 2017 Singapore (69)	Case-control	To determine the reliability of telehealth aphasia assessment based on measures that examine aphasia across the ICF spectrum within the home environment using Access2Aphasia.	30 participants Stroke onset: chronic stroke (stroke onset 6–79 months). Stroke severity: mild to severe aphasia	A: Assessment aphasia	Setting: at-home (online-led group) or rehabilitation centre (face-2-face group) Duration:1-time	A moderate to almost perfect agreement between online and face-to-face speech-pathologist assessment was found.
16	Jang S.H. & Jang W.H., 2016 Korea (19)	RCT	To investigate the effect of a finger training application program using a tablet PC in chronic hemiparetic stroke patients.	21participants Stroke onset: chronic stroke (more than 6 months). Stroke severity: mild-moderate (incomplete weakness of hand and fingers, MMSE ≥ 24, modified Ashworth scale score < 2)	C: Fine motor training	Setting: not reported Duration: application training was conducted 1 session/day, 6 days per week for 4 weeks.	Using application with a tablet PC for 4 weeks, hemiparetic Stroke patients achieved functional recovery of the hand as well as motor recovery of the wrist and hand.
17	Joo S et al., 2019 USA (71)	Comparative study	To (1) quantify the reliability and validity of SGA (smartphone game-based assessment) for pulmonary function and (2) to assess the validity of SGA in comparison to spirometry.	34 participants Stroke onset: chronic stroke (more than 6 months) Stroke severity: mild stroke (Mini-Mental State Examination – Korean version (MMSE-K) score > 24)	A: Assessment of pulmonary function	Setting: rehabilitation centre Duration: 3 spirometer assessments that were compared with the SGA (14 games, to guide in- and expiration)	The SGA data were statistically significant and reliable for pulmonary function assessment in stroke patients.
18	Kang Y-N et al., 2019 Taiwan (22)	RCT	To compare the effectiveness of a stroke health education booklet and a stroke health-education mobile app (SHEMA) with same stroke-related health information, in improving patients' knowledge of stroke risk factors and health-related quality of life compared with stroke health-education booklet.	63 stroke survivors with experience in using apps (30 in intervention and 33 control group) Stroke onset: acute Stroke severity: Mild –moderate stroke (Mini-Mental State Examination ≥ 24, mRS (modified Rankin scale) 1–4	E: Health information education	Setting: at-home Duration: The patients read the booklet or the SHEMA content at home for 7–14 days, and 5 min per day.	No significant effect between groups on improving patients' knowledge of stroke risk factors after a short-term intervention. A larger number of younger patients (aged ≤55 years) tend to improve their stroke-knowledge in SHEMA group than the control group.

(Continued)

Table II. (*contd.*)

Study	First author, year, country	Study design	Objective/aim	Participants	Target of app	Context	Study
19	Kim JH et al., 2012 Korea (25)	RCT	To examine the effects of Rhythmic auditory stimulation (RAS) gait training using a smartphone metronome application on balance and gait abilities of stroke patients	20 patients RAS group, $n=10$ Control group $n=10$ Stroke onset: subacute stroke (stroke onset < 6 months) Stroke severity: mild stroke (Mini-Mental State Examination score ≥ 24)	C: Gait training	Setting: at the hospital Duration: 30-min sessions conventional physical therapy. In addition, RAS group received 30-min gait training 3 times per week for 5 weeks.	Increases in dynamic balance and spatiotemporal gait parameters were observed in both groups. Compared with the control group, the RAS group showed significant improvements in scores on the ABC scale, DGI, TUG, and Up stair and Down stair times.
20	Kizoni et al., 2016 Israel (44)	Feasibility study	To compare performance of app-based hand activities in individuals without a disability from 3-age groups and to assess the feasibility of using tablet apps in individuals with stroke.	20 stroke survivors Stroke onset: subacute (<4 months) and chronic (<12 months) Stroke severity: Mild–moderate upper extremity impairment (Fugl-Meyer assessment)	A: Fine motor assessment ScribbleKid app PegLight app Tap it app Bowling App.	Setting: rehabilitation centre Duration: 1-time assessment	PegLight was the only app that all participants were able to complete. And 15 out of 20 stroke survivors were able to complete the 2 trials of the tapping task. All participants enjoyed using the apps.
21	Ko BW et al., 2016 Korea (51)	Pilot study	To investigate the effects of various rhythmic auditory stimulation (RAS) tempos on stroke gait pattern, when training patients with a smartphone-based rhythmic auditory stimulation application.	15 patients Stroke onset: chronic stroke (>6 months since stroke). Stroke severity: Mild stroke (able to walk >10 m independently, MMSE= 28,4	C: Gait training Each RAS tempo practiced for 10 min. Sensors were attached to ankle joints.	Setting: rehabilitation centre Duration: 1-time assessment and training	Stride length increased significantly after training. Rhythmical tempo variations have a positive impact on the gait of stroke patients.
22	Kondapalli A et al., 2016 USA (52)	Pilot study	To study the use of a phonology-free communication mobile app.	1 participant Stroke onset: not reported Stroke severity: Severe expressive/ Broca's aphasia	C: Aphasia training	Setting: not reported Duration: not reported	The app may be of great benefit to patients with aphasia and other communication disorders. The nurse also found the app helpful for communicating with the patient.
23	Kringle EA et al., 2019 USA (65)	Case study	To describe lessons learned through adaptation of a complex intervention, strategy training, for delivery via mHealth technology (the iADAPTS application).	5 participants who had previously engaged in strategy training Stroke onset: chronic stroke (1–3 years earlier) Stroke severity: Mild–moderate (intact language, Self-Awareness of Deficits Inventory >2, motor impairment 16–36 (Chedoke McMaster Stroke Assessment)	C: Cognition training (strategy training)	Setting: at-home Duration: 5-week intervention period, 3 times per week	Overall high satisfaction with this approach to intervention measured on CSQ-8. It may be feasible to adapt the strategy training protocol for delivery using mobile health technology.
24	Lavoie M et al., 2019 Canada (66)	Case study	To investigate the efficacy of a self-administered treatment using a smart tablet to improve naming of functional words in post-stroke anomia.	4 participants with aphasia Stroke onset: chronic stroke (>1-year post-stroke) Stroke severity: mild to moderate deficits for oral sentence comprehension.	C: Aphasia training	Setting: at-home Duration: 4 times a week for 4 weeks	Significant improvement for both sets of trained words that was maintained at 2 months follow-up. Moreover, in 2 participants, evidence of generalization to conversation was found.

(Continued)

Table II. (*contd.*)

Study	First author, year, country	Study design	Objective/aim	Participants	Target of app	Context	Study
25	Lawson et al., 2017 USA (53)	Pilot study	To investigate the utility of a mobile app to improve motor control for stroke survivors by examining changes in motor ability and participation.	6 participants Stroke onset: chronic stroke Stroke severity: Mild stroke (independently use of app).	C: Upper extremity therapy	Setting: at-home Duration: daily home-training for 6 weeks.	Statistically significant changes were not obtained with this pilot study. However, improvements were found in many assessments posttreatment.
26	Li et al., 2020 China (62)	RCT	To evaluate the feasibility, validity, and reliability of functional assessments administered through the videoconference function of a mobile phone-based app compared with administration through the telephone function in post-stroke patients after rehabilitation hospitalization	120 participants (videoconference follow-up ($n=60$) or a telephone follow-up ($n=60$) group Stroke onset: acute stroke Stroke severity: mild stroke (normal cognitive function according to the Montreal Cognitive Assessment with a cut-off score > 26)	A: motor assessment	Setting: at-home Duration: assessment made 2 times (functional status of participants using modified Barthel Index scores were measured in each group at 2-week and 3-month follow-up periods.	The agreements between video-conference follow-up and home visit assessments was higher than that between telephone follow-up and home visit assessments at all follow-up periods.
27	Mallet 2016 Canada (54)	Pilot study	To assess the feasibility of a tablet-based speech-language therapy for patients with communication deficits following acute stroke	30 participants Stroke onset: acute stroke Stroke severity: mild to moderate communication Deficits (AlphaFim 47–90).	C: Aphasia therapy 7 apps tested	Setting: acute care setting (hospital) Duration: 1 h per day (median time with the tablet was 10 days).	97% of patients scored the mobile tablet-based communication therapy as at least moderately convenient 3/5 or better with 5/5 being most convenient.
28	Markle-Reid et al., 2020 Canada (49)	Feasibility	To examine the feasibility of a 6-month integrated transitional care stroke intervention using the "My Stroke Team (MyST)" app and explored its effects on health outcomes, patient and provider experience, and cost. (Team: occupational therapist, physio-therapist, speech language pathologist, nurse, social worker).	30 participants with stroke and multimorbidity (≥ 2 chronic conditions) newly discharged from hospital and referred to an outpatient stroke rehabilitation setting Stroke onset: sub-acute (average of 2 months post-stroke) Stroke severity: not reported	A-B-E MyST app: a secure space for *(i)* patient information; *(ii)* documenting and sharing of visits, conference records, screening tool scores, client goals, and follow-up items; *(iii)* posting "alerts" for individuals or the team; and *(iv)* resource links (stroke guidelines, educational materials, health and social services in the community).	Setting: at-home Duration: Monthly 1-h visits from a member of the team for 6 months (average 4.73 home visits). The team met once per month for 6 months to discuss patient-identified goals and develop a person-centred and evidence-based plan of care for each participant. Only the team were able to add content or communicate through the MyST app	The intervention was feasible and acceptable to both stroke survivors and providers. From baseline to 6 months, there was no statistically significant difference in health outcomes. However, there was a significant reduction in the total per person use and costs of health services.
29	Meeuwis et al., 2020 Netherland (68)	Case study	To present how WhatsApp Messenger may support a modern neurological examination	3 participants (2 with stroke and 1 with TIA) Stroke onset: acute Stroke severity: mild	A: Assessment of deficits	Setting: acute care setting Duration: 1-times assessment	This easy and useful evaluation of the ability to use a text message program should be incorporated in standard neurological history taking and examination
30	Nef et al., 2018 Switzerland (conference abstract) (55)	Pilot study	To develop and test an aphasia tele-rehabilitation application (Bern Aphasia App) to increase training frequency and duration for patients.	The app was tested on 25 healthy participants. Later 134 patients trained with the app. Stroke onset: not reported Stroke severity: not reported	C: Aphasia training	Setting: not reported Duration: not reported	134 aphasia patients trained for 197.2 h. System Usability Scores (SUS) 94.5 for healthy participants and 93.2 for aphasia patients (maximum 100).

(Continued)

Table II. (*contd.*)

Study	First author, year, country	Study design	Objective/aim	Participants	Target of app	Context	Study
31	Park et al., 2017 Korea (56)	Pilot study	To assess whether MMSE-K (Korean version) can be applied effectively using a smartphone to evaluate cognitive function in stroke patients.	30 participants Stroke onset: acute stroke Stroke severity: mild-moderate aphasia	A: Cognition assessment MMSE (live assessment though the facetime app -30 questions)	Setting: at the hospital Duration: 1-time face-to-face assessment, and 1-time virtual assessment	Good agreement between face-to-face and remote assessments on total MMSE-K score and sub-scores for each domain were found (orientation, memory, attention/calculation, language, and visuospatial function).
32	Paul L et al., 2016 Scotland (28)	Case-control study	To evaluate the potential effectiveness of STARFISH a mobile app- based intervention in stroke survivors	23 participants (15 in intervention group (IG), 8 in control group (CG)) Stroke onset: chronic stroke (4.2±4.0 years) Stroke severity: mild (the ability to walk independently, and comprehend instructions)	C: Physical activity (measuring steps)	Setting: at-home Duration: IG: STARFISH program for 6 weeks. CG: usual care, (no rehabilitation, only appointments with health professionals as required).	A 6-week intervention using the STARFISH app, can increase steps/day (39.3%), walking time, and reduce fatigue in community-dwelling stroke survivors.
33	Polese JC et al., 2019 Brazil (72)	Comparative study	To examine the validity of the GT3X®ActiGraph accelerometer and the Google Fit® smartphone application in estimating stepping activity in people with chronic stroke, compared with the number of steps counted by the examiner.	37 participants Stroke onset: chronic stroke Stroke severity: mild-moderate stroke (no cognitive impairments and the ability to walk independently)	A: Physical activity (measuring stepping activity)	Setting: rehabilitation centre Duration: 1-time assessment	The Google Fit smartphone application showed higher associations with that determined by the examiner than the data estimated by the GT3X® ActiGraph accelerometer.
34	Sarfo FS et al., 2018 Ghana (57)	Pilot study	To assess the feasibility of and satisfaction with a physical therapy tele-rehabilitation intervention (9zest Stroke Rehab App®)	20 participants Stroke onset: chronic stroke Stroke severity: mild-moderate stroke (mRs 1–4)	C: Physical therapy	Setting: at-home Duration: 5-days-a-week exercise program (30–60 min) for 12 weeks.	The intervention was associated with improvements in baseline motor deficits over the 12 weeks of intervention.
35	Segura et al., 2021 Spain (48)	Feasibility	To test the feasibility of the novel app solution; Music-supported Therapy (MST) intervention to support upper limb motor function recovery	5 participants Stroke onset: chronic stroke Stroke severity: mild-moderate (Mini-Mental State Examination > 24) (4 participants with mild strength deficits and 1 patient with moderate strength deficits)	C: Upper-motor therapy	Setting: at-home (3 participants were trained initially at the rehabilitation centre) Duration: 10-week intervention program of 3 sessions per week	Patients clinically improved in upper limb motor function achieving the minimal detectable change or minimal clinically important difference in most of motor tests. The app received high usability ratings post-intervention.
36	Shin DC 2020 Korea (63)	RCT	To investigate the role of smartphone-based visual feedback trunk control training (SPVFTCT) for improvement of trunk control and spatio-temporal gait parameters in stroke patients.	24 participants (12 in each group.) Stroke onset: chronic stroke Stroke severity: mild stroke (sit independently > 30 min.; the ability to walk for 10 min; minimal cognitive deficits (MMSE-K > 24).	C: Physical therapy	Setting: rehabilitation centre Duration: 4 weeks conventional therapy. The SPVFTCT group additionally received SPVFTCT 4 weeks.	The trunk impairment scale in the experimental group was significantly improved compared with the control group ($p < 0.05$). The spatio-temporal gait parameters (stride and step length and step width) are significantly more different in the experimental group than in the control group ($p < 0.05$).

(Continued)

Table II. (*contd.*)

Study	First author, year, country	Study design	Objective/aim	Participants	Target of app	Context	Study
37	Shin and Song. 2016 Korea (61)	RCT	To assess the preliminary efficacy and feasibility of smartphone-based visual feedback trunk control training (SPVFTCT) for improving balance and trunk performance in stroke patients.	24 participants (12 in each group.) Stroke onset: chronic stroke Stroke severity: mild stroke (sit independently > 30 min.; the ability to walk for 10 min; minimal cognitive deficits (MMSE-K > 24).	C: Physical therapy	Setting: rehabilitation centre Duration: 4 weeks conventional therapy. The SPVFTCT group additionally received 3 20-min sessions per week of SPVFTCT for 4 weeks.	All measures in the SPVFTCT group improved significantly compared with the control group. In the SPVFTCT group, retention and adherence rates were 100% and 97%, respectively. The SPVFTCT approach is a feasible method to improve balance and trunk performance in stroke patients.
38	Siegel J et al., 2016 USA (67)	Case study	To examine ischemic stroke patients who used the PHA app (personal health assistant). The hypotheses were that the PHA would improve post-discharge satisfaction and decrease rehospitalization for patients recovering from stroke.	2 participants Stroke onset: acute stroke Stroke severity: mild stroke (sufficient dexterity and cognition to utilize the intervention app)	E: Seamless communication intervention and discharge support	Setting: at-home Duration: 30 days of PHA service though the app (to facilitate communication with healthcare after discharge).	Both patients who used the app were very satisfied with the PHA and their post-hospital care coordination. One patient was readmitted for stroke within 30 days.
39	Silveira TM et al., 2018 Australia (24)	Case study	To examine the effect of combining Functional Electric Stimulation (FES), music therapy, and tablet technology for upper limb stroke rehabilitation	1 participant Stroke onset: acute stroke Stroke severity: Moderate stroke (assistance on transfers and ADL, Minimal hand function on paretic side)	C: Motor training (upper limb) Combining a FES protocol with an iPad application.	Setting: rehabilitation setting Duration: during 9-weeks patient received FES 1–2 × per week, and music therapy 1 × per week.	Improvements in all areas of assessment at discharge, which had continued to improve at 7 months post-discharge.
40	Simpson DB et al., 2020 Australia (45)	Feasibility study	1) To investigate the feasibility of delivering a 4-week remotely prescribed functional exercise program using a tablet, app, and sensor system, and 2) provide an estimate of the effect of the intervention on functional outcomes.	10 participants Stroke onset: acute, subacute, and chronic (29–296 days) Stroke severity: mild-moderate (Short performance physical battery test 3–11 points (max. is 12)).	C: Physical activity	Setting: at-home Duration: 4-weeks	It was feasible to prescribe and monitor exercises using an app and sensor-based system. Participants performed 125% and 104% of prescribed sessions and repetitions. Participants rated the system usability (78%), enjoyment (70%) and system benefit (80%).
41	Stark and Warburton, 2016 England (58)	Pilot study	To investigate the effectiveness and feasibility of self-delivered and directed iPad-based speech therapy in patients with chronic aphasia following a left middle cerebral artery-territory stroke.	10 participants Stroke onset: Chronic stroke Stroke severity: Severe aphasia (using the CAT score)	C: Aphasia training Speech therapy app compared with mind-game app	Setting: at-home Duration: Recommended use of 20 min per day, for 4 weeks.	There was significant post-therapy improvement, but no significant improvement after the mind-game intervention and they were maintained at 6-month follow-up.
42	Sureshkumar K et al., 2016 India (76)	Mixed methods	To evaluate the feasibility and acceptability of the intervention "care for stroke app" (information about stroke and the ways to manage post-stroke disabilities (text and videos), home-based exercises, functional skills training, ADL, and assistive devices).	60 participants (stroke survivors and their caregivers) Stroke onset: acute stroke (stroke onset < 6 weeks). Stroke severity: mild and moderate stroke (NIH Stroke Scale 1–15)	C + E: Education/ information and physical activity app	Setting: at-home Duration: Participants were asked to use this intervention at home for 2 weeks during the field-testing phase and for 4 weeks during the pilot-testing.	53% ($n = 16$) of stroke survivors and 67% ($n = 20$) of carers rated "Care for Stroke" as excellent. The remaining participants rated the intervention as very useful

(Continued)

Table II. (*contd.*)

Study	First author, year, country	Study design	Objective/aim	Participants	Target of app	Context	Study
43	Szalados et al., 2021England (59)	Cohort	To investigate the clinical effectiveness of Eye-Search, a web-based therapy app designed to improve visual search times, in patients with either hemianopia, neglect or both.	426 participants (84% with stroke) Stroke onset: subacute-chronic (44–210 days) Stroke severity: not reported	C. visual training	Setting: at-home Duration: 5–20 days	Eye-search therapy improved search times to the affected visual field of patients with either hemianopia alone or neglect and hemianopia, but not those with neglect alone
44	Verna V et al.,2020(64)	RCT	The aim of this study was to investigate the effectiveness of a negative mismatch-based therapy on the disability and quality of life in patients with stroke in sub-acute phase (using the "Temporal Musical Patterns Organisation (Te.M.P.O)" app.	30 participants randomized to Mismatch group (Mg) or Control group (CTRLg) (receiving Sham) Stroke onset: sub-acute Stroke severity: mild stroke (Mini-Mental State Examination < 24)	C. Cognitive training	Setting: not reported Duration: 20 min/3 times a week for 4 weeks, in addition to the standard cognitive and physical rehabilitation.	The results show a major improvement of the Mg with respect to the CTRLg in all clinical scales score (disability and quality of life)
45	Wall KJ et al.2018 Australia (73)	Comparative study	To develop and explore the feasibility and user acceptance of the Cognitive Assessment for Aphasia App (C3A): a cognitive assessment for stroke survivors, designed to be inclusive of individuals with aphasia	64 participants Stroke onset: acute and chronic stroke (35 with aphasia, 29 without aphasia). Stroke severity: Mild to very severe language deficits.	A: Cognitive assessment	Setting: acute, rehabilitation centre and community settings Duration:1-time assessment	The app seems to be feasible for cognitive assessment for stroke survivors with and without aphasia.
46	Woodhead et al. 2018 England (77)	Crossover design	To test the impact of a novel training app, "iReadMore", and transcranial direct current stimulation (tDCS) of the left inferior frontal gyrus, on word reading ability in central alexia.	21 participants Stroke onset: chronic stroke and with central alexia (more than 1-year post-stroke) Stroke severity: not reported	C: Aphasia (for promoting reading ability)	Setting: rehabilitation centre Duration: 4-week blocks of iReadMore therapy, accompanied with either real or sham tDCS (3 40-min face-to-face sessions/ week). In addition, independently using the app at-home at least 35 h in total.	iReadMore training in post-stroke central alexia resulted in an 8.7% improvement in reading accuracy for trained words, but did not generalize to untrained words. Reading accuracy improved by 2.6% more during anodal tDCS than sham. Reaction times also improved. Reading accuracy gains were still significant (but reduced) 3 months after training cessation.

the most studies were conducted were Korea ($n=8$) and the USA ($n=7$). Pilot studies ($n=10$), feasibility studies ($n=9$) and RCTs ($n=8$) were the most common research designs. The designs of the remaining 19 studies were case ($n=5$), survey ($n=1$), mixed method ($n=1$), cross-over ($n=1$), cohort ($n=2$), case-control ($n=3$), qualitative design ($n=1$), or comparative design ($n=5$).

The target area of the 46 included studies were assessment support ($n=16$), training ($n=25$), discharge ($n=2$), and 2 studies were targeted at supporting both discharge/education/information and training. Only 1 study targeted goal-setting, in addition to assessment and discharge support. No studies were related to the element "participation" using Wade's model of rehabilitation.

The number of participants varied from 1 to 426 stroke survivors, and the app solutions usage was between 1 occasion (assessment support) and 12 weeks (from 1–7 days a week, training support). A total of 23 studies included participants with chronic stroke, 16 studies included participants with acute and/or subacute stroke, while 5 studies included participants with 2 or 3 different onset of stroke. In 2 studies, stroke onset was not reported.

Thirty-three studies included participants with mild-moderate stroke, and only 4 studies included participants with severe stroke (2 studies included participants with severe aphasia, and 2 studies included participants with mild-moderate-severe aphasia). None of the studies included participants with severe motor deficits. In 9 studies the stroke severity was not reported.

JRM Journal of Rehabilitation Medicine JRM Journal of Rehabilitation Medicine JRM

The key characteristics of the included studies are shown in Tables I and II.

App solutions to support assessment (n = 16)
Of the 16 studies that investigated apps used in assessment support, 7 apps targeted physical activity (lower limb and mobility ($n=5$), upper limb ($n=1$) and pulmonary function ($n=1$)), 6 stroke severity, dependency, and cognitive function, 2 speech (aphasia) and 1 study addressed identification of unmet needs. Nine apps were used in a hospital or rehabilitation setting, 3 in both a rehabilitation setting and at-home, 3 were used at-home and for 1 app solution the setting was not reported. Five studies included participants with acute stroke, 7 studies included participants with chronic stroke, and 4 studies included participants from 2 or all phases. Participants with mild stroke were included in 7 studies, participants with mild-moderate stroke were included in 3 studies and participants with mild to severe stroke were included in 2 studies. Stroke severity was not reported in 4 studies.

Assessments of physical activity (n = 7). Capela et al. (41) developed a smartphone app to detect physical activity in stroke survivors. They found that the app was able to differentiate between lying, sitting, standing, and walking, but could not detect smaller movements and stair-climbing. Faria et al. (70) showed that neither an Actigraph accelerometer app nor the Google Fit app could provide valid measurements of energy expenditure among chronic stroke survivors during walking. However, Polese et al. (72) compared the same 2 app solutions and found that the Google Fit app could more accurately measure steps taken by chronic stroke survivors. Costa et al. (47), tested different app solutions (Google Fit, Health, STEPZ, Pacer, and Fitbit Ultra) ability to detect steps taken by chronic stroke, and found the solutions were reliable and not influenced by the paretic limb. Fell et al. (30) developed an app solution to support remote assessment of physical function supported by sensors. The authors concluded that the app was feasible and significant correlations were found between clinician and mobile app scores in all 4 tests, i.e. arm function and leg function of the National Institutes of Health Stroke Scale (NIHHS); Functional Reach test; and 10-meter walk test. However, in all the above studies, only participants with mild stroke and the ability to walk were included. Only 1 of the included studies, Kizoni et al. (46), targeted the assessment of upper limb deficits through app solutions supporting hand activity among stroke survivors. One of the included studies showed that an app solution was reliable to support the measurement of pulmonary function among chronic stroke survivors (71).

Assessments of stroke severity and cognitive function (n = 6). Six studies developed app solutions to support assessment of stroke severity, dependency and/or cognitive function. Li et al. (62) found agreement on ADL assessments (using the Modified Barthel Index), between home visit assessment and video-conference assessment though the mobile WeChat app (62). Meeuwis et al. (68) concluded that including the WhatsApp Messenger (a text messaging program) may be a useful contribution in the modern neurological examination to detect cognitive deficits (68). Chang et al. (43) and Cooray et al. (40) developed app solutions for assessment of disability/dependency after stroke, using the modified Ranking Scale (mRS). In both studies there was substantial consistency between the stroke survivors'/caregivers' and health professionals' assessments. In both studies, however, caregivers played an important role in supporting the stroke survivor to perform the assessment, because of cognitive challenges and/or inability to handle a mobile phone. Park et al. (56) found that it was feasible to assess cognitive deficits through the Facetime app, and Wall et al. (73) found that an app solution, used together with a virtual reality device, supported cognitive assessment among stroke survivors with and without aphasia.

Assessment of speech-related deficits (n = 2). Two studies developed app solutions to support the assessment of speech-related deficits. Choi et al. (27) and Guo et al. (69) found that app solutions were valid and reliable for remote assessment of aphasia in stroke survivors.

Assessment of unmet needs (n = 1). De Bartolo et al. (75) developed an app-based questionnaire to support the identification of unmet needs among chronic stroke survivors and to identify changes since discharge or last check-up. They concluded that the app was feasible and that it improved communication between stroke survivors and health professionals after discharge. Most participants were able to use the app by themselves, without support from a health professional or caregiver.

App solutions to support training/intervention (n = 27) Of the 27 studies that included apps aimed at supporting training, 9 targeted lower limb and balance, 5 upper limb or fine motor training, 8 speech training, 1 visual therapy, 2 targeted cognitive therapy, and 2 studies targeted both exercises and health-related information/education. Fourteen interventions took place in the home setting, 8 in a rehabilitation setting, and in 5 studies the setting was not reported. The apps were used for participants with acute stroke ($n=4$), subacute stroke ($n=2$), chronic stroke ($n=16$), subacute-chronic stroke ($n=1$), and acute-subacute-chronic stroke ($n=2$). For 2 studies, the stroke onset was not reported. The study participants' stroke severity was mild ($n=8$),

JRM Journal of Rehabilitation Medicine JRM Journal of Rehabilitation Medicine JRM

mild-moderate ($n=14$), moderate ($n=1$), and severe ($n=1$). In 3 studies stroke severity was not reported.

App solutions to support lower limb and balance training ($n=11$). Nine studies were targeted at supporting physical training/intervention of lower limb/gait. Ciou et al. (50) found that drop-foot training using an app improved participants' ankle strength and dorsiflexion angle and their Timed up and Go score. Grau-Pellicer et al. (60) showed that an app improved the level of daily activity. The authors pointed out that the support of caregivers or researchers was of significance in achieving adherence to training. Kim et al. (25) showed that a smartphone-based metronome app improved gait, balance and stair climbing. Ko et al. (51) included the same metronome app to support gait training and found positive effects on gait after just 1 training session. In the studies by Shin & Song (61) and Shin (63) an app to support trunk and balance control among chronic stroke survivors was included in addition to conventional therapy, and improvements were found post-treatment in comparison with the control group. Paul et al. (28) found that chronic stroke survivors who used a tablet-based app at home increased their numbers of steps/day and reduced their level of fatigue. Also Sarfo et al. (57) found that physical activity through an app used at home resulted in improvements, although non-significant, in motor and cognitive recovery in chronic stroke survivors. Furthermore, the participants were satisfied with the intervention; however, internet connectivity and stability were a challenge in the Ghanaian context. Simpson et al. (45) found that training at home using a tablet-based app with prescribed exercises was feasible and safe, and participants adhered to both prescribed exercises and repetitions. Finally, 2 studies included app solutions that supported 2 elements of the rehabilitation process; Sureshkumar et al. (76) showed that a smartphone-based app containing both health-related information and prescribed exercises (using both videos and text) was found feasible and useful by acute stroke survivors and caregivers. Epalte et al. (78) also found the app useful in their home environment; however, the participants expressed the need for greater variety of exercises in the app, as well as a need for support in using the app solution.

App solutions to support upper limb training ($n=5$). Three included studies were targeted at supporting upper limb training. Jang & Jang (19) showed that chronic stroke survivors who trained with an app improved their hand and wrist function of the affected upper limb. Also, Lawson et al. (53) showed that chronic stroke survivors who trained at home with an app improved in many assessments post-treatment, though the results were not statistically significant. Chae et al. (74) included 17 participants with chronic stroke and found that a smart watch and a smartphone app can facilitate participation in upper-limb home rehabilitation and improve the functional score of the Wolf-Motor-Function-Test and shoulder ROM of flexion and internal rotation. Segura et al. (48) included 5 participants with chronic stroke and found a minimal clinically important improvement on upper limb motor function recovery when using a music-supported therapy app. Finally, Silveira et al. (24) included 1 stroke survivor who received a combination of functional electric stimulation (FES), music therapy and a tablet-based app. Authors found improvements in all assessments at discharge, which were maintained at 7 months follow-up.

App solutions to support speech training ($n=8$). Eight studies had developed app solutions to support training in relation to speech-related deficits. In a study by Ballard et al. (42) chronic stroke survivors improved their word production accuracy after having trained their speech using an app in addition to face-to-face visits. Participants expressed that they enjoyed the app training, but that regular contact with a speech pathologist was also important. Kondapalli et al. (52) included only 1 participant with severe aphasia in their study. Authors found improvements in communication speed and the app supported staff-patient communication. However, the setting and duration of training was not reported. Four chronic participants with aphasia also trained using an app at home in the study by Lavoie et al. (66). Significant improvements were found in naming of functional words, which were maintained 2 months post-training. In a study by Stark & Warburton (58) chronic stroke survivors with severe aphasia improved significantly after training with a tablet-based language therapy app. Woodhead et al. (77) showed that chronic stroke survivors who trained with an app at home improved reading accuracy of trained words, although not generalization to other words. Mallet et al. (54) included acute stroke survivors who used several tablet-based apps to train their speech. Most participants found the tablet-based therapy convenient. Cock et al. (46) also found a speech therapy app feasible to be used independently alongside conventional therapy by acute stroke survivors. Nef et al. (55) found high usability scores for an app with the aim to increase frequency and duration in training of stroke-related aphasia. However, the stroke severity, onset, and the setting were not reported.

App solutions to support cognitive therapy ($n=2$). Only 2 studies were targeted at supporting training of cognitive function. Kringle et al. (65) and Verna et al. (64) concluded that an app to support cognitive training was feasible. The setting was home training in the study of Kringle et al. (65), but not reported in Verna et al. (64).

JRM Journal of Rehabilitation Medicine JRM Journal of Rehabilitation Medicine JRM

App solutions to support discharge, knowledge of stroke and goal-setting (n = 5)

Five studies developed an app specifically aimed at supporting discharge and knowledge of stroke (22, 49, 67, 76, 78). In a study by Sureshkumar et al. (76) approximately half of the stroke survivors and caregivers found that the app provided very useful information about their problems, causes of the stroke, and ways to manage their recovery independently. Also, Epalte (78) concluded that the Vigo app was handy, provided understandable information, and supported participation in therapeutic activities. However, Kang et al. (22) found no significant difference between knowledge provided through an app or conventional stroke health education through a booklet. Siegel et al. (67) and Makle-Reid et al. (49) explored how an app could support a seamless communication between healthcare professionals and stroke survivors after discharge. In the study of Siegel et al. (67) only 2 patients were included, of which 1 was readmitted to hospital within 30 days after discharge. Nonetheless, both participants found the app solution useful in improving the coordination of healthcare after discharge from hospital. Markle-Reid et al. (49) included 30 participants with stroke and other comorbidities, and found the app solution to be feasible for both stroke survivors and the healthcare team that delivered outpatient rehabilitation, but found no statistically significant difference in patient health outcomes after 6 months' intervention. However, there was a significant reduction in the total per person use and costs of health services, especially a great reduction in hospital readmissions. Furthermore, this study used the app solution for several purposes; assessment, goalsetting, documentation, and communication within the health professional team as well as for providing health-related and educational material for both the health professional team and the stroke survivors/caregivers.

DISCUSSION

The aim of this scoping review was to describe and review the evidence regarding mobile/tablet and web-based apps used to support rehabilitation after stroke. A further aim was to describe participants' stroke severity and the use of apps in relation to setting and phase of the rehabilitation process. Of the 46 included studies, 16 included apps that would support assessment (stroke severity, language, motor deficits, etc.), and 25 studies supported training (lower limb and balance, upper limb, visual, cognitive and speech) among stroke survivors; however, none of these studies focused on compensatory strategies. Five studies included an app that supported health-related information/education and discharge (22, 49, 67, 76, 78). None of the 46 studies focused on supporting activities related to participation (e.g. social participation) and only 1 study focused on setting goals and continuously evaluating these (49). However, Lawson et al. (53) had a secondary goal focused on increasing participation through upper motor recovery. Likewise, optimizing the patients' language and motor abilities may have a positive effect on quality of life and social participation (28, 66). Even though app solutions supporting stroke rehabilitation are emerging, there is still a lack of research on the clinical effects of apps within post-stroke rehabilitation, which is reflected in the small number of studies, small sample sizes and study designs that involve mainly preliminary and feasibility results. In a review by Zhou et al. (31), 12 studies that integrated apps in post-stroke rehabilitation were included, with 5–48 participants. Only 2 of the 12 studies used a RCT design, and 9 studies showed positive effects regarding physical, cognitive and language function. However, neither the stroke onset or severity, nor the phase or setting in which the app was used was reported. In our scoping review, 23 studies included participants with chronic stroke, 16 studies involved participants with acute and/or subacute stroke, and 5 studies included participants with acute and/or subacute and/or chronic stroke. Two studies did not report stroke onset.

Of the 46 included studies, 21 took place in a rehabilitation setting, 17 in the participants' homes, and 3 in both settings. In 5 studies the setting was not reported. Of the 27 studies focusing on training, 14 studies were carried out in the home setting. With a shorter time on inpatient rehabilitation and reduced support from the health professionals, rehabilitation in the home setting has increased (79). Solutions that have shown positive effects on compliance and adherence to home training after discharge (45, 53) would be highly relevant and contribute to person-centred stroke rehabilitation. The use of apps can thus support stroke rehabilitation, promote self-management and empowerment in the later stages of the rehabilitation process (80). However, in the majority of apps targeting training in a home-setting, a health professional continuously supported/facilitated/adjusted/evaluated the home-training intervention using either face-to-face visits (66), and/or phone/Skype (42, 45, 48, 53, 57, 65, 74). In 2 studies support was supplied by significant others (58, 76) and in 1 study there was a lack of information regarding support (59). Since lack of continuous personalization of content and exercises may negatively affect adherence and motivation (78), it seems to be important that stroke survivors receive some kind of continuously customized support to maintain motivation, self-management and empowerment (45, 78) when interventions are delivered through an app.

JRM Journal of Rehabilitation Medicine JRM Journal of Rehabilitation Medicine JRM

Stroke rehabilitation based on mobile apps has the advantage of availability, convenience, and low cost (31). Several studies point out that apps are accepted among stroke survivors and significant others, even older patients without previous experience of using a tablet could utilize app-supported therapy (27, 44, 58). However, patients unable to use technologies due to motor or cognitive deficits are often excluded from studies (67) as shown in our scoping review. Given that 25% of stroke survivors are at risk of a new stroke or in life-long need of support (81), technologies supporting activity and participation should be taken into account (6, 7, 82). Stroke survivors who are cognitively challenged may require more training (69) and/or assistance from a significant other (54) to use the app independently. It is important to develop user-friendly apps with large text size, minimum user options, and step-by-step voice-guided directions (27). Furthermore, it should be considered how patients with limited hand function can make use of the app solutions, since double-clicking and zooming may be difficult (5, 44), although using touch screen-compatible gloves might assist.

Currently available apps have limited usability and focus, providing only one aspect of support in the rehabilitation process, such as clinical assessment or a narrow focus on exercise (30). Only three studies included in this scoping review aimed to support several components of the rehabilitation process (not only information regarding how to manage stroke-related deficits (video/text/pictures), but also discharge support, in addition to providing exercise programmes (videos and text) (76) and goal-setting (49). Stroke survivors, their significant others, and health professionals, have expressed a need for more timely information, a more coordinated cross-sectional transition from inpatient to outpatient rehabilitation, a better overview of the entire rehabilitation process, and improved follow-up and contact with health professionals after discharge (6, 7, 42, 82). An app solution aimed at supporting people with chronic diseases and accommodating patients' needs for a more comprehensive solution and a greater overview of the rehabilitation process has been tested in patients newly diagnosed with osteoporosis in Denmark (83). The app was shown to empower the patients by increasing adherence to medicine and training, and self-management of the disease (84). However, the app has not yet been tested within stroke rehabilitation and across different phases of the rehabilitation process. Future apps should accommodate stroke survivors' and significant others' needs for solutions, regardless of stroke severity and throughout the entire rehabilitation process (82).

Strengths and limitations

One strength of this scoping review was the broad literature search that resulted in a final total of 46 studies. The literature search was systematically conducted using selected databases based on relevant search terms. Although the database search was done with assistance from a librarian expert in the field, it is possible that some studies might have been missed due to lack of search terms. For example, 10 apps to support speech-related deficits (aphasia) was identified in the search. Knowing that a great research focus exists on this specific deficit, relevant studies may have been missed. However, searching all different stroke-related deficits was not the intention of this study. To get the latest published studies, a search alert was created for PubMed and CINAHL, which yielded only one further study. An updated search was therefore performed before submission, which resulted in an additional 11 studies to this scoping review. However, studies written in other languages than English, as well as studies that could not be retrieved, constitute a selection bias. A further strength was that 2 authors and a research assistant determined which studies to include, and 2 authors extracted chosen variables on the data-charting form, while bearing in mind that the extracted data was to be presented in a structured way. Disagreements on inclusion or exclusion of an abstract or full text were resolved by discussion and reaching a consensus (33). These steps increased the reliability of the study. Another strength was that transparency of the literature search and selection process throughout the study was ensured, rendering it replicable (32). The studies included had a large variation in study designs. Therefore, it was not possible to give a more specific summary of the results. However, this was not the intention; the aim was broad: we wanted to find numerous studies to present the state-of the art regarding the current use of apps in stroke rehabilitation. A further limitation, often mentioned when doing scoping reviews, is that studies with few participants are included. Three studies in this scoping review concluded that the app solution was feasible, based on only 1 or 2 participants (24, 52, 67). However, due to the heterogeneity of included studies in a scoping review, assessment of the quality of included studies was not the aim. Also, in some of the included studies stroke severity (mild, moderate, severe) was either not mentioned, or lacked detail on how it was defined or measured. A final limitation of this scoping review was the lack of a prespecified study protocol.

Conclusion

This scoping review provides an overview of studies that include mobile/tablet and web-based app solutions to support stroke rehabilitation. In all but one study,

JRM Journal of Rehabilitation Medicine JRM Journal of Rehabilitation Medicine JRM

the app targeted only one element of the rehabilitation process: usually assessment or intervention/training. Future app solutions to support stroke rehabilitation should also include participants with severe cognitive and motor deficits and should support stroke survivors and their significant others throughout the entire rehabilitation process.

Knowledge from this scoping review may have important implications for researchers and policymakers in their consideration of how and which app solutions may support stroke rehabilitation. In future studies, stroke survivors and significant others should be involved in the development of person-centred app solutions.

Rehabilitation implications

- Stroke rehabilitation can be supported by using mobile/tablet and web-based applications.
- Most developed applications focus on either assessments, supporting training or discharge from hospital.
- Applications to support the stroke survivors and the significant others throughout the entire rehabilitation process are needed.
- Most of the applications supporting stroke rehabilitation are addressed to people with mild to moderate stroke.
- People with severe stroke should be included in future research on the use of applications to support stroke rehabilitation.

ACKNOWLEDGEMENTS

The authors thank Anne-Dorte Tækker Pedersen, librarian at University College Southern Denmark (UC Syd), for valuable help with the systematic literature search. We also thank the health research departments at UC Syd and University of Southern Denmark (SDU), for time resources, and the SDU library for publication resources, so that this scoping review could be performed.

The authors have no conflicts of interest to declare.

REFERENCES

1. Katan M, Luft A. Global burden of stroke. Semin Neurol 2018; 38: 208–211.
2. Camak DJ. Addressing the burden of stroke caregivers: a literature review. J Clin Nurs 2015; 24: 2376–2382.
3. Sumathipala K, Radcliffe E, Sadler E, Wolfe CD, McKevitt C. Identifying the long-term needs of stroke survivors using the International Classification of Functioning, Disability and Health. Chronic Illn 2012; 8: 31–44.
4. Donnan GA, Fisher M, Macleod M, Davis SM. Stroke. Lancet 2008; 371: 1612–1623.
5. Kotzian ST, Haider S, Grabovac I, Schubert N, Josef S. Successful performance of basic gestures in tablet technology in post stroke patients: a feasibility study. Technol Health Care 2019; 27: 613–622.
6. Gustavsson M, Ytterberg C, Nabsen Marwaa M, Tham K, Guidetti S. Experiences of using information and communication technology within the first year after stroke – a grounded theory study. Disabil Rehabil 2018; 40: 561–568.
7. Marwaa MN, Ytterberg C, Guidetti S. Significant others' perspectives on person-centred information and communication technology in stroke rehabilitation – a grounded theory study. Disabil Rehabil 2020; 42: 2115–2122.
8. Lyden P. Using the National Institutes of Health Stroke Scale: a cautionary tale. Stroke 2017; 48: 513–519.
9. Wade DT. What is rehabilitation? An empirical investigation leading to an evidence-based description. Clin Rehabil 2020; 34: 571–583.
10. Wildevuur SE, Simonse LWL. Information and communication technology-enabled person-centered care for the "big five" chronic conditions: scoping review. J Med Internet Res 2015; 17: e77.
11. Kristensen HK, Tistad M, Koch L von, Ytterberg C. The importance of patient involvement in stroke rehabilitation. PLOS One 2016; 11: e0157149.
12. Brouns B, Meesters JJL, Wentink MM, de Kloet AJ, Arwert HJ, Vliet Vlieland TPM, Boyce LW, van Bodegom-Vos L. Why the uptake of eRehabilitation programs in stroke care is so difficult – a focus group study in the Netherlands. Implement Sci 2018. Available from: https: //doi.org/10.1186/s13012-018-0827-5
13. Ekstam L, Johansson U, Guidetti S, Eriksson G, Ytterberg C. The combined perceptions of people with stroke and their carers regarding rehabilitation needs 1 year after stroke: a mixed methods study. BMJ Open 2015; 5: e006784–e006784.
14. Bergström A, Koch L, Andersson M, Tham K, Eriksson G. Participation in everyday life and life satisfaction in persons with stroke and their caregivers 3–6 months after onset. J Rehabil Med 2015; 47: 508–515.
15. Wade DT. Describing rehabilitation interventions. Clin Rehabil 2005; 19: 811–818.
16. Wade D. Rehabilitation – a new approach. Overview and Part One: the problems. Clin Rehabil 2015; 29: 1041–1050.
17. Lindberg B, Nilsson C, Zotterman D, Söderberg S, Skär L. Using information and communication technology in home care for communication between patients, family members, and healthcare professionals: a systematic review. Int J Telemed Appl 2013; 2013: 461829
18. Iosa M, Morone G, Fusco A, Bragoni M, Coiro P, Multari M, et al. Seven capital devices for the future of stroke rehabilitation. Stroke Res Treat 2012; 2012: 187965.
19. Jang SH, Jang WH. The effect of a finger training application using a tablet PC in chronic hemiparetic stroke patients. Somatosens Mot Res 2016; 33: 124–129.
20. Tielbur BR, Rice Cella DE, Currie A, Roach JD, Mattingly B, Boone J, et al. Discharge huddle outfitted with mobile technology improves efficiency of transitioning stroke patients Into Follow-Up Care. Am J Med Qual 2015; 30: 36–44.
21. Blake H. Innovation in practice: mobile phone technology in patient care. Br J Community Nurs 2008; 13: 160–165.
22. Kang Y-N, Shen H-N, Lin C-Y, Elwyn G, Huang S-C, Wu T-F, Hou W-H. Does a mobile app improve patients' knowledge of stroke risk factors and health-related quality of life in patients with stroke? A randomized controlled trial. BMC Med Inform Decis Mak 2019; 19: 282.
23. Dubey D, Amritphale A, Sawhney A, Amritphale N, Dubey P, Pandey A. Smart phone applications as a source of information on stroke. J Stroke 2014; 16: 86.
24. Silveira TM, Tamplin J, Dorsch S, Barlow A. Let's Improvise! iPad-based music therapy with functional electrical stimulation for upper limb stroke rehabilitation. Aust J Music Ther 2018; 29.
25. Kim J, Park S, Lim H, Park G, Kim M, Lee B. Effects of the combination of rhythmic auditory Stimulation and task-oriented training on functional recovery of subacute stroke patients. J Phys Ther Sci 2012; 24: 1307–1313.

JRM Journal of Rehabilitation Medicine JRM Journal of Rehabilitation Medicine JRM

26. Tistad M, von Koch L, Sjöstrand C, Tham K, Ytterberg C. What aspects of rehabilitation provision contribute to self-reported met needs for rehabilitation one year after stroke - amount, place, operator or timing? Health Expect 2013; 16: e24–e35.
27. Choi Y-H, Park HK, Ahn K, Son Y, Paik N-J. A telescreening tool to detect aphasia in patients with stroke. Telemed E-Health 2015; 21: 729–734.
28. Paul L, Wyke S, Brewster S, Sattar N, Gill JMR, Alexander G, et al. Increasing physical activity in stroke survivors using STARFISH, an interactive mobile phone application: a pilot study. Top Stroke Rehabil 2016; 23: 170–177.
29. McKay FH, Cheng C, Wright A, Shill J, Stephens H, Uccellini M. Evaluating mobile phone applications for health behaviour change: a systematic review. J Telemed Telecare 2018; 24: 22–30.
30. Fell N, True HH, Allen B, Harris A, Cho J, Hu Z, et al. Functional measurement post-stroke via mobile application and body-worn sensor technology. mHealth 2019; 5: 47–47.
31. Zhou X, Du M, Zhou L. Use of mobile applications in post-stroke rehabilitation: a systematic review. Top Stroke Rehabil 2018; 25: 489–499.
32. Levac D, Colquhoun H, O'Brien KK. Scoping studies: advancing the methodology. Implement Sci 2010; 5: 69.
33. Munn Z, Peters MDJ, Stern C, Tufanaru C, McArthur A, Aromataris E. Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. BMC Med Res Methodol 2018; 18: 143.
34. Arksey H, O'Malley L. Scoping studies: towards a methodological framework. Int J Soc Res Methodol 2005; 8: 19–32.
35. Joanna Briggs Institute Introduction to Scoping reviews. Why a scoping review? (accessed 30 Mar 2020). Available from: https: //wiki.joannabriggs.org/display/MANUAL/11.1+Introduction+to+Scoping+reviews
36. Cronin P, Ryan F, Coughlan M. Undertaking a literature review: a step-by-step approach. Br J Nurs 2008; 17: 38–43.
37. Graneheim UH, Lindgren B-M, Lundman B. Methodological challenges in qualitative content analysis: a discussion paper. Nurse Educ Today 2017; 56: 29–34.
38. National Institute for Health and Care Excellence. Stroke rehabilitation, NICE Pathways 2021 (accessed 01 Jun 2021).Available from: http://pathways.nice.org.uk/pathways/stroke.
39. Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): checklist and explanation. Ann Intern Med 2018; 169: 467–473.
40. Cooray C, Matusevicius M, Wahlgren N, Ahmed N. Mobile phone-based questionnaire for assessing 3 months modified Rankin Score after acute atroke: a pilot study. Circ Cardiovasc Qual Outcomes 2015; 8: S125–S130.
41. Capela NA, Lemaire ED, Baddour N, Rudolf M, Goljar N, Burger H. Evaluation of a smartphone human activity recognition application with able-bodied and stroke participants. J NeuroEngineering Rehabil 2016; 13: 5.
42. Ballard KJ, Etter NM, Shen S, Monroe P, Tien Tan C. Feasibility of automatic speech recognition for providing feedback during tablet-based treatment for apraxia of speech plus aphasia. Am J Speech Lang Pathol 2019; 28: 818–834.
43. Chang H, Zhao J, Qiao Y, Yao H, Wang X, Li J, Liu J. Mobile phone application for self-assessment of acute stroke patients: a tool for extended care and follow-up. Medicine (Baltimore) 2018; 97: e11263.
44. Kizony R, Zeilig G, Dudkiewicz I, Schejter-Margalit T, Rand D. Tablet apps and eexterity: comparison between 3 age groups and proof of concept for stroke Rehabilitation. J Neurol Phys Ther 2016; 40: 31–39.
45. Simpson DB, Bird M-L, English C, Gall SL, Breslin M, Smith S, et al. Connecting patients and therapists remotely using technology is feasible and facilitates exercise adherence after stroke. Top Stroke Rehabil 2020; 27: 93–102.
46. De Cock E, Batens K, Feiken J, Hemelsoet D, Oostra K, De Herdt V. The feasibility, usability and acceptability of a tablet-based aphasia therapy in the acute phase following stroke. J Commun Disord 2021; 89: 106070.
47. Costa PHV, de Jesus TPD, Winstein C, Torriani-Pasin C, Polese JC. An investigation into the validity and reliability of mHealth devices for counting steps in chronic stroke survivors. Clin Rehabil 2020; 34: 394–403.
48. Segura E, Grau-Sánchez J, Sanchez-Pinsach D, De la Cruz M, Duarte E, Arcos JL, et al. Designing an app for home-based enriched music-supported therapy in the rehabilitation of patients with chronic stroke: a pilot feasibility study. Brain Inj 2021; 35: 1585–1597.
49. Markle-Reid M, Valaitis R, Bartholomew A, Fisher K, Fleck R, Ploeg J, Salerno J. An integrated hospital-to-home transitional care intervention for older adults with stroke and multimorbidity: a feasibility study. J Comorbidity 2020; 10: 2235042X1990045.
50. Ciou S-H, Hwang Y-S, Chen C-C, Luh J-J, Chen S-C, Chen Y-L. Football APP based on smart phone with FES in drop foot rehabilitation. Technol Health Care 2017; 25: 541–555.
51. Ko B-W, Lee H-Y, Song W-K. Rhythmic auditory stimulation using a portable smart device: short-term effects on gait in chronic hemiplegic stroke patients. J Phys Ther Sci 2016; 28: 1538–1543.
52. Kondapalli A, Zhang LR, Patel S, Han X, Kim HJ, Li X, Altschuler EL. A phonology-free mobile communication app. Disabil Rehabil Assist Technol 2016; 11: 678–682.
53. Lawson S, Tang Z, Feng J. Supporting stroke motor recovery through a mobile application: A pilot study. Am J Occup Ther 2017; 71: 7103350010p1.
54. Mallet KH, Shamloul RM, Corbett D, et al. RecoverNow: feasibility of a mobile tablet-based rehabilitation intervention to treat post-stroke communication deficits in the acute care setting. PLOS One 2016; 11: e0167950.
55. Nef T, Wyss P, Gerber S, et al. WCNR 2018 oral abstracts. Neurorehabil Neural Repair 2018; 32: 317–362.
56. Park H, Jeon S, Lee J-Y, Cho A-R, Park JH. Korean version of the mini-mental state examination using smartphone: a validation study. Telemed E-Health 2017; 23: 815–821.
57. Sarfo FS, Adusei N, Ampofo M, Kpeme FK, Ovbiagele B. Pilot trial of a tele-rehab intervention to improve outcomes after stroke in Ghana: a feasibility and user satisfaction study. J Neurol Sci 2018; 387: 94–97.
58. Stark BC, Warburton EA. Improved language in chronic aphasia after self-delivered iPad speech therapy. Neuropsychol Rehabil 2018; 28: 818–831.
59. Szalados R, Leff AP, Doogan CE. The clinical effectiveness of eye-search therapy for patients with hemianopia, neglect or hemianopia and neglect. Neuropsychol Rehabil 2021; 31: 971–982.
60. Grau-Pellicer M, Lalanza J, Jovell-Fernández E, Capdevila L. Impact of mHealth technology on adherence to healthy PA after stroke: a randomized study. Top Stroke Rehabil 2020; 27: 354–368.
61. Shin DC, Song CH. Smartphone-based visual feedback trunk control training using a gyroscope and mirroring technology for stroke patients: single-blinded, randomized clinical trial of efficacy and feasibility. Am J Phys Med Rehabil 2016; 95: 319–329.
62. Li L, Huang J, Wu J, Jiang C, Chen S, Xie G, et al. A mobile health app for the collection of functional outcomes after inpatient stroke rehabilitation: pilot randomized controlled trial. JMIR MHealth UHealth 2020; 8: e17219.
63. Shin D-C. Smartphone-based visual feedback trunk control training for gait ability in stroke patients: a single-blind randomized controlled trial. Technol Health Care 2020; 28: 45–55.
64. Verna V, De Bartolo D, Iosa M, Fadda L, Pinto G, Caltagirone C, et al. Te.M.P.O., an app for using temporal musical mismatch in post-stroke neurorehabilitation: a preliminary randomized controlled study. NeuroRehabilitation 2020; 47: 201–208.

JRM Journal of Rehabilitation Medicine JRM Journal of Rehabilitation Medicine JRM

65. Kringle EA, Setiawan IMA, Golias K, Parmanto B, Skidmore ER. Feasibility of an iterative rehabilitation intervention for stroke delivered remotely using mobile health technology. Disabil Rehabil Assist Technol 2020; 15: 908–916.
66. Lavoie M, Bier N, Macoir J. Efficacy of a self-administered treatment using a smart tablet to improve functional vocabulary in post-stroke aphasia: a case-series study: Using a smart tablet to improve functional vocabulary in post-stroke aphasia. Int J Lang Commun Disord 2019; 54: 249–264.
67. Siegel J, Edwards E, Mooney L, Smith C, Peel JB, Dole A, et al. A feasibility pilot using a mobile personal health assistant (PHA) app to assist stroke patient and caregiver communication after hospital discharge. mHealth 2016; 2: 31–31.
68. Meeuwis IHM, van Rooijen EM, Verhagen WIM. The use of WhatsApp Messenger: an important tool in modern neurological examination. Acta Neurol Belg 2020; 120: 1045–1049.
69. Guo YE, Togher L, Power E, Hutomo E, Yang Y-F, Tay A, et al. Assessment of aphasia across the International Classification of Functioning, Disability and Health Using an iPad-based application. Telemed E-Health 2017; 23: 313–326.
70. Faria GS, Polese JC, Ribeiro-Samora GA, Scianni AA, Faria CDCM, Teixeira-Salmela LF. Validity of the accelerometer and smartphone application in estimating energy expenditure in individuals with chronic stroke. Braz J Phys Ther 2019; 23: 236–243.
71. Joo S, Lee K, Song C. A comparative study of smartphone game with spirometry for pulmonary function assessment in stroke patients. BioMed Res Int 2018; 1–6.
72. Polese JC, e Faria GS, Ribeiro-Samora GA, Lima LP, Coelho de Morais Faria CD, Scianni AA, Teixeira-Salmela LF. Google Fit smartphone application or Gt3X Actigraph: which is better for detecting the stepping activity of individuals with stroke? A validity study. J Bodyw Mov Ther 2019; 23: 461–465.
73. Wall KJ, Cumming TB, Koenig ST, Pelecanos AM, Copland DA. Using technology to overcome the language barrier: the Cognitive Assessment for Aphasia App. Disabil Rehabil 2018; 40: 1333–1344.
74. Chae SH, Kim Y, Lee K-S, Park H-S. Development and clinical evaluation of a web-based upper limb home rehabilitation system using a smartwatch and machine learning model for chronic stroke survivors: prospective comparative study. JMIR MHealth UHealth 2020; 8: e17216.
75. De Bartolo D, Morone G, Lupo A, Aloise F, Baricich A, Di Francesco D, et al. From paper to informatics: the Post Soft Care-App, an easy-to-use and fast tool to help therapists identify unmet needs in stroke patients. Funct Neurol 2018; 33: 200–205.
76. Sureshkumar K, Murthy GVS, Munuswamy S, Goenka S, Kuper H. 'Care for Stroke', a web-based, smartphone-enabled educational intervention for management of physical disabilities following stroke: feasibility in the Indian context. BMJ Innov 2015; 1: 127–136.
77. Woodhead ZVJ, Kerry SJ, Aguilar OM, Ong Y-H, Hogan JS, Pappa K, Leff AP, Crinion JT. Randomized trial of iReadMore word reading training and brain stimulation in central alexia. Brain 2018; 141: 2127–2141.
78. Epalte K, Tomsone S, Vētra A, Bērziņa G. Patient experience using digital therapy "Vigo" for stroke patient recovery: a qualitative descriptive study. Disabil Rehabil Assist Technol 2020; 1–10.
79. Mawson S, Nasr N, Parker J, Zheng H, Davies R, Mountain G. Developing a personalised self-management system for post stroke rehabilitation; utilising a user-centred design methodology. Disabil Rehabil Assist Technol 2014; 9: 521–528.
80. Sit JWH, Chair SY, Yip CWH Chan, Tang SW, Choi KC, Lee DTF, et al. Effect of health empowerment intervention for stroke self-management on behaviour and health in stroke rehabilitation patients. Hong Kong Med J 2018; 24 Suppl 2: 12–15.
81. The Brain Damage Association. (hjerneskader i tal). (accessed 18 Dec 2018). Available from: https://hjerneskadet.dk/om-hjerneskader/tal-og-fakta-om-hjerneskader/).
82. Marwaa MN, Kristensen HK, Guidetti S, Ytterberg C. Physiotherapists' and occupational therapists' perspectives on information and communication technology in stroke rehabilitation. Disabil Rehabil 2020: 15: e0236831.
83. Region Syddanmark. My hospital in the region of Southern Denmark [Mit Sygehus i region Syddanmark]. Accessed 23 Aug 2019). Avalilable from: https://www.regionsyddanmark.dk/wm509838
84. Ravn Jakobsen P, Hermann AP, Søndergaard J, Wiil UK, Clemensen J. Help at hand: women's experiences of using a mobile health application upon diagnosis of asymptomatic osteoporosis. SAGE Open Med 2018; 6:205031211880761.

Research Involvement and Engagement

RESEARCH

Open Access

Using experience-based co-design to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation

Mille Nabsen Marwaa[1,2,3*], Susanne Guidetti[2,4], Charlotte Ytterberg[2,4,5*] and Hanne Kaae Kristensen[3]

Abstract

Aim To investigate and describe the process of using experience-based co-design (EBCD) to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation.

Setting Two cross-sectoral stroke rehabilitation settings in Denmark comprising six rehabilitation units.

Participants Stroke survivors (n = 23), significant others (n = 18), occupational therapists (n = 12), physiotherapists (n = 9), representative of a patient organization (n = 1), application developers (n = 3) and researchers (n = 2).

Method A structured, facilitated EBCD approach comprising six stages was used to co-design a service that aimed to address the priorities and needs of all relevant end-users. Data were collected by interviews, participant observations, notes on "flip sheets" and written feedback on the content in the apps and on the instruction pages. Data were analyzed descriptively and with a constructivist grounded theory analysis.

Results The content in the application solutions "Mit Sygehus" and "Genoptræn.dk" were co-designed to support the needs identified by all end-users. Relevant evidence-based knowledge, person-centred exercises and guidelines using video recordings were the most important among the developed content in the applications to support person-centred and empowering stroke rehabilitation. Furthermore, easy, and seamless communication were considered important.

Conclusions EBCD facilitated the development of content in the applications to support a person-centred and empowering stroke rehabilitation. Participants experienced that their contribution was considered important and valued.

Plain English summary

Many app-based solutions exist to support stroke rehabilitation; however, stroke survivors and significant others are seldom involved in the design of these solutions, which often reduces the uptake in clinical setting.

*Correspondence:
Mille Nabsen Marwaa
mnma@ucsyd.dk
Charlotte Ytterberg
charlotte.ytterberg@ki.se
Full list of author information is available at the end of the article

© The Author(s) 2023. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/. The Creative Commons Public Domain Dedication waiver (http://creativecommons.org/publicdomain/zero/1.0/) applies to the data made available in this article, unless otherwise stated in a credit line to the data.

By involving stroke survivors, significant others, therapists from diverse rehabilitation settings, patient organizations, app-designers, and researchers, we co-designed the content in two app-solutions to support a person-centred and empowering stroke rehabilitation, based on the needs and wishes of all stakeholders. Stroke survivors and significant others experienced that their voices were heard throughout the co-design process, and that their contribution was considered important and valued.

Implications

- Relevant content of applications to support a person-centred and empowering stroke rehabilitation can be identified and developed using EBCD.
- In depth understanding of stroke survivors, significant others and therapists needs and perspectives are valuable and important in an EBCD process to promote development of new useable solutions.
- Stroke survivors, significant others and therapists perceived that participation in an EBCD process and sharing experiences were valuable.

Introduction

Stroke is recognized as a major cause of disability worldwide [1] and is often associated with long-term physical, cognitive, social and emotional disabilities [2–4], that can result in dependence and reduced participation in everyday life [3, 5, 6]. To respond to these challenges, priority must be given to person-centred and empowering rehabilitation strategies [7, 8]. It is suggested that empowerment can be achieved through person-centred rehabilitation [9, 10], provided in collaboration between stroke survivors and health professionals, based on the needs, preferences and goals identified by the stroke survivors and their significant others [7, 8]. There are several synonyms for person-centred initiatives, such as client-centred, user-centred and individual focus [11]. In this study we describe person-centred rehabilitation as initiatives/interventions that supports the individual stroke survivors' and significant others' needs, preferences and goals for the rehabilitation e.g., therapy, interventions, exercises, and knowledge [7, 8]. Furthermore, a person-centred rehabilitation should start as early as possible, be coherent (i.e., coordinated with smooth transitions between different rehabilitation settings) and should contain goal setting, intervention/training, support for social participation, discharge support and continuous evaluation [7, 12].

Nonetheless, studies show that stroke survivors, significant others, occupational therapists (OTs) and physiotherapists (PTs) working in stroke rehabilitation often experience challenges related to cross-sectoral information- and communication, and gaining an overview of the stroke rehabilitation process and call for a more person-centred and tailored stroke rehabilitation [5, 6, 13, 14]. Several studies suggest that information and communication technologies (ICT) such as mobile phone and tablet applications (apps) have the potential to support person-centred stroke rehabilitation and empower stroke survivors and significant others throughout the rehabilitation process. ICT and apps (e.g. phone calls, messages, reminders, remote monitoring or intervention, videos and images) may support the delivery of relevant, evidence-based health information [15–17], support continuity and reduce gaps in the rehabilitation process [18], and support the feeling of being connected with the therapists [17]. Furthermore, stroke survivors' motivation, engagement and empowerment to independently perform person-centred exercises post-discharge [17, 19], and to become more active in managing their own health may be supported by ICT and apps [15–17, 20–24]. However, because of the lack of tailoring to individual needs, many app solutions do not support person-centred rehabilitation [13]. Furthermore, lack of end-user involvement in developing the solutions reduces the uptake [25, 26]. A recent scoping review also shows that many app solutions developed to support stroke rehabilitation are narrowly focused on only one or few parts of the entire rehabilitation process e.g., information, exercises, or single health related measurements (weight loss, or blood glucose management), and do not support a person-centred and empowering stroke rehabilitation for stroke survivors and significant others [27].

There is a need to involve end-users in the development of content in the app solutions. "Mit Sygehus" [in English, My hospital] is a knowledge-based and safe app solution implemented in all hospitals in the southern region of Denmark. "Mit Sygehus" is currently implemented in over 300 diagnoses in several departments (e.g. gynecology, oncology, orthopedics, surgery, endocrinology, rheumatology, etc.) [28]. The existing modules in the app have shown potential to accommodate many of the above-mentioned unmet needs expressed by stroke survivors, significant others, and therapists in relation to the stroke rehabilitation process, however, "Mit Sygehus" has not yet been designed, and thus has

no content, to support stroke rehabilitation. "Genoptræn.dk" [in English, Rehab.dk], is a training-based app-solution already integrated with "Mit Sygehus" as a "training" module. "Genoptræn.dk" already contains more than 700 generic video-recorded exercises, that can be assigned by the therapists to the patients' personal app [29]. "Genoptræn.dk" is currently being used in both hospital and municipal rehabilitation settings; however, the content in the app has not yet been developed to support person-centred-stroke rehabilitation.

Stroke survivors represent a heterogeneous group and often have complex needs for person-centred rehabilitation processes. Therefore, stroke survivors, significant others, therapists, app developers and researchers all need to be involved in a co-design process using some of the modules in "Mit Sygehus" and "Genoptræn.dk" and in developing new functions to match the needs of the end-users.

Experience-based co-design

Patient-involvement to improve health care has been applied since 1993 [30], and experience-based co-design (EBCD) since 2006 [31]. However, EBCD literature within stroke rehabilitation is lacking, and have mainly been focused on improving the activity level of stroke survivors in acute care settings [32–34]. EBCD has often been used as a quality improvement methodology [35], to address the priorities, needs, and wishes of all relevant end-users in a collaborative process [31, 36]. EBCD approaches draw upon participatory action research (AR) design and user experience design [25, 31]. AR principles support the co-design actions in EBCD and recognize that especially qualitative data may more often lead to actions taken to improve patient experiences and healthcare services [31], since quantitative methods do not indicate what needs to be done to improve any situation [25]. EBCD represents a shift in the perception of patients and significant others—from a role characterized by passivity and dependency to participation as more active, empowered and autonomous individuals [31, 35, 37]. The novelty of EBCD is thus involving end-users throughout all research phases and reflecting on the actions and activities from each phase to support the next [25, 35]. Ideally, this results in greater uptake of the service and increased engagement throughout the process [25, 35], provided that equal power dynamics are achieved between stakeholders [30, 32, 38]. There has been limited scientific evaluation of the process and outcomes of the EBCD approach [39], especially within stroke rehabilitation [32]. Therefore, the overall aim of this study was to investigate and describe the process of using EBCD to develop the content in two mobile and tablet apps to support a person-centred and empowering stroke rehabilitation. This included describing how user involvement was achieved and perceived by participants throughout the EBCD process and how the reflections and learnings from each stage of the EBCD cycle supported the actions in the next stage.

Methods and results

Experience-based co-design

EBCD involves a structured, facilitated two-phased process that comprises six stages (see Table 1). The discovery phase of EBCD involves the first three stages: setting up the project (stage 1), observations of a particular rehabilitation pathway and engaging patients, significant others, and staff concerning their experiences of rehabilitation (stage 2 and 3). The co-design phase also comprises three stages. In this phase, end-user experiences are analyzed to identify "emotional touchpoints" in healthcare services, where something could have been done better, or which exemplify a good experience. Identified "touchpoints", supported by existing evidence on the topic, are then presented to all end-users to trigger discussion about local quality issues and agree on a set of improvement priorities, through co-design groups (stage 4). "Touchpoints" can be presented by using "trigger films" or "touchpoint lists", experience maps, list of improvements or interview quotes [38]. Smaller co-design working groups (stage 5) and a celebration event brings the co-design phase to an end (stage 6) [37]. Reflective questions such as "what did we learn from this stage", "how was the power relation between participants and how can we make it more equal" and "how useful was this stage for the research process" support the next stages and actions in the EBCD process [25, 35]. The GRIPP2 reporting checklist on patient and public involvement in research was reported in Additional file 1 [40].

Discovery phase

Stage 1: setting up the study

In Denmark stroke rehabilitation may take place in different stroke rehabilitations settings. Initially, the stroke survivor is admitted to an acute hospital stroke unit. When required, rehabilitation may continue in a subacute hospital stroke unit and/or rehabilitation continues in the municipalities (i.e., home-rehabilitation or in rehabilitation centres). Participants in need of rehabilitation in all rehabilitation settings most often have more complex difficulties (i.e. cognitive, emotional, mental and/or physical) and need individualized and intensive interdisciplinary rehabilitation delivered in acute and subacute stroke units, before they can continue their rehabilitation in the municipalities [41]. The rehabilitation process when in need of all three rehabilitation settings may

Table 1 Overview of purposes and activities in the EBCD process

	EBCD phases	Activities		Purpose
Discovery phase	*Stage 1*	*Engaging stroke rehabilitation settings (n = 6)*		Engage different stroke rehabilitation settings
	Setting up the project 2014–2017	Acute stroke units (n = 2)		
		Subacute stroke units (n = 2)		
		Stroke units in the municipalities (n = 2)		
	Stage 2 + 3	*Interviews (focus group or individual)*		Explore experiences of stroke rehabilitation across sectors, identification of unmet needs and use of ICT in the rehabilitation process and everyday life
	Engaging stroke survivors, significant others, and therapists 2014–2017	Stroke survivors (n = 18)		
		Significant others (n = 13)		
		PTs (n = 4)		
		OTs (n = 5)		
	2021	*Observations of stroke rehabilitation settings (5 settings)*		
Co-design phase	*Stage 4*	*Workshop 1 (Jutland) (n = 14)*	*Workshop 1 (Funen) (n = 11)*	Present "emotional touchpoints" and the modules in "Mit Sygehus". Additionally, to generate concrete input on the content of each module, to meet stakeholders' needs and to prioritize the modules
	Co-design meetings 2021	Stroke survivors (n = 2)	Stroke survivors (n = 2)	
		Significant others (n = 2)	Significant others (n = 2)	
		PTs (n = 3)	PTs (n = 2)	
		OTs (n = 3)	OT (n = 1)	
		App developers (n = 2)	App developers (n = 2)	
		Researchers (n = 2)	Researchers (n = 2)	
		Scoping review on existing app solutions to support stroke rehabilitation		
	Stage 5	*Step 1: Co-design meetings in Jutland working with "knowledge" module (n = 4)*	*Step 1: Co-design meetings on Funen working with "knowledge" module (n = 3)*	Co-design content in "Mit Sygehus" to meet stakeholders' prioritized needs. Furthermore, to assess if the "training module" was suitable to meet stakeholders' needs and applicable in the different rehabilitation settings
	Small co-design meetings 2021	PT (n = 1)	PT (n = 1)	
		OT (n = 1)	Speech therapist (n = 1)	
		Head therapist (n = 1)	Physician (n = 1)	
		Representative from patient organization (n = 1)		
		Step 2: Co-design meetings in Jutland working with "training" module (n = 4)	*Step 2: Co-design meetings on Funen working with "training" module (n = 5)*	
		PT (n = 1)	PT (n = 1)	
		OT (n = 1)	OT (n = 1)	
		App developer (n = 1)	App developers (n = 2)	
		Researcher (n = 1)	Researcher (n = 1)	

Table 1 (continued)

EBCD phases	Activities		Purpose
Stage 6	*Workshop 2 (Jutland) (n = 10)*	*Workshop 2 (Funen) (n = 9)*	Presentation of app solution and "hands-on" test of all the content in the app, as well as feedback on written instructions to be used in later a testing period
Celebration event 2021	PTs (n = 2)	Stroke survivor (n = 1)	
	OTs (n = 3)	Significant other (n = 1)	
	Representative from a patient organization (n = 1)	PT (n = 1)	
	App developers (n = 2) -one attended online	OTs (n = 2)	
	Researchers (n = 2)	App developers (n = 2)	
		Researchers (n = 2)	
		Workshop 2 (Funen) (n = 4) Online	
		PT (n = 1)	
		OT (n = 1)	
		App developer (n = 1)	
		Researcher (n = 1)	

range from 1 to 4 months. To have representation from all stroke rehabilitation settings, two acute stroke units, two subacute stroke units (i.e., the hospitals responsible for in-ward stroke rehabilitation in the region of Southern Denmark) and two larger stroke rehabilitation units in the municipalities were engaged in stage 1.

Stage 2 and 3: engaging patients, significant others, and staff and performing observations

Participants The head therapist of each rehabilitation setting identified two therapists (one OT and one PT) who voluntarily agreed to participate in focus group interviews in this stage. Stroke survivors living at home who had just ended their rehabilitation process throughout all rehabilitation settings (i.e., participants with more complex difficulties) and significant others (i.e., someone who was close to the stroke survivor and offered support) were identified by the municipal therapists to participate in interviews in this stage. The inclusion criteria for stroke survivors were (a) ability to understand and answer interview questions and (b) experience of ICT before and after stroke. For therapists, criteria were that they should (a) work in one of the stroke rehabilitation settings, (b) as either a PT or OT.

Method To capture experiences in relation to the stroke rehabilitation process, unmet needs, and the integration of ICT as part of the rehabilitation process and in everyday life, individual or focus group interviews were held with stroke survivors (n = 18), significant others (n = 13), and therapists (n = 9) from different settings of the Danish stroke rehabilitation process. Stroke survivors and significant others were interviewed 6–12 months post-stroke. Analysis of the transcribed interviews was performed using a constructivist grounded theory (GT) approach comprising open and focused coding, constant comparative method and theoretical sensitivity [42].

Additionally, prior to stage 4, non-participant and participant observations were performed by the first author in five of the six stroke rehabilitation settings included in this study for a total of five full days. The observations had several purposes and benefits: (1) as an "outsider" in all rehabilitation settings, the obvious benefit was that relations and connections were made with all participating therapists, (2) therapists used the opportunity to ask more about the project and get a greater insight to the purpose of the next stage of the research process, and (3) observations supported and contributed to the identified "touchpoints". The municipality in Jutland agreed to participate in this study just before stage 4 of the EBCD process, and therefore there was not time enough to perform observations in this municipality.

Insight to similarities and differences in local organizational processes, context, interventions, and interactions among therapists, stroke survivors and significant others was valuable knowledge.

For example, knowledge regarding the short hospitalization and the number of therapists involved in the stroke survivors' rehabilitation process made it clear that especially in acute rehabilitation settings continuity was difficult to achieve, which may challenge the implementation of the apps. The insights from the observations were also used to facilitate discussions in the workshops, regarding which apps and which content that would fit the different settings.

Findings The results from the interviews have been reported elsewhere [5, 6, 13]. In summary, findings indicated that stroke survivors and significant others welcomed the use of ICT and apps to support stroke rehabilitation, since they may promote activity, participation in everyday life, independency, and adherence to perform person-centred exercises [5, 6]. Therapists highlighted that ability to use ICT and apps are crucial for stroke survivors in today's society and that ICT and apps may support continuity and coherence of the rehabilitation process as well as support a person-centred rehabilitation e.g., through videorecorded exercise therapy and guidelines [13].

A cross-analysis was furthermore performed by the first author (and discussed with the last author), using a constructivist GT analysis to find "emotional touchpoints" across interviews to inform stage 4. The purpose of a cross-analysis was to make sure to bring participants' experiences and voices to the next research stage and to transform these into actions.

Co-design phase

Stage 4: co-design meetings

Workshop 1 *Purpose* The purpose of the first workshops was to generate concrete input on the content of each module of the apps, to meet the needs of all participants and to prioritize the modules. Additionally, to understand which mechanisms in stroke rehabilitation that were perceived important by the stakeholders to support empowerment and how participating in the EBCD process was perceived.

The workshops were planned and led by the first and last author in the spring of 2021 and took place in the Region of Southern Denmark (one in Jutland, the western part of Denmark, and one on Funen, the central part of Denmark).

Participants The head therapists (mentioned in stage 1) identified one OT and one PT from each rehabilitation

setting to participate in the workshops. Therapists from the municipalities identified stroke survivors living at home that had either recently terminated or were about to terminate their rehabilitation process and had received rehabilitation in all rehabilitation settings (acute, subacute and rehabilitation in the municipalities, i.e., having complex difficulties, however being able to participate for 2 h and without severe communication deficits) and a significant other to participate in the workshops. In the Jutland workshop, 14 participated and 11 on Funen (see Table 2 for participant characteristics). Of the 12 therapists participating in stage 4, three had also participated in the interviews in stage 2 and 3. Each workshop lasted for 2 h.

Method The workshop was started by participants giving written informed consent to participate, and consent to allow photographs to be taken during the workshops that could be used for presenting the results. Prior to attending the workshop, the participants had received written information about the project and the purpose of the workshops. The first author introduced the background for conducting the workshops. "Emotional touchpoints" identified from stage 2 and 3 (Table 3) were also presented i.e., the need for easier access to knowledge about stroke and the option to have the text read out loud. Furthermore, the need for more person-centred rehabilitation was presented, i.e., exercises that would be tailored to a person's specific motor and/or cognitive difficulties, video-recorded guidelines for transfers, etc. Also presented were the need to minimize the gaps when a patient transfer between rehabilitation settings, through easier documentation and communication with significant others and colleagues across rehabilitation settings. Other important "touchpoints" presented were a need for follow-up when rehabilitation terminates, continuous support from therapists throughout rehabilitation and support to learn to use ICT and apps that may support rehabilitation and everyday life, thereby relieving strains on significant others. Also, a need for support to establish contact with peers/support groups was presented. Accommodating the expressed needs identified in Table 3 was intended to lead to a more person-centred stroke rehabilitation and increased empowerment in stroke survivors and significant others [7, 9, 10, 12].

Additionally, knowledge from a recent scoping review on existing apps used to support stroke rehabilitation [27] supported the list of "touchpoints" and the content of the workshops. The review showed that app solutions to support stroke rehabilitation can be used in different rehabilitation settings, however most existing app solutions only support a limited aspect of the rehabilitation process, e.g., assessment or training, and do not

Table 2 Characteristics of participants in the two workshops

Participants	Age, years	Gender	Origin	Workshop 1	Workshop 2	Rehabilitation setting	Work experience in neurological rehabilitation, years
Stroke survivor and husband	63/66	Female/male	Funen	X			
Stroke survivor and wife	85/79	Male/female	Funen	X			
Stroke survivor and wife	57/55	Male/female	Funen		X		
Stroke survivor and cousin	37/35	Female/female	Jutland	X			
Stroke survivor and husband	69/74	Female/male	Jutland	X			
OT	44	Female	Funen		X	Acute	18
OT	47	Female	Funen		X (online)	Subacute	19
OT	38	Female	Funen		X	Municipality	10
OT	46	Female	Funen	X		Municipality	13
OT	46	Female	Jutland	X	X	Acute	4
OT	29	Male	Jutland	X	X	Subacute	6
OT	36	Female	Jutland	X	X	Municipality	3
PT	35	Female	Funen	X	X (online)	Subacute	11
PT	37	Female	Funen	X	X	Municipality	6
PT	43	Female	Jutland	X		Acute	14
PT	25	Female	Jutland	X	X	Subacute	2
PT	36	Female	Jutland	X	X	Municipality	8
Representative from patient organization	48	Female	Jutland		X		

Table 3 Emotional touchpoints

	Emotional touchpoints—Needs and areas for improvement identified			
	Acute and subacute care in specialized stroke units (inpatient) Typically, 0–4 weeks	**Transition to home-rehabilitation**	**Rehabilitation in the municipalities Typically, 1–3 months**	**Maintenance phase (rehabilitation terminated)**
Stroke survivors	Easy access to relevant information (diagnosis, prognosis, and consequences (brochures, app solution) and "read out loud" function An overview of the goals and planned activities (rehabilitation plan) Assessments of the ability to use ICT and need to learn to use ICT (mobile phone, tablet, computer, apps) Need for person-centred guidelines and exercises	Need for overview and transparency	Managing everyday life (learning what deficits influence daily living and participation) Tracking (phone calls to feel safe/physical activity tracking) Need for support to learn to manage ICT Staying connected through ICT	Access to health professionals continuously (new questions arise, a need for answers) Support groups (peers)
Significant others	Easy access to relevant information (diagnosis, prognosis, and consequences (brochures, app solution) and "read out loud" function Need for insight into rehabilitation plan Need for therapists to support person-centred daily training (relieving the significant others)	Involvement No gaps Skills to manage new role	Insight into rehabilitation plan Therapists to support daily training (relieving the significant others) Stroke survivor being able to manage ICT—training/skills (to feel safe/being able to participate in daily activities, gaining overview (work, hobbies, etc.)	Lack of follow-up A "life-line" to reduce strain (responsibility shared with health professionals regarding adherence to exercises/activities)—reminders, phone calls, text messages, app solution) Peers to share experiences with
Therapists	Assessment, goal setting and training of cognitive and physical deficits/resources (using ICT when relevant) Teaching skills to manage ICT and how to comprehend/get an overview of a complex rehabilitation process Easy and simple ICT solutions for stroke survivors Person-centred guidelines for transfers, exercises, and daily activities (pictures, documents, videos, "read out loud" function) Easier documentation/communication in and across sectors (also with significant others)	Sharing information across sectors (pictures, videos) Different IT systems in sectors is a barrier Fewer places to document	Training gross motor skills, fine motor skills and cognitive deficits Involving caregivers even more Learning to manage new life conditions—using ICT to compensate (calendar, reminders, dictate messages)	Monthly/3-monthly phone calls from health professionals Access/contact to patient organizations (peers)

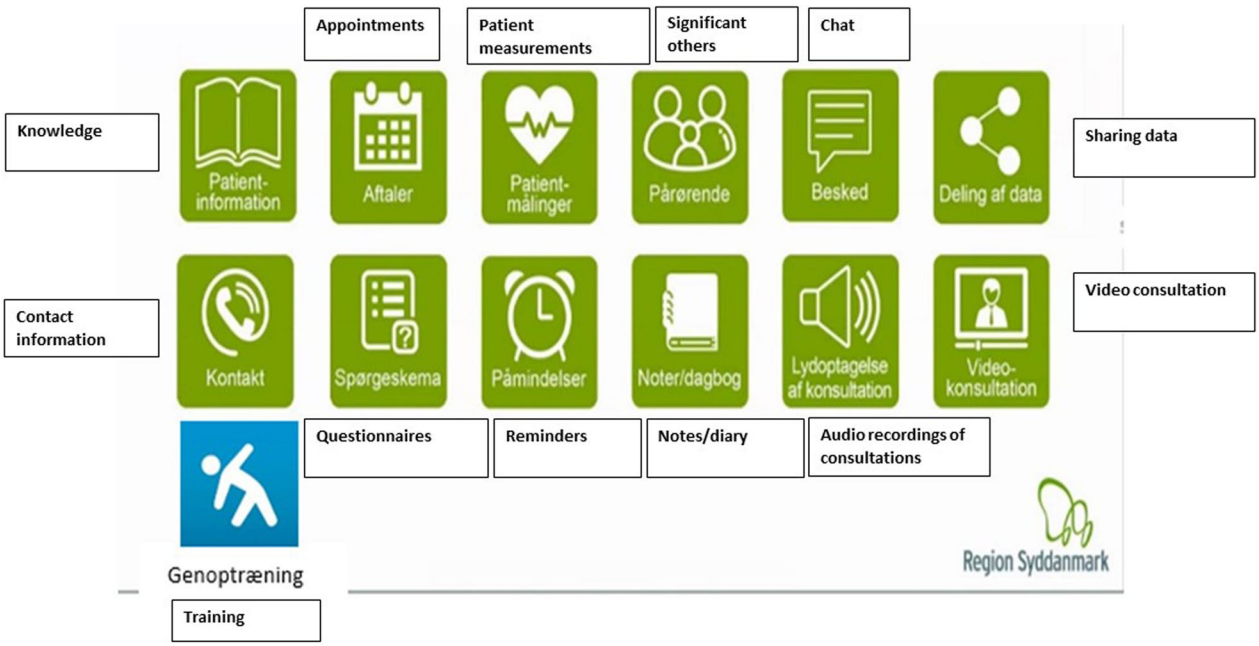


Fig. 1 Available modules in Mit Sygehus. The white boxes are English translations of the modules in Mit Sygehus

accommodate end-users' need for more comprehensive person-centred solutions.

Next, participants were given an insight into existing modules in "Mit Sygehus" (see Fig. 1). "Mit Sygehus" encompasses 12 different modules: (1) knowledge module (evidence-based information about the diagnosis, treatment, rehabilitation etc.), (2) an overview of all appointments, (3) patient measurements (e.g., weight, blood pressure, etc.), (4) significant others (i.e., information that is specifically important for significant others, such as peer support groups), (5) communication with health professionals through a chat function, (6) sharing patient data with health professionals, (7) contact information for relevant rehabilitation settings, (8) option to fill out relevant data or questionnaires prior to consultations, (9) reminders, (10) personal notes/diary, (11) audio recordings of consultations, and (12) video consultation providing significant others, cross-sectoral colleagues or the patient themselves the opportunity to participate in consultation without being present at the hospital. These modules are to be seen as headlines without any content, yet.

"Genoptræn.dk", which can work as an integrated "training" module in "Mit Sygehus", was also included for this study, to meet end-users' need for easy access to all relevant content related to the stroke rehabilitation in one place [13]. However, "Genoptræn.dk" only contain generic exercises. Therefore, to accommodate earlier identified needs for more person-centred exercises and guidelines [5, 6, 13], prior to the workshop, a "video recording" function was developed. The intention of this "video recording" function was that therapists during interventions, recorded for example specific exercises while verbally supporting and guiding the stroke survivor. The person-centred recordings are then to be stored safely in the stroke survivors' personalized app solution serving also as documentation and communication within and across rehabilitation settings. If the stroke survivor consents, significant others have the same access to the contents of the app.

The workshop was then divided in two stations in each end of the room, and participants were split into two groups. After consulting with the therapists, the stroke survivors and significant others were paired with a therapist they knew, to create a safe space where they would feel comfortable and motivated to equally contribute to the workshops [43]. Each workshop comprised three or four big "flip sheets" with seven modules from "Mit Sygehus": "knowledge", "training", "reminders", "chat", "contact information", "notes/diary", and "video consultation/audio recordings of consultation". Statements/suggestions from the identified "touchpoint" were added to each headline (as small written sentences), to support the discussions in the workshops (see Fig. 2 and Table 4). The



Fig. 2 Input on the content in "Mit Sygehus" and prioritizing modules using "flip sheets"

module's "appointments" and "significant others" were integrated in the "knowledge" "flip sheet". The activities at each station lasted for 30 min, with a 15-min break in between.

The first and last author who attended the workshop, made individual observational notes.

Findings Lessons learned from the first workshops were that presenting the "touchpoint" analysis from earlier studies had a validating effect, since participants agreed on the needs identified. They were also asked to supplement the list of "touchpoints" on the flip sheets. Several inputs to make the "training" module more useful were identified. Also, recordings of the stroke survivors' current functioning were mentioned, to make it easier to follow progression for all end-users:

> *I would have liked that access to the exercises (significant other, Funen)*

> *Insight to what exercises and the stroke survivors' abilities before arriving here (PT, municipality rehabilitation setting, Funen)*

One stroke survivor furthermore pointed out that exercise therapy through video recordings should be available also after rehabilitation has ended in the municipality to continuously support adherence. Additionally, a PT suggested that providing relevant person-centred information through an app could support coping with the new situation stroke survivors and significant others find themselves in:

> *Relevant information may facilitate coping (PT, subacute rehabilitation setting, Funen)*

Each end-user also graded each module to be: (1) not important, (2) important, or (3) very important after working on each station (see Table 4). For example, the "diary" module was graded "not important" by three participants while 14 did not grade this module and the "chat" module was only graded important by one therapist. The "training" module was graded "very important" by most participants, especially because it included the chance to make exercises more person-centred than was the case in existing known solutions. Also, the idea that gait function, transfer guidelines or instructions for daily activities could be recorded and stored as documentation and shared for communication was highly welcomed and considered to be necessary for person-centred and empowering stroke rehabilitation. Significant others were also excited about the idea that exercise support was provided through the app, with reminders as additional support (see Table 4).

Table 4 Input on the content and prioritizing the modules

Modules discussed in workshops	Input on the content in the modules (n = 17)	Grade 1: not important	Grade 2: important	Grade 3: very important	Not reported
Knowledge module	Stroke information and treatment Information about cognitive deficits The rehabilitation process and phases The maintenance phase (phase 4) Support group/peers Daily appointment/programme Video/audio of the information			17	
Training module	Step-by-step guidance about transfers, daily activities—also as communication Personalized video recordings of exercises and progress Easily accessible Available when rehabilitation determines	1	5	10	1
Reminder function	Reminders to exercise and appointments Must be personalized	2	2	5	8
Chat module	Worried that it will escalate Who will answer the questions and when? How long is this option open?	5		1	11
Note/diary module		3			14
Contact module	Pictures of staff Contact information for all relevant departments and treating staff		1	7	9
Video consultation	Participation from other stakeholders (social worker, coordinator, significant others, therapists from other rehabilitation setting, etc.) Video consultation/follow-ups better than phone calls	2	7	7	1

Empowerment exercise Method As a final exercise, based on theory and literature on empowerment [9, 10, 44, 45] participants marked on an empowerment model (constructed by the first author) which key elements they felt were important for their experience of empowerment in stroke rehabilitation. The model of empowerment contained six headings with supporting sentences: (1) communication (to and between health professionals and rehabilitation settings), (2) knowledge (about stroke, prognosis, rehabilitation plan), (3) support from health professionals, (4) involvement (in goal setting and in rehabilitation plan and activities), (5) control and overview, and (6) mastery (to act on their own, to participate in valued activities, to find the knowledge needed).

Findings The results from this exercise indicated that mechanisms such as better communication between health professionals (e.g., on the stroke survivors' goals, progress, and status) when a patient transfer between rehabilitation settings, is important to experience empowerment. Sufficient knowledge about stroke and the rehabilitation plan, active involvement and greater overview of own rehabilitation, and skills to participate in valued activities in everyday life were also mentioned as important mechanisms. Additionally, most therapists wrote elaborating sentences, stating that they considered individualizing the rehabilitation to the needs of stroke survivors and involving them would be empowering. Participants were also asked to give written feedback on participating in the workshop, and all participants valued it as a great experience. Therapists found it rewarding to get an insight to greater part of the rehabilitation process from the perspectives of stroke survivors and significant others, and to discuss the rehabilitation content with colleagues in other rehabilitation settings.

> *It has been rewarding to get knowledge about the experiences from the patient and significant other related to the entire rehabilitation process and to have the opportunity to discuss this with cross-sectoral colleagues (OT, subacute rehabilitation setting, Funen)*

One PT even suggested that more stroke survivors and significant others could have been included to get more perspectives. Stroke survivors and significant others were positively surprised about the opportunities in the apps, and on how their experiences and opinions were valued in the workshops. Also, they appreciated the opportunity to refresh their rehabilitation experiences.

> *I was surprised to see the many opportunities in the apps, and I felt that the others listened to what I said (stroke survivor, Jutland)*

Reflections on the first workshops Data analysis from participants' inputs on "flip-sheets", and their prioritization, facilitated the discussion on how to proceed to the next stage of EBCD.

In the workshops all participants' experiences and views were valued by each other. Bearing in mind that both researchers (respectively trained PT and OT) had not been working in stroke rehabilitation settings for several years, open-ended and elaborating questions during facilitation were used. Questions from the therapists and/or the two facilitators (researchers), such as: "what content would have supported you in this module?", "would you use this module?" and "if we need to learn from this, what would you suggest be done differently" made it possible to show sincere interest in making stroke rehabilitation better for participants and supported their active engagement. Also, being paired with a therapist they knew had a positive effect on sharing their experiences.

To make sure that participants discussed all the modules in both sessions within the time they were given, more facilitation was provided in the workshop on Jutland. Since stroke survivors expressed that attending the workshop for 2 h were cognitively fatiguing, extending the time of the workshops was not an option. Prioritizing the modules was less strenuous than anticipated. Despite different perspectives and experiences participants co-constructed their prioritization of each module. For example, if a stroke survivor initially scored a module "not important", and the significant other explained the reason for a higher prioritization, they could agree on either disagreeing or changing one of the scores when receiving insights to new perspectives. These argumentations between all participants, showed that there existed an open, equal, and respectful collaboration between participants.

Stage 5: small co-design meetings

Small co-design groups worked on the most relevant modules prior to the second workshop.

Step 1 Based on participants' joint priorities, the first author initially worked with the two modules that were prioritized the highest, i.e., the "knowledge" module and "training" module. Relevant evidence-based information was added into "Mit Sygehus" using a "test patient". Then, access to the app was shared with health professionals from two different sub-acute stroke units, given that longer duration in this rehabilitation setting may give the stroke survivors the time to learn how to use the app. One OT, two PTs, one head therapist, one physician, one speech therapist and one representative from the patient organization "Hjerneskadeforeningen" [in English, the Brain Injury Association] (see Table 1) gave their feedback to the content in the app. Mostly, the feedback implied having more reader friendly language, being more concise in the language choice, giving more examples and having more information on cognitive deficits. Also, the representative from the patient organization suggested creating links to the municipal "brain coordinators" and the "significant other supervisor" (i.e., a health professional that may support significant others) in the app.

Step 2 The newly developed recording function in the "training" module was tested in two different stroke rehabilitation settings (subacute stroke unit and municipality rehabilitation) together with two OTs and two PTs. The purpose was to test if the recording function was easy to use for the therapists, to assess the quality of the recordings made and to assess if the time needed to upload the recordings to the patient's app was acceptable. Both tests revealed the need for smaller adjustments; for example, the duration of the recorded videos was made visible, and all engaged therapists had access to the stroke survivors' "training" module, to be able to access assigned exercises and adjust content according to current needs of the patient (i.e., person-centred rehabilitation). The app developers and the first author worked with the feedback prior to the second workshop in stage 6. Furthermore, written instructions on how to download the apps "Mit Sygehus" and "Genoptræn.dk", as well as how to turn on a "read-aloud" function on the mobile phone or tablet were developed, prior to the second workshop.

Step 1 and 2 in stage 5 of the EBCD showed to be valuable for bringing more useable app solutions to the next stage. Especially, the need for written instruction on how to download and use the apps were valuable input to make implementation more successful.

Stage 6: celebration event

The last stage of the EBCD process was held as part of the second workshop to recognize the achievements of the co-design process and the participants taking part [33, 37] by introducing the pilot version of the co-designed app solutions. Experiences and feedback were sought from participants at this stage.

Workshop 2 *Purpose* The program and purpose for the workshop were presented, which was to test all modules in "Mit Sygehus" using participants' own mobile phones/tablets and to give feedback on the written instructions to use the app-solutions. For that reason, this workshop had a more practical hands-on focus than the first workshop.

Method The second 2-h workshop was planned and led by the first and last author and carried out with participants in Jutland (n = 10) and Funen (n = 9). Additionally, an online workshop was held with two therapists due to high

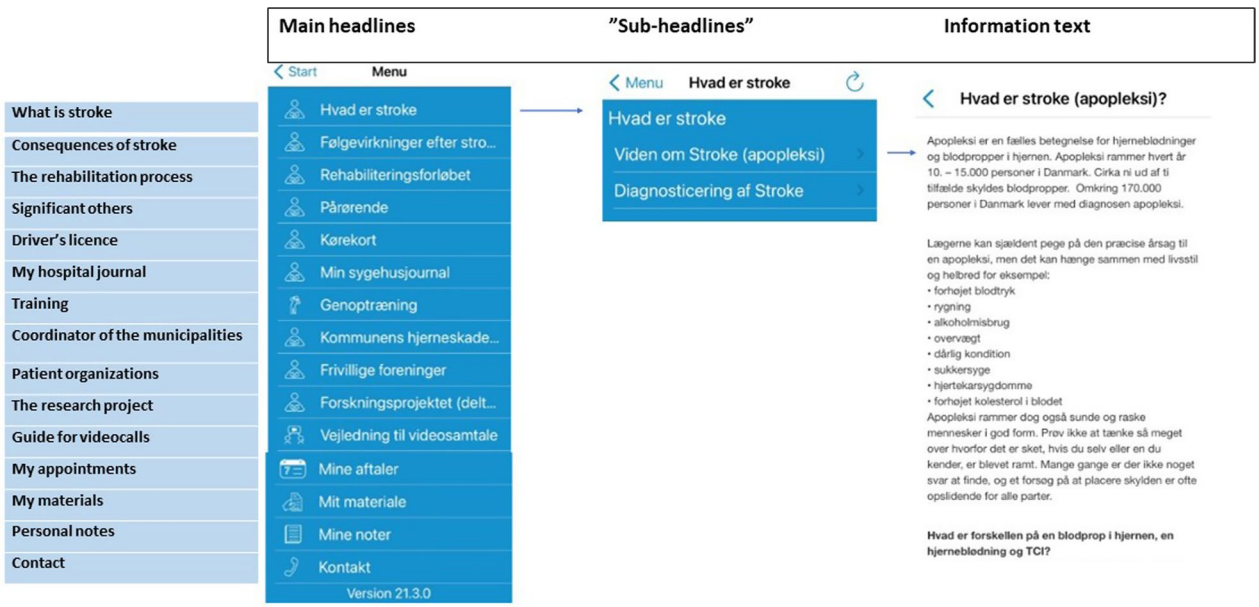
workload and COVID-19-related isolation (see Table 1). None of the stroke survivors and significant others participating in the first workshop (stage 4) were able to attend the second workshop due to other engagements or COVID-19-related isolation. Therefore, the second workshop in Jutland was performed without stroke survivors and significant others; however, a representative of the patient organization participated and spoke on their behalf.

The group was divided into two sub-groups. After consulting with the therapists, the stroke survivors and significant others were paired with a therapist they knew, in the workshop on Funen, to create a safe space where they would feel comfortable and motivated to equally contribute to the workshops. Both groups received written instructions on how to download and use "Mit Sygehus" and on how to install a "read-aloud"-function on their own mobile phone/tablet. They were instructed to visit all modules in the pilot version of the app (see Fig. 3), except the "training" module, which was to be the next task in the workshop. They were also asked to give feedback on the headings and "sub-headings" chosen for each module. They worked with the task for 30 min and, after a 15-min break, they tested the "training" module. Participants were asked to download "Genoptræn.dk" and make a recording of an exercise using an iPad, upload it to the test-patient's app and watch the exercises on the app afterwards using their own mobile phones/tablets. The participants were asked to give feedback on the written instruction on how to download "Genoptræn.dk" and use of the "training" module.

Findings Participants in stage 6 unanimously found the app solutions relevant to support stroke rehabilitation and empowerment. In particular, the option to read/listen to relevant content in the app when needed by the stroke survivor and/or significant other, and the option for person-centred rehabilitation using the "training" module were mentioned as potentially benefitting stroke rehabilitation.

The second workshop only gave minor adjustment to the content of the app solutions and the written instructions. It was suggested to change the heading "consequences of stroke" to "effects of stroke" since this was perceived by the therapists as less confronting and persistent. Changes to the written instruction were mostly about clarifying the different steps in the instruction by using numbers and circles to highlight.

Reflections on the second workshop In the second workshops all participants' experiences and views were valued by each other. Questions from the therapists, the app developers or the two facilitators (researchers), such as: "What do you think about this module?", "Could this module have been of importance in your rehabilitation process?" and "Would you use this function?" supported participants' active engagement. Statements such as: *"This function could have benefitted our rehabilitation"* (significant other, workshop 2, Funen) and *"This function is really great and smart, I would have liked this in my rehabilitation"* (stroke survivor, workshop 2, Funen), also showed how the participants were actively engaged throughout the workshops.



The boxes on the left are English translations of the modules in Mit Sygehus

Fig. 3 Content of "Mit Sygehus" to support person-centred and empowering stroke rehabilitation

Discussion

In this study end-users were involved throughout the process of developing the content in two app solutions to enable a person-centred and empowering stroke rehabilitation. In the development process, the participants prioritized the "knowledge" module and "training" module as of greatest importance. Tailored app solutions to meet end users' needs and support stroke survivors' deficits were also found important elsewhere [46]. The need for general information and education about stroke, patient organizations, where to find peers, links to relevant webpages, and the option to have text read aloud were also found important in another study [47]. Patient education is a known predictor for increased engagement, satisfaction, and treatment adherence. Given that patients may only remember a limited amount of information delivered in a short time, apps are suggested to be useful in providing education and information that can be accessed as often as liked, at any place and time [48].

Stroke-related deficits are both complex and heterogeneous [41]. The "training" module with the newly developed recording function with audio support allows for exercises and guidelines to be personalized to end-users' specific needs (e.g., cognitive, speech, facial and motor exercises and/or guidelines for supporting transfers or other daily activities), thus promoting adherence and empowering behaviors. Person-centred exercises/guidelines with audio instructions may be important support tools, especially for stroke survivors who suffer from cognitive or speech-related deficits [47]. Furthermore, the "training" module may support, guide, and inform significant others and colleagues within and across rehabilitation settings. Finally, therapists across rehabilitation settings have the same access to content in the app, allowing them to see, follow-up and adjust the stroke survivors' exercises, thereby promoting coherent person-centred and empowering stroke rehabilitation. Feedback on exercises to increase motivation and adherence is suggested to support the stroke survivor (e.g., symbols, sounds, etc.) [47]. In the "training module" stroke survivors click "done" when each exercise has been performed; however personalized feedback on performance is not an option. The therapists, however, can access statistics on the exercises that the stroke survivor has/has not performed, and follow-up in the next session, if needed.

The reminder function in "Mit Sygehus" was valued important/very important by seven participants in our study. This potentially empowers stroke survivors and relieve strain on significant others. The reminder function to complete exercises or attend appointments was also valued important in another study [46]. Although not found important in this study, chat functions have been found important to promote communication and a feeling of connectedness to the health professionals from a distance [47], thereby promoting stroke survivors' confidence, empowerment, motivation and satisfaction with the rehabilitation process [46]. Furthermore, easy access to support from health professionals prevents the experience of being left on their own after discharge from hospital [47]. Digital solutions have been suggested to support stroke rehabilitation through easier access to the rehabilitation plan, goals, and agreements made during consultations, because of the fact that information may be forgotten or misunderstood [47]. This was also mentioned by end-users of the current study, and since much information in the "knowledge" module is of a generic character, it was suggested in the second workshop that individualized plans and agreements could be uploaded in the "training" module, with a heading indicating the content (e.g., goals, status, rehabilitation plan, etc.). This would make it easy for stroke survivors, significant others, and therapists within and across rehabilitation settings to access this information, thus facilitating a person-centred and empowering stroke rehabilitation. Literature about empowerment [9, 10, 44, 45], states that sufficient knowledge, easy and seamless communication within and between rehabilitation settings, overview of rehabilitation plan and person-centred approaches (e.g., involvement in plan and goals, exercises, guidelines etc.) are important to the experience of empowerment, which aligns well with the content developed for the app solutions in our current study.

Several studies have shown that a majority of stroke survivors have positive attitudes towards including ICT and app solutions in the rehabilitation process and everyday life (e.g., exercises, social participation and/or communication) [5, 49–51]. However, to increase acceptability and usability, it is recommended that app solutions to support stroke rehabilitation are easy and simple. This involves limited clicks and limited amount of information. Furthermore, support on how to use ICT and apps has been mentioned as crucial to successful uptake [6, 13, 47, 51]. However, this may be no different from healthy older adults who needs to learn to use ICT or apps [50]. Thus, "Mit Sygehus" and "Genoptræn.dk" are integrated app solutions that provide everything "in one place" for the stroke survivor and the significant other. Also, written instructions were developed, and all participants had "hands-on" in the second workshop, to test the app solutions and to identify the elements that require more support from therapists, when these apps are to be implemented in stroke rehabilitation in a later stage. It is furthermore recommended that new interventions developed through co-design are evaluated upon implementation, and that participants are involved in identifying relevant outcomes [39]. In the final exercise

of the second workshop, participants gave their insights on components that are of importance to experience empowerment, which will be valuable when performing an evaluation on the use of the app in the next stage of the research project [39]. The app solutions will be implemented in 8–10 cases (one case being a stroke survivor, a significant other and an OT and PT). As recommended by the Good Practice Guidelines on Health Apps and Smart Devices evaluation on user acceptability and usability [52] and experienced empowerment will be performed in the next phase of this research project. Qualitative interviews (dyadic interviews with stroke survivors and significant others and focus group interviews with therapists) as well descriptive statistics on the use of the app solutions are planned.

Reflections on EBCD

This study showed that it was possible to use a structured EBCD approach to develop the content in two app solutions to support person-centred stroke rehabilitation (i.e., develop solutions that support stroke survivors' and significant others' various needs, preferences, and goals). Despite knowledge on unequal power dynamics in co-production designs [35, 53], statements from the first workshop such as "*it was valuable to receive different perspectives*" (OT, first workshop), "*it has been rewarding to refresh the rehabilitation process*" (significant other) and "*I felt heard*" (stroke survivor) supported that this development process was associated with positive experiences, including learning together and a feeling of being listened to, similar to findings in another study [39]. Sincere interest in lived experiences, creating a safe space to share these experiences and participation of several stroke survivors and significant others may facilitate and motivate stroke survivors and significant others to share both positive and negative experiences in hopes of that other can learn from them [43]. Accomplishing a more equal power relation through the use of EBCD was also seen in a recent review [38]. The results indicate that being engaged in co-design research may have facilitated a sense of accomplishment and empowerment [54]. Another review on experiences of co-design research also found an increase in participants' self-esteem, and experience of being educated in relation to user participation [30].

We learned that facilitation is important to maintain momentum during the workshops' different stages, actions and activities, which is supported in previous studies [32, 38]. The implementation and uptake of the solution may have been facilitated by structured user-involvement in all stages and having a "hands-on-focus" in the second workshop, which gave all participants the opportunity to test the written instructions and the pilot version of the apps [25, 55]. Furthermore, the following drivers of importance for developed co-designed solutions to be sustainable have been identified: continuing the work of improving the solutions in each context (i.e., local implementation/quality improvement teams that prompts follow-up actions), mobilization of narratives of being heard and seen, and management systems (i.e., involvement of leaders and decision makers) [25], which is essential knowledge when implementing the app solutions in the next phase of the research project.

Strengths and limitations

The engagement of end-users throughout the process of developing the content in the two app-solutions to support stroke rehabilitation and empowerment is a strength of this study. One known limitation of EBCD is that it is time-consuming for both end-users and researchers. Furthermore, end-users may be heterogeneous and experience a lack of consensus [25, 55]. However, despite the heterogeneity of end-users involved in this study, all participants agreed that the "knowledge" and "training" module were of greatest importance in the app solutions. The lack of stroke survivors and significant others in the second workshop is a limitation. Finally, even though end-users participated throughout the EBCD process, it is unlikely that all stroke survivors and significant others may benefit from the app solutions, due to the heterogeneity of stroke survivors and their deficits. However, it was not our intention that one or two app solutions should be applicable for all stroke survivors and in all rehabilitation settings and contexts. Nevertheless, by involving six different stroke rehabilitation settings in the development of the content in the apps, some context-specific considerations have been considered.

Conclusion

Using a structured and facilitated six-stages EBCD approach supported the development of the content in two app solutions "Mit Sygehus" and "Genoptræn.dk" to support a person-centred and empowering stroke rehabilitation. Using open-ended questions, pairing stroke survivors and significant others with a therapist they knew, and ongoing facilitation during workshops, showed to have a positive impact on participants' sharing, engaging, learning, and collaboration in the EBCD approach. Using an EBCD approach may ultimately also support the subsequent implementation.

Abbreviations

OTs	Occupational therapists
PTs	Physiotherapists
ICT	Information and communication technologies

Apps	Mobile phone and tablet applications
EBCD	Experience-based co-design
GT	Grounded theory

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40900-023-00472-z.

Additional file 1. The GRIPP2 reporting checklist on patient and public involvement in research.

Acknowledgements
The authors thank all stroke survivors, their significant others, the therapists, the representative from the patient organization and the app developers for participating in this project.

Author contributions
MNM and HKK completed the data generation and analysis throughout the EBCD process. MNM wrote the main body of this manuscript, including developing figures and tables. All authors participated in the ongoing writing and reviewing of this manuscript and the final version was approved by all authors.

Funding
Open access funding provided by Karolinska Institute. We thank the health research departments at University College Southern Denmark and University of Southern Denmark for time resources and the Danish Association of Physiotherapists for funding resources to publish this article (Grant Number R39-A1042).

Availability of data and materials
Not applicable.

Declarations

Ethics approvals and consent to participate
Participants received written information about the project and the purpose of the workshops and gave written informed consent to participate, and for photographs being taken during the workshops that could be used for presenting the results. Ethical approval was acquired from the Danish Data Protection Agency (18/60280).

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

Author details
[1]Department of Physiotherapy Education, University College Southern Denmark, Esbjerg, Denmark. [2]Department of Neurobiology Care Sciences and Society, Karolinska Institutet, Huddinge, Sweden. [3]Center for Innovative Medical Technology, Department of Clinical Research, University of Southern Denmark, Odense, Denmark. [4]Women's Health and Allied Health Professionals Theme, Karolinska University Hospital, Stockholm, Sweden. [5]Geriatric Research Unit, Department of Clinical Research University Hospital Odense, University of Southern Denmark, Odense, Denmark.

Received: 17 June 2023 Accepted: 18 July 2023
Published online: 24 August 2023

References

1. Katan M, Luft A. Global burden of stroke. Semin Neurol. 2018;38(02):208–11.
2. Camak DJ. Addressing the burden of stroke caregivers: a literature review. J Clin Nurs. 2015;24(17–18):2376–82.
3. Sumathipala K, Radcliffe E, Sadler E, Wolfe CD, McKevitt C. Identifying the long-term needs of stroke survivors using the international classification of functioning, disability and health. Chronic Illn. 2012;8(1):31–44.
4. Donnan GA, Fisher M, Macleod M, Davis SM. Stroke. Lancet. 2008;371(9624):1612–23.
5. Gustavsson M, Ytterberg C, Nabsen Marwaa M, Tham K, Guidetti S. Experiences of using information and communication technology within the first year after stroke—a grounded theory study. Disabil Rehabil. 2016;40:1–8.
6. Marwaa MN, Ytterberg C, Guidetti S. Significant others' perspectives on person-centred information and communication technology in stroke rehabilitation—a grounded theory study. Disabil Rehabil. 2019;42:1–8.
7. Wade DT. What is rehabilitation? An empirical investigation leading to an evidence-based description. Clin Rehabil. 2020;34(5):571–83.
8. Bergström A, Koch L, Andersson M, Tham K, Eriksson G. Participation in everyday life and life satisfaction in persons with stroke and their caregivers 3–6 months after onset. J Rehabil Med. 2015;47(6):508–15.
9. Barr PJ, Scholl I, Bravo P, Faber MJ, Elwyn G, McAllister M. Assessment of patient empowerment—a systematic review of measures. Bond K redaktør. PLoS ONE. 2015;10(5):e0126553.
10. Zimmerman MA. Psychological empowerment: issues and illustrations. Am J Community Psychol. 1995;23(5):581–99.
11. Kaae Kristensen H, Pallesen H. Personcentreret neurorehabilitering—viden og praksis [Person-centred neurorehabilitation—knowledge and practice]. Copenhagen: Gad; 2022. p. 19–32.
12. Wade DT. Describing rehabilitation interventions. Clin Rehabil. 2005;19(8):811–8.
13. Marwaa MN, Kristensen HK, Guidetti S, Ytterberg C. Physiotherapists' and occupational therapists' perspectives on information and communication technology in stroke rehabilitation. Disabil Rehabil. 2020;15:e0236831.
14. Tistad M, von Koch L, Sjöstrand C, Tham K, Ytterberg C. What aspects of rehabilitation provision contribute to self-reported met needs for rehabilitation one year after stroke—Amount, place, operator or timing? Health Expect. 2013;16(3):e24-35.
15. Dubey D, Amritphale A, Sawhney A, Amritphale N, Dubey P, Pandey A. Smart phone applications as a source of information on stroke. J Stroke. 2014;16(2):86.
16. Kang YN, Shen HN, Lin CY, Elwyn G, Huang SC, Wu TF. Does a Mobile app improve patients' knowledge of stroke risk factors and health-related quality of life in patients with stroke? A randomized controlled trial. BMC Med Inform Decis Mak. 2019;19(1):282.
17. Simpson DB, Bird ML, English C, Gall SL, Breslin M, Smith S. Connecting patients and therapists remotely using technology is feasible and facilitates exercise adherence after stroke. Top Stroke Rehabil. 2020;27(2):93–102.
18. Brouns B, Meesters JJL, Wentink MM, de Kloet AJ, Arwert HJ, Vliet Vlieland TPM. Why the uptake of eRehabilitation programs in stroke care is so difficult—a focus group study in the Netherlands. Implement Sci. 2018. https://doi.org/10.1186/s13012-018-0827-5.
19. Silveira TM, Tamplin J, Dorsch S, Barlow A. Let's Improvise! iPad-based music therapy with functional electrical stimulation for upper limb stroke rehabilitation. Aust J Music Ther. 2018;29. Tilgængelig hos: Retrieved from https://www.austmta.org.au/journal/article/let%E2%80%99s-improvise-ipad-based-music-therapyfunctional-electrical-stimulation-upper.
20. Choi YH, Park HK, Ahn K, Son Y, Paik NJ. A telescreening tool to detect aphasia in patients with stroke. Telemed E-Health. 2015;21(9):729–34.
21. Fell N, True HH, Allen B, Harris A, Cho J, Hu Z. Functional measurement post-stroke via mobile application and body-worn sensor technology. mHealth. 2019;5:47–47.
22. McKay FH, Cheng C, Wright A, Shill J, Stephens H, Uccellini M. Evaluating mobile phone applications for health behaviour change: a systematic review. J Telemed Telecare. 2018;24(1):22–30.
23. Paul L, Wyke S, Brewster S, Sattar N, Gill JMR, Alexander G. Increasing physical activity in stroke survivors using STARFISH, an interactive mobile phone application: a pilot study. Top Stroke Rehabil. 2016;23(3):170–7.
24. Zhou X, Du M, Zhou L. Use of mobile applications in post-stroke rehabilitation: a systematic review. Top Stroke Rehabil. 2018;25(7):489–99.
25. Bate P, Robert G. Experience-based design: from redesigning the system around the patient to co-designing services with the patient. Qual Saf Health Care. 2006;15(5):307–10.

26. Ramey L, Osborne C, Kasitinon D, Juengst S. Apps and mobile health technology in rehabilitation. Phys Med Rehabil Clin N Am. 2019;30(2):485–97.
27. Marwaa MN, Guidetti S, Ytterberg C, Kristensen HK. The use of mobile and web-based applications to support rehabilitation after stroke: a scoping review. J Rehabil Med. 2022 [henvist 4. april 2022]; Tilgængelig hos: https://medicaljournalssweden.se/jrm/article/view/452.
28. Region Syddanmark. MitSygehus i region Syddanmark. 2019. Tilgængelig hos: https://www.regionsyddanmark.dk/wm509838.
29. genoptræn.dk. Genoptræn|dk—din digitale træningsmakker. 2019. Tilgængelig hos: https://www.syddansksundhedsinnovation.dk/projekter/genoptraen-dk-din-digitale-traeningsmakker/.
30. Bombard Y, Baker GR, Orlando E, Fancott C, Bhatia P, Casalino S. Engaging patients to improve quality of care: a systematic review. Implement Sci. 2018;13(1):98.
31. Donetto S, Pierri P, Tsianakas V, Robert G. Experience-based co-design and healthcare improvement: realizing participatory design in the public sector. Des J. 2015;18(2):227–48.
32. Clarke D, Gombert-Waldron K, Honey S, Cloud G, Harris R, Macdonald A. Co-designing organisational improvements and interventions to increase inpatient activity in four stroke units in England: a mixed-methods process evaluation using normalisation process theory. BMJ Open. 2021;11(1):e042723.
33. Donetto S, Jones F, Clarke DJ, Cloud GC, Gombert-Waldron K, Harris R. Exploring liminality in the co-design of rehabilitation environments: the case of one acute stroke unit. Health Place. 2021;72:102695.
34. Jones F, Gombert- K, Honey S, Cloud G, Harris R, Macdonald A. Addressing inactivity after stroke: the collaborative rehabilitation in acute stroke (CREATE) study. Int J Stroke Off J Int Stroke Soc. 2021;16(6):669–82.
35. Gustavsson SMK, Andersson T. Patient involvement 2.0: experience-based co-design supported by action research. Action Res. 2019;17(4):469–91.
36. Launsø L, Rieper O, Olsen L (2011) Forskning om og med mennesker: forskningstyper og forskningsmetoder i samfundsforskning. 6. udgave. Kbh.: Nyt Nordisk Forlag; 2011. 250 sider.
37. Fylan B, Tomlinson J, Raynor DK, Silcock J. Using experience-based co-design with patients, carers and healthcare professionals to develop theory-based interventions for safer medicines use. Res Soc Adm Pharm. 2021;17(12):2127–35.
38. Green T, Bonner A, Teleni L, Bradford N, Purtell L, Douglas C. Use and reporting of experience-based codesign studies in the healthcare setting: a systematic review. BMJ Qual Saf. 2020;29(1):64–76.
39. Slattery P, Saeri AK, Bragge P. Research co-design in health: a rapid overview of reviews. Health Res Policy Syst. 2020;18(1):17.
40. Staniszewska S, Brett J, Simera I, Seers K, Mockford C, Goodlad S. GRIPP2 reporting checklists: tools to improve reporting of patient and public involvement in research. Res Involv Engagem. 2017;3(1):13.
41. Danish Health Authority. Anbefalinger for tværsektorielle forløb for voksne med erhvervet hjerneskade - apopleksi og transitorisk cerebral iskæmi (TCI)—traume, infektion, tumor, subarachnoidalblødning og encephalopati [Internet]. 2020. Tilgængelig hos: https://www.sst.dk/-/media/Udgivelser/2020/Hjerneskade/Anbefalinger-forloeb-hjerneskade.ashx?la=da&hash=A7A96AC766D6AA68D26F32B96C0015BF828C93FF.
42. Charmaz K. Constructing grounded theory. 2nd ed. London: Sage; 2014. p. 388s (**Introducing qualitative methods**).
43. Richards DP, Poirier S, Mohabir V, Proulx L, Robins S, Smith J. Reflections on patient engagement by patient partners: How it can go wrong. Res Involv Engagem. 2023;9(1):41.
44. Deyhoul N, Vasli P, Rohani C, Shakeri N, Hosseini M. The effect of family-centered empowerment program on the family caregiver burden and the activities of daily living of Iranian patients with stroke: a randomized controlled trial study. Aging Clin Exp Res. 2020;32(7):1343–52.
45. Johnsen AT, Eskildsen NB, Thomsen TG, Grønvold M, Ross L, Jørgensen CR. Conceptualizing patient empowerment in cancer follow-up by combining theory and qualitative data. Acta Oncol. 2017;56(2):232–8.
46. Hughes CML, Padilla A, Hintze A, Raymundo TM, Sera M, Weidner S. Developing an mHealth app for post-stroke upper limb rehabilitation: feedback from US and Ethiopian rehabilitation clinicians. Health Inform J. 2019. https://doi.org/10.1177/1460458219868356.
47. Wentink M, van Bodegom-Vos L, Brouns B, Arwert H, Houdijk S, Kewalbansing P. How to improve eRehabilitation programs in stroke care? A focus group study to identify requirements of end-users. BMC Med Inform Decis Mak. 2019;19(1):145.
48. Timmers T, Janssen L, Kool RB, Kremer JA. Educating patients by providing timely information using smartphone and tablet apps: systematic review. J Med Internet Res. 2020;22(4):e17342.
49. Kamwesiga JT, Tham K, Guidetti S. Experiences of using mobile phones in everyday life among persons with stroke and their families in Uganda—a qualitative study. Disabil Rehabil. 2017;39(5):438–49.
50. Kim H, Kim GJ. Attitudes and use patterns for mobile technology and upper extremity home exercises in stroke survivors in the United States. Br J Occup Ther. 2022. https://doi.org/10.1177/03080226211070564.
51. Sureshkumar K, Murthy G, Natarajan S, Naveen C, Goenka S, Kuper H. Evaluation of the feasibility and acceptability of the 'Care for Stroke' intervention in India, a smartphone-enabled, carer-supported, educational intervention for management of disability following stroke. BMJ Open februar. 2016;6(2):e009243.
52. Haute Autorité de Santé. Good Practice Guidelines on Health Apps and Smart Devices (Mobile Health or mHealth) [Internet]. Haute Autorité de Santé; 2016. Tilgængelig hos: https://www.has-sante.fr/upload/docs/application/pdf/2017-03/dir1/good_practice_guidelines_on_health_apps_and_smart_devices_mobile_health_or_mhealth.pdf.
53. Lindblom S, Flink M, Elf M, Laska AC, von Koch L, Ytterberg C. The manifestation of participation within a co-design process involving patients, significant others and health-care professionals. Health Expect. 2021;24(3):905–16.
54. Gustafsson S, Falk C, Tillman S, Holtz L, Lindahl L. Life filming as a means of participatory approach together with older community-dwelling persons regarding their local environment. Scand J Occup Ther. 2018;25(5):347–57.
55. Hawkins J, Madden K, Fletcher A, Midgley L, Grant A, Cox G. Development of a framework for the co-production and prototyping of public health interventions. BMC Public Health. 2017;17(1):689.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Ready to submit your research? Choose BMC and benefit from:

- fast, convenient online submission
- thorough peer review by experienced researchers in your field
- rapid publication on acceptance
- support for research data, including large and complex data types
- gold Open Access which fosters wider collaboration and increased citations
- maximum visibility for your research: over 100M website views per year

At BMC, research is always in progress.

Learn more biomedcentral.com/submissions



Title page

Acceptability of two mobile applications to support cross-sectoral, person-centred and empowering stroke rehabilitation – a process evaluation

Authors Names

Mille Nabsen Marwaa[a,c*], Susanne Guidetti[b,d], Charlotte Ytterberg[b,d,e], Hanne Kaae Kristensen[c]

Institutions of origin

[a]Department of Physiotherapy Education, University College Southern Denmark, 6705 Esbjerg, Denmark; [b]Department of Neurobiology Care Sciences and Society, Karolinska Institutet, 141 83 Huddinge, Sweden; [c]Center for Innovative Medical Technologies, Department of Clinical Research University of Southern Denmark, 5000 Odense, Denmark; [d]Women's Health and Allied Health Professionals Theme, Karolinska University Hospital, 107 90 Stockholm, Sweden; [e]Geriatric Research Unit, University Hospital Odense, Department of Clinical Research University of Southern Denmark, Odense, Denmark

***Corresponding author**

Charlotte Ytterberg, corresponding author

Docent, Karolinska Institutet
E-mail: charlotte.ytterberg@ki.se
Telephone: 070-899 13 30
Orcid: https://orcid.org/0000-0003-3704-8887.

Acknowledgements

The authors thank the stroke survivors, significant others, and therapists for participating, engaging, and sharing their experiences throughout the research process.

Funding

This study was supported by University College South Denmark and the Danish organization for physiotherapists (grant number R39-A1042).

Conflicts of interest

The authors have no conflicts of interest to declare.

Acceptability of two mobile applications to support cross-sectoral, person-centred and empowering stroke rehabilitation – a process evaluation

Abstract

Aim

To evaluate the acceptability of two co-designed mobile applications *Mit Sygehus* [a knowledge-based solution] and *Genoptræn.dk* [a self-training solution] to support a cross-sectoral person-centred and empowering stroke rehabilitation.

Setting

The applications were implemented and tested throughout two stroke rehabilitation trajectories in Southern Denmark, comprising two acute, two sub-acute and two municipal stroke rehabilitation settings.

Methods, participants and analysis

A process evaluation focusing on acceptability was conducted. Individual and dyadic interviews were performed with ten stroke survivors (three women and seven men, aged 50-84) with moderate stroke and seven significant others (five women and two men, aged 50-78) post-rehabilitation.
A constructivist Grounded Theory analysis was used to explore what, why, when, and how the apps worked or did not work throughout the stroke rehabilitation trajectory and if adaptions were needed.

Results

Participants found that *Mit Sygehus* provided adequate and sufficient knowledge and was easy to use, however, acceptability of *Mit Sygehus* declined throughout the rehabilitation process. Also, knowledge on "return-to-work" and "re-gaining driver's license/permission to drive" needed to be developed. The content in *Genoptræn.dk* was perceived as acceptable, through content being person-centred, motivating and meaningful. *Genoptræn.dk* furthermore, supported the transfer between rehabilitation settings, provided a sense of progress throughout the rehabilitation process, facilitated positive habits regarding self-training, and relieved the burden on significant others. *Genoptræn.dk* was perceived most acceptable in the sub-acute rehabilitation setting and declined when rehabilitation continued in the municipal setting.

Conclusion

Stroke survivors and their significant others found *Mit Sygehus* and *Genoptræn.dk* acceptable to support cross-sectoral, person-centred and empowering stroke rehabilitation, however acceptability declined

throughout the rehabilitation process. Further investigations are required to determine how cognitive rehabilitation can play a greater role in app-supported stroke rehabilitation and how the need for more long-term follow-up can be supported.

Implications for rehabilitation

- By using process evaluation, the user acceptability of newly developed and complex interventions can be evaluated.
- ICT and app-based solutions can support cross-sectoral person-centred and empowering stroke rehabilitation.
- Therapists play an important role in implementing and supporting stroke survivors using app-based solutions to support person-centred and empowering stroke rehabilitation.

Keywords:

stroke rehabilitation; person-centered rehabilitation; cross-sectoral rehabilitation; mobile applications; mobile apps; empowerment; occupational therapy, physiotherapy.

Abbreviations

Apps: applications

ICT: information and communication technology

MRC: Medical Research Council

NIHSS: National Institutes of Health Stroke Scale

OT: occupational therapist

PT: physiotherapist

SSS: Scandinavian Stroke Scale

Background

Rehabilitation after stroke should be person-centred, that is, based on the stroke survivors' and significant others' needs, goals, preferences, and resources [1–4], which supports empowerment [5,6]. In the rehabilitation process, empowerment may be seen as an enabling process in which health professionals collaborate with stroke survivors and significant others, where the outcome is a stroke survivor and significant other with greater ability to exercise control, manage his/her condition, and make informed decisions [1,4,6,7]. To be empowered, three components must be achieved: **intrapersonal** *(*perceives

control, mastery, competence, being motivated, and showing initiative), **interactional** (has adequate relevant skills and knowledge to navigate in a specific context), and **behavioral** (empowered behaviors and actions taken to master or influence specific outcomes) [8]. Significant others are important because stroke might also affect their physical and emotional health [6,9–11]. Therefore, person-centred and empowering rehabilitation must be continuously co-constructed in a partnership consisting of shared power, responsibility, and ownership for the rehabilitation process and outcomes [1]. Person-centred and empowering stroke rehabilitation should start as early as possible; be coherent and coordinated with smooth transitions between different rehabilitation settings; include goal-setting, intervention/training, support for social participation, discharge support, and continuous evaluation [1,2,6]. However, a well-known challenge to the quality of person-centred stroke rehabilitation is lack of coherence in cross-sectoral coordination and communication [1,2,12] and lack of person-centred initiatives [1,11,13,14], leading to dis-empowered stroke survivors and significant others [5,8,11–13]. New technologies co-designed with all relevant stakeholders and rehabilitation settings to support these challenges are in demand [15,16].

ICT in stroke rehabilitation

The integration of information and communication technologies (ICT) and mobile-based applications (apps) are seen as both local and national visions and strategies to achieve equality and quality of health care, complement usual rehabilitation, deliver person-centred rehabilitation, and enhance independence and participation in everyday life [15–17]. ICT and apps (e.g., phone calls, messages, reminders, remote monitoring or intervention, text, videos, and images) may support the timing and delivery of relevant evidence-based health information [18–20], improve communication [21], support coordination, continuity, and coherence in the rehabilitation process [21,22], increase accessibility to rehabilitation, reduce travel time and costs, and support follow-up after discharge [23]. Furthermore, stroke survivors' motivation, engagement, and empowerment to independently perform person-centred exercises [19,20,23–26], and becoming more active in managing their own health and participating in everyday life and society may be supported by ICT and apps [18,19,24,26–29] as well as reducing significant others' strains and burden [30].

However, in Denmark, there are no overarching ICT/app-based solutions supporting cross-sectoral rehabilitation, which accommodates the need for coherent person-centred and empowering initiatives for stroke survivors and significant others [31]. In addition, existing ICT and app-based solutions focus mainly on isolated parameters of stroke rehabilitation [31], and there has been a lack of end-user involvement in the development, which reduces the acceptability and implementation of the solutions [21,32–34].

Aim

This study aimed to evaluate the acceptability of two co-designed app-solutions, *Mit Sygehus* and *Genoptræn.dk* to support cross-sectoral person-centred and empowering stroke rehabilitation.

Design and methods

We drew on the framework of the Medical Research Council (MRC) [35], who recommend that a process evaluation follows the development and testing of complex interventions before continuing to a full-scale implementation and evaluation [35,36]. Qualitative results are recommended being published to support other researchers performing similar interventions [37]. The Standards for Reporting Qualitative Research (SRQR) guidelines were applied in this study [38] (see Supplementary Material).

The rehabilitation settings

Stroke rehabilitation is recommended to take place in interdisciplinary specialized stroke units [12,39]. In Denmark, stroke survivors are initially admitted to an acute hospital stroke unit (typically for a few days) [40,41]. When required, rehabilitation continues in a sub-acute hospital stroke rehabilitation unit and/or in municipalities, that is, home-based or rehabilitation centers. Stroke survivors with more complex cognitive, emotional, mental, and/or physical difficulties most often need individualized and intensive interdisciplinary rehabilitation delivered in all three settings [39]. Municipal rehabilitation must be initiated within seven days after discharge from acute or sub-acute rehabilitation and typically continuous for three months [39]. Stroke severity is often defined as mild, moderate, or severe, using the Scandinavian Stroke Scale (SSS) and/or National Institutes of Health Stroke Scale (NIHSS) [39]. This study will mainly focus on stroke survivors in need of cross-sectoral rehabilitation in all three rehabilitation settings, that is, moderate to severe stroke.

The intervention: *Mit Sygehus* and *Genoptræn.dk*

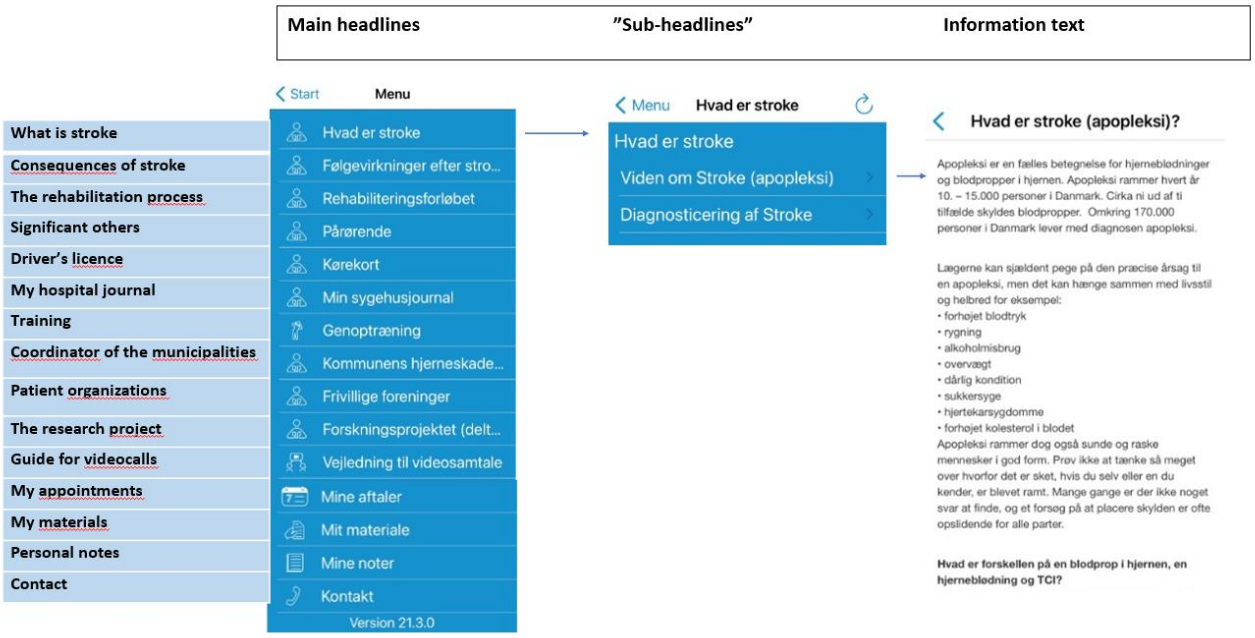
Mit Sygehus [in English: My Hospital] and *Genoptræn.dk* [in English: Rehab.dk] which are aimed at supporting cross-sectoral rehabilitation and involving multiple therapists, may be perceived as a complex intervention [36]. *Mit Sygehus* and *Genoptræn.dk* have been implemented in other patient trajectories in Southern Denmark; but have not yet been developed and implemented to support coherent person-centred and empowering stroke rehabilitation. Therefore, we co-designed the content of the two apps together with stroke survivors, significant others, physiotherapists (PTs) and occupational therapists (OTs) from all rehabilitation settings, one physician, one speech therapist, one representative from a patient

organization, and three app developers [42] to accommodate end-user needs and to promote the uptake and implementation of the solutions. The developed content is described briefly below.

Mit Sygehus

Mit Sygehus [in English, My hospital] is a free-of-charge app used in hospitals in southern Denmark to provide information related to the specific diagnosis. In an earlier co-design process [42] fourteen modules/headlines to support stroke rehabilitation were developed in *Mit Sygehus* each with "sub-headlines" and information text: 1) what is stroke, 2) consequences of stroke, 3) the rehabilitation process, 4) significant others, 5) driver's license, 6) my hospital journal, 7) training, 8) coordinator of the municipalities, 9) patient organizations, 10) the research project, 11) guide for videocalls, 12) my appointments, 13) personal notes and 14) contact (i.e. to the different rehabilitation settings) (see Figure 1).

FIGURE 1: The content and interface of Mit Sygehus

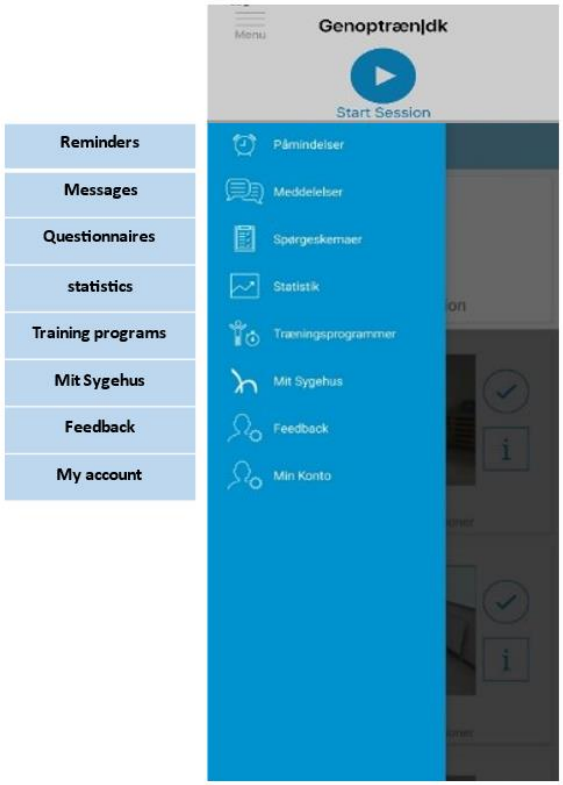


Genoptræn.dk

Genoptræn.dk [in English, Rehab.dk] is currently being used in both hospital and municipal rehabilitation settings in Southern Denmark; however, the app has not previously been designed to support stroke rehabilitation. *Genoptræn.dk* is already integrated with *Mit Sygehus* meaning that the stroke survivor has all the content "in one place" or can use the apps separately, depending on preference (see Figure 2). The

app contains more than 700 generic video-recorded exercises that can be assigned by therapists to the personal app for stroke survivors. Content in the app is accessible to therapists in all rehabilitation settings, thus supporting cross-sectoral coherence, communication and documentation [43]. We additionally developed a "video recording" function in *Genoptræn.dk* to accommodate earlier identified needs for more person-centred exercises and guidelines [11,13,32]. Further details of the co-design development process can be found elsewhere [42].

FIGURE 2: The content and interface of Genoptræn.dk



Implementation of *Mit Sygehus* and *Genoptræn.dk* and participants

The two app-solutions were implemented and tested through two stroke rehabilitation trajectories, comprising two acute stroke units (hospitals X and Y), two sub-acute stroke units (hospitals X1 and Y1), and two larger municipal stroke rehabilitation settings (municipalities X2 and Y2) in Southern Denmark. In each rehabilitation setting, one PT and one OT who also participated in the co-design process were engaged in the implementation. The PTs and OTs from the acute rehabilitation settings were responsible for identifying and including stroke survivors who were: 1) diagnosed with stroke; 2) living in one of the municipalities included in this study (municipality X2 or Y2); 3) in need of rehabilitation throughout all rehabilitation settings, that is, participants with more complex difficulties; 4) owning and using ICT such as a smartphone,

tablet, and/or computer; and 5) preferably having a significant other to participate, that is, someone close to the stroke survivor offering support. Since acute stroke rehabilitation is often limited to a few days [40,41], inclusion could also occur in sub-acute rehabilitation settings.

The PT or OT supported the stroke survivor and significant others to download and install both app solutions. Written instructions facilitated this. The therapists created access to the content in the apps by registering the stroke survivors' social security numbers in the web-based back-end of the apps. If the stroke survivor consented, significant others could also receive access to the content of the apps on their own device.

Across the stroke rehabilitation settings, the therapists included apps to support the usual rehabilitation. *Mit Sygehus* was included if the stroke survivor and/or significant other expressed a need for supplementary information through the app. *Genoptræn.dk* was included when exercises, self-training, activities and/or guidelines were part of the intervention. Furthermore, if their performance status needed to be recorded, for example, gait function, ability to climb stairs, etc., and to support cross-sectoral communication. All included therapists had access to the stroke survivors' content in *Genoptræn.dk*, which was adjusted throughout the rehabilitation process according to the stroke survivors' goals, needs, and physical/cognitive functioning.

During the testing phase, the first author made participant observations during each stroke survivor's rehabilitation process. The observations had three purposes:1) to gain ongoing knowledge in each context and continue to maintain a relationship with the participating therapists; 2) to facilitate implementation of the two apps, for example, supporting the use of the apps and providing education when required; and 3) to get acquainted with the stroke survivor and significant others and obtain knowledge on their current use of the apps. Field notes on the therapists' and stroke survivors' use of the apps were made immediately after observation and were used to support post-intervention interviews.

The PTs and OTs from the acute setting delivered information to the first author on the participants' stroke severity on admission (SSS or NIHHS). If multiple SSS scores were available, the first score was used (Table 1). In one case, the NIHHS score was provided and converted to SSS, as instructed by Gray et al. [44]. Therapists provided data on the length of stay in each rehabilitation setting and prints of the stroke survivors' journals regarding the content and dates of the therapist interventions. Additionally, PTs and OTs informed the receiving stroke rehabilitation setting through secure crypted e-mails of their use or non-use of the two app-solutions.

Sampling and included participants

A convenient sampling strategy was used [45,46] with attention to variations in physical and cognitive deficits, as diversity is important when testing new solutions [37]. We planned to include 10 stroke survivors and 10 significant others, since 10 users would identify a minimum of 80% of the problems with the technology during testing and 20 users would identify 95% [37]. Ten stroke survivors and seven significant others were consecutively included from January 2022 to October 2022. Three stroke survivors, who were 1) not married, 2) divorced, or 3) widowers, did not wish to have a significant included. All invited stroke survivors and significant others participated in the study.

Figure 3 illustrates the cross-sectoral transfers when the stroke survivors needed rehabilitation in all three settings, that is, due to more complex declined functioning after stroke and the number of stroke survivors included from the acute and sub-acute rehabilitation settings. The participants' characteristics are shown in Table 1.

Figure 3: Included rehabilitation settings and included stroke survivors

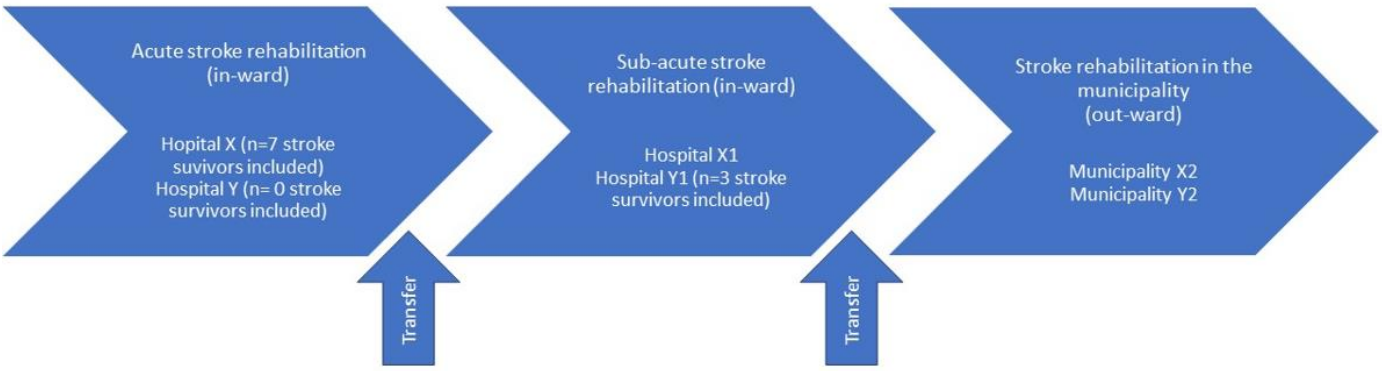


TABLE 1: participant characteristics

	Patient, age (years)	**Diagnoses and SSS**	**Significant other, age (years), relation**	**Duration of rehabilitation in each setting**	**Interview type and duration**
1. Setting X	Beate, 65	Right side infarct SSS 46	Jonas, 72 Husband	Acute: 9 days Sub-acute: 7 days Municipal: 149 days (30 interventions)	Dyadic interview 97 minutes
2. Setting Y	Peter, 74	Left side infarct SSS 39*	Ella, 72 Wife	Acute: 10 days Sub-acute: 20 days Municipal: 91 days (3 interventions)	Dyadic interview 99 minutes

3. Setting X	Liam, 86	Right side infarct SSS: 43	Jane, 71 Wife	Acute: 2 days Sub-acute: 14 days Municipal: 123 days (2 interventions)	Dyadic interview 49 minutes
4: Setting Y	Jamie, 61	Left side cerebellum infarct SSS: 46	Betty, 56 Wife	Acute: 6 days Sub-acute: 14 days Municipal: 212 days (79 interventions)	Dyadic interview 47 minutes
5. Setting X	Robert, 58	Right side infarct SSS: 42	Ida, 61 Wife	Acute: 4 days Sub-acute: 15 days Municipal: 8 days (5 interventions)	Dyadic interview 83 minutes
6. Setting X	David, 50	Right side infarct SSS: 47	None Divorced	Acute: 7 days Sub-acute: 10 days Municipal: 28 July – **	Individual interview 49 minutes
7. Setting X	Erin, 80	Right side haemorrhage SSS: 29	Berry, 78 Husband	Acute: 6 days Sub-acute: 24 days Municipal: 29 days (unknown number of interventions)	Dyadic interview 17 minutes (telephone)
8. Setting X	Noah, 63	Right side infarct SSS: 31	None Divorced	Acute: 10 days Sub-acute: 57 days Municipal: continued in a private rehabilitation clinic	Individual interview 70 minutes (online)
9. Setting X	Edvin, 54	Right side infarct SSS: 54	Carol, 50 Wife	Acute: 8 days Sub-acute: 14 days Municipal: 211 days (39 interventions)	Dyadic interview 62 minutes
10. Setting Y	Anne, 84	Right side infarct SSS:49	None Widower	Acute: 3 days Sub-acute: 21 days Municipal: unknown duration (21 interventions)	Individual interview 44 minutes

*NIHSS score converted to SSS score [44]

**The stroke survivor had a second stroke just after being interviewed and just before being terminated from municipal rehabilitation, and thus continued his rehabilitation. Duration in municipal rehabilitation was therefore not obtained.

Process evaluation

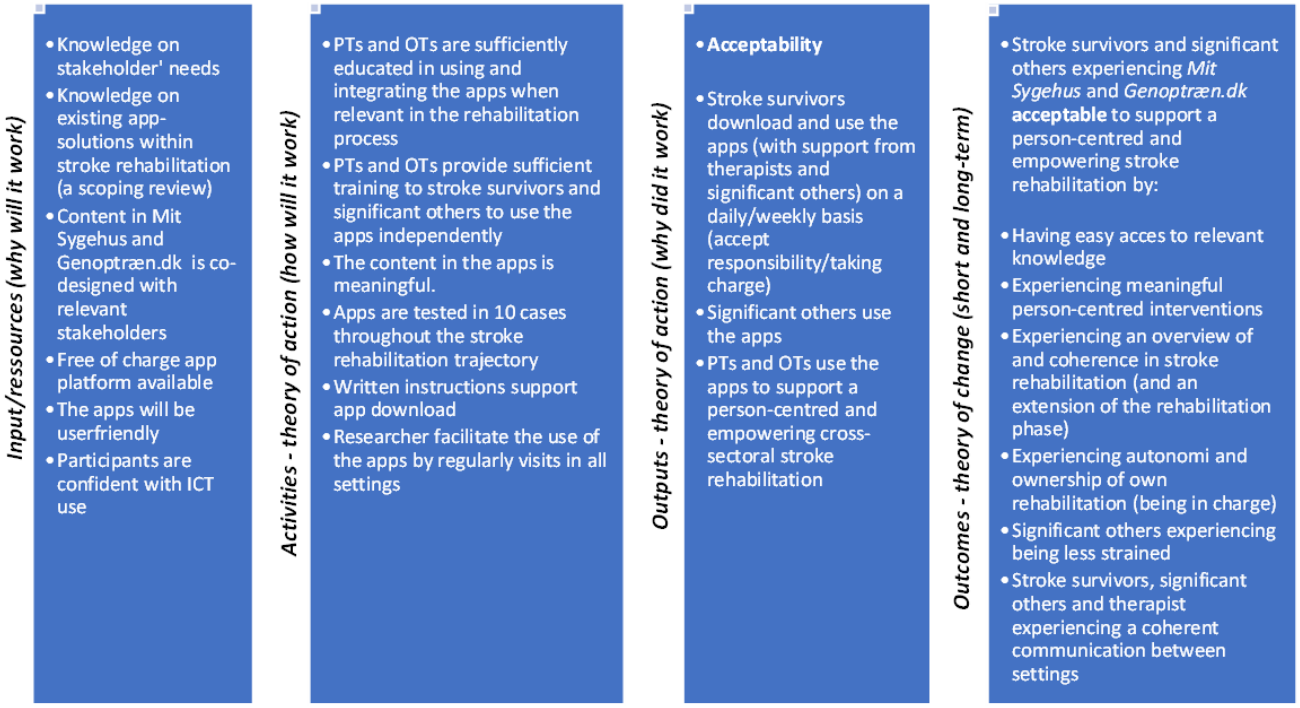
Process evaluations are increasingly used to evaluate complex interventions and are less concerned about effectiveness outcomes but more concerned with understanding ***what***, ***why, how, for whom,*** and under ***what*** circumstances and in which contexts an intervention works, unexpected consequences, and whether adaptions are required [35,36,47,48]. The evaluation of user ***acceptability*** is essential when conducting process evaluations of complex interventions [35,49]. Acceptability, according to Sekhon et al. (2017), is a multi-faceted construct that reflects not only quantitative measures (if and how much the interventions were used), but also qualitative measures that capture the extent to which people delivering or receiving a healthcare intervention consider it appropriate to address their needs in a specific context [49], which

impacts intervention implementation, uptake, adherence, intended outcomes, and overall effectiveness [50].

Logic model developed for this study

A program theory (i.e., a theory of action and theory of change) was developed and visualized in a logic model to provide a better understanding of which mechanisms [8,35,36,47,51] that were expected to facilitate stroke survivors and significant others acceptability of *Mit Sygehus* and *Genoptræn.dk* to support person-centred and empowering stroke rehabilitation, which was the overall main outcome chosen for this study. Previous knowledge on stakeholders' needs [11,13,32] and on existing app-solutions [31], having co-designed the content in the apps with stakeholders [42] were considered important mechanisms for achieving acceptability of the apps (see Figure 4).

FIGURE 4: The final logic model



Ethical considerations

Written information in *Mit Sygehus* and oral information were delivered to stroke survivors and significant others by the PT and/or OT responsible for inclusion. Consent from all participants to participate were obtained in agreement with the Danish code of conduct and the Helsinki declaration. Ethical approval was obtained from the Danish Data Protection Agency (18/60280).

Data generation

Based on the logic model, the following measures were chosen to evaluate the potential of *Mit Sygehus* and *Genoptræn.dk* to support a person-centred and empowering stroke rehabilitation: 1) acceptability, that is, how the intervention was perceived in relation to practicality, barriers, facilitators of using the intervention, and how it supported the rehabilitation process, adaptions required and any unexpected consequences, and 2) what/which part of the apps were used. Table 2 presents the related research questions.

TABLE 2: Acceptability questions

Research questions		
	Theory of action - *how* and *why* it worked/not worked	**Theory of change** - *outcomes*
Acceptability assessed qualitatively	Which inputs and actions were required for the apps to be used throughout the cross-sectoral rehabilitation processes? To what extent do the apps need to be refined or adapted to make them more acceptable, more relevant, or more useful to the specific contexts in which they are delivered? Were there any positive or negative unexpected consequences? Has implementation varied by settings/ contextual factors that needs to be considered?	What are the stroke survivors´ and significant others' perspectives of the content in *Mit Sygehus* and *Genoptræn.dk*? What were stroke survivors´ and significant others' experiences of *Mit Sygehus* and *Genoptræn.dk* to support a person-centred and empowering stroke rehabilitation?
Acceptability assessed quantitatively	Which parts of the apps were used by participants	

The concepts of person-centred rehabilitation [6] and empowerment [5,8] supported the qualitative interviews, that is, elaborating questions on motivation, experienced competences, skills, knowledge, and behaviors in relation to the rehabilitation process were asked [5,8]. Interviews were conducted by the first author in the stroke survivors' home or in the rehabilitation setting of the municipality, in agreement with the participants' preferences, when rehabilitation in the municipality had ended or was about to end. Data were gathered through dyadic interviews (stroke survivors and significant others). In three cases, individual interviews with the stroke survivors were performed, as they did not have a significant other included in the study. Open-ended questions were asked, for example, "*tell me about your rehabilitation process*",

"*how and when was Mit Sygehus and Genoptræn.dk included in the rehabilitation process*", "*what are your thoughts about the content in the apps*?" and *"how did the apps support your rehabilitation process?*". Field notes from the observations made during the testing phase also supported the interviews. Interviews lasted 17-99 minutes.

Data on the quantitative use of the app were collected from the web-based back-end of the app, that is, which modules were used in the app and how many times.

Data from the OTs and PTs delivering the intervention, that is, acceptability, recruitment capability, implementation, adaptions needed and unexpected consequences [37], were gathered in focus group interviews after the testing period had ended and will be reported elsewhere.

Data analysis

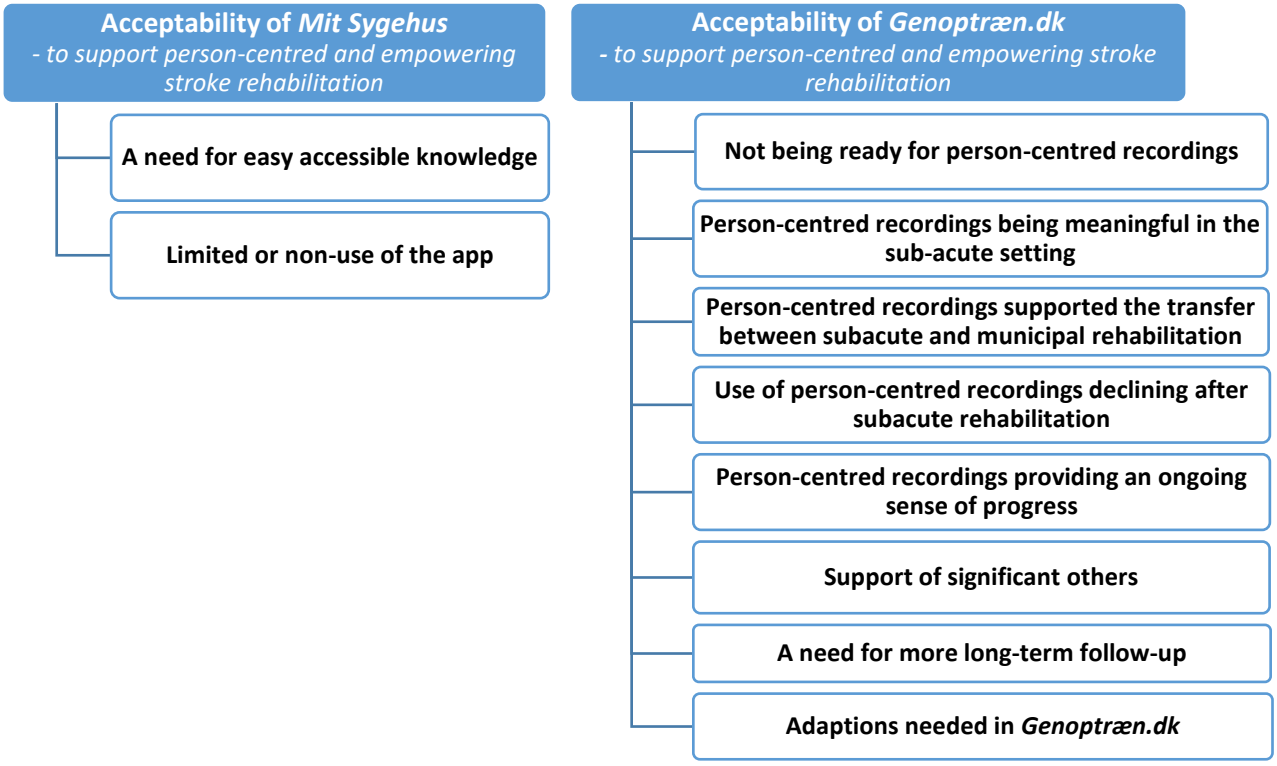
It is suggested that qualitative data used in process evaluations be analyzed continuously to allow learning to be successively implemented [36,45], which was possible due to participants being consecutively included from January 2022 to October 2022. Thus, a constructivist Grounded Theory (GT) approach to analysis was appropriate [37,45], as data generation and analysis should be carried out simultaneously. Conducting a constant comparison method, keeping memos, using theoretical sensitivity, and theoretical sampling are the steps used in GT [45,46]. In a constructivist GT approach, data between the researcher and participant are constructed rather than theories being discovered [45].

The interviews were transcribed verbatim by the first author. The next step of the analysis was open coding using the qualitative analysis tool NVivo [52]. Codes in this step were kept close to the data, sentence-by-sentence, or incident-by-incident and were made in Danish to make quick and spontaneous codes [53]. This was done by asking "*what is going on here?*", "*what does this mean*?" [45], and more processual questions like "*who* does what, *when*, *where, how* and with *what consequences?"* [54]. The codes were then compared and organized into code groups (focused coding). A total of 20 code groups were constructed. After this stage, categories were developed by identifying the most significant content and the relationships between categories (theoretical coding). At this stage, the code groups were translated into English. Following this stage, the concepts of person-centred and empowering rehabilitation [1,4–8] was included to support development of sub-themes describing the construct of ***acceptability*** of *Mit Sygehus* and *Genoptræn.dk* to support person-centred and empowering stroke rehabilitation, from the perspective of stroke survivors and significant others (Figure 5). Comparing, contrasting and (re)organizing the code groups, categories, themes and sub-themes was an iterative process. Throughout the data analysis process, the findings were discussed and reviewed by all authors.

Results

Acceptability of the two apps *Mit Sygehus* and *Genoptræn.dk* to support person-centred and empowering stroke rehabilitation, from the perspectives of stroke survivors and significant others, is presented in two overall themes: "***acceptability*** of *Mit Sygehus"* and "**acceptability** of *Genoptræn.dk"* to support person-centred and empowering stroke rehabilitation. Mechanisms facilitating acceptability as well as contextual factors (i.e., rehabilitation phases) affecting acceptability, reasons for "non-use" and suggestions for adaptions are elaborated on in respectively two and eight sub-themes (Figure 5).

FIGURE 5: Results - acceptability of Mit Sygehus and Genoptræn.dk



Acceptability of Mit Sygehus

Mit Sygehus was the most used app among the significant others, especially in the acute setting. Use of *Mit Sygehus* decreased when rehabilitation continued in the subacute and municipal setting. In the following sub-theme "a need for easy accessible knowledge" mechanisms for acceptability of *Mit Sygehus* are elaborated on, as well as reasons for limited or non-use.

A need for easy accessible knowledge

Having all relevant knowledge/information in one place was valued by all participants using *Mit Sygehus*,

346 although significant others seemed to use it more than the stroke survivors. "*I read it all, not just some of it,*
347 *since it interested me… it is well explained in the app, and all is located in one place… it is ten times easier*"
348 (Jonas). The participants used *Mit Sygehus* to get informed, share knowledge, prepare for meetings, and re-
349 read when needed. Participants explained mechanisms such as confidence in using ICT, the apps being
350 user-friendly, containing relevant and evidence-based knowledge that was "just at hand" and could be
351 accessed when needed as crucial for acceptability. Most stroke survivors received support from therapists
352 to install the two apps on their mobile phones. Four significant others (Jonas, Jane, Ella, Betina) had
353 installed *Mit Sygehus* on their own mobile phone, only supported by written instructions.
354

355 ## Limited or non-use of the app

356 *Mit Sygehus* was mostly used in the acute (Carol, Jonas) and sub-acute (Jane) rehabilitation settings
357 according to significant others. Carol explained that she gradually forgot about *Mit Sygehus* and instead
358 used the Internet to gain the required knowledge. Statistical data from the back-end of *Mit Sygehus*
359 showed that she had used the app on the day of installation and not again before the day of the interview.
360 Significant others, furthermore, called for information on "return-to-work" and how to "regain driver's
361 license", especially during the later phases of rehabilitation.
362 Some stroke survivors explained they had only visited the *Mit Sygehus* a few times (Beate, Robert, Jamie,
363 Noah) or not at all (Edvin, Erin, Liam, Peter, Anne). Reasons for limited or non-use were different, among
364 others it had to do with the content being "confronting" and reminding them of deficits they had/could
365 have had. Lack of reading abilities, fine motor difficulties, and/or cognitive deficits, such as trouble
366 remembering codes, were only mentioned in one case as barriers for using the app (Peter). Stroke survivors
367 instead explained that they gained the needed knowledge through a significant other (Edvin, Beate, Peter,
368 Jamie) or by receiving sufficient knowledge from health professionals (Liam, Edvin, Beate, Anne, Erin,
369 Robert, Noah). However, statistical data from the back-end of *Mit Sygehus* showed that some stroke
370 survivors who explained the non-use or limited use of *Mit Sygehus* during interviews had actually accessed
371 the app a number of times throughout the rehabilitation process (Beate, Peter) or just a few times (Jamie,
372 Robert, Noah) typically in the acute phase of rehabilitation (Table 3). Three stroke survivors (Liam, Erin,
373 Edvin) had no longer *Mit Sygehus* installed on their mobile phone when the interviews were conducted,
374 thus their "ID-number" could not be identified, and eventual statistical data on the use of *Mit Sygehus*
375 could not be identified from the back-end of the app (Table 3). Three significant others (Betty, Ida, and
376 Berry) had either never heard about *Mit Sygehus* (Betty), failed to install the app (Ida) or did not wish to use
377 apps (Berry). They explained gaining knowledge from the Internet, a booklet received during acute
378 rehabilitation, through health professionals or through having previous knowledge about stroke. The "read-

out-load" function was not mentioned being used in any of the cases, most participant did not know it was an option, and/or did not perceive it as relevant.

TABLE 3: Stroke survivors' and significant others' use of Mit Sygehus and Genoptræn.dk

Modules / participants	Beate	Jonas husband	Peter	Ella wife	Liam	Jane wife	Jamie	Betty wife	Robert	Ida wife	David	Erin	Berry husband	Noah	Edvin	Carol wife	Anne
What is stroke	4				-	3	2	-		-	3	-	-		-	2	3
Consequences of stroke	26	14	8	2	-	7	1	-	1	-	3	-	-		-	7	
The rehabilitation process	22	13	10	2	-	5		-	3	-	7	-	-	2	-	1	
Significant others	10	23	1	13	-	4	1	-	5	-	10	-	-	2	-	1	1
Driver's licence	16	4	4	3	-	5	1	-	3	-	5	-	-	5	-	2	
My hospital jounal	48	4	4	2	-	2		-	6	-	12	-	-	3	-	2	2
Training	35	8	0	0	-	0	0	-	13	-	19	-	-	11	-	1	0
Genoptræn.dk app	50	-	10	-	8	-	15	-	69	-	2	17	-	15	1	-	15
Coordinator of the municipality	23	15		2	-	4	1	-		-	7	-	-	5	-	1	1
Patient organizations		5			-	2		-		-	1	-	-		-		
The research project	22	12	4		-	3	3	-		-	4	-	-		-		
Guide for videocalls	4			2	-	2		-		-		-	-		-		
My appointments	45	4		4	-	1	1	-	2	-	28	-	-	1	-	1	1
Personal notes	3	2	1	2	-	1	1	-	1	-	3	-	-		-		
Contact	4	2		1	-	1		-		-		-	-		-	1	

The numbers indicate number of times participants accessed the different modules
"-" indicates that participants' ID-number was in-retrievable, and therefore eventual use of *Mit Sygehus* could not be identified

Acceptability of *Genoptræn.dk*

Genoptræn.dk was the most used app among the stroke survivors, especially in the subacute setting. App installation mostly occurred in the acute setting of hospital X or hospital Y1, as hospital Y did not include any participants. Use of *Genoptræn.dk* decreased when rehabilitation continued in the municipal setting and was used even less when municipal rehabilitation had ended. In the following sub-themes acceptability of *Genoptræn.dk* in relation to the rehabilitation phases are elaborated on, as well as reasons for non-use and adaptions needed.

Not being ready for person-centred recordings

Mostly, stroke survivors did not perceive *Genoptræn.dk* as acceptable in the acute rehabilitation setting because of the short admission in this setting and the focus being more on acute care: "*In the beginning you use a lot of energy on just being there, just sleeping and surviving…you need to be able to handle self-training before it makes sense*" (Noah). In only one case content in *Genoptræn.dk* was included in the acute setting (hospital X), where the stroke survivor had herself (Beate) initiated to do her self-training (facial exercises) and shared the videos with her husband, children, and grandchildren the same day they were provided in the app. This could have to do with this stroke survivor being the first participant in the study

(and the therapist practicing using the app) and due to Beate experiencing a longer admission in the acute setting (nine days). Two other stroke survivors (David and Robert), perceived *Genoptræn.dk* could have been acceptable in the acute setting, to support daily training and waiting time before being transferred to a sub-acute setting, or before sub-acute therapists engaged in using the app to support the rehabilitation. In order for *Genoptræn.dk* to be implemented in the acute setting, therapists are required to collaborate with the stroke survivor and the significant other regarding their wishes, needs and "readiness" to use person-centred recordings to support early rehabilitation.

Person-centred recordings being meaningful in the sub-acute setting

Providing person-centred recordings in *Genoptræning.dk* to support self-training, based on individual and co-constructed goals, were overall meaningful and valued by the participants. The content in *Genoptræn.dk* were applied by the OT and or PT in the sub-acute setting and mostly contained video-recorded exercises and activities performed by stroke survivors themselves, such as gait, balance, strength (upper and lower limbs), fine motor, and facial exercises. In some cases, the gait performance was also recorded for status purposes: "*Exercises were added in my app in the sub-acute setting…. It shows me in detail what I must do and how many times … it´s important that it is yourself on the recordings and not someone who has no problems*" (Robert). By using person-centred recordings, the stroke survivor was provided with a clear picture of "why, what and how" to perform self-training which subsequently supported motivation: *"It made it very clear for me which activities that were difficult, and thus needed to be trained. That motivated me"* (Edvin). In addition, the therapists in the sub-acute rehabilitation setting adjusted the content gradually to match individual goals and functioning: "*They made the video recordings to support me … and when I didn´t need them anymore they were deleted or adjusted*" (Noah).

Participants explained that confidence in using ICT, the apps being user-friendly, and the content in *Genoptræn.dk* being relevant and meaningful as an important mechanism for acceptability: *"It was easy to access and use… I just pushed there, and I am in*" (Beate). Most therapists instructed stroke survivors to perform daily self-training between other planned activities. However, some stroke survivors took initiative to perform the exercises also as compensation for cancelled training or if they felt restless and needed to do even more on their own: "*I performed the exercises once a day, sometimes more, but the time was still long… sometimes I also went for a walk*" (Edvin).

Therapist-support, through education, facilitation and frequent integration of *Genoptræn.dk* during sub-acute rehabilitation setting also seemed to be important mechanisms, for stroke survivors' motivation, active engagement and ownership of the content in the app. Frequent use also supported creating habits

to perform daily self-training, and along the way it was not necessary to enter the app every time to perform self-training, since they could gradually be remembered and was made a habit: "*I think it is easier to make a habit of it during a 3-4 week stay, than if he had been discharged after 4 days* (Ida).

To additionally support self-training, therapists could use verbal instructions during person-centred recordings. Peter reported that audio instructions such as "*slowly slowly*" from the PT during a "getting up and down from a chair" exercise were stored and remembered long after rehabilitation had ended. Audio support also facilitated good quality exercises according to his wife Ella, especially if reading instructions were difficult.

Person-centred recordings supported the transfer between subacute and municipal rehabilitation

According to the participants, *Genoptræn.dk* supported the transfer from sub-acute to municipal setting, as well as the cross-sectoral communication between therapists and the "waiting time" until municipal rehabilitation was initiated. One stroke survivor (Robert) experienced the therapist being prepared before the first municipal visit by beforehand watching the recordings made in sub-acute setting; and during the first visit, she made some adjustments according to his actual physical functioning. Robert imagined that watching the recording beforehand also supported the therapists: "*I imagine they visit many patients during the day, and if they just watched the recordings beforehand, they would be prepared and maybe even save some reading time*" (Robert).

Use of person-centred recordings deteriorating after subacute rehabilitation

Stroke survivors and significant others explained that person-centred self-training were used less during municipal rehabilitation than in sub-acute rehabilitation: "*I used to perform them a lot, now it is less frequent"* (Jamie). When discharged to the home setting, participants explained that everyday life activities took up more time, such as emptying the dishwasher, sweeping the terrace, cooking, fixing things in the garage, or gradually returning to work, and thus partially substituted using *Genoptræn.dk.* In addition, *Genoptræn.dk* appeared to be included less by therapists in municipal rehabilitation. Most stroke survivors did not remember having received any instructions on frequency regarding self-training during municipal rehabilitation, however, some stroke survivors found it supportive to receive a reminder, others performed exercises when it suited their everyday life or had made it a habit to perform them on specific weekdays: "*I just performed them without reminders. I have made a habit out of doing my self-training on Tuesdays, Wednesdays, Fridays, Saturdays, and Sundays*" (David).

When performing person-centred self-training during municipal rehabilitation, stroke survivors explained that they did only access *Genoptræn.dk* occasionally to ensure that all exercises were remembered, and that the quality of the performance was in line with the recordings and guidelines given by the therapists: "*Even after doing my exercises many times, doubt can still appear, and it is important for me to know if I perform the exercises correct. Then this solution is easy"* (Jamie). One stroke survivor explained that having self-training through *Genoptræn.dk* supported his engagement in everyday life, and that he and his wife could go camping or fishing when they wished to as the self-training was "along with him" and could be performed everywhere (Robert). It was however also suggested that *Genoptræn.dk* was used more often during municipal rehabilitation and was continuously adjusted by the therapists to actual abilities to continuously support motivation and a sense of ongoing progress (Jamie and Robert).

Only two stroke survivors could imagine using the generic content in *Genoptræn.dk* (Noah and Jamie): *"The better you get, maybe then it does not need to be yourself on the recordings"* (Jamie). Another stroke survivor declined municipal rehabilitation after the first visit because he felt that the oral information and guidelines received from hospital X1 were sufficient to manage on his own and without app support (Liam).

Person-centred recordings providing an ongoing sense of progress

The person-centred recordings seemed to provide an important ongoing "sense of progress" for stroke survivors. And even though less frequent, many stroke survivors still occasionally performed their exercises also after municipal rehabilitation had ended, mostly to keep going, to ensure that the quality of the exercises was in line with recordings, and to perceive and watch progress made. Having ongoing access to the person-centred recordings in *Genoptræn.dk* and to "how it looked back then" seemed to be important and continuously supported motivation, adherence and provided a sense of accomplishment for stroke survivors: "I*t gives you small updates, otherwise it could be hard to perceive progress, and you get small victories continuously, and not only at the end… also it is often my wife that thinks I have progressed; however, it is important that I myself get that experience… the recordings help me to experience that*" (Robert). Another stroke survivor explained: "*I could see that it was myself on the recordings, and there was progress. Had there not been any progress I would have thrown the phone away*" (Edvin). Also, if motivation was low one day or for a period, watching the recording would remind them that things had progressed and that effort was worthwhile: "*If you have an off day, you can access the app... it could be motivating"* (Beate). Also, it seemed that some stroke survivors were on their own able to adapt the use of *Genoptræn.dk* to match their current needs, for example, Robert had himself increased the number of repetitions on each exercise: "*I still do them once in a while, but I have adjusted the number of repetitions, since I have progressed that much since they were recorded*".

Support of significant others

Most significant others found it rewarding to have the opportunity to watch person-centred recordings in *Genoptræn.dk*: "*I watched the exercises on her mobile phone-…otherwise, I would never have had the insight to what was relevant for her. It made me capable of supporting her better*" (Jonas). One significant other explained that she and their children felt more prepared when her husband came home for the weekends: "F*rom these recording we prepared which tools were needed for him to perform his self-training at home…it was a kind of communication that helped me to prepare for the upcoming weekends*" (Carol). Significant others only had access to *Genoptræn.dk* on their own ICT devices when using the stroke survivor's login. This was only done by Ella, whose husband struggled with using ICT and apps. The others found it sufficient to have the opportunity to watch the content on the stroke survivors' mobile phone, which together with provided reminders through the app also seemed to relieve the significant others of "being in charge": "*With the app, he just did it, no days of… it is his responsibility, I just ask about the app*" (Ida).

A need for more long-term follow-up

Even though participant expressed being well informed and receiving sufficient support from therapists during the rehabilitation process, several stroke survivors still called for follow-up support after municipal rehabilitation had ended. Some suggested a visit or just a phone call 3-6 months post-rehabilitation to support an ongoing sense of progress and status of performance: "*I could imagine that a phone call after 6 months could get your thought back on how things actually are going… maybe some kind of re-test, or you do your self-training to get an idea of how things are actually going before all access is deleted in the app…that would be great"*(Robert). Some significant others confirmed a need for a more long-term follow-up and even though they knew that municipal rehabilitation would end after typically three months, they still experienced a sudden end of the rehabilitation process: "*it would be great if the therapist returned or if we could go back… but it feels more like "three months has passed, over and out, you don't have problems any longer*""(Jonas).

Adaptions needed in *Genoptræn.dk*

Two stroke survivors explained that they had not used *Genoptræn.dk* after discharge from sub-acute rehabilitation (Liam and Erin). Liam explained that he could remember the instructions received by the therapists, and thus he performed self-training without any other support. Erin explained that there were

so many other things going on, and that fatigue and pain in her knee prevented her from performing self-training (waiting for knee replacement). She also explained that there were some challenges with *Genoptræn.dk* during the sub-acute rehabilitation, such as when performing the first exercise, it "jumped" back to the first exercise instead of the next exercise in the app: "*It annoyed me, and then I stopped using it"* (Erin). Two other stroke survivors (David and Jamie) also mentioned that obtaining an overview of how far they were in their self-training program was challenging. However, this was adjusted before their rehabilitation ended by supplying each exercise with a number, which solved the issue for them. However, some participants expressed a need for more flexibility in the exercises they wished to perform on that day. *Genoptræn.dk* supplies the exercises in the same order each time, and stroke survivors had to "tick-off" each exercise before they could continue to the next or shift to another exercise program (if they had two or more programs, e.g., balance, lower limb strength, facial, upper motor exercises, etc.).

Discussion

It is suggested that to make app-based solutions acceptable for supporting an empowering stroke rehabilitation, the developed solutions must be co-designed with relevant stakeholders, address the needs of stroke survivors and significant others [21,55], and support a greater part of stroke rehabilitation and not only isolated parameters [31]. Furthermore, not only should the content of the app be evaluated but also the participants' experiences with the developed solutions [56]. Therefore, this study aimed to evaluate the acceptability of two co-designed app-solutions, *Mit Sygehus* and *Genoptræn.dk* to support person-centred and empowering stroke rehabilitation by asking *what, why*, *how, for whom*, and under what circumstances and in which contexts the intervention works, as well as unexpected consequences and adaptions required [57].

Overall, the results showed that *Mit Sygehus* and *Genoptræn.dk* were acceptable and supported person-centred and empowering stroke rehabilitation; however, *Genoptræn.dk* was less acceptable in the acute setting, and the acceptability of *Mit Sygehus* declined throughout the rehabilitation process.

Important mechanisms for acceptability to support person-centred and empowering stroke rehabilitation were apps that were accessible, free-of-charge, user-friendly, participants being confident in ICT use, and content being relevant and meaningful. This is in line with previous studies showing that co-designed, evidence-based, and free-of-charge information and educational material delivered through an app [18] can support patient empowerment [19]. In this process evaluation, it appeared that the content in *Mit Sygehus* supported in gaining relevant knowledge throughout the rehabilitation process. It especially seemed to support significant others who, by reading and re-reading prepared for meetings with health professionals, supported their memory and were able to better support their stroke survivors. However,

some significant others were not introduced to or insufficiently supported in using *Mit Sygehus.* If *Mit Sygehus* was only introduced and integrated in acute rehabilitation, it could seem to be a barrier for ongoing use throughout the rehabilitation process and thus a possible implementation failure. Also, a lack of knowledge on "return-to work" and "re-gaining driver's license/permission to drive car" were mentioned as areas of improvement. The involvement of other sectors, such as the job sector, when trying to bridge "return-to-work" is mentioned as challenging by the Danish whitepaper on rehabilitation [2]. Providing a designated "contact person" that supports the coordination of post-stroke services and provides educational material is suggested to support the continuity of care [1,58]. One initiative in Denmark, in trying to bridge this gap, is by employing "coordinators" in municipalities [39]. Knowledge from the web-based *back-end* of *Mit Sygehus* identified that the module on "coordinator of the municipality" had been used. However, this module primarily contained contact information to the coordinator and currently no detailed information regarding "return-to-work".

Stroke survivors were less explicit about reading and using the content in *Mit Sygehus* even though statistical data from the "back-end" of the app showed that some stroke survivors had accessed the content throughout the rehabilitation process. It seemed that stroke survivors' need for information was lower than that of significant others, perhaps because significant others and therapists continuously supported them to obtain the needed information. Thus, the intrapersonal and interactional component of stroke survivor empowerment may have been supported through adequate significant other and therapist support. It is also likely that, at the time of the interview, they had forgotten when and for what they had used *Mit Sygehus*.

Stroke survivors seemed to gain more support from *Genoptræn.dk*. Stroke rehabilitation is complex, coordinated stroke rehabilitation is challenging and stroke survivors and their significant others have diverse individual needs regarding the rehabilitation process. [1,2,6]. Involving *Genoptræn.dk* in the stroke rehabilitation trajectory in this study seemed to support bridging the transfer between rehabilitation settings, especially between the sub-acute and municipal settings. Stroke survivors mostly did not feel ready to initiate self-training in the acute setting, although a few did, if the duration in the acute setting was prolonged more than the usual 2-3 days or if self-training was initiated late in the sub-acute setting. The inability to engage in and manage self-training in the very early stages after stroke has also been reported elsewhere [59,60]. One explanation could be the short length of hospital stay, which challenges the identification of stroke survivors' needs, goals and resources [59,61]. However, this may negatively affect stroke survivors and significant others' motivation, adherence [56] and empowerment [31]. Through person-centred rehabilitation, health professionals may be able to identify when stroke survivors are ready

595 for engaging in co-constructed goal-setting and person-centred self-training to support stroke survivors'
596 empowerment in an acute setting.

597 Having person-centred exercises recorded in the sub-acute setting seemed to support motivation,
598 ownership, and communication with significant others and therapists in the municipality. To our
599 knowledge, there is limited use of person-centred video recordings to support self-training during stroke
600 rehabilitation. Prescribed video recorded exercises are to our knowledge mostly generic recordings
601 performed by healthy individuals without any deficits, which can affect stroke survivors' motivation and
602 make it hard to relate to as mentioned in this study. Although, a recent study protocol describe the use of
603 people with stroke-related disabilities in their prescribed instructional generic video recordings to promote
604 physical activity among stroke survivors [62], the results from our study indicate that recordings of the
605 stroke survivor themselves support the experience of "clarity", "necessity", "ableness", and a sense of
606 progress.

607 The ongoing follow-up from therapists on the use of *Genoptræn.dk* during the rehabilitation process,
608 including adjusting the content to match needs, goals, and stroke survivors' abilities, seemed to be an
609 important mechanism for their engagement, motivation, and adherence regarding person-centred self-
610 training and seemed to support both intrapersonal, interactional and behavioural components of
611 empowerment**.** Continuous co-constructed person-centred goal-setting and evaluation are an important
612 part of rehabilitation [6,65]; however, it is well known that stroke survivors struggle with setting goals and
613 understanding their problems, especially in the acute setting [61], as mentioned above. A review on goal-
614 setting found that stroke survivors tend to set broad long-term goals on an activity/participation level,
615 whereas health professionals tend to set short-term and specific goals at the impairment level. Thus,
616 conflicting priorities were mentioned as barriers to co-constructed person-centred goal-setting, in addition
617 to limited participation from stroke survivors, and health professionals' skills in goal-setting [59]. It seemed
618 in this study that the person-centred recordings supported the co-constructed goal-setting process by
619 making it clearer which activities and exercises were challenging and thus needed to improve both
620 impairment and activity levels. Therefore, stroke survivors were supported in understanding their activity
621 problems they needed to work on and how, which seemed to motivate them and supported the
622 intrapersonal and interactional component of empowerment. Ongoing adjustment of the content in
623 *Genoptræn.dk* to match current activity levels, new person-centred goals based on needs and preferences
624 are known facilitators for co-construction and participation in the ongoing goal-setting process, which
625 subsequently facilitates motivation and adherence to the chosen activities [59]. Additionally, experiencing a

sense of forward momentum during stroke rehabilitation was important for stroke survivors' motivation and engagement in this study, which was also found important in a review by Lloyd et al. from 2018 [64].

Therapists play an important role in teaching stroke survivors how to perform their self-training [24], and the connection with and encouragement from therapists has been shown to promote rehabilitation satisfaction [1], adherence [24], and confidence in their own abilities [61]. Although the duration of sub-acute rehabilitation was only 3-4 weeks, the daily and intensive use of *Genoptræn.dk* seemed to facilitate the development of positive habits regarding self-training, through support of intrapersonal and interactional component of empowerment, that continued in the municipality and post-rehabilitation where empowered behaviors were reported by the stroke survivors. In addition, the flexibility to perform self-training when fitted in daily life was valued by the participants in this study. Stabile habits can be created within a month through repeated active engagement in desirable behaviors such as self-training under consistent contextual circumstances [67], comprising tailored approaches aligned with specific and achievable person-centred goals [60]. Active engagement and participation in everyday life are often highly prioritized goals for stroke survivors [60]. Performing activities that are defined by the stroke survivor and not for the stroke survivor, which is not too easy or too difficult and can be achieved within a restricted time frame, positively affects the stroke survivors' perceived competency in carrying out self-training [66]. Person-centred activities, exercises and guidelines also supported an overview of the progress made during the rehabilitation process in this study, when therapists across rehabilitation settings adjusted and deleted irrelevant content to continuously align with stroke survivors' needs, wishes, and goals. This also meant that the content in *Genoptræn.dk* was diverse and matched individual stroke survivors' abilities, which has been a challenge in another app-based study [56]. Continuous customized support to maintain motivation, self-management, and empowerment has been found to be important elsewhere [24,56] when interventions are delivered through an app [31].

Although it is increasingly known that apps may support repetitive task-specific exercises and self-training in stroke rehabilitation [26], little evidence exists on apps supporting cognitive rehabilitation. In our study, *Genoptræn.dk* was mainly used to support physical function and facial deficits; however, a review by Zhou (2018) also found evidence for cognitive rehabilitation through apps, although this conclusion was based on only two studies [26]. Furthermore, a recent review concluded that ICT use could support cognitive function among older adults, such as memory, executive functioning and processing speed. Also, participants experienced that using ICT to support cognitive function drew less attention to their deficits, which made them feel less stigmatized [67].

Having therapists continuously follow-up seemed important for the participants, which is also supported elsewhere [29,60]. However, stroke survivors in this study and in other studies [11,13] also called for a follow-up three to six month post-rehabilitation to maintain a sense of progress and connectedness with therapists, which may also prevent a feeling of abandonment [1]. It is well known that rehabilitation needs may persist for at least one year after stroke [9] and that participation and engagement in everyday life is challenging to maintain when organized rehabilitation and support from health professionals ends [9,11,13,14,23], typically three months post-stroke [11,13,14]. Receiving reminders has been shown to support continued engagement in valued activities post-rehabilitation [68], and SMS reminders or phone calls until a year post-stroke may optimize satisfaction with the healthcare system, reduce significant others' burden [30,69], and decrease readmission rates [58]. In an ongoing study, weekly support through SMS for three months post-rehabilitation is currently being evaluated [69]. Thus, more long-term support for continued participation in meaningful activities after organized rehabilitation termination needs to be further investigated.

In this study, most participants who wished to use apps to support stroke rehabilitation found it easy to engage with *Mit Sygehus* and *Genoptræn.dk*. Although, patients with severe cognitive or physical disabilities have been reported to have trouble using apps independently [20,58]. However, patient characteristics have not conclusively been able to explain differences in the perception of satisfaction with including technological solutions to support rehabilitation from the perspectives of health professionals [20] and stroke survivors [61]. Facilitators for using technologies to support rehabilitation are solutions that are simple and easy to use [20,58], provide relevant information and insight into rehabilitation goals [20], can be customized to the patient's needs, support self-training/home exercises and can be continued independently after discharge, [17,20]. All these aspects have been complied with in this study and have been discussed above. In addition, contextual factors such as quality of coordinated communication and information [61], easy adoption in existing routines [1,17], and sufficient technological knowledge and support for health professionals [21,55] may facilitate satisfaction and uptake. These organizational factors will be discussed in a later manuscript describing the therapists' acceptability.

In summary lessons learned from this process evaluation were that *Mit Sygehus* may be acceptable to support a person-centred and empowering stroke rehabilitation, however content on "return-to-work" and "re-gaining driver's license/permission to drive" needs to be developed and more stringent implementation is needed in the acute setting, that is continued in the other settings. *Genoptræn.dk* was perceived as acceptable to support person-centred and empowering stroke rehabilitation; through containing relevant, meaningful and motivating content and providing a sense of progress. However, integration already in an

acute setting for participants expressing being ready for this could support person-centred and empowering stroke rehabilitation even more.

Strengths and limitations of this study

A strength of this study was the developed logic model that described the mechanisms of actions needed for the acceptability of the app-solutions to support person-centred and empowering stroke rehabilitation. The logic model was continuously adjusted throughout the research process, which may have benefitted the participants. In addition, using dyadic interviews when possible was perceived as a strength, since the significant other and stroke survivors could collaborate on giving their insights, jogged each other's memories, and together constructed suggestions for improvements. Furthermore, the fact that participants were interviewed around the time of post-rehabilitation was perceived as a strength regarding their willingness to be honest about *what* worked, *whe*n, and *how* it came to work. Insight into overall experienced empowerment would have been harder to understand if the timing of the interview was earlier. On the other hand, a later interview or a 3–6-month follow-up interview could potentially have elaborated on experienced empowerment in everyday life post-rehabilitation.

Making observations during the testing phase may be an important strength when evaluating new interventions to gain insight into what, how and why they worked, and what required further enhancements [36]. However, performing only one processual observational data collection for each included stroke survivor might also be perceived as a limitation because areas that may have needed further improvement may have been overlooked because they were forgotten at the time of the interview. However, PTs and OTs were also interviewed in focus groups, which provided insight into further adjustments needed before advancing to implementation and testing on a larger scale. In addition, more than one observational visit could have further facilitated the implementation of *Mit Sygehus* throughout the rehabilitation process by providing more support and education to therapists.

Conclusion

Existing app solutions to support stroke rehabilitation mainly focus on isolated parameters of complex stroke rehabilitation, usually only assessment, or intervention/training. This study showed that *Mit Sygehus* and *Genoptræn.dk* may be acceptable from the perspectives of stroke survivors and significant others by:1) focusing on a greater part of person-centred and empowering stroke rehabilitation; 2) providing easy and sufficient access to relevant and evidence-based information and education regarding stroke, deficits, and

721 rehabilitation, and how to navigate within the healthcare system; 3) providing easy and meaningful person-
722 centred intervention/training/activities/self-training based on stroke survivors' and significant others'
723 actual and changing needs and goals; 4) supporting coherence between different rehabilitation sectors; and
724 5) continuous support from therapists. However, content on "return-to-work" and "re-gaining driver's
725 license/permission to drive" needs to be developed in *Mit Sygehus*. Also, more stringent implementation of
726 *Genoptræn.dk* is needed in the acute setting as well as an investigation of how to provide cognitive
727 rehabilitation in the app.
728
729

730 ## Acknowledgements

731 The authors thank the stroke survivors, significant others, and therapists for participating, engaging, and
732 sharing their experiences throughout the research process.

733

734 ## Funding

735 This study was supported by the University College of South Denmark and the Danish Organization for
736 Physiotherapists (grant number R39-A1042).

737

738 ## Conflicts of interest

739 The authors have no conflicts of interest to declare.
740
741

742 ## Data availability

743 Can be shared upon request.
744
745

746 ## Author contributions

747 All authors have contributed to the conception and design, analysis, and interpretation of the data, drafting
748 and critically reviewing the content of the paper, and have given final approval for the version to be
749 published. All authors agree to be accountable for all aspects of the work.

750
751

References

1. Jesus TS, Papadimitriou C, Bright FA, Kayes NM, Pinho CS, Cott CA. Person-Centered Rehabilitation Model: Framing the Concept and Practice of Person-Centered Adult Physical Rehabilitation Based on a Scoping Review and Thematic Analysis of the Literature. Archives of Physical Medicine and Rehabilitation. 2022 Jan;103(1):106–20.

2. Maribo T, Ibsen C, Thuesen J, Nielsen CV, Johansen CV, Vind AB. Hvidbog om rehabilitering. Aarhus, Denmark: Rehabiliteringsforum Danmark; 2022.

3. Wade D. Rehabilitation – a new approach. Overview and Part One: the problems. Clin Rehabil. 2015 Nov;29(11):1041–50.

4. Bergström A, Koch L, Andersson M, Tham K, Eriksson G. Participation in everyday life and life satisfaction in persons with stroke and their caregivers 3-6 months after onset. Journal of Rehabilitation Medicine. 2015;47(6):508–15.

5. Barr PJ, Scholl I, Bravo P, Faber MJ, Elwyn G, McAllister M. Assessment of Patient Empowerment - A Systematic Review of Measures. Bond K, editor. PLoS ONE. 2015 May 13;10(5):e0126553.

6. Wade DT. What is rehabilitation? An empirical investigation leading to an evidence-based description. Clin Rehabil. 2020 May;34(5):571–83.

7. Cerezo PG, Juvé-Udina ME, Delgado-Hito P. Concepts and measures of patient empowerment: a comprehensive review. Rev esc enferm USP. 2016 Aug;50(4):667–74.

8. Zimmerman MA. Psychological empowerment: Issues and illustrations. American Journal of Community Psychology. 1995 Oct;23(5):581–99.

9. Ekstam L, Johansson U, Guidetti S, Eriksson G, Ytterberg C. The combined perceptions of people with stroke and their carers regarding rehabilitation needs 1 year after stroke: a mixed methods study. BMJ Open. 2015 Feb 12;5(2):e006784–e006784.

10. Ytterberg C, von Koch L, Erikson A. Abandoned to the strains of daily life: a qualitative study of the long-term experiences in partners to persons after a mild to moderate stroke. Disability and Rehabilitation. 2017 Nov 13;1–7.

11. Marwaa MN, Ytterberg C, Guidetti S. Significant others' perspectives on person-centred information and communication technology in stroke rehabilitation – a grounded theory study. Disability and Rehabilitation. 2019 Jan 16;1–8.

12. Wissel J, Olver J, Sunnerhagen KS. Navigating the Poststroke Continuum of Care. Journal of Stroke and Cerebrovascular Diseases. 2013 Jan;22(1):1–8.

13. Gustavsson M, Ytterberg C, Nabsen Marwaa M, Tham K, Guidetti S. Experiences of using information and communication technology within the first year after stroke – a grounded theory study. Disability and Rehabilitation. 2016 Dec 15;1–8.

14. Tistad M, von Koch L, Sjöstrand C, Tham K, Ytterberg C. What aspects of rehabilitation provision contribute to self-reported met needs for rehabilitation one year after stroke - amount, place, operator or timing? Health Expectations. 2013 Sep;16(3):e24–35.

15. Bech M, Bonde Klausen M, Buch MS. Tværsektoriel sundhedsforskning i Danmark: aktører, fokusområder og forskningsbehov med udgangspunkt i Region Syddanmark. VIVE - Det Nationale Forsknings- og Analysecenter for Velfærd; 2022.

16. DANSKE REGIONER. EN KLAR RETNING FOR SUNDHEDSFORSKNING I DANMARK [Internet]. 2023. Available from: https://www.regioner.dk/media/23170/udspil-en-klar-retning-for-sundhedsforskning-i-danmark.pdf

17. Bezuidenhout L, Joseph C, Thurston C, Rhoda A, English C, Conradsson DM. Telerehabilitation during the COVID-19 pandemic in Sweden: a survey of use and perceptions among physiotherapists treating people with neurological diseases or older adults. BMC Health Serv Res. 2022 Dec;22(1):555.

18. Dubey D, Amritphale A, Sawhney A, Amritphale N, Dubey P, Pandey A. Smart Phone Applications as a Source of Information on Stroke. J Stroke. 2014;16(2):86.

19. Kang YN, Shen HN, Lin CY, Elwyn G, Huang SC, Wu TF, Hou WH. Does a Mobile app improve patients' knowledge of stroke risk factors and health-related quality of life in patients with stroke? A randomized controlled trial. BMC Med Inform Decis Mak. 2019 Dec;19(1):282.

20. Brouns B, van Bodegom-Vos L, de Kloet AJ, Vliet Vlieland TPM, Gil ILC, Souza LMN, Braga LW, Meesters JJL. Differences in factors influencing the use of eRehabilitation after stroke; a cross-sectional comparison between Brazilian and Dutch healthcare professionals. BMC Health Serv Res. 2020 Dec;20(1):488.

21. Nyman A, Zingmark M, Lilja M, Guidetti S. Information and communication technology in home-based rehabilitation – a discussion of possibilities and challenges. Scandinavian Journal of Occupational Therapy. 2023 Jan 2;30(1):14–20.

22. Brouns B, Meesters JJL, Wentink MM, de Kloet AJ, Arwert HJ, Vliet Vlieland TPM, Boyce LW, van Bodegom-Vos L. Why the uptake of eRehabilitation programs in stroke care is so difficult—a focus group study in the Netherlands. Implementation Science. 2018 Dec;13(1).

23. Saywell N, Taylor D. Focus group insights assist trial design for stroke telerehabilitation: a qualitative study. Physiotherapy Theory and Practice. 2015 Apr 3;31(3):160–5.

24. Simpson DB, Bird ML, English C, Gall SL, Breslin M, Smith S, Schmidt M, Callisaya ML. "Connecting patients and therapists remotely using technology is feasible and facilitates exercise adherence after stroke." Topics in Stroke Rehabilitation. 2020 Feb 17;27(2):93–102.

25. Silveira TM, Tamplin J, Dorsch S, Barlow A. Let's Improvise! iPad-based music therapy with functional electrical stimulation for upper limb stroke rehabilitation. Australian Journal of Music Therapy [Internet]. 2018;29. Available from: Retrieved from https:// www.austmta.org.au/journal/article/let%E2%80%99s-improvise-ipad-based-music-therapyfunctional-electrical-stimulation-upper

26. Zhou X, Du M, Zhou L. Use of mobile applications in post-stroke rehabilitation: a systematic review. Topics in Stroke Rehabilitation. 2018 Oct 3;25(7):489–99.

27. Fell N, True HH, Allen B, Harris A, Cho J, Hu Z, Sartipi M, Place KK, Salstrand R. Functional measurement post-stroke via mobile application and body-worn sensor technology. mHealth. 2019 Oct;5:47–47.

28. McKay FH, Cheng C, Wright A, Shill J, Stephens H, Uccellini M. Evaluating mobile phone applications for health behaviour change: A systematic review. J Telemed Telecare. 2018 Jan;24(1):22–30.

29. Paul L, Wyke S, Brewster S, Sattar N, Gill JMR, Alexander G, Rafferty D, McFadyen AK, Ramsay A, Dybus A. Increasing physical activity in stroke survivors using STARFISH, an interactive mobile phone application: a pilot study. Topics in Stroke Rehabilitation. 2016 Apr 2;23(3):170–7.

30. Bakas T, Clark PC, Kelly-Hayes M, King RB, Lutz BJ, Miller EL. Evidence for Stroke Family Caregiver and Dyad Interventions: A Statement for Healthcare Professionals From the American Heart Association and American Stroke Association. Stroke. 2014 Sep;45(9):2836–52.

31. Marwaa MN, Guidetti S, Ytterberg C, Kristensen HK. The use of mobile and web-based applications to support rehabilitation after stroke: a scoping review. JRM [Internet]. 2022 Feb 17 [cited 2022 Apr 4]; Available from: https://medicaljournalssweden.se/jrm/article/view/452

32. Marwaa MN, Kristensen HK, Guidetti S, Ytterberg C. Physiotherapists' and occupational therapists' perspectives on information and communication technology in stroke rehabilitation. Disability and Rehabilitationdo. 2020;

33. Bate P, Robert G. Experience-based design: from redesigning the system around the patient to co-designing services with the patient. Quality and Safety in Health Care. 2006 Oct 1;15(5):307–10.

34. Ramey L, Osborne C, Kasitinon D, Juengst S. Apps and Mobile Health Technology in Rehabilitation. Physical Medicine and Rehabilitation Clinics of North America. 2019 May;30(2):485–97.

35. Skivington K, Matthews L, Simpson SA, Craig P, Baird J, Blazeby JM, Boyd KA, Craig N, French DP, McIntosh E, Petticrew M, Rycroft-Malone J, White M, Moore L. Framework for the development and evaluation of complex interventions: gap analysis, workshop and consultation-informed update. Health Technol Assess. 2021 Sep;25(57):1–132.

36. Moore GF, Audrey S, Barker M, Bond L, Bonell C, Hardeman W, Moore L, O'Cathain A, Tinati T, Wight D, Baird J. Process evaluation of complex interventions: Medical Research Council guidance. BMJ. 2015 Mar 19;350(mar19 6):h1258–h1258.

37. O'Cathain A, Hoddinott P, Lewin S, Thomas KJ, Young B, Adamson J, Jansen YJFM, Mills N, Moore G, Donovan JL. Maximising the impact of qualitative research in feasibility studies for randomised controlled trials: guidance for researchers. Pilot Feasibility Stud. 2015 Dec;1(1):32.

38. O'Brien BC, Harris IB, Beckman TJ, Reed DA, Cook DA. Standards for Reporting Qualitative Research: A Synthesis of Recommendations. Academic Medicine. 2014 Sep;89(9):1245–51.

39. Danish Health Authority. Anbefalinger for tværsektorielle forløb for voksne med erhvervet hjerneskade - apopleksi og transitorisk cerebral iskæmi (TCI) - traume, infektion, tumor, subarachnoidalblødning og encephalopati [Internet]. 2020. Available from: https://www.sst.dk/-/media/Udgivelser/2020/Hjerneskade/Anbefalinger-forloeb-hjerneskade.ashx?la=da&hash=A7A96AC766D6AA68D26F32B96C0015BF828C93FF

40. Tistad M, Ytterberg C, Sjöstrand C, Widén Holmqvist L, von Koch L. Shorter Length of Stay in the Stroke Unit: Comparison Between the 1990s and 2000s. Topics in Stroke Rehabilitation. 2012 Mar;19(2):172–81.

41. Stroke sengeafsnit. Aarhus Universitetshospital [Internet]. 2023. Available from: https://www.auh.dk/afdelinger/neurologi/stroke-sengeafsnit/

42. Marwaa MN, Guidetti S, Ytterberg C, Kristensen HK. Using Experience-based Co-design to develop mobile/tablet applications to support a person-centred and empowering stroke rehabilitation [Internet]. In Review; 2023 Jun [cited 2023 Jul 5]. Available from: https://www.researchsquare.com/article/rs-3075410/v1

43. genoptræn.dk. Genoptræn|dk – din digitale træningsmakker [Internet]. 2019. Available from: https://www.syddansksundhedsinnovation.dk/projekter/genoptraen-dk-din-digitale-traeningsmakker/

44. Gray LJ, Ali M, Lyden PD, Bath PMW. Interconversion of the National Institutes of Health Stroke Scale and Scandinavian Stroke Scale in Acute Stroke. Journal of Stroke and Cerebrovascular Diseases. 2009 Nov;18(6):466–8.

45. Charmaz K. Constructing grounded theory. 2nd edition. London ; Thousand Oaks, Calif: Sage; 2014. 388 p. (Introducing qualitative methods).

46. Qureshi HA, Ünlü Z. Beyond the Paradigm Conflicts: A Four-Step Coding Instrument for Grounded Theory. International Journal of Qualitative Methods. 2020 Jan 1;19:160940692092818.

47. Funnell SC, Rogers PJ. Purposeful program theory: effective use of theories of change and logic models. 1st ed. San Francisco, CA: Jossey-Bass; 2011. 550 p.

48. Schultz Petersen K, Maindal HT, Ledderer L, Overgaard C. Komplekse interventioner: Udvikling, test, evaluering og implementering [in english: Complex Interventions: Development, test, evaluation and implementation]. Aalborg Universitetsforlag; 2022.

49. Sekhon M, Cartwright M, Francis JJ. Acceptability of healthcare interventions: an overview of reviews and development of a theoretical framework. BMC Health Serv Res. 2017 Dec;17(1):88.

50. Sekhon M, Cartwright M, Francis JJ. Development of a theory-informed questionnaire to assess the acceptability of healthcare interventions. BMC Health Serv Res. 2022 Dec;22(1):279.

51. Fumagalli LP, Radaelli G, Lettieri E, Bertele' P, Masella C. Patient Empowerment and its neighbours: Clarifying the boundaries and their mutual relationships. Health Policy. 2015 Mar;119(3):384–94.

52. Lumivero. NVIVO 13 [Internet]. 2018. Available from: www.lumivero.com

53. Charmaz K. Grounded Theory in Global Perspective: Reviews by International Researchers. Qualitative Inquiry. 2014 Nov;20(9):1074–84.

54. Timmermans S, Tavory I. Data analysis in qualitative research: theorizing with abductive analysis. Chicago: University of Chicago Press; 2022.

55. Brouns B, Meesters J, Wentink M, Kloet A, Arwert H, Boyce L, Vlieland T, Bodegom-Vos L. Factors associated with willingness to use eRehabilitation after stroke: A cross-sectional study among patients, informal caregivers and healthcare professionals. J Rehabil Med. 2019;51(9):665–74.

56. Epalte K, Tomsone S, Vētra A, Bērziņa G. Patient experience using digital therapy “Vigo” for stroke patient recovery: a qualitative descriptive study. Disability and Rehabilitation: Assistive Technology. 2020 Nov 6;1–10.

57. Moore GF, Evans RE. What theory, for whom and in which context? Reflections on the application of theory in the development and evaluation of complex population health interventions. SSM - Population Health. 2017 Dec;3:132–5.

58. Deutschbein J, Grittner U, Schneider A, Schenk L. Community care coordination for stroke survivors: results of a complex intervention study. BMC Health Serv Res. 2020 Dec;20(1):1143.

59. Plant SE, Tyson SF, Kirk S, Parsons J. What are the barriers and facilitators to goal-setting during rehabilitation for stroke and other acquired brain injuries? A systematic review and meta-synthesis. Clin Rehabil. 2016 Sep;30(9):921–30.

60. Warner G, Packer T, Villeneuve M, Audulv A, Versnel J. A systematic review of the effectiveness of stroke self-management programs for improving function and participation outcomes: self-management programs for stroke survivors. Disability and Rehabilitation. 2015 Nov 6;37(23):2141–63.

61. Lindblom S, Flink M, Sjöstrand C, Laska AC, von Koch L, Ytterberg C. Perceived Quality of Care Transitions between Hospital and the Home in People with Stroke. Journal of the American Medical Directors Association. 2020 Dec;21(12):1885–92.

62. Thurston C, Bezuidenhout L, Humphries S, Johansson S, von Koch L, Häger CK, Holmlund L, Sundberg CJ, Garcia-Ptacek S, Kwak L, Nilsson M, English C, Conradsson DM. Mobile health to promote physical activity in people post stroke or transient ischemic attack – study protocol for a feasibility randomised controlled trial. BMC Neurol. 2023 Mar 28;23(1):124.

63. Rehabilitation International Denmark. Hvidbog om rehabilitering. 2022.

64. Lloyd A, Bannigan K, Sugavanam T, Freeman J. Experiences of stroke survivors, their families and unpaid carers in goal setting within stroke rehabilitation: a systematic review of qualitative evidence. JBI Database of Systematic Reviews and Implementation Reports. 2018 Jun;16(6):1418–53.

65. Beshears J, Lee HN, Milkman KL, Mislavsky R, Wisdom J. Creating Exercise Habits Using Incentives: The Trade-off Between Flexibility and Routinization. Management Science. 2021 Jul;67(7):4139–71.

66. Maes S, Karoly P. Self-Regulation Assessment and Intervention in Physical Health and Illness: A Review. Applied Psychology. 2005 Apr;54(2):267–99.

67. Wilson SA, Byrne P, Rodgers SE, Maden M. A Systematic Review of Smartphone and Tablet Use by Older Adults With and Without Cognitive Impairment. Pak R, editor. Innovation in Aging. 2022 Feb 1;6(2):igac002.

68. Lindqvist E, Borell L. Computer-based assistive technology and changes in daily living after stroke. Disability and Rehabilitation: Assistive Technology. 2012 Sep;7(5):364–71.

69. Danske Fysioterapeuter. Træning skal skræddersyes [in English: Training must be individualized]. 8–2022:17–22.